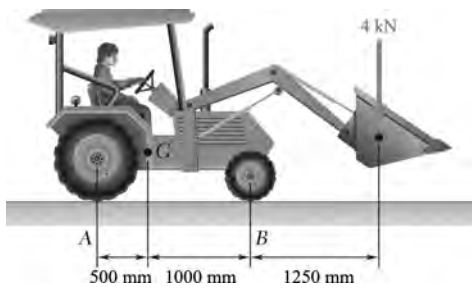


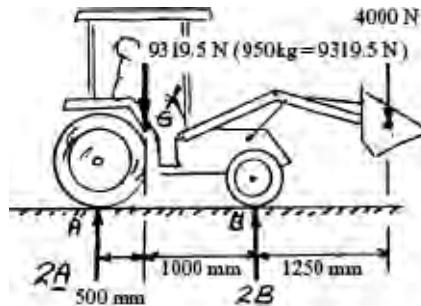
CHAPTER 4



PROBLEM 4.1

A tractor of mass 950 kg is used to lift gravel weighing 4 kN. Determine the reaction at each of the two (a) rear wheels A, (b) front wheels B.

SOLUTION



(a) Rear wheels $\quad \overset{+}{\curvearrowright} \Sigma M_B = 0: \quad + (9319.5 \text{ N})(1000 \text{ mm}) - (4000 \text{ N})(1250 \text{ mm}) - 2A(1500 \text{ mm}) = 0$
 $A = +1439.83 \text{ N} \quad \quad \quad \mathbf{A = 1440 \text{ N} \uparrow \blacktriangleleft}$

(b) Front wheels $\quad \overset{+}{\curvearrowright} \Sigma M_A: \quad - (9319.5 \text{ N})(500 \text{ mm}) - (4000 \text{ N})(2750 \text{ mm}) + 2B(1500 \text{ mm}) = 0$
 $B = +5219.92 \text{ N} \quad \quad \quad \mathbf{B = 5220 \text{ N} \uparrow \blacktriangleleft}$

Check: $\quad +\Sigma F_y = 0: \quad 2A + 2B - 9320 \text{ N} - 4000 \text{ N} = 0$
 $2(1440) + 2(5220) - 9320 - 4000 = 0$
 $0 = 0 \quad (\text{Checks})$

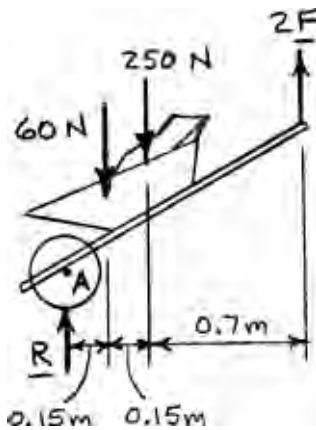
PROBLEM 4.2

A gardener uses a 60-N wheelbarrow to transport a 250-N bag of fertilizer. What force must she exert on each handle?



SOLUTION

Free-Body Diagram:



$$\sum M_A = 0: (2F)(1 \text{ m}) - (60 \text{ N})(0.15 \text{ m}) - (250 \text{ N})(0.3 \text{ m}) = 0$$

$$F = 42.0 \text{ N} \uparrow \blacktriangleleft$$



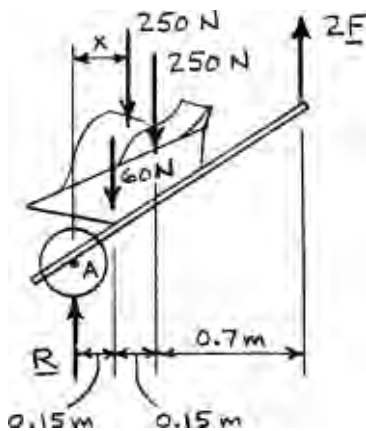
PROBLEM 4.3

The gardener of Problem 4.2 wishes to transport a second 250-N bag of fertilizer at the same time as the first one. Determine the maximum allowable horizontal distance from the axle A of the wheelbarrow to the center of gravity of the second bag if she can hold only 75 N with each arm.

PROBLEM 4.2 A gardener uses a 60-N wheelbarrow to transport a 250-N bag of fertilizer. What force must she exert on each handle?

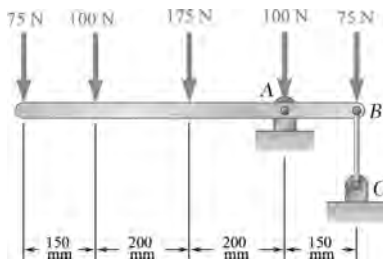
SOLUTION

Free-Body Diagram:



$$\begin{aligned} \sum M_A = 0: & \quad 2(75 \text{ N})(1 \text{ m}) - (60 \text{ N})(0.15 \text{ m}) \\ & \quad - (250 \text{ N})(0.3 \text{ m}) - (250 \text{ N})x = 0 \end{aligned}$$

$$x = 0.264 \text{ m} \quad \blacktriangleleft$$

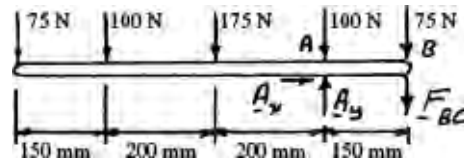


PROBLEM 4.4

For the beam and loading shown, determine (a) the reaction at A, (b) the tension in cable BC.

SOLUTION

Free-Body Diagram:



(a) Reaction at A $\Sigma F_x = 0: A_x = 0$

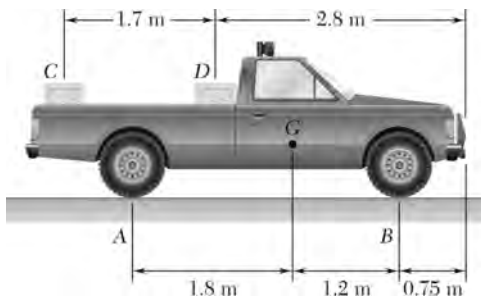
$$\begin{aligned} \curvearrowright \Sigma M_B = 0: & (75 \text{ N})(700 \text{ mm}) + (100 \text{ N})(550 \text{ mm}) + (175 \text{ N})(350 \text{ mm}) \\ & + (100 \text{ N})(150 \text{ mm}) - A_y(150 \text{ mm}) = 0 \\ & A_y = +1225 \text{ N} \quad A = 1225 \text{ N} \uparrow \blacktriangleleft \end{aligned}$$

(b) Tension in BC $\curvearrowright \Sigma M_A = 0: (75 \text{ N})(550 \text{ mm}) + (100 \text{ N})(400 \text{ mm}) + (175 \text{ N})(200 \text{ mm})$

$$\begin{aligned} & - (75 \text{ N})(150 \text{ mm}) - F_{BC}(150 \text{ mm}) = 0 \\ & F_{BC} = +700 \text{ N} \quad F_{BC} = 700 \text{ N} \blacktriangleleft \end{aligned}$$

Check: $+\uparrow \Sigma F_y = 0: -75 \text{ N} - 100 \text{ N} - 175 \text{ N} - 100 \text{ N} - 75 \text{ N} + A - F_{BC} = 0$

$$\begin{aligned} & -525 + 1225 - 700 = 0 \\ & 0 = 0 \quad (\text{Checks}) \end{aligned}$$

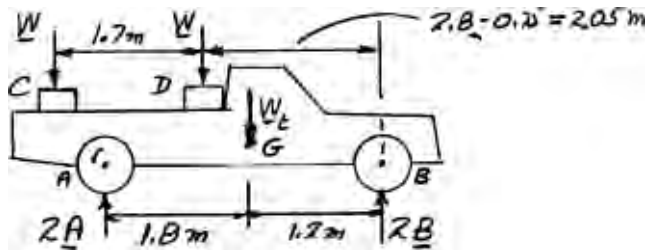


PROBLEM 4.5

Two crates, each of mass 350 kg, are placed as shown in the bed of a 1400-kg pickup truck. Determine the reactions at each of the two (a) rear wheels A, (b) front wheels B.

SOLUTION

Free-Body Diagram:



$$W = (350 \text{ kg})(9.81 \text{ m/s}^2) = 3.434 \text{ kN}$$

$$W_t = (1400 \text{ kg})(9.81 \text{ m/s}^2) = 13.734 \text{ kN}$$

(a) Rear wheels $\curvearrowright \Sigma M_B = 0: W(1.7 \text{ m} + 2.05 \text{ m}) + W(2.05 \text{ m}) + W_t(1.2 \text{ m}) - 2A(3 \text{ m}) = 0$

$$(3.434 \text{ kN})(3.75 \text{ m}) + (3.434 \text{ kN})(2.05 \text{ m}) + (13.734 \text{ kN})(1.2 \text{ m}) - 2A(3 \text{ m}) = 0$$

$$A = +6.0663 \text{ kN}$$

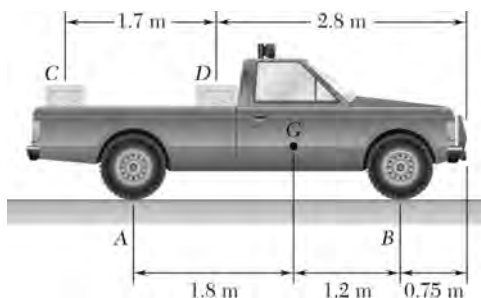
$$\mathbf{A = 6.07 \text{ kN} \uparrow \blacktriangleleft}$$

(b) Front wheels $\uparrow \Sigma F_y = 0: -W - W - W_t + 2A + 2B = 0$

$$-3.434 \text{ kN} - 3.434 \text{ kN} - 13.734 \text{ kN} + 2(6.0663 \text{ kN}) + 2B = 0$$

$$B = +4.2347 \text{ kN}$$

$$\mathbf{B = 4.23 \text{ kN} \uparrow \blacktriangleleft}$$



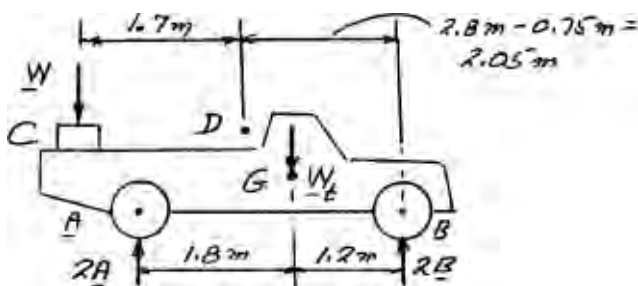
PROBLEM 4.6

Solve Problem 4.5, assuming that crate D is removed and that the position of crate C is unchanged.

PROBLEM 4.5 Two crates, each of mass 350 kg, are placed as shown in the bed of a 1400-kg pickup truck. Determine the reactions at each of the two (a) rear wheels A , (b) front wheels B .

SOLUTION

Free-Body Diagram:



$$W = (350 \text{ kg})(9.81 \text{ m/s}^2) = 3.434 \text{ kN}$$

$$W_t = (1400 \text{ kg})(9.81 \text{ m/s}^2) = 13.734 \text{ kN}$$

(a) Rear wheels $\quad +\curvearrowright \Sigma M_B = 0: \quad W(1.7 \text{ m} + 2.05 \text{ m}) + W_t(1.2 \text{ m}) - 2A(3 \text{ m}) = 0$
 $(3.434 \text{ kN})(3.75 \text{ m}) + (13.734 \text{ kN})(1.2 \text{ m}) - 2A(3 \text{ m}) = 0$

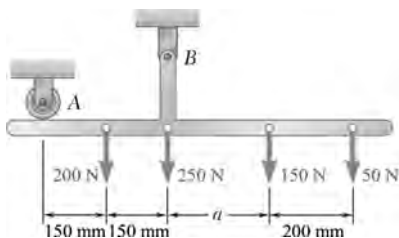
$$A = +4.893 \text{ kN}$$

$$\mathbf{A} = 4.89 \text{ kN} \uparrow \blacktriangleleft$$

(b) Front wheels $\quad +\uparrow \Sigma M_y = 0: \quad -W - W_t + 2A + 2B = 0$
 $-3.434 \text{ kN} - 13.734 \text{ kN} + 2(4.893 \text{ kN}) + 2B = 0$

$$B = +3.691 \text{ kN}$$

$$\mathbf{B} = 3.69 \text{ kN} \uparrow \blacktriangleleft$$



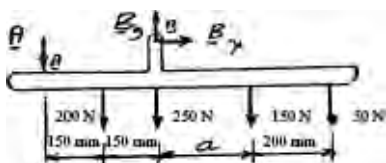
PROBLEM 4.7

A T-shaped bracket supports the four loads shown. Determine the reactions at A and B (a) if $a = 250$ mm, (b) if $a = 175$ mm. Neglect the weight of the bracket.

SOLUTION

Free-Body Diagram: $\rightarrow \Sigma F_x = 0: B_x = 0$

$$\curvearrowright \Sigma M_B = 0: (200 \text{ N})(150 \text{ mm}) - (250 \text{ N})a - (50 \text{ N})(a + 200 \text{ mm}) + (300 \text{ mm})A = 0$$



$$A = \frac{(200a - 20000)}{300} \quad (1)$$

$$\curvearrowright \Sigma M_A = 0: -(200 \text{ N})(150 \text{ mm}) - (250 \text{ N})(300 \text{ mm}) - (150 \text{ N})(a + 300 \text{ mm}) - (50 \text{ N})(a + 500 \text{ mm}) + (300 \text{ mm})B_y = 0$$

$$B_y = \frac{(175000 + 200a)}{300}$$

Since

$$B_x = 0 \quad B = \frac{(175000 + 200a)}{300} \quad (2)$$

(a) For $a = 250$ mm

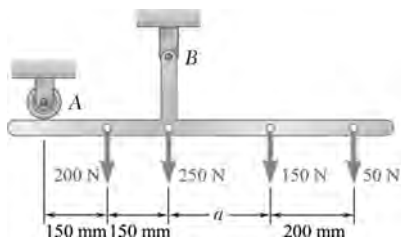
$$\text{Eq. (1):} \quad A = \frac{(200 \times 250 - 20000)}{300} = +100 \text{ N} \quad \mathbf{A = 100.0 \text{ N} \downarrow \blacktriangleleft}$$

$$\text{Eq. (2):} \quad B = \frac{(175000 + 200 \times 250)}{300} = +750 \text{ N} \quad \mathbf{B = 750 \text{ N} \downarrow \blacktriangleleft}$$

(b) For $a = 175$ mm

$$\text{Eq. (1):} \quad A = \frac{(200 \times 175 - 20000)}{300} = +50 \text{ N} \quad \mathbf{A = 50.0 \text{ N} \downarrow \blacktriangleleft}$$

$$\text{Eq. (2):} \quad B = \frac{(175000 + 200 \times 175)}{300} = +700 \text{ N} \quad \mathbf{B = 700 \text{ N} \uparrow \blacktriangleleft}$$



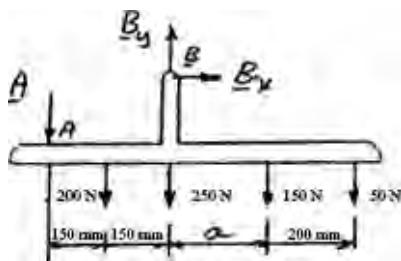
PROBLEM 4.8

For the bracket and loading of Problem 4.7, determine the smallest distance a if the bracket is not to move.

PROBLEM 4.7 A T-shaped bracket supports the four loads shown. Determine the reactions at A and B (a) if $a = 250$ mm, (b) if $a = 175$ mm. Neglect the weight of the bracket.

SOLUTION

Free-Body Diagram:

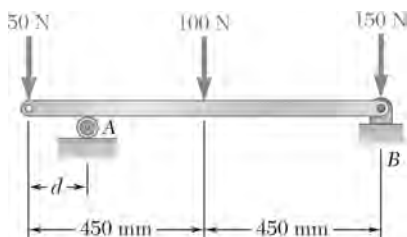


For no motion, reaction at A must be downward or zero; smallest distance a for no motion corresponds to $A = 0$.

$$\sum M_B = 0: (200 \text{ N})(150 \text{ mm}) - (150 \text{ N})a - (50 \text{ N})(a + 200 \text{ mm}) + (300 \text{ mm})A = 0$$

$$A = \frac{(200a - 20000)}{300}$$

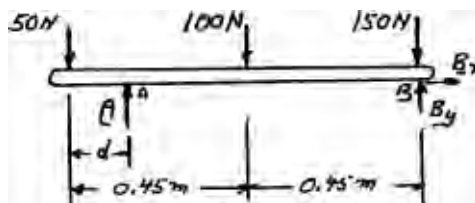
$$A = 0: (200a - 20000) = 0 \qquad a = 100.0 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.9

The maximum allowable value of each of the reactions is 180 N. Neglecting the weight of the beam, determine the range of the distance d for which the beam is safe.

SOLUTION



$$\Sigma F_x = 0: \quad B_x = 0$$

$$B = B_y$$

$$\curvearrowright \Sigma M_A = 0: \quad (50 \text{ N})d - (100 \text{ N})(0.45 \text{ m} - d) - (150 \text{ N})(0.9 \text{ m} - d) + B(0.9 \text{ m} - d) = 0$$

$$50d - 45 + 100d - 135 + 150d + 0.9B - Bd$$

$$d = \frac{180 \text{ N} \cdot \text{m} - (0.9 \text{ m})B}{300A - B} \quad (1)$$

$$\curvearrowright \Sigma M_B = 0: \quad (50 \text{ N})(0.9 \text{ m}) - A(0.9 \text{ m} - d) + (100 \text{ N})(0.45 \text{ m}) = 0$$

$$45 - 0.9A + Ad + 45 = 0$$

$$d = \frac{(0.9 \text{ m})A - 90 \text{ N} \cdot \text{m}}{A} \quad (2)$$

Since $B \leq 180 \text{ N}$, Eq. (1) yields

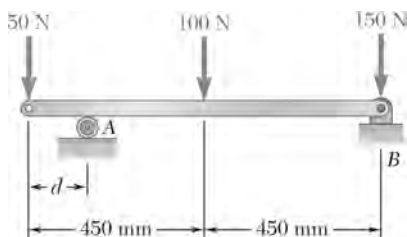
$$d \geq \frac{180 - (0.9)180}{300 - 180} = \frac{18}{120} = 0.15 \text{ m} \quad d \geq 150.0 \text{ mm} \quad \triangleleft$$

Since $A \leq 180 \text{ N}$, Eq. (2) yields

$$d \leq \frac{(0.9)180 - 90}{180} = \frac{72}{180} = 0.40 \text{ m} \quad d \leq 400 \text{ mm} \quad \triangleleft$$

Range:

$$150.0 \text{ mm} \leq d \leq 400 \text{ mm} \quad \blacktriangleleft$$

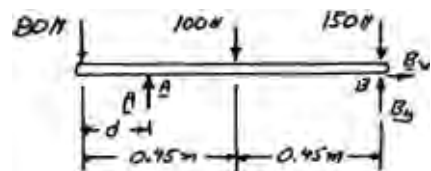


PROBLEM 4.10

Solve Problem 4.9 if the 50-N load is replaced by an 80-N load.

PROBLEM 4.9 The maximum allowable value of each of the reactions is 180 N. Neglecting the weight of the beam, determine the range of the distance d for which the beam is safe.

SOLUTION



$$\Sigma F_x = 0: \quad B_x = 0$$

$$B = B_y$$

$$\curvearrowleft \Sigma M_A = 0: \quad (80 \text{ N})d - (100 \text{ N})(0.45 \text{ m} - d) - (150 \text{ N})(0.9 \text{ m} - d) + B(0.9 \text{ m} - d) = 0$$

$$80d - 45 + 100d - 135 + 150d + 0.9B - Bd = 0$$

$$d = \frac{180 \text{ N} \cdot \text{m} - 0.9B}{330 \text{ N} - B} \quad (1)$$

$$\curvearrowleft \Sigma M_B = 0: \quad (80 \text{ N})(0.9 \text{ m}) - A(0.9 \text{ m} - d) + (100 \text{ N})(0.45 \text{ m}) = 0$$

$$d = \frac{0.9A - 117}{A} \quad (2)$$

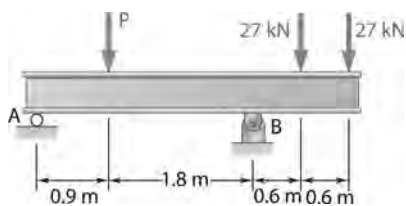
Since $B \leq 180 \text{ N}$, Eq. (1) yields

$$d \geq (180 - 0.9 \times 180) / (330 - 180) = \frac{18}{150} = 0.12 \text{ m} \quad d = 120.0 \text{ mm} \quad \triangleleft$$

Since $A \leq 180 \text{ N}$, Eq. (2) yields

$$d \leq (0.9 \times 180 - 117) / 180 = \frac{45}{180} = 0.25 \text{ m} \quad d = 250 \text{ mm} \quad \triangleleft$$

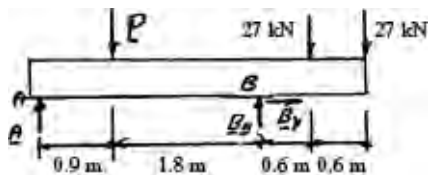
Range: $120.0 \text{ mm} \leq d \leq 250 \text{ mm} \quad \blacktriangleleft$



PROBLEM 4.11

For the beam of Sample Problem 4.2, determine the range of values of P for which the beam will be safe, knowing that the maximum allowable value of each of the reactions is 135 kN and that the reaction at A must be directed upward.

SOLUTION



$$\Sigma F_x = 0: \quad B_x = 0$$

$$\mathbf{B} = B_y \uparrow$$

$$\overset{+}{\curvearrowleft} \Sigma M_A = 0: \quad -P(0.9 \text{ m}) + B(2.7 \text{ m}) - (27 \text{ kN})(3.3 \text{ m}) - (27 \text{ kN})(3.9 \text{ m}) = 0$$

$$P = 3B - 216 \text{ kN} \quad (1)$$

$$\overset{+}{\curvearrowleft} \Sigma M_B = 0: \quad -A(2.7 \text{ m}) + P(1.8 \text{ m}) - (27 \text{ kN})(0.6 \text{ m}) - (27 \text{ kN})(1.2 \text{ m}) = 0$$

$$P = 1.5A + 27 \text{ kN} \quad (2)$$

Since $B \leq 135 \text{ kN}$, Eq. (1) yields

$$P \leq (3)(135) - 216 \text{ kN} \quad P \leq 189.0 \text{ kN} \quad \triangleleft$$

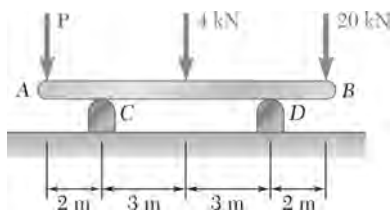
Since $0 \leq A \leq 135 \text{ kN}$, Eq. (2) yields

$$0 + 27 \text{ kN} \leq P \leq (1.5)(135) + 27$$

$$27 \text{ kN} \leq P \leq 229.5 \text{ kN} \quad \triangleleft$$

Range of values of P for which beam will be safe:

$$27.0 \text{ kN} \leq P \leq 189.0 \text{ kN} \quad \blacktriangleleft$$

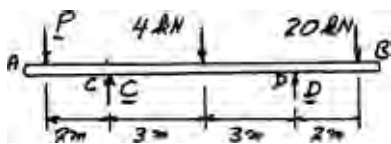


PROBLEM 4.12

The 10-m beam AB rests upon, but is not attached to, supports at C and D . Neglecting the weight of the beam, determine the range of values of P for which the beam will remain in equilibrium.

SOLUTION

Free-Body Diagram:



$$\begin{aligned} \sum M_C = 0: \quad & P(2 \text{ m}) - (4 \text{ kN})(3 \text{ m}) - (20 \text{ kN})(8 \text{ m}) + D(6 \text{ m}) = 0 \\ & P = 86 \text{ kN} - 3D \end{aligned} \quad (1)$$

$$\begin{aligned} \sum M_D = 0: \quad & P(8 \text{ m}) + (4 \text{ kN})(3 \text{ m}) - (20 \text{ kN})(2 \text{ m}) - C(6 \text{ m}) = 0 \\ & P = 3.5 \text{ kN} + 0.75C \end{aligned} \quad (2)$$

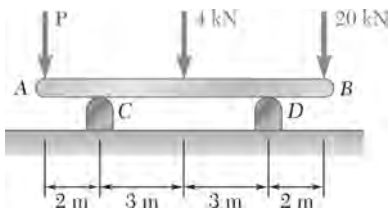
For no motion $C \geq 0$ and $D \geq 0$

For $C \geq 0$ from (2) $P \leq 3.50 \text{ kN}$

For $D \geq 0$ from (1) $P \leq 86.0 \text{ kN}$

Range of P for no motion:

$$3.50 \text{ kN} \leq P \leq 86.0 \text{ kN} \quad \blacktriangleleft$$

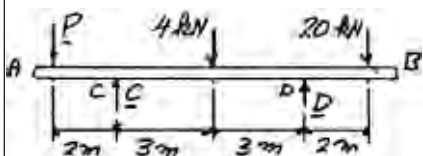


PROBLEM 4.13

The maximum allowable value of each of the reactions is 50 kN, and each reaction must be directed upward. Neglecting the weight of the beam, determine the range of values of P for which the beam is safe.

SOLUTION

Free-Body Diagram:



$$\begin{aligned} \sum M_C = 0: \quad & P(2 \text{ m}) - (4 \text{ kN})(3 \text{ m}) - (20 \text{ kN})(8 \text{ m}) + D(6 \text{ m}) = 0 \\ & P = 86 \text{ kN} - 3D \end{aligned} \quad (1)$$

$$\begin{aligned} \sum M_D = 0: \quad & P(8 \text{ m}) + (4 \text{ kN})(3 \text{ m}) - (20 \text{ kN})(2 \text{ m}) - C(6 \text{ m}) = 0 \\ & P = 3.5 \text{ kN} + 0.75C \end{aligned} \quad (2)$$

$$\text{For } C \geq 0, \text{ from (2):} \quad P \geq 3.50 \text{ kN} \quad \triangleleft$$

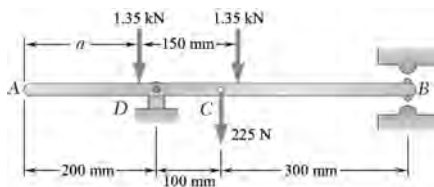
$$\text{For } D \geq 0, \text{ from (1):} \quad P \leq 86.0 \text{ kN} \quad \triangleleft$$

$$\begin{aligned} \text{For } C \leq 50 \text{ kN, from (2):} \\ & P \leq 3.5 \text{ kN} + 0.75(50 \text{ kN}) \\ & P \leq 41.0 \text{ kN} \end{aligned} \quad \triangleleft$$

$$\begin{aligned} \text{For } D \leq 50 \text{ kN, from (1):} \\ & P \geq 86 \text{ kN} - 3(50 \text{ kN}) \\ & P \geq -64.0 \text{ kN} \end{aligned} \quad \triangleleft$$

Comparing the four criteria, we find

$$3.50 \text{ kN} \leq P \leq 41.0 \text{ kN} \quad \blacktriangleleft$$

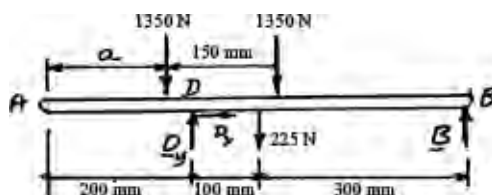


PROBLEM 4.14

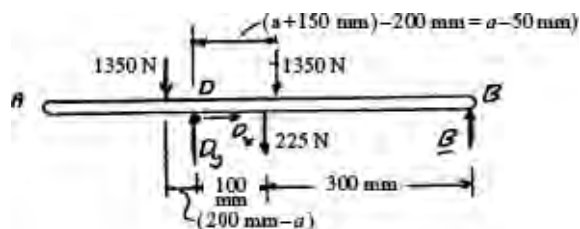
For the beam and loading shown, determine the range of the distance a for which the reaction at B does not exceed 450-N downward or 900-N upward.

SOLUTION

Assume B is positive when directed $\uparrow$



Sketch showing distance from D to forces.



$$\sum M_D = 0: (1350 \text{ N})(200 \text{ mm} - a) - (1350 \text{ N})(a - 50) - (225 \text{ N})(100 \text{ mm}) + (400)B = 0$$

$$-2700a + 315000 + 400B = 0$$

$$a = \frac{(315000 + 400B)}{2700} \quad (1)$$

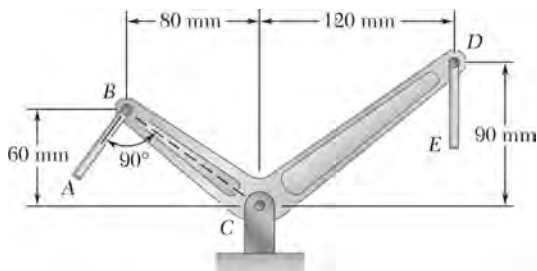
For $B = 450 \text{ N} \downarrow = -450 \text{ N}$, Eq. (1) yields:

$$a \geq \frac{[315000 + 400(-450)]}{2700} \quad a \geq 50.0 \text{ mm} \quad \triangleleft$$

For $B = 900 \text{ N} \uparrow = +900 \text{ N}$, Eq. (1) yields:

$$a \leq \frac{[315000 + 400(900)]}{2700} \quad a \leq 250 \text{ mm} \quad \triangleleft$$

Required range: $50.0 \text{ mm} \leq a \leq 250 \text{ mm}$ ◀

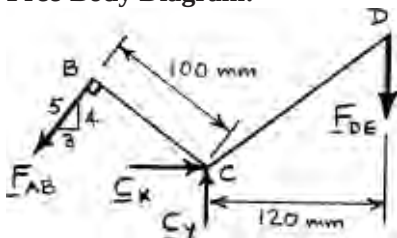


PROBLEM 4.15

Two links AB and DE are connected by a bell crank as shown. Knowing that the tension in link AB is 720 N, determine (a) the tension in link DE , (b) the reaction at C .

SOLUTION

Free-Body Diagram:



$$\sum M_C = 0: F_{AB}(100 \text{ mm}) - F_{DE}(120 \text{ mm}) = 0$$

$$F_{DE} = \frac{5}{6} F_{AB} \quad (1)$$

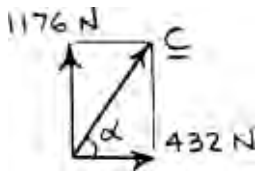
(a) For

$$F_{AB} = 720 \text{ N}$$

$$F_{DE} = \frac{5}{6}(720 \text{ N})$$

$$F_{DE} = 600 \text{ N} \quad \blacktriangleleft$$

(b)



$$\sum F_x = 0: -\frac{3}{5}(720 \text{ N}) + C_x = 0$$

$$C_x = +432 \text{ N}$$

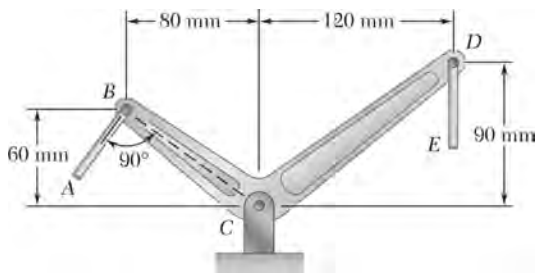
$$\sum F_y = 0: -\frac{4}{5}(720 \text{ N}) + C_y - 600 \text{ N} = 0$$

$$C_y = +1176 \text{ N}$$

$$C = 1252.84 \text{ N}$$

$$\alpha = 69.829^\circ$$

$$C = 1253 \text{ N} \quad \nearrow 69.8^\circ \quad \blacktriangleleft$$



PROBLEM 4.16

Two links AB and DE are connected by a bell crank as shown. Determine the maximum force that may be safely exerted by link AB on the bell crank if the maximum allowable value for the reaction at C is 1600 N.

SOLUTION

See solution to Problem 4.15 for F. B. D. and derivation of Eq. (1)

$$F_{DE} = \frac{5}{6} F_{AB} \quad (1)$$

$$+\rightarrow \Sigma F_x = 0: -\frac{3}{5} F_{AB} + C_x = 0 \quad C_x = \frac{3}{5} F_{AB}$$

$$+\uparrow \Sigma F_y = 0: -\frac{4}{5} F_{AB} + C_y - F_{DE} = 0$$

$$-\frac{4}{5} F_{AB} + C_y - \frac{5}{6} F_{AB} = 0$$

$$C_y = \frac{49}{30} F_{AB}$$

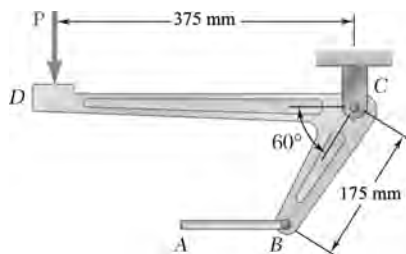
$$\begin{aligned} C &= \sqrt{C_x^2 + C_y^2} \\ &= \frac{1}{30} \sqrt{(49)^2 + (18)^2} F_{AB} \\ C &= 1.74005 F_{AB} \end{aligned}$$



For

$$C = 1600 \text{ N}, \quad 1600 \text{ N} = 1.74005 F_{AB}$$

$$F_{AB} = 920 \text{ N} \quad \blacktriangleleft$$

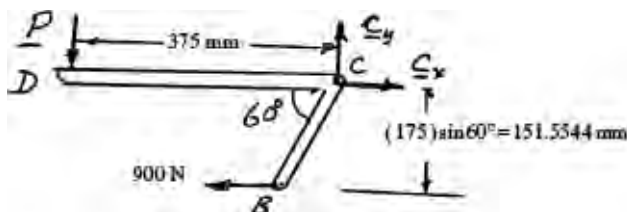


PROBLEM 4.17

The required tension in cable AB is 900 N. Determine (a) the vertical force P that must be applied to the pedal, (b) the corresponding reaction at C .

SOLUTION

Free-Body Diagram:



$$BC = 175 \text{ mm}$$

$$(a) \quad \sum M_C = 0: P(375 \text{ mm}) - (900 \text{ N})(151.5544 \text{ mm}) = 0$$

$$P = 363.731 \text{ N}$$

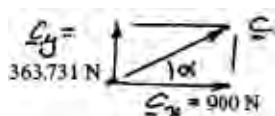
$$P = 364 \text{ N} \downarrow \blacktriangleleft$$

$$(b) \quad \sum F_x = 0: C_x - 900 \text{ N} = 0$$

$$C_x = 900 \text{ N} \rightarrow$$

$$\sum F_y = 0: C_y - P = 0 \quad C_y - 363.731 \text{ N} = 0$$

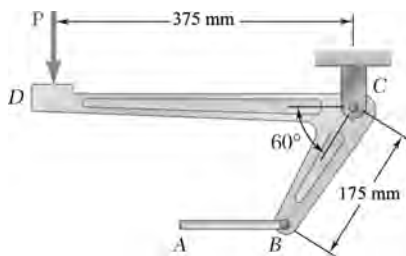
$$C_y = 364 \text{ N} \uparrow$$



$$\alpha = 22.0^\circ$$

$$C = 970.722 \text{ N}$$

$$C = 971 \text{ N} \nearrow 22.0^\circ \blacktriangleleft$$

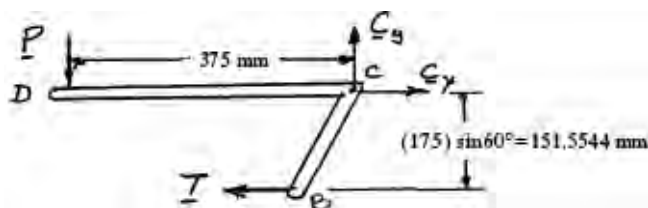


PROBLEM 4.18

Determine the maximum tension that can be developed in cable AB if the maximum allowable value of the reaction at C is 1125 N.

SOLUTION

Free-Body Diagram:



$$BC = 175 \text{ mm}$$

$$\curvearrowleft \Sigma M_C = 0: P(375 \text{ mm}) - T(151.5544 \text{ mm}) = 0 \quad P = 0.40415T$$

$$+\uparrow \Sigma F_y = 0: -P + C_y = 0 \quad -0.40415T + C_y = 0$$

$$C_y = 0.40415T$$

$$+\rightarrow \Sigma F_x = 0: -T + C_x = 0$$

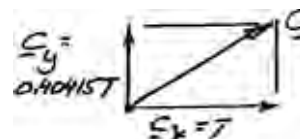
$$C_x = T$$

$$C = \sqrt{C_x^2 + C_y^2} = \sqrt{T^2 + (0.40415T)^2}$$

$$C = 1.0786T$$

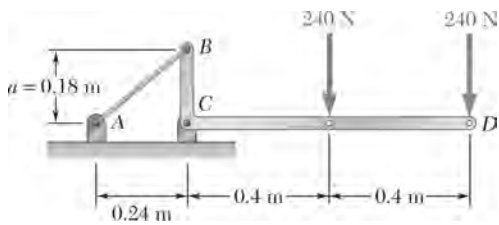
$$C = 1125 \text{ N}$$

$$1125 \text{ N} = 1.0786T \Rightarrow T = 1043.02 \text{ N}$$



$$T = 1043 \text{ N} \quad \blacktriangleleft$$

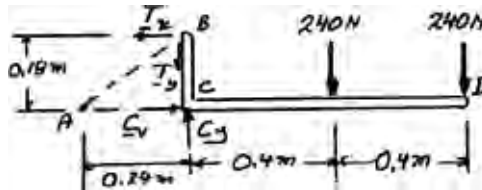
For



PROBLEM 4.19

The bracket BCD is hinged at C and attached to a control cable at B . For the loading shown, determine (a) the tension in the cable, (b) the reaction at C .

SOLUTION



At B:

$$\frac{T_y}{T_x} = \frac{0.18 \text{ m}}{0.24 \text{ m}}$$

$$T_y = \frac{3}{4} T_x \quad (1)$$

(a) $\sum M_C = 0: T_x(0.18 \text{ m}) - (240 \text{ N})(0.4 \text{ m}) - (240 \text{ N})(0.8 \text{ m}) = 0$

$$T_x = +1600 \text{ N}$$

Eq. (1) $T_y = \frac{3}{4}(1600 \text{ N}) = 1200 \text{ N}$

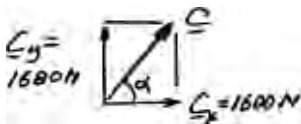
$$T = \sqrt{T_x^2 + T_y^2} = \sqrt{1600^2 + 1200^2} = 2000 \text{ N} \quad T = 2.00 \text{ kN} \quad \nwarrow$$

(b) $\sum F_x = 0: C_x - T_x = 0$

$$C_x - 1600 \text{ N} = 0 \quad C_x = +1600 \text{ N} \quad C_x = 1600 \text{ N} \rightarrow$$

$$\sum F_y = 0: C_y - T_y - 240 \text{ N} - 240 \text{ N} = 0$$

$$C_y - 1200 \text{ N} - 480 \text{ N} = 0$$

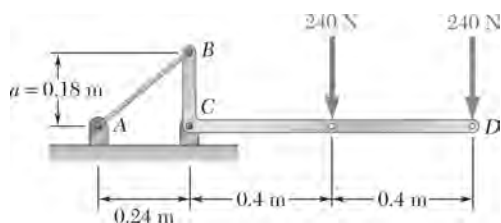


$$C_y = +1680 \text{ N} \quad C_y = 1680 \text{ N} \uparrow$$

$$\alpha = 46.4^\circ$$

$$C = 2320 \text{ N}$$

$$C = 2.32 \text{ kN} \nearrow 46.4^\circ \quad \nwarrow$$

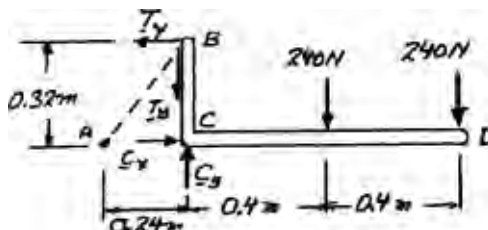


PROBLEM 4.20

Solve Problem 4.19, assuming that $a = 0.32$ m.

PROBLEM 4.19 The bracket BCD is hinged at C and attached to a control cable at B . For the loading shown, determine (a) the tension in the cable, (b) the reaction at C .

SOLUTION



At B:

$$\frac{T_y}{T_x} = \frac{0.32 \text{ m}}{0.24 \text{ m}}$$

$$T_y = \frac{4}{3} T_x$$

$$\sum M_C = 0: T_x(0.32 \text{ m}) - (240 \text{ N})(0.4 \text{ m}) - (240 \text{ N})(0.8 \text{ m}) = 0$$

$$T_x = 900 \text{ N}$$

Eq. (1)

$$T_y = \frac{4}{3}(900 \text{ N}) = 1200 \text{ N}$$

$$T = \sqrt{T_x^2 + T_y^2} = \sqrt{900^2 + 1200^2} = 1500 \text{ N}$$

$$T = 1.500 \text{ kN} \quad \nwarrow$$

$$\sum F_x = 0: C_x - T_x = 0$$

$$C_x - 900 \text{ N} = 0 \quad C_x = +900 \text{ N}$$

$$C_x = 900 \text{ N} \rightarrow$$

$$\sum F_y = 0: C_y - T_y - 240 \text{ N} - 240 \text{ N} = 0$$

$$C_y - 1200 \text{ N} - 480 \text{ N} = 0$$

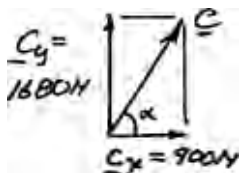
$$C_y = +1680 \text{ N}$$

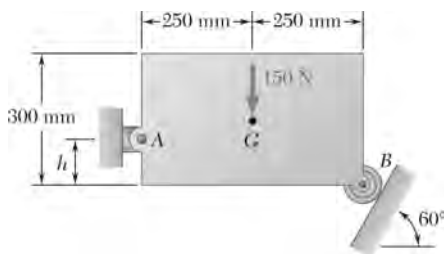
$$C_y = 1680 \text{ N} \uparrow$$

$$\alpha = 61.8^\circ$$

$$C = 1906 \text{ N}$$

$$C = 1.906 \text{ kN} \nearrow 61.8^\circ \quad \nwarrow$$



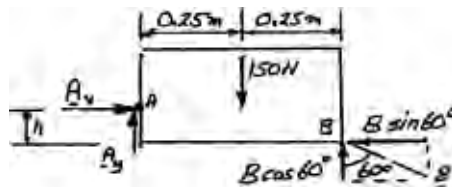


PROBLEM 4.21

Determine the reactions at A and B when (a) $h = 0$, (b) $h = 200$ mm.

SOLUTION

Free-Body Diagram:



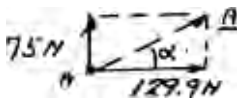
$$\begin{aligned} \sum M_A = 0: & (B \cos 60^\circ)(0.5 \text{ m}) - (B \sin 60^\circ)h - (150 \text{ N})(0.25 \text{ m}) = 0 \\ B = & \frac{37.5}{0.25 - 0.866h} \end{aligned} \quad (1)$$

(a) When $h = 0$:

$$\text{Eq. (1):} \quad B = \frac{37.5}{0.25} = 150 \text{ N} \quad \mathbf{B = 150.0 N \nearrow 30.0^\circ \blacktriangleleft}$$

$$\begin{aligned} \sum F_x = 0: & A_x - B \sin 60^\circ = 0 \\ A_x = & (150) \sin 60^\circ = 129.9 \text{ N} \quad \mathbf{A_x = 129.9 N \rightarrow} \end{aligned}$$

$$\begin{aligned} \sum F_y = 0: & A_y - 150 + B \cos 60^\circ = 0 \\ A_y = & 150 - (150) \cos 60^\circ = 75 \text{ N} \quad \mathbf{A_y = 75 N \uparrow} \end{aligned}$$



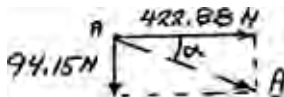
$$\begin{aligned} \alpha = & 30^\circ \\ \mathbf{A = 150.0 N \nearrow 30.0^\circ \blacktriangleleft} \end{aligned}$$

(b) When $h = 200 \text{ mm} = 0.2 \text{ m}$

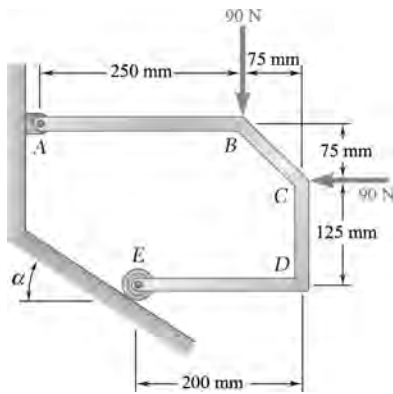
$$\text{Eq. (1):} \quad B = \frac{37.5}{0.25 - 0.866(0.2)} = 488.3 \text{ N} \quad \mathbf{B = 488 N \nearrow 30.0^\circ \blacktriangleleft}$$

$$\begin{aligned} \sum F_x = 0: & A_x - B \sin 60^\circ = 0 \\ A_x = & (488.3) \sin 60^\circ = 422.88 \text{ N} \quad \mathbf{A_x = 422.88 N \rightarrow} \end{aligned}$$

$$\begin{aligned} \sum F_y = 0: & A_y - 150 + B \cos 60^\circ = 0 \\ A_y = & 150 - (488.3) \cos 60^\circ = -94.15 \text{ N} \quad \mathbf{A_y = 94.15 N \downarrow} \end{aligned}$$



$$\begin{aligned} \alpha = & 12.55^\circ \\ \mathbf{A = 433 N \searrow 12.6^\circ \blacktriangleleft} \end{aligned}$$

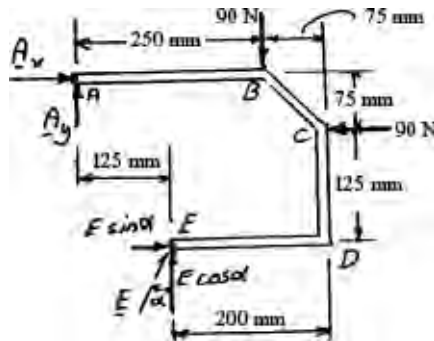


PROBLEM 4.22

For the frame and loading shown, determine the reactions at A and E when (a) $\alpha = 30^\circ$, (b) $\alpha = 45^\circ$.

SOLUTION

Free-Body Diagram:

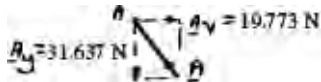


$$\begin{aligned} \sum M_A = 0: & (E \sin \alpha)(200 \text{ mm}) + (E \cos \alpha)(125 \text{ mm}) \\ & - (90 \text{ N})(250 \text{ mm}) - (90 \text{ N})(75 \text{ mm}) = 0 \end{aligned}$$

$$E = \frac{29250}{200 \sin \alpha + 125 \cos \alpha}$$

(a) When $\alpha = 30^\circ$: $E = \frac{29250}{200 \sin 30^\circ + 125 \cos 30^\circ} = 140.454 \text{ N}$ $E = 140.5 \text{ N} \nearrow 60.0^\circ \blacktriangleleft$

$$\sum F_x = 0: A_x - 90 \text{ N} + (140.454 \text{ N}) \sin 30^\circ = 0$$



$$A_x = +19.773 \text{ N}$$

$$A_x = 19.77 \text{ N} \rightarrow$$

$$\sum F_y = 0: A_y - 90 \text{ N} + (140.454 \text{ N}) \cos 30^\circ = 0$$

$$A_y = -31.637 \text{ N}$$

$$A_y = 31.7 \text{ N} \downarrow$$

$$A = 37.3 \text{ N} \searrow 58.0^\circ \blacktriangleleft$$

PROBLEM 4.22 (Continued)

(b) When $\alpha = 45^\circ$:

$$E = \frac{29250}{200 \sin 45^\circ + 125 \cos \alpha} = 127.279 \text{ N} \quad \mathbf{E} = 127.3 \text{ N} \nearrow 45.0^\circ \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: \quad A_x - 90 \text{ N} + (127.279 \text{ N}) \sin 45^\circ = 0$$

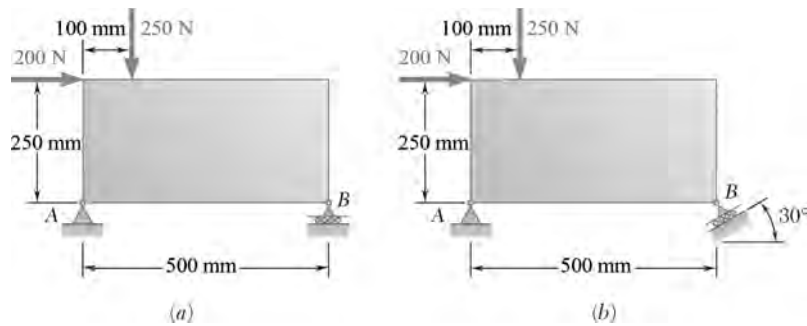
$$A_x = 0 \quad \mathbf{A}_x = 0$$

$$+\uparrow \Sigma F_y = 0: \quad A_y - 90 \text{ N} + (127.279) \cos 45^\circ = 0$$

$$A_y = 0 \quad A_y = 0 \quad \mathbf{A} = 0 \blacktriangleleft$$

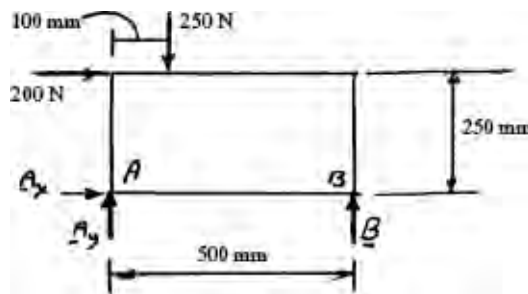
PROBLEM 4.23

For each of the plates and loadings shown, determine the reactions at A and B.



SOLUTION

(a) Free-Body Diagram:



$$\curvearrowright \Sigma M_A = 0: B(500 \text{ mm}) - (250 \text{ N})(100 \text{ mm}) - (200 \text{ N})(250 \text{ mm}) = 0$$

$$B = +150 \text{ N}$$

$$\mathbf{B} = 150.0 \text{ N} \uparrow \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: A_x + 200 \text{ N} = 0$$

$$A_x = -200 \text{ N}$$

$$\mathbf{A}_x = 200 \text{ N} \leftarrow$$

$$\uparrow \Sigma F_y = 0: A_y + B - 250 \text{ N} = 0$$

$$A_y + 150 \text{ N} - 250 \text{ N} = 0$$

$$A_y = +100 \text{ N}$$

$$\mathbf{A}_y = 100.0 \text{ N} \uparrow$$

$$\alpha = 26.56^\circ$$

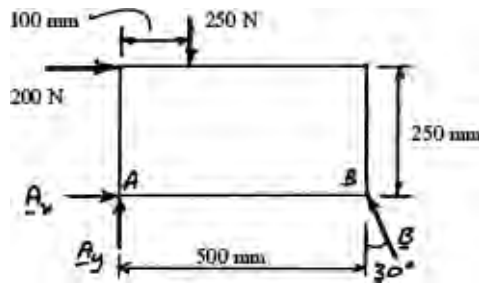
$$A = 223.607 \text{ N}$$

$$\mathbf{A} = 224 \text{ N} \nearrow 26.6^\circ \blacktriangleleft$$



PROBLEM 4.23 (Continued)

(b) Free-Body Diagram:



$$\curvearrowright \Sigma M_A = 0: (B \cos 30^\circ)(500 \text{ mm}) - (200 \text{ N})(250 \text{ mm}) - (250 \text{ N})(100 \text{ mm}) = 0$$

$$B = 173.205 \text{ N}$$

$$\mathbf{B} = 173.2 \text{ N} \nearrow 60.0^\circ \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: A_x - B \sin 30^\circ + 200 \text{ N} = 0$$

$$A_x - (173.205 \text{ N}) \sin 30^\circ + 200 \text{ N} = 0$$

$$A_x = -113.3975 \text{ N}$$

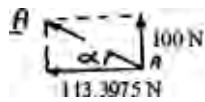
$$\mathbf{A}_x = 113.4 \text{ N} \leftarrow$$

$$\uparrow \Sigma F_y = 0: A_y + B \cos 30^\circ - 250 \text{ N} = 0$$

$$A_y + (173.205) \cos 30^\circ - 250 \text{ N} = 0$$

$$A_y = +100 \text{ N}$$

$$\mathbf{A}_y = 100.0 \text{ N} \uparrow$$



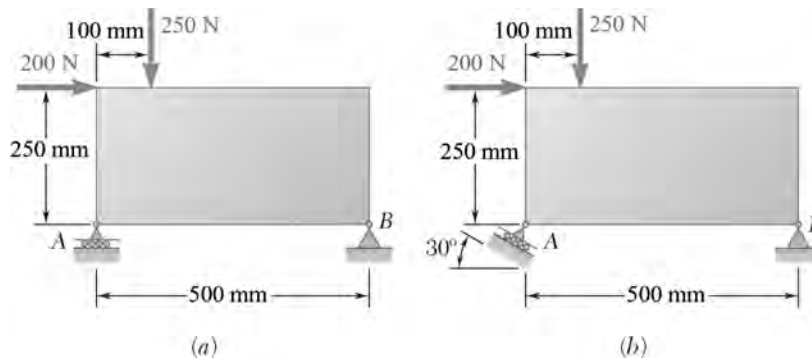
$$\alpha = 41.4^\circ$$

$$A = 151.192 \text{ N}$$

$$\mathbf{A} = 151.2 \text{ N} \nearrow 41.4^\circ \blacktriangleleft$$

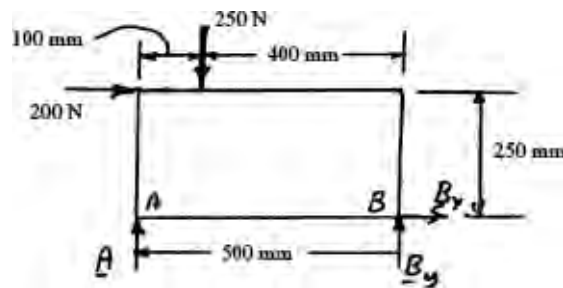
PROBLEM 4.24

For each of the plates and loadings shown, determine the reactions at A and B .



SOLUTION

(a) Free-Body Diagram:



$$\curvearrowleft \Sigma M_B = 0: -A(500 \text{ mm}) + (250 \text{ N})(400 \text{ mm}) - (200 \text{ N})(250 \text{ mm}) = 0$$

$$A = +100 \text{ N}$$

$$\mathbf{A} = 100.0 \text{ N } \uparrow \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: 200 \text{ N} + B_x = 0$$

$$B_x = -200 \text{ N}$$

$$\mathbf{B}_x = 200 \text{ N } \leftarrow$$

$$\uparrow \Sigma F_y = 0: A + B_y - 250 \text{ N} = 0$$

$$100 \text{ N} + B_y - 250 \text{ N} = 0$$

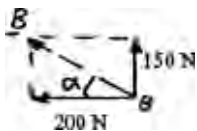
$$B_y = +150 \text{ N}$$

$$\mathbf{B}_y = 150.0 \text{ N } \uparrow$$

$$\alpha = 36.87^\circ$$

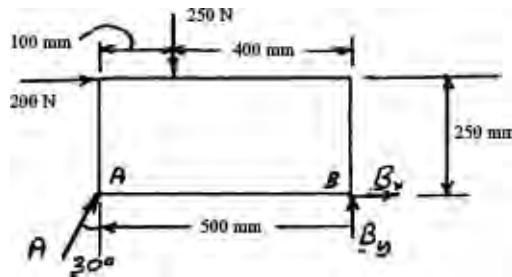
$$B = 250 \text{ N}$$

$$\mathbf{B} = 250 \text{ N } \nearrow 36.9^\circ \blacktriangleleft$$



PROBLEM 4.24 (Continued)

(b)



$$\curvearrowleft \Sigma M_A = 0: -(A \cos 30^\circ)(500 \text{ mm}) - (200 \text{ N})(250 \text{ mm}) + (250 \text{ N})(400 \text{ mm}) = 0$$

$$A = 115.47 \text{ N}$$

$$\mathbf{A} = 115.5 \text{ N} \nearrow 60.0^\circ \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: A \sin 30^\circ + 200 \text{ N} + B_x = 0$$

$$(115.47 \text{ N}) \sin 30^\circ + 200 \text{ N} + B_x = 0$$

$$B_x = -257.735 \text{ N}$$

$$\mathbf{B}_x = 258 \text{ N} \leftarrow$$

$$\uparrow \Sigma F_y = 0: A \cos 30^\circ + B_y - 250 \text{ N} = 0$$

$$(115.47 \text{ N}) \cos 30^\circ + B_y - 250 \text{ N} = 0$$

$$B_y = +150 \text{ N}$$

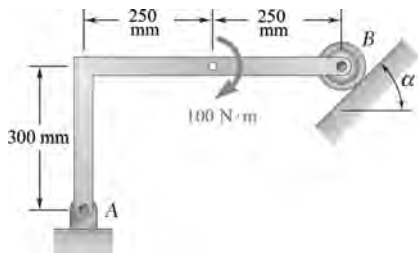
$$\mathbf{B}_y = 150.0 \text{ N} \uparrow$$

$$\alpha = 30.2^\circ$$

$$B = 298.207 \text{ N}$$

$$\mathbf{B} = 298 \text{ N} \nearrow 30.2^\circ \blacktriangleleft$$



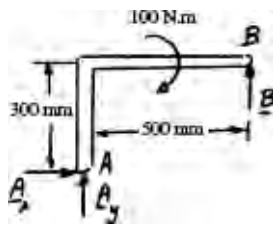


PROBLEM 4.25

Determine the reactions at A and B when (a) $\alpha = 0^\circ$, (b) $\alpha = 90^\circ$, (c) $\alpha = 30^\circ$.

SOLUTION

(a) $\alpha = 0^\circ$



$$\curvearrowright \Sigma M_A = 0: B(0.5 \text{ m}) - 100 \text{ N} \cdot \text{m} = 0$$

$$B = 200 \text{ N}$$

$$\rightarrow \Sigma F_x = 0: A_x = 0$$

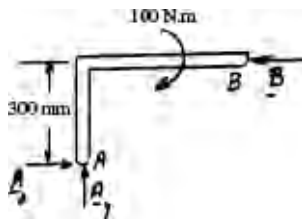
$$\uparrow \Sigma F_y = 0: A_y + 200 = 0$$

$$A_y = -200 \text{ N}$$

$$\mathbf{A} = 200 \text{ N} \downarrow$$

$$\mathbf{B} = 200 \text{ N} \uparrow \blacktriangleleft$$

(b) $\alpha = 90^\circ$



$$\curvearrowright \Sigma M_A = 0: B(0.3 \text{ m}) - 100 \text{ N} \cdot \text{m} = 0$$

$$B = \frac{1000}{3} \text{ N} = 333.33 \text{ N}$$

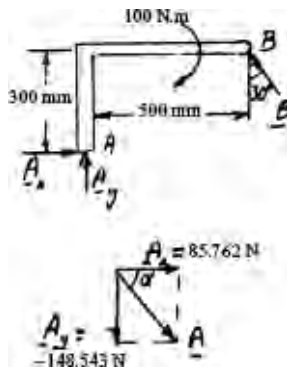
$$\rightarrow \Sigma F_x = 0: A_x - 333.33 \text{ N} = 0, \quad A_x = 333.33 \text{ N}$$

$$\uparrow \Sigma F_y = 0: A_y = 0$$

$$\mathbf{A} = 333 \text{ N} \rightarrow$$

$$\mathbf{B} = 333 \text{ N} \leftarrow \blacktriangleleft$$

(c) $\alpha = 30^\circ$



$$\curvearrowright \Sigma M_A = 0: (B \cos 30^\circ)(0.5 \text{ m}) + (B \sin 30^\circ)(0.3 \text{ m}) - 100 \text{ N} \cdot \text{m} = 0$$

$$B = 171.523 \text{ N}$$

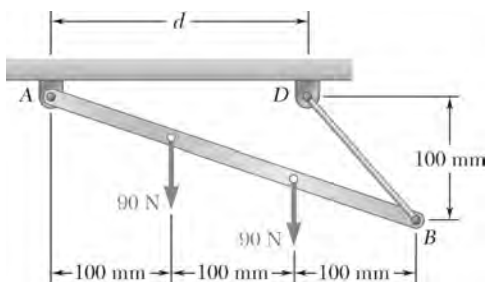
$$\rightarrow \Sigma F_x = 0: A_x - (171.523 \text{ N}) \sin 30^\circ = 0$$

$$A_x = 85.762 \text{ N}$$

$$\uparrow \Sigma F_y = 0: A_y + (171.523 \text{ N}) \cos 30^\circ = 0 \quad A_y = -148.543 \text{ N}$$

$$A = 171.523 \text{ N} \quad \alpha = 60.0^\circ$$

$$\mathbf{A} = 171.5 \text{ N} \searrow 60.0^\circ \quad \mathbf{B} = 171.5 \text{ N} \nearrow 60.0^\circ \blacktriangleleft$$

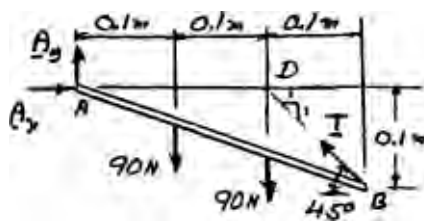


PROBLEM 4.26

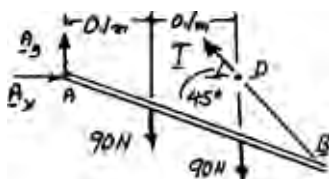
A rod AB hinged at A and attached at B to cable BD supports the loads shown. Knowing that $d = 200$ mm, determine (a) the tension in cable BD , (b) the reaction at A .

SOLUTION

Free-Body Diagram:



(a) Move T along BD until it acts at Point D .



$$\sum M_A = 0: (T \sin 45^\circ)(0.2 \text{ m}) + (90 \text{ N})(0.1 \text{ m}) + (90 \text{ N})(0.2 \text{ m}) = 0$$

$$T = 190.92 \text{ N}$$

$$T = 190.9 \text{ N} \quad \blacktriangleleft$$

(b)

$$\sum F_x = 0: A_x - (190.92 \text{ N}) \cos 45^\circ = 0$$

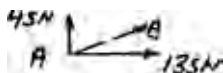
$$A_x = +135 \text{ N}$$

$$A_x = 135.0 \text{ N} \rightarrow$$

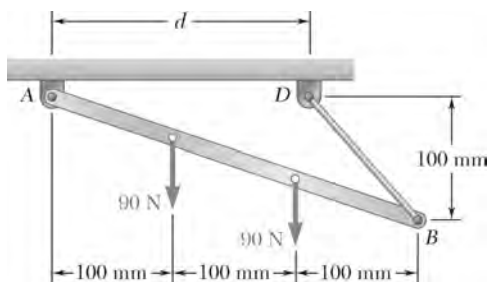
$$\sum F_y = 0: A_y - 90 \text{ N} - 90 \text{ N} + (190.92 \text{ N}) \sin 45^\circ = 0$$

$$A_y = +45 \text{ N}$$

$$A_y = 45.0 \text{ N} \uparrow$$



$$A = 142.3 \text{ N} \quad \nearrow 18.43^\circ \quad \blacktriangleleft$$

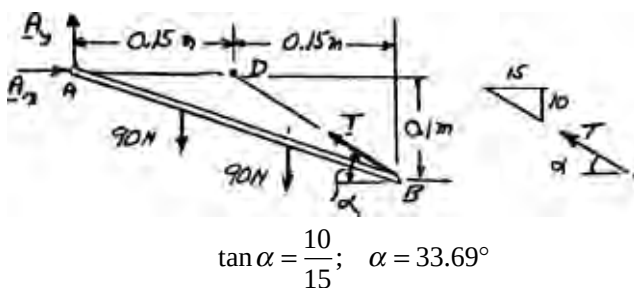


PROBLEM 4.27

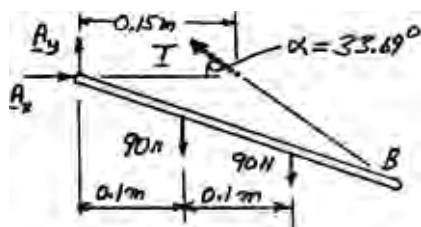
A rod AB hinged at A and attached at B to cable BD supports the loads shown. Knowing that $d = 150$ mm, determine (a) the tension in cable BD , (b) the reaction at A .

SOLUTION

Free-Body Diagram:



(a) Move T along BD until it acts at Point D .



$$\sum M_A = 0: (T \sin 33.69^\circ)(0.15 \text{ m}) - (90 \text{ N})(0.1 \text{ m}) - (90 \text{ N})(0.2 \text{ m}) = 0$$

$$T = 324.5 \text{ N}$$

$$T = 324 \text{ N} \quad \blacktriangleleft$$

(b) $\sum F_x = 0: A_x - (324.99 \text{ N}) \cos 33.69^\circ = 0$

$$A_x = +270 \text{ N}$$

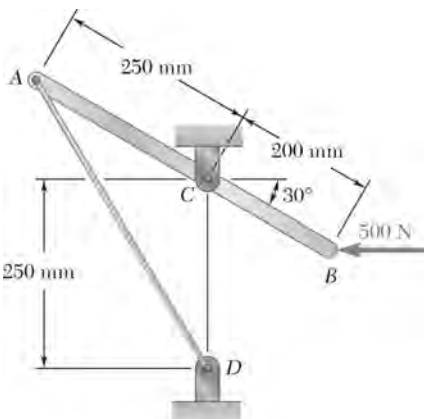
$$A_x = 270 \text{ N} \quad \rightarrow$$

$$\sum F_y = 0: A_y - 90 \text{ N} - 90 \text{ N} + (324.5 \text{ N}) \sin 33.69^\circ = 0$$

$$A_y = 0$$

$$A_y = 0$$

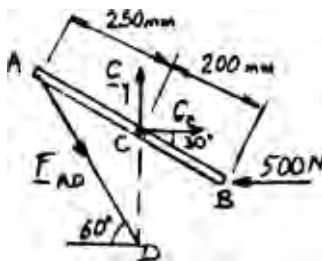
$$A = 270 \text{ N} \quad \rightarrow \blacktriangleleft$$



PROBLEM 4.28

A lever AB is hinged at C and attached to a control cable at A . If the lever is subjected to a 500-N horizontal force at B , determine (a) the tension in the cable, (b) the reaction at C .

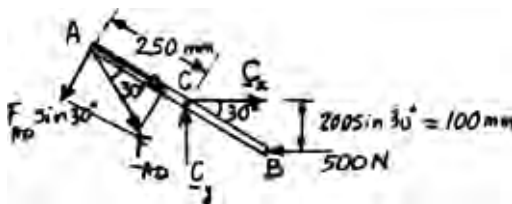
SOLUTION



Triangle ACD is isosceles with $\angle C = 90^\circ + 30^\circ = 120^\circ$ $\angle A = \angle D = \frac{1}{2}(180^\circ - 120^\circ) = 30^\circ$

Thus DA forms angle of 60° with horizontal.

(a) We resolve $\mathbf{F}_{AD}$ into components along AB and perpendicular to AB .

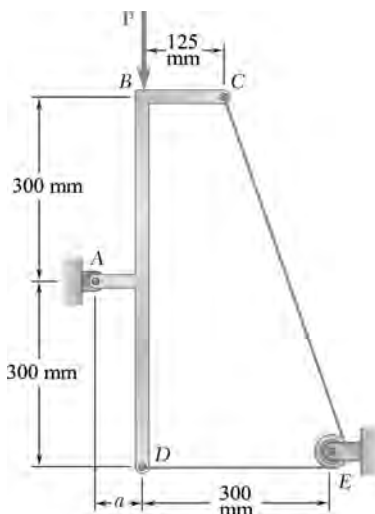


$$\sum M_C = 0: (F_{AD} \sin 30^\circ)(250 \text{ mm}) - (500 \text{ N})(100 \text{ mm}) = 0 \quad F_{AD} = 400 \text{ N} \quad \blacktriangleleft$$

$$\sum F_x = 0: -(400 \text{ N}) \cos 60^\circ + C_x - 500 \text{ N} = 0 \quad C_x = +300 \text{ N}$$

$$\sum F_y = 0: -(400 \text{ N}) \sin 60^\circ + C_y = 0 \quad C_y = +346.4 \text{ N}$$

$$\mathbf{C} = 458 \text{ N} \quad \nearrow 49.1^\circ \quad \blacktriangleleft$$

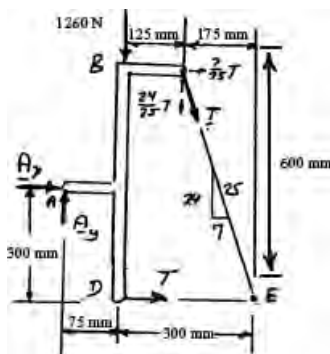


PROBLEM 4.29

A force $\mathbf{P}$ of magnitude 1260 N is applied to member $ABCD$, which is supported by a frictionless pin at A and by the cable CED . Since the cable passes over a small pulley at E , the tension may be assumed to be the same in portions CE and ED of the cable. For the case when $a = 75$ mm, determine (a) the tension in the cable, (b) the reaction at A .

SOLUTION

Free-Body Diagram:



$$(a) \quad \sum M_A = 0: \quad -(1260 \text{ N})(75 \text{ mm})$$

$$T(300 \text{ mm}) - \frac{7}{25}T(300 \text{ mm})$$

$$- \frac{24}{25}T(200 \text{ mm}) = 0$$

$$24T = 94500$$

$$T = 3937.5 \text{ N}$$

$$T = 3940 \text{ N} \quad \swarrow$$

$$(b) \quad \sum F_x = 0: \quad \frac{7}{25}(3937.5) + 3937.5 + A_x = 0$$

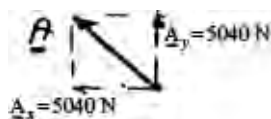
$$A_x = -5040 \text{ N} \quad \mathbf{A}_x = 5040 \text{ N} \quad \leftarrow$$

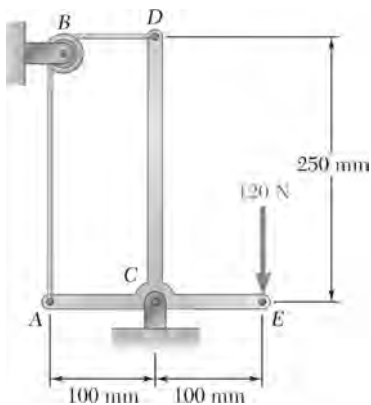
$$\sum F_y = 0: \quad A_y - 1260 \text{ N} - \frac{24}{25}(3937.5 \text{ N}) = 0$$

$$A_y = +5040 \text{ N}$$

$$\mathbf{A}_y = 5040 \text{ N} \quad \uparrow$$

$$\mathbf{A} = 7128 \text{ N} \quad \searrow 45.0^\circ \quad \swarrow$$



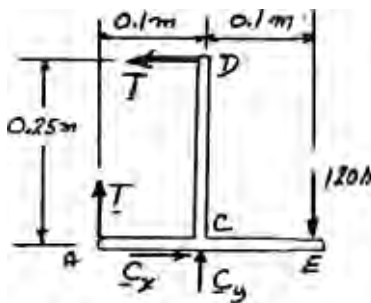


PROBLEM 4.30

Neglecting friction, determine the tension in cable ABD and the reaction at support C .

SOLUTION

Free-Body Diagram:



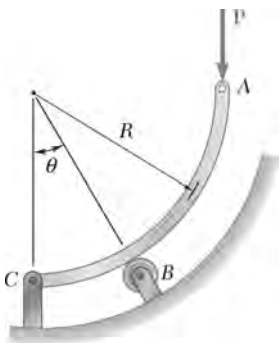
$$+\circlearrowleft \Sigma M_C = 0: T(0.25 \text{ m}) - T(0.1 \text{ m}) - (120 \text{ N})(0.1 \text{ m}) = 0 \quad T = 80.0 \text{ N} \quad \blacktriangleleft$$

$$+\rightarrow \Sigma F_x = 0: C_x - 80 \text{ N} = 0 \quad C_x = +80 \text{ N} \quad C_x = 80.0 \text{ N} \rightarrow$$

$$+\uparrow \Sigma F_y = 0: C_y - 120 \text{ N} + 80 \text{ N} = 0 \quad C_y = +40 \text{ N} \quad C_y = 40.0 \text{ N} \uparrow$$



$$C = 89.4 \text{ N} \quad \angle 26.6^\circ \quad \blacktriangleleft$$



PROBLEM 4.31

Rod ABC is bent in the shape of an arc of circle of radius R . Knowing the $\theta = 30^\circ$, determine the reaction (a) at B, (b) at C.

SOLUTION

Free-Body Diagram:

$$\curvearrowleft \Sigma M_D = 0: C_x(R) - P(R) = 0$$

$$C_x = +P$$

$$\rightarrow \Sigma F_x = 0: C_x - B \sin \theta = 0$$

$$P - B \sin \theta = 0$$

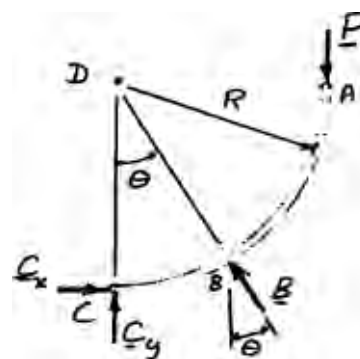
$$B = P / \sin \theta$$

$$\mathbf{B} = \frac{P}{\sin \theta} \nearrow \theta$$

$$\uparrow \Sigma F_y = 0: C_y + B \cos \theta - P = 0$$

$$C_y + (P / \sin \theta) \cos \theta - P = 0$$

$$C_y = P \left(1 - \frac{1}{\tan \theta} \right)$$



For $\theta = 30^\circ$:

(a) $B = P / \sin 30^\circ = 2P$

$\mathbf{B} = 2P \nearrow 60.0^\circ \blacktriangleleft$

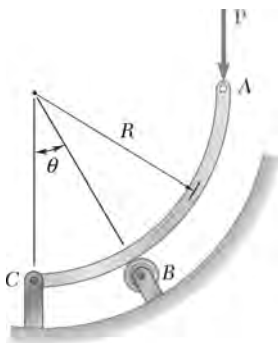
(b) $C_x = +P \quad C_x = P \rightarrow$

$C_y = P(1 - 1/\tan 30^\circ) = -0.732/P$

$C_y = 0.7321P \downarrow$

$\mathbf{C} = 1.239P \searrow 36.2^\circ \blacktriangleleft$





PROBLEM 4.32

Rod ABC is bent in the shape of an arc of circle of radius R . Knowing the $\theta = 60^\circ$, determine the reaction (a) at B , (b) at C .

SOLUTION

See the solution to Problem 4.31 for the free-body diagram and analysis leading to the following expressions:

$$C_x = +P$$

$$C_y = P \left(1 - \frac{1}{\tan \theta} \right)$$

$$B = \frac{P}{\sin \theta}$$

For $\theta = 60^\circ$:

(a)

$$B = P / \sin 60^\circ = 1.1547P$$

$$\mathbf{B} = 1.155P \nearrow 30.0^\circ \blacktriangleleft$$

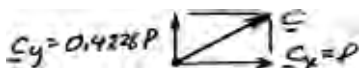
(b)

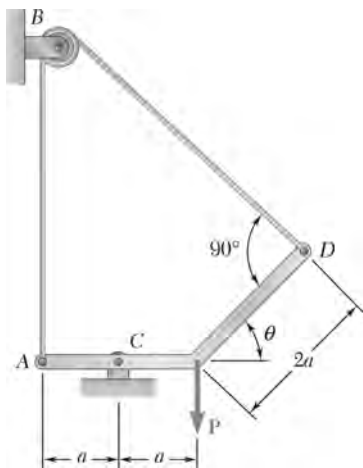
$$C_x = +P \quad C_x = P \rightarrow$$

$$C_y = P(1 - 1/\tan 60^\circ) = +0.4226P$$

$$\mathbf{C}_y = 0.4226P \downarrow$$

$$\mathbf{C} = 1.086P \nearrow 22.9^\circ \blacktriangleleft$$

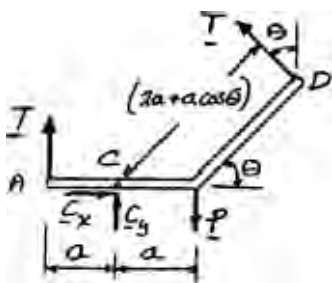




PROBLEM 4.33

Neglecting friction, determine the tension in cable ABD and the reaction at C when $\theta = 60^\circ$.

SOLUTION



$$\overset{+}{\curvearrowright} \Sigma M_C = 0: T(2a + a \cos \theta) - Ta + Pa = 0$$

$$T = \frac{P}{1 + \cos \theta} \quad (1)$$

$$\overset{+}{\rightarrow} \Sigma F_x = 0: C_x - T \sin \theta = 0$$

$$C_x = T \sin \theta = \frac{P \sin \theta}{1 + \cos \theta}$$

$$\overset{+}{\curvearrowright} \Sigma F_y = 0: C_y + T + T \cos \theta - P = 0$$

$$C_y = P - T(1 + \cos \theta) = P - P \frac{1 + \cos \theta}{1 + \cos \theta}$$

$$C_y = 0$$

$$C_y = 0, \quad C = C_x$$

$$C = P \frac{\sin \theta}{1 + \cos \theta} \rightarrow (2)$$

Since

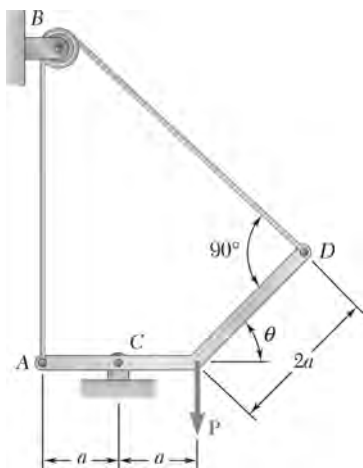
For $\theta = 60^\circ$:

Eq. (1):

$$T = \frac{P}{1 + \cos 60^\circ} = \frac{P}{1 + \frac{1}{2}} \quad T = \frac{2}{3}P \quad \blacktriangleleft$$

Eq. (2):

$$C = P \frac{\sin 60^\circ}{1 + \cos 60^\circ} = P \frac{0.866}{1 + \frac{1}{2}} \quad C = 0.577P \rightarrow \blacktriangleleft$$

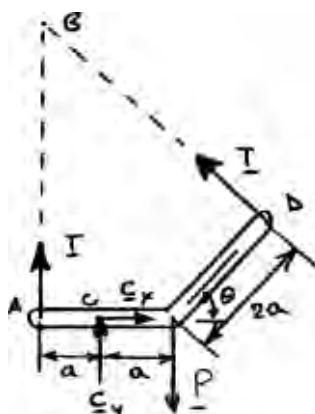


PROBLEM 4.34

Neglecting friction, determine the tension in cable ABD and the reaction at C when $\theta = 45^\circ$.

SOLUTION

Free-Body Diagram:



Equilibrium for bracket:

$$\begin{aligned} \curvearrowright \Sigma M_C = 0: & -T(a) - P(a) + (T \sin 45^\circ)(2a \sin 45^\circ) \\ & + (T \cos 45^\circ)(a + 2a \cos 45^\circ) = 0 \end{aligned}$$

$$T = 0.58579$$

$$\text{or } T = 0.586P \quad \blacktriangleleft$$

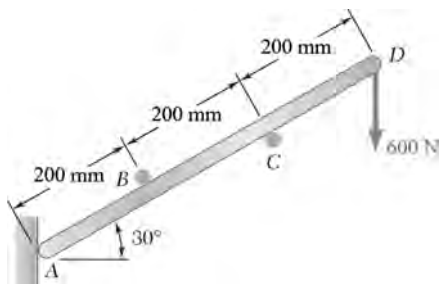
$$\rightarrow \Sigma F_x = 0: C_x + (0.58579P) \sin 45^\circ = 0$$

$$C_x = 0.41422P$$

$$\uparrow \Sigma F_y = 0: C_y + 0.58579P - P + (0.58579P) \cos 45^\circ = 0$$

$$C_y = 0$$

$$\text{or } C = 0.414P \rightarrow \blacktriangleleft$$

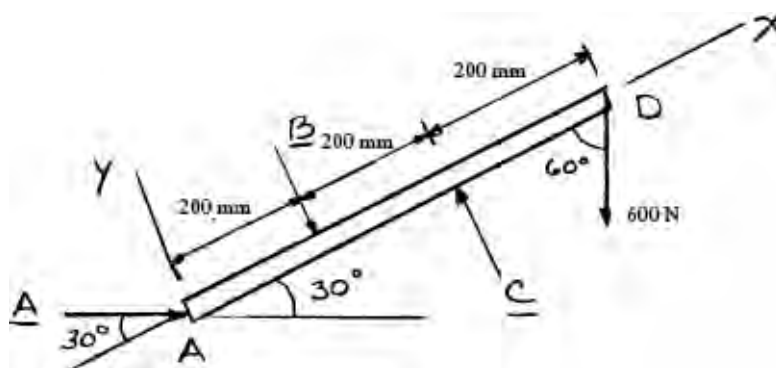


PROBLEM 4.35

A light rod AD is supported by frictionless pegs at B and C and rests against a frictionless wall at A . A vertical 600 N force is applied at D . Determine the reactions at A , B , and C .

SOLUTION

Free-Body Diagram:



$$\sum F_x = 0: A \cos 30^\circ - (600\text{ N}) \cos 60^\circ = 0$$

$$A = 346.4102\text{ N}$$

$$\mathbf{A} = 346\text{ N} \rightarrow \blacktriangleleft$$

$$\begin{aligned} \sum M_B = 0: & C(200\text{ mm}) - (600\text{ N})(400\text{ mm}) \cos 30^\circ \\ & + (346.4102\text{ N})(200\text{ mm}) \sin 30^\circ = 0 \end{aligned}$$

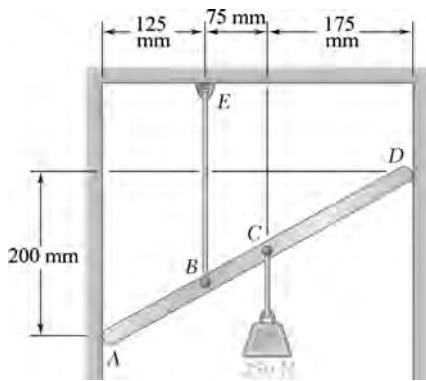
$$C = 866.0253\text{ N}$$

$$\mathbf{C} = 866\text{ N} \nearrow 60.0^\circ \blacktriangleleft$$

$$\begin{aligned} \sum M_C = 0: & B(200\text{ mm}) - (600\text{ N})(200\text{ mm}) \cos 30^\circ \\ & + (346.4102\text{ N})(400\text{ mm}) \sin 30^\circ = 0 \end{aligned}$$

$$B = 173.205\text{ N}$$

$$\mathbf{B} = 173.2\text{ N} \searrow 60.0^\circ \blacktriangleleft$$

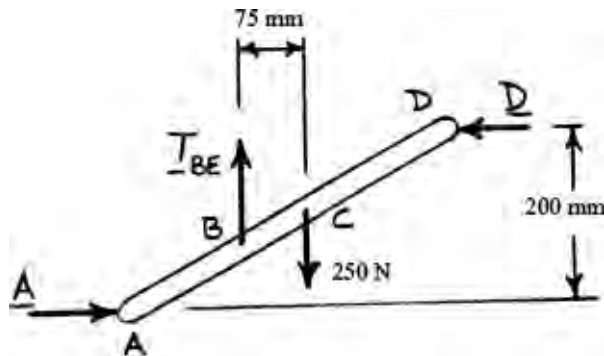


PROBLEM 4.36

A light bar AD is suspended from a cable BE and supports a 250-N block at C . The ends A and D of the bar are in contact with frictionless vertical walls. Determine the tension in cable BE and the reactions at A and D .

SOLUTION

Free-Body Diagram:



$$\Sigma F_x = 0: \quad A = D$$

$$\Sigma F_y = 0: \quad T_{BE} = 250 \text{ N} \quad \blacktriangleleft$$

We note that the forces shown form two couples.

$$+\circlearrowleft \Sigma M = 0: \quad A(200 \text{ mm}) - (250 \text{ N})(75 \text{ mm}) = 0$$

$$A = 93.75 \text{ N}$$

$$A = 93.75 \text{ N} \rightarrow \quad D = 93.75 \text{ N} \leftarrow \quad \blacktriangleleft$$



SOLUTION



$$(a) \quad \overset{\curvearrowright}{+}\Sigma M_A = 0: T_x(0.25 \text{ m}) - \left(\frac{0.075}{0.35}T_x\right)(0.5 \text{ m}) - (400 \text{ N})(0.1 \text{ m}) - (400 \text{ N})(0.4 \text{ m}) = 0$$

$$T_x = 1400 \text{ N}$$

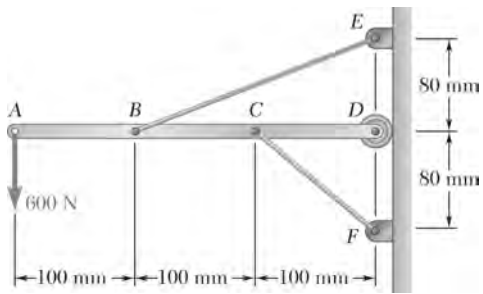
$$T_y = \frac{0.075}{0.35}(1400 \text{ N})$$
$$= 300 \text{ N}$$

$$T = 1432 \text{ lb} \quad \blacktriangleleft$$

42

PROBLEM 4.37 (Continued)

(b)	$+\uparrow \Sigma F_y = 0: A - 300 \text{ N} - 400 \text{ N} - 400 \text{ N} = 0$	
	$A = +1100 \text{ N}$	$\mathbf{A = 1100 \text{ N} \uparrow \blacktriangleleft}$
(c)	$\rightarrow \Sigma F_x = 0: -C + 1400 \text{ N} = 0$	
	$C = +1400 \text{ N}$	$\mathbf{C = 1400 \text{ N} \leftarrow \blacktriangleleft}$



PROBLEM 4.38

Determine the tension in each cable and the reaction at D.

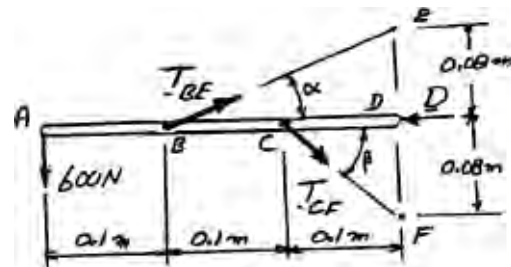
SOLUTION

$$\tan \alpha = \frac{0.08 \text{ m}}{0.2 \text{ m}}$$

$$\alpha = 21.80^\circ$$

$$\tan \beta = \frac{0.08 \text{ m}}{0.1 \text{ m}}$$

$$\beta = 38.66^\circ$$



$$\curvearrowright \Sigma M_B = 0: (600 \text{ N})(0.1 \text{ m}) - (T_{CF} \sin 38.66^\circ)(0.1 \text{ m}) = 0$$

$$T_{CF} = 960.47 \text{ N}$$

$$T_{CF} = 96.0 \text{ N} \quad \blacktriangleleft$$

$$\curvearrowright \Sigma M_C = 0: (600 \text{ N})(0.2 \text{ m}) - (T_{BE} \sin 21.80^\circ)(0.1 \text{ m}) = 0$$

$$T_{BE} = 3231.1 \text{ N}$$

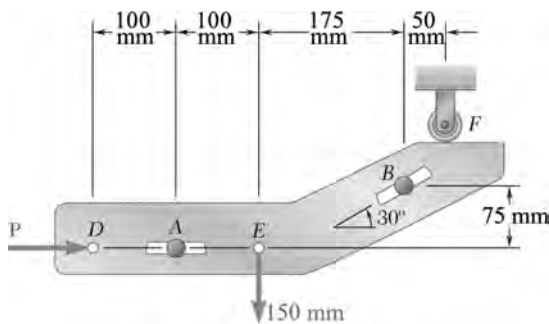
$$T_{BE} = 3230 \text{ N} \quad \blacktriangleleft$$

$$\rightarrow \Sigma F_y = 0: T_{BE} \cos \alpha + T_{CF} \cos \beta - D = 0$$

$$(3231.1 \text{ N}) \cos 21.80^\circ + (960.47 \text{ N}) \cos 38.66^\circ - D = 0$$

$$D = 3750.03 \text{ N}$$

$$D = 3750 \text{ N} \quad \blacktriangleleft$$

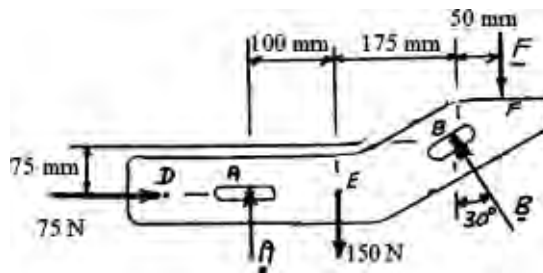


PROBLEM 4.39

Two slots have been cut in plate *DEF*, and the plate has been placed so that the slots fit two fixed, frictionless pins *A* and *B*. Knowing that $P = 75 \text{ N}$, determine (a) the force each pin exerts on the plate, (b) the reaction at *F*.

SOLUTION

Free-Body Diagram:



$$\rightarrow \Sigma F_x = 0: 75 \text{ N} - B \sin 30^\circ = 0 \quad \mathbf{B = 150 \text{ N} \nearrow 60.0^\circ \blacktriangleleft}$$

$$\curvearrowright \Sigma M_A = 0: -(150 \text{ N})(100 \text{ mm}) + B \sin 30^\circ(75 \text{ mm}) + B \cos 30^\circ(275 \text{ mm}) - F(325 \text{ mm}) = 0$$

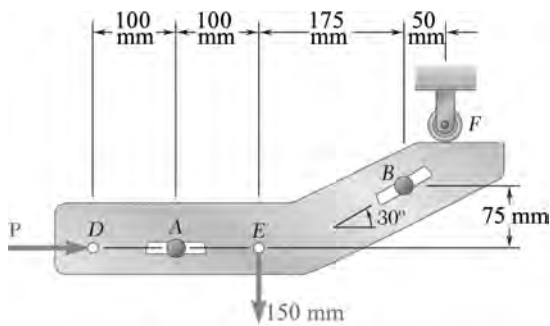
$$-15000 + 5625 + 35723.55 - F(325) = 0$$

$$F = +81.0725 \text{ N} \quad \mathbf{F = 81.1 \text{ N} \downarrow \blacktriangleleft}$$

$$\uparrow \Sigma F_y = 0: A - 150 \text{ N} + B \cos 30^\circ - F = 0$$

$$A - 150 \text{ N} + (150 \text{ N}) \cos 30^\circ - 81.0725 \text{ N} = 0$$

$$A = +101.169 \text{ N} \quad \mathbf{A = 101.2 \text{ N} \uparrow \blacktriangleleft}$$



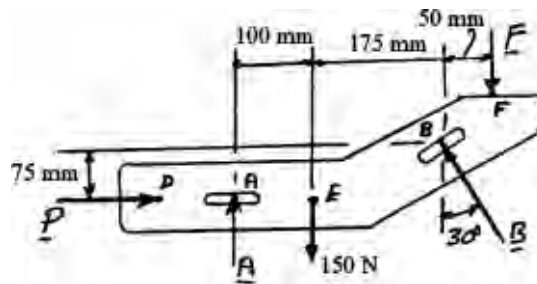
PROBLEM 4.40

For the plate of Problem 4.39 the reaction at F must be directed downward, and its maximum allowable value is 100 N. Neglecting friction at the pins, determine the required range of values of P .

PROBLEM 4.39 Two slots have been cut in plate DEF , and the plate has been placed so that the slots fit two fixed, frictionless pins A and B . Knowing that $P = 75$ N, determine (a) the force each pin exerts on the plate, (b) the reaction at F .

SOLUTION

Free-Body Diagram:



$$\rightarrow \Sigma F_x = 0: P - B \sin 30^\circ = 0$$

$$B = 2P \nearrow 60^\circ$$

$$\curvearrowleft \Sigma M_A = 0: -(150 \text{ N})(100 \text{ mm}) + B \sin 30^\circ(75 \text{ mm}) + B \cos 30^\circ(275 \text{ mm}) - F(325 \text{ mm}) = 0$$

$$-15000 \text{ N} \cdot \text{mm} + 2P \sin 30^\circ(75 \text{ mm}) + 2P \cos 30^\circ(275 \text{ mm}) - F(325 \text{ mm}) = 0$$

$$-15000 + 75P + 476.314P - 325F = 0$$

$$P = \frac{325F + 15000}{551.314} \quad (1)$$

For $F = 0$:

$$P = \frac{15000}{551.314} = 27.208 \text{ N}$$

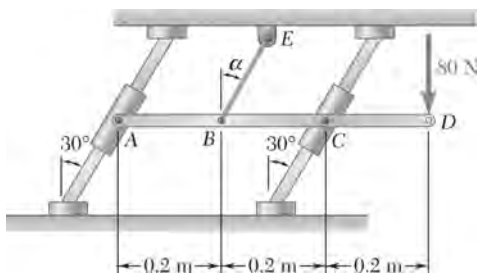
For $P = 100$ N:

$$P = \frac{32500 + 15000}{551.314} = 86.158 \text{ N}$$

For

$$0 \leq F \leq 100 \text{ N}$$

$$27.2 \text{ N} \leq P \leq 86.2 \text{ N} \quad \blacktriangleleft$$

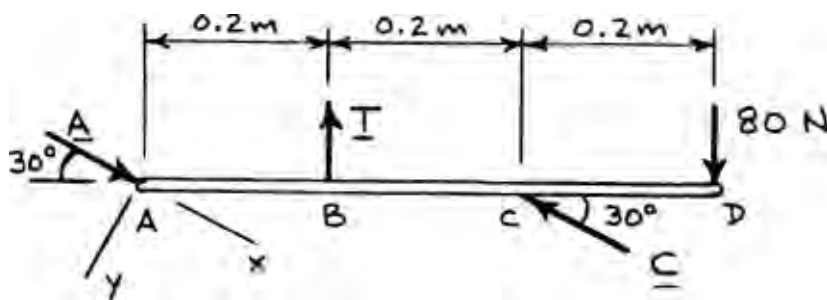


PROBLEM 4.41

Bar AD is attached at A and C to collars that can move freely on the rods shown. If the cord BE is vertical ($\alpha = 0$), determine the tension in the cord and the reactions at A and C .

SOLUTION

Free-Body Diagram:



$$\uparrow \Sigma F_y = 0: -T \cos 30^\circ + (80 \text{ N}) \cos 30^\circ = 0$$

$$T = 80 \text{ N}$$

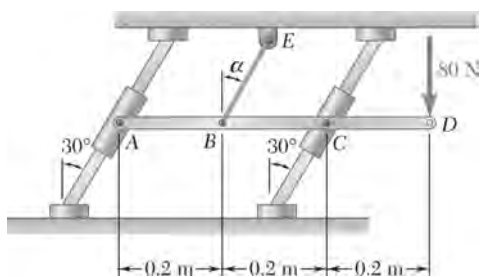
$$T = 80.0 \text{ N} \quad \blacktriangleleft$$

$$\curvearrowright \Sigma M_C = 0: (A \sin 30^\circ)(0.4 \text{ m}) - (80 \text{ N})(0.2 \text{ m}) - (80 \text{ N})(0.2 \text{ m}) = 0$$

$$A = +160 \text{ N} \quad A = 160.0 \text{ N} \quad \searrow 30.0^\circ \quad \blacktriangleleft$$

$$\curvearrowright \Sigma M_A = 0: (80 \text{ N})(0.2 \text{ m}) - (80 \text{ N})(0.6 \text{ m}) + (C \sin 30^\circ)(0.4 \text{ m}) = 0$$

$$C = +160 \text{ N} \quad C = 160.0 \text{ N} \quad \swarrow 30.0^\circ \quad \blacktriangleleft$$



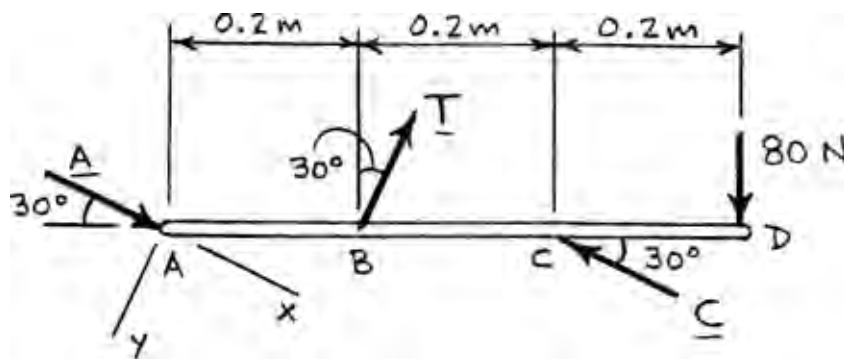
PROBLEM 4.42

Solve Problem 4.41 if the cord BE is parallel to the rods ($\alpha = 30^\circ$).

PROBLEM 4.41 Bar AD is attached at A and C to collars that can move freely on the rods shown. If the cord BE is vertical ($\alpha = 0$), determine the tension in the cord and the reactions at A and C.

SOLUTION

Free-Body Diagram:



$$\uparrow \Sigma F_y = 0: -T + (80 \text{ N}) \cos 30^\circ = 0$$

$$T = 69.282 \text{ N}$$

$$T = 69.3 \text{ N} \quad \blacktriangleleft$$

$$\begin{aligned} \curvearrowright \Sigma M_C = 0: & -(69.282 \text{ N}) \cos 30^\circ (0.2 \text{ m}) \\ & -(80 \text{ N})(0.2 \text{ m}) + (A \sin 30^\circ)(0.4 \text{ m}) = 0 \end{aligned}$$

$$A = +140.000 \text{ N}$$

$$A = 140.0 \text{ N} \quad \nearrow 30.0^\circ \quad \blacktriangleleft$$

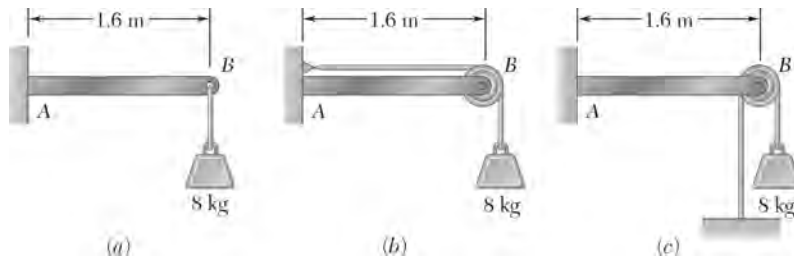
$$\begin{aligned} \curvearrowright \Sigma M_A = 0: & +(69.282 \text{ N}) \cos 30^\circ (0.2 \text{ m}) \\ & -(80 \text{ N})(0.6 \text{ m}) + (C \sin 30^\circ)(0.4 \text{ m}) = 0 \end{aligned}$$

$$C = +180.000 \text{ N}$$

$$C = 180.0 \text{ N} \quad \nearrow 30.0^\circ \quad \blacktriangleleft$$

PROBLEM 4.43

An 8-kg mass can be supported in the three different ways shown. Knowing that the pulleys have a 100-mm radius, determine the reaction at A in each case.



SOLUTION

$$W = mg = (8 \text{ kg})(9.81 \text{ m/s}^2) = 78.48 \text{ N}$$

$$(a) \quad \Sigma F_x = 0: A_x = 0$$

$$+\uparrow \Sigma F_y = 0: A_y - W = 0$$

$$A_y = 78.48 \text{ N} \uparrow$$

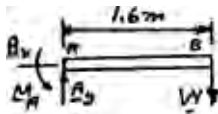
$$+\circlearrowleft \Sigma M_A = 0: M_A - W(1.6 \text{ m}) = 0$$

$$M_A = +(78.48 \text{ N})(1.6 \text{ m})$$

$$M_A = 125.56 \text{ N} \cdot \text{m} \curvearrowright$$

$$A = 78.5 \text{ N} \uparrow$$

$$M_A = 125.6 \text{ N} \cdot \text{m} \curvearrowleft$$



$$(b) \quad \rightarrow \Sigma F_x = 0: A_x - W = 0$$

$$A_x = 78.48 \text{ N} \rightarrow$$

$$+\uparrow \Sigma F_y = 0: A_y - W = 0$$

$$A_y = 78.48 \text{ N} \uparrow$$

$$A = (78.48 \text{ N})\sqrt{2} = 110.99 \text{ N} \nearrow 45^\circ$$

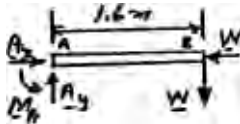
$$+\circlearrowleft \Sigma M_A = 0: M_A - W(1.6 \text{ m}) = 0$$

$$M_A = +(78.48 \text{ N})(1.6 \text{ m})$$

$$M_A = 125.56 \text{ N} \cdot \text{m} \curvearrowright$$

$$A = 111.0 \text{ N} \nearrow 45^\circ$$

$$M_A = 125.6 \text{ N} \cdot \text{m} \curvearrowleft$$



$$(c) \quad \Sigma F_x = 0: A_x = 0$$

$$+\uparrow \Sigma F_y = 0: A_y - 2W = 0$$

$$A_y = 2W = 2(78.48 \text{ N}) = 156.96 \text{ N} \uparrow$$

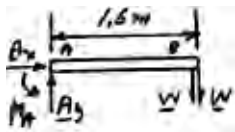
$$+\circlearrowleft \Sigma M_A = 0: M_A - 2W(1.6 \text{ m}) = 0$$

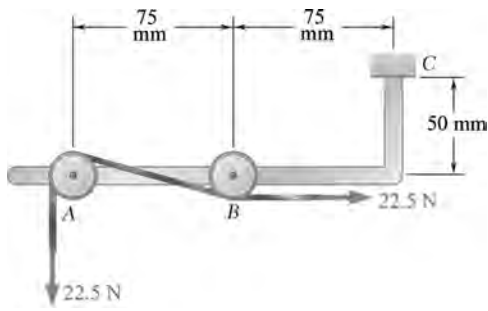
$$M_A = +2(78.48 \text{ N})(1.6 \text{ m})$$

$$M_A = 125.1 \text{ N} \cdot \text{m} \curvearrowright$$

$$A = 157.0 \text{ N} \uparrow$$

$$M_A = 125 \text{ N} \cdot \text{m} \curvearrowleft$$





PROBLEM 4.44

A tension of 22.5 N is maintained in a tape as it passes through the support system shown. Knowing that the radius of each pulley is 10 mm, determine the reaction at C.

SOLUTION

From f.b.d. of system

$$\rightarrow \Sigma F_x = 0: C_x + (22.5 \text{ N}) = 0$$

$$C_x = -22.5 \text{ N}$$

$$+\uparrow \Sigma F_y = 0: C_y - (22.5 \text{ N}) = 0$$

$$C_y = 22.5 \text{ N}$$

Then

$$\begin{aligned} C &= \sqrt{(C_x)^2 + (C_y)^2} \\ &= \sqrt{(22.5)^2 + (22.5)^2} \\ &= 31.8198 \text{ N} \end{aligned}$$

and

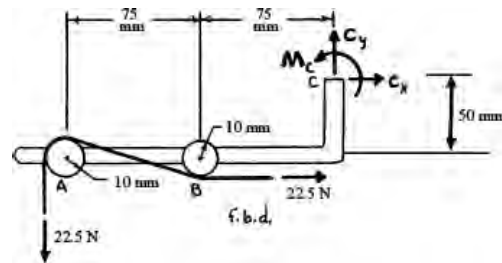
$$\theta = \tan^{-1} \left(\frac{+22.5}{-22.5} \right) = -45^\circ \quad \text{or} \quad \mathbf{C} = 31.8 \text{ N} \nearrow 45.0^\circ \blacktriangleleft$$

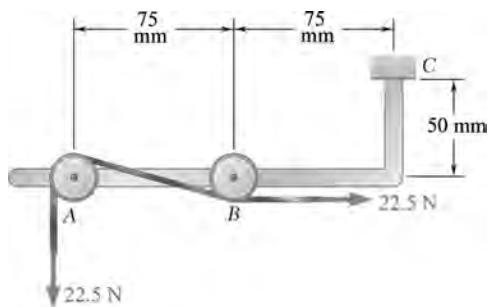
$$\curvearrowright \Sigma M_C = 0: M_C + (22.5 \text{ N})(160 \text{ mm}) + (22.5 \text{ N})(60 \text{ mm}) = 0$$

$$M_C = -4950 \text{ N} \cdot \text{mm}$$

or

$$\mathbf{M}_C = 4950 \text{ N} \cdot \text{mm} \curvearrowright \blacktriangleleft$$





PROBLEM 4.45

Solve Problem 4.44, assuming that 15 mm radius pulleys are used.

PROBLEM 4.44 A tension of 22.5 N is maintained in a tape as it passes through the support system shown. Knowing that the radius of each pulley is 10 mm, determine the reaction at C.

SOLUTION

From f.b.d. of system

$$\rightarrow \Sigma F_x = 0: C_x + (22.5 \text{ N}) = 0$$

$$C_x = -22.5 \text{ N}$$

$$\uparrow \Sigma F_y = 0: C_y - (22.5 \text{ N}) = 0$$

$$C_y = 22.5 \text{ N}$$

Then

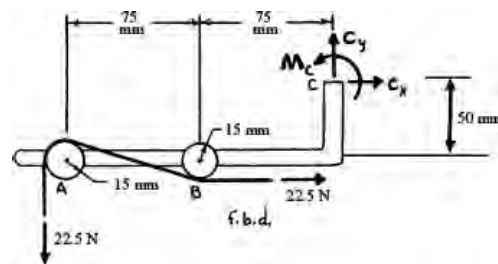
$$\begin{aligned} C &= \sqrt{(C_x)^2 + (C_y)^2} \\ &= \sqrt{(22.5)^2 + (22.5)^2} \\ &= 31.8198 \text{ N} \end{aligned}$$

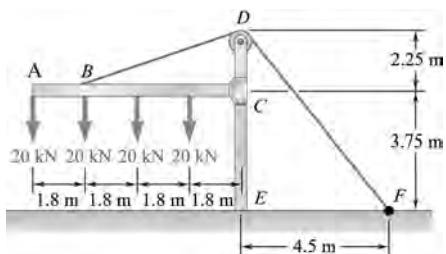
and

$$\theta = \tan^{-1} \left(\frac{22.5}{-22.5} \right) = -45.0^\circ \text{ or } \mathbf{C} = 31.8 \text{ N } \nearrow 45.0^\circ \blacktriangleleft$$

$$\curvearrowright \Sigma M_C = 0: M_C + (22.5 \text{ N})(165 \text{ mm}) + (22.5 \text{ N})(65 \text{ mm}) = 0$$

$$M_C = -5175 \text{ N} \cdot \text{mm} \quad \text{or} \quad \mathbf{M}_C = 5180 \text{ N} \cdot \text{mm } \curvearrowright \blacktriangleleft$$



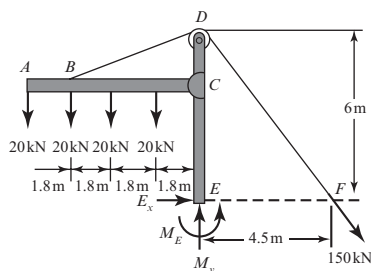


PROBLEM 4.46

The frame shown supports part of the roof of a small building. Knowing that the tension in the cable is 150 kN, determine the reaction at the fixed end E .

SOLUTION

Free-Body Diagram:



Equilibrium Equations $DF = \sqrt{(4.5 \text{ m})^2 + (6 \text{ m})^2} = 7.5 \text{ m},$

$$\rightarrow \Sigma F_x = 0: E_x + \frac{4.5}{7.5}(150 \text{ kN}) = 0$$

$$E_x = -90.0 \text{ kN}$$

$$E_x = 90.0 \text{ kN} \leftarrow \blacktriangleleft$$

$$\uparrow \Sigma F_y = 0: E_y - 4(20 \text{ kN}) - \frac{6}{7.5}(150 \text{ kN}) = 0$$

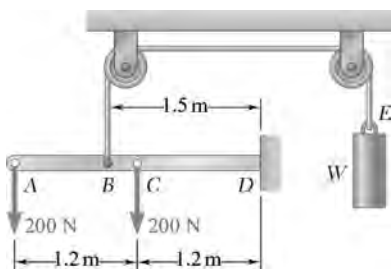
$$E_y = +200 \text{ kN}$$

$$E_y = 200 \text{ kN} \uparrow \blacktriangleleft$$

$$\begin{aligned} \curvearrowright \Sigma M_E = 0: & (20 \text{ kN})(7.2 \text{ m}) + (20 \text{ kN})(5.4 \text{ m}) + (20 \text{ kN})(3.6 \text{ m}) \\ & + (20 \text{ kN})(1.8 \text{ m}) - \frac{6}{7.5}(150 \text{ kN})(4.5 \text{ m}) + M_E = 0 \end{aligned}$$

$$M_E = +180.0 \text{ kN} \cdot \text{m}$$

$$M_E = 180.0 \text{ kN} \cdot \text{m} \curvearrowright \blacktriangleleft$$



PROBLEM 4.47

Beam AD carries the two 200-N loads shown. The beam is held by a fixed support at D and by the cable BE that is attached to the counterweight W . Determine the reaction at D when (a) $W = 500$ N, (b) $W = 450$ N.

SOLUTION

(a)

$$W = 500 \text{ N}$$

From f.b.d. of beam AD

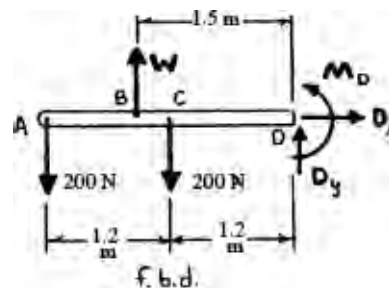
$$\rightarrow \Sigma F_x = 0: D_x = 0$$

$$\uparrow \Sigma F_y = 0: D_y - 200 \text{ N} - 200 \text{ N} + 500 \text{ N} = 0$$

$$D_y = -100 \text{ N} \quad \text{or} \quad \mathbf{D} = 100.0 \text{ N} \downarrow \blacktriangleleft$$

$$\curvearrowright \Sigma M_D = 0: M_D - (500 \text{ N})(1.5 \text{ m}) + (200 \text{ N})(2.4 \text{ m}) + (200 \text{ N})(1.2 \text{ m}) = 0$$

$$M_D = 30 \text{ N} \cdot \text{m} \quad \text{or} \quad \mathbf{M}_D = 30.0 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$$



(b)

$$W = 450 \text{ N}$$

From f.b.d. of beam AD

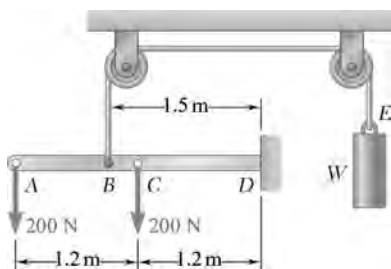
$$\rightarrow \Sigma F_x = 0: D_x = 0$$

$$\uparrow \Sigma F_y = 0: D_y + 450 \text{ N} - 200 \text{ N} - 200 \text{ N} = 0$$

$$D_y = -50 \text{ N} \quad \text{or} \quad \mathbf{D} = 50.0 \text{ N} \downarrow \blacktriangleleft$$

$$\curvearrowright \Sigma M_D = 0: M_D - (450 \text{ N})(1.5 \text{ m}) + (200 \text{ N})(2.4 \text{ m}) + (200 \text{ N})(1.2 \text{ m}) = 0$$

$$M_D = -45 \text{ N} \cdot \text{m} \quad \text{or} \quad \mathbf{M}_D = -45.0 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$$



PROBLEM 4.48

For the beam and loading shown, determine the range of values of W for which the magnitude of the couple at D does not exceed $60 \text{ N} \cdot \text{m}$.

SOLUTION

For $W_{\min}$,

$$M_D = -60 \text{ N} \cdot \text{m}$$

From f.b.d. of beam AD $\quad \curvearrowleft \Sigma M_D = 0: (200 \text{ N})(2.4 \text{ m}) - W_{\min}(1.5 \text{ m}) + (200 \text{ N})(1.2 \text{ m}) - 60 \text{ N} \cdot \text{m} = 0$

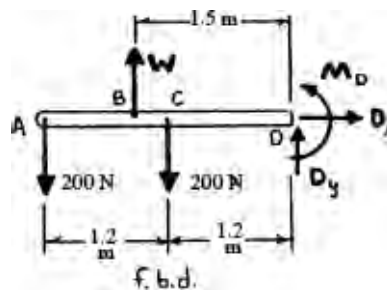
$$W_{\min} = 440 \text{ N}$$

For $W_{\max}$,

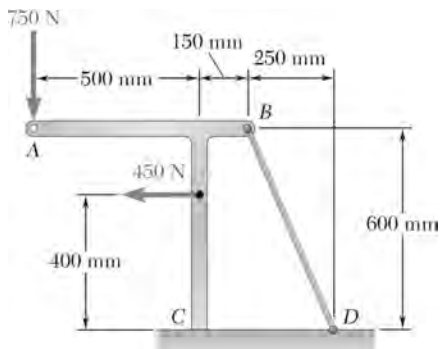
$$M_D = 60 \text{ N} \cdot \text{m}$$

From f.b.d. of beam AD $\quad \curvearrowleft \Sigma M_D = 0: (200 \text{ N})(2.4 \text{ m}) - W_{\max}(1.5 \text{ m}) + (200 \text{ N})(1.2 \text{ m}) + 60 \text{ N} \cdot \text{m} = 0$

$$W_{\max} = 520 \text{ N}$$



$$\text{or } 440 \text{ N} \leq W \leq 520 \text{ N} \quad \blacktriangleleft$$



PROBLEM 4.49

Knowing that the tension in wire BD is 1300 N, determine the reaction at the fixed support C of the frame shown.

SOLUTION

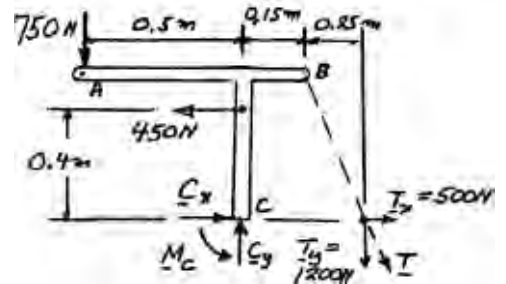
$$T = 1300 \text{ N}$$

$$T_x = \frac{5}{13} T$$

$$= 500 \text{ N}$$

$$T_y = \frac{12}{13} T$$

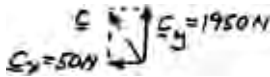
$$= 1200 \text{ N}$$



$$\rightarrow \Sigma M_x = 0: C_x - 450 \text{ N} + 500 \text{ N} = 0 \quad C_x = -50 \text{ N} \quad C_x = 50 \text{ N} \leftarrow$$

$$+\uparrow \Sigma F_y = 0: C_y - 750 \text{ N} - 1200 \text{ N} = 0 \quad C_y = +1950 \text{ N} \quad C_y = 1950 \text{ N} \uparrow$$

$$C = 1951 \text{ N} \nearrow 88.5^\circ \blacktriangleleft$$

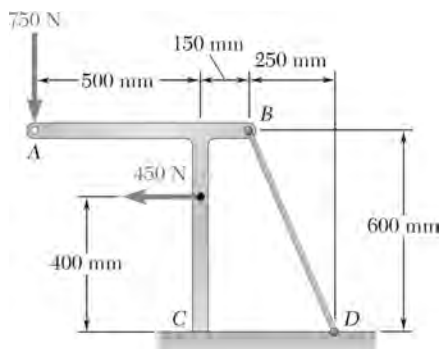


$$\curvearrowright \Sigma M_C = 0: M_C + (750 \text{ N})(0.5 \text{ m}) + (4.50 \text{ N})(0.4 \text{ m})$$

$$- (1200 \text{ N})(0.4 \text{ m}) = 0$$

$$M_C = -75.0 \text{ N} \cdot \text{m}$$

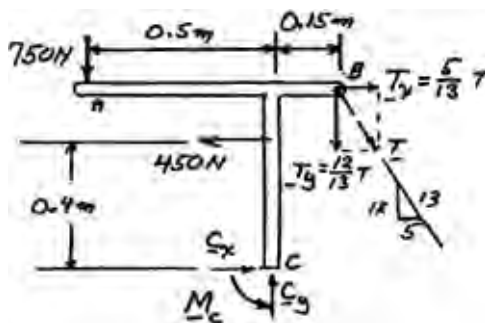
$$M_C = 75.0 \text{ N} \cdot \text{m} \curvearrowright \blacktriangleleft$$



PROBLEM 4.50

Determine the range of allowable values of the tension in wire BD if the magnitude of the couple at the fixed support C is not to exceed $100 \text{ N} \cdot \text{m}$.

SOLUTION



$$\begin{aligned} \uparrow \Sigma M_C = 0: & (750 \text{ N})(0.5 \text{ m}) + (450 \text{ N})(0.4 \text{ m}) - \left(\frac{5}{13}T\right)(0.6 \text{ m}) \\ & - \left(\frac{12}{13}T\right)(0.15 \text{ m}) + M_C = 0 \end{aligned}$$

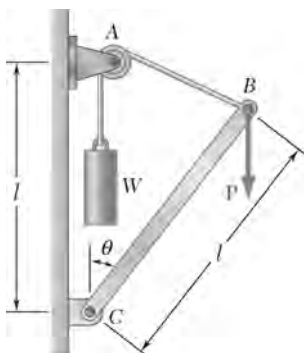
$$375 \text{ N} \cdot \text{m} + 180 \text{ N} \cdot \text{m} - \left(\frac{4.8}{13} \text{ m}\right)T + M_C = 0$$

$$T = \frac{13}{4.8}(555 + M_C)$$

For $M_C = -100 \text{ N} \cdot \text{m}$: $T = \frac{13}{4.8}(555 - 100) = 1232 \text{ N}$

For $M_C = +100 \text{ N} \cdot \text{m}$: $T = \frac{13}{4.8}(555 + 100) = 1774 \text{ N}$

For $|M_C| \leq 100 \text{ N} \cdot \text{m}$: $1.232 \text{ kN} \leq T \leq 1.774 \text{ kN} \quad \blacktriangleleft$



PROBLEM 4.51

A vertical load $\mathbf{P}$ is applied at end B of rod BC . (a) Neglecting the weight of the rod, express the angle θ corresponding to the equilibrium position in terms of P , l , and the counterweight W . (b) Determine the value of θ corresponding to equilibrium if $P = 2W$.

SOLUTION

(a) Triangle ABC is isosceles.

We have $CD = (BC) \cos \frac{\theta}{2} = l \cos \frac{\theta}{2}$

$$\sum M_C = 0: W \left(l \cos \frac{\theta}{2} \right) - P(l \sin \theta) = 0$$

Setting $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$: $Wl \cos \frac{\theta}{2} - 2Pl \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 0$

$$W - 2P \sin \frac{\theta}{2} = 0$$

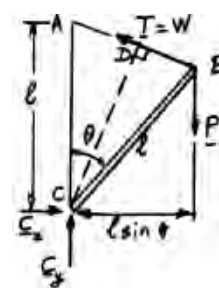
(b) For $P = 2W$:

$$\sin \frac{\theta}{2} = \frac{W}{2P} = \frac{W}{4W} = 0.25$$

$$\frac{\theta}{2} = 14.5^\circ$$

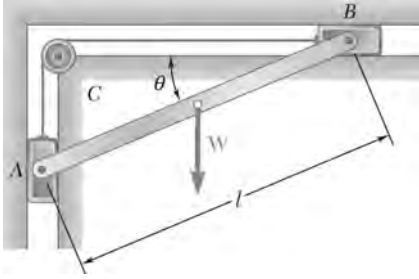
or

$$\frac{\theta}{2} = 165.5^\circ \quad \theta = 331^\circ (\text{discard})$$



$$\theta = 2 \sin^{-1} \left(\frac{W}{2P} \right) \quad \blacktriangleleft$$

$$\theta = 29.0^\circ \quad \blacktriangleleft$$



PROBLEM 4.52

A slender rod AB , of weight W , is attached to blocks A and B , which move freely in the guides shown. The blocks are connected by an elastic cord that passes over a pulley at C . (a) Express the tension in the cord in terms of W and θ . (b) Determine the value of θ for which the tension in the cord is equal to $3W$.

SOLUTION

(a) From f.b.d. of rod AB

$$\sum M_C = 0: T(l \sin \theta) + W \left[\left(\frac{1}{2} \right) \cos \theta \right] - T(l \cos \theta) = 0$$

$$T = \frac{W \cos \theta}{2(\cos \theta - \sin \theta)}$$

Dividing both numerator and denominator by $\cos \theta$,

$$T = \frac{W}{2} \left(\frac{1}{1 - \tan \theta} \right)$$

$$\text{or } T = \frac{\left(\frac{W}{2} \right)}{(1 - \tan \theta)} \blacktriangleleft$$

(b) For $T = 3W$,

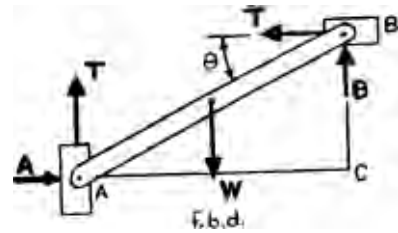
$$3W = \frac{\left(\frac{W}{2} \right)}{(1 - \tan \theta)}$$

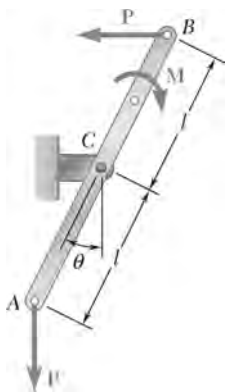
$$1 - \tan \theta = \frac{1}{6}$$

or

$$\theta = \tan^{-1} \left(\frac{5}{6} \right) = 39.806^\circ$$

$$\text{or } \theta = 39.8^\circ \blacktriangleleft$$





PROBLEM 4.53

Rod AB is acted upon by a couple $\mathbf{M}$ and two forces, each of magnitude P . (a) Derive an equation in θ , P , M , and l that must be satisfied when the rod is in equilibrium. (b) Determine the value of θ corresponding to equilibrium when $M = 1500 \text{ N}\cdot\text{m}$, $P = 200 \text{ N}$, and $l = 600 \text{ mm}$.

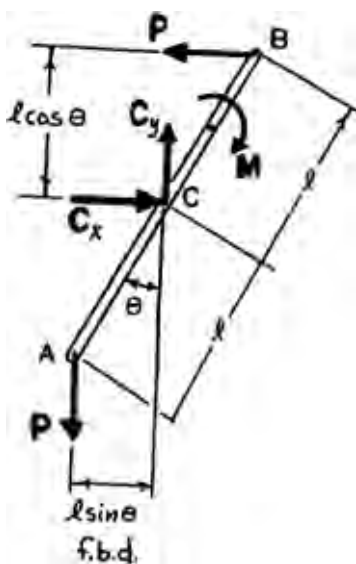
SOLUTION

Free-Body Diagram:

(a) From free-body diagram of rod AB

$$\sum M_C = 0: P(l \cos \theta) + P(l \sin \theta) - M = 0$$

$$\text{or } \sin \theta + \cos \theta = \frac{M}{Pl} \quad \blacktriangleleft$$



(b) For $M = 1500 \text{ N}\cdot\text{m}$, $P = 200 \text{ N}$, and $l = 600 \text{ mm}$

$$\sin \theta + \cos \theta = \frac{1500 \text{ N}\cdot\text{m}}{(200 \text{ N})(600 \text{ mm})} = \frac{5}{4} = 1.25$$

Using identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\sin \theta + (1 - \sin^2 \theta)^{1/2} = 1.25$$

$$(1 - \sin^2 \theta)^{1/2} = 1.25 - \sin \theta$$

$$1 - \sin^2 \theta = 1.5625 - 2.5 \sin \theta + \sin^2 \theta$$

$$2 \sin^2 \theta - 2.5 \sin \theta + 0.5625 = 0$$

Using quadratic formula

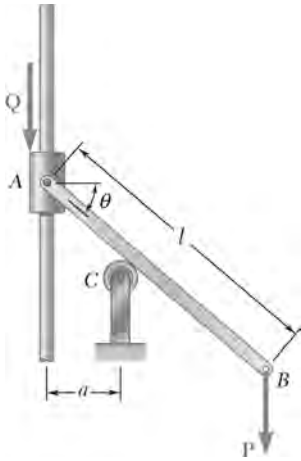
$$\begin{aligned} \sin \theta &= \frac{-(-2.5) \pm \sqrt{(625) - 4(2)(0.5625)}}{2(2)} \\ &= \frac{2.5 \pm \sqrt{1.75}}{4} \end{aligned}$$

$$\text{or } \sin \theta = 0.95572 \quad \text{and} \quad \sin \theta = 0.29428$$

$$\theta = 72.886^\circ \quad \text{and} \quad \theta = 17.1144^\circ$$

$$\text{or } \theta = 17.11^\circ \quad \text{and} \quad \theta = 72.9^\circ \quad \blacktriangleleft$$

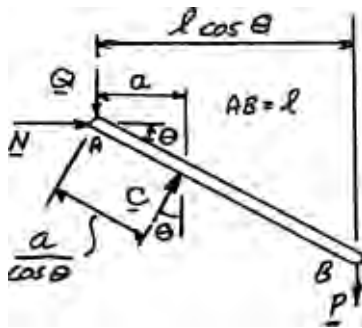
PROBLEM 4.54



Rod AB is attached to a collar at A and rests against a small roller at C . (a) Neglecting the weight of rod AB , derive an equation in P , Q , a , l , and θ that must be satisfied when the rod is in equilibrium. (b) Determine the value of θ corresponding to equilibrium when $P = 80 \text{ N}$, $Q = 60 \text{ N}$, $l = 500 \text{ mm}$, and $a = 125 \text{ mm}$.

SOLUTION

Free-Body Diagram:



$$+\uparrow \Sigma F_y = 0: C \cos \theta - P - Q = 0$$

$$C = \frac{P + Q}{\cos \theta}$$

$$(a) \quad \curvearrowleft \Sigma M_A = 0: C \frac{a}{\cos \theta} - Pl \cos \theta = 0$$

$$\frac{P + Q}{\cos \theta} \cdot \frac{a}{\cos \theta} - Pl \cos \theta = 0$$

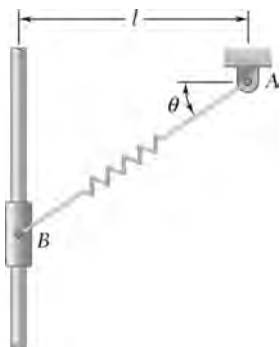
$$\cos^3 \theta = \frac{a(P + Q)}{Pl} \quad \blacktriangleleft$$

$$(b) \quad \text{For } P = 80 \text{ N}, \quad Q = 60 \text{ N}, \quad l = 500 \text{ mm}, \quad \text{and } a = 125 \text{ mm}$$

$$\begin{aligned} \cos^3 \theta &= \frac{(125 \text{ mm})(80 \text{ N} + 60 \text{ N})}{(80 \text{ N})(500 \text{ mm})} \\ &= 0.4375 \end{aligned}$$

$$\cos \theta = 0.75915$$

$$\theta = 40.6^\circ \quad \blacktriangleleft$$



PROBLEM 4.55

A collar B of weight W can move freely along the vertical rod shown. The constant of the spring is k , and the spring is unstretched when $\theta = 0$. (a) Derive an equation in θ , W , k , and l that must be satisfied when the collar is in equilibrium. (b) Knowing that $W = 300 \text{ N}$, $l = 500 \text{ mm}$, and $k = 800 \text{ N/m}$, determine the value of θ corresponding to equilibrium.

SOLUTION

First note

$$T = ks$$

where

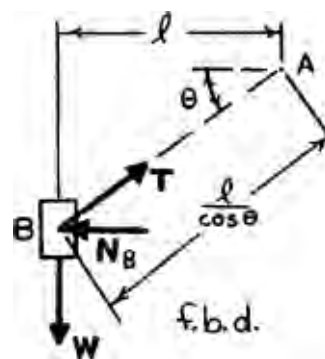
k = spring constant

s = elongation of spring

$$= \frac{l}{\cos \theta} - l$$

$$= \frac{l}{\cos \theta} (1 - \cos \theta)$$

$$T = \frac{kl}{\cos \theta} (1 - \cos \theta)$$



(a) From f.b.d. of collar B

$$+\uparrow \sum F_y = 0: T \sin \theta - W = 0$$

or

$$\frac{kl}{\cos \theta} (1 - \cos \theta) \sin \theta - W = 0$$

$$\text{or } \tan \theta - \sin \theta = \frac{W}{kl} \quad \blacktriangleleft$$

(b) For

$$W = 300 \text{ N}$$

$$l = 500 \text{ mm}$$

$$k = 800 \text{ N/m}$$

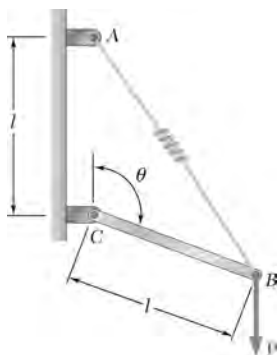
$$l = \frac{500 \text{ mm}}{2400 \text{ N/m}} = 0.5 \text{ m}$$

$$\tan \theta - \sin \theta = \frac{300 \text{ N}}{(800 \text{ N/m})(0.5 \text{ m})} = 0.75$$

Solving numerically,

$$\theta = 57.957^\circ$$

$$\text{or } \theta = 58.0^\circ \quad \blacktriangleleft$$



PROBLEM 4.56

A vertical load P is applied at end B of rod BC . The constant of the spring is k , and the spring is unstretched when $\theta = 90^\circ$. (a) Neglecting the weight of the rod, express the angle θ corresponding to equilibrium in terms of P , k , and l . (b) Determine the value of θ corresponding to equilibrium when $P = \frac{1}{4}kl$.

SOLUTION

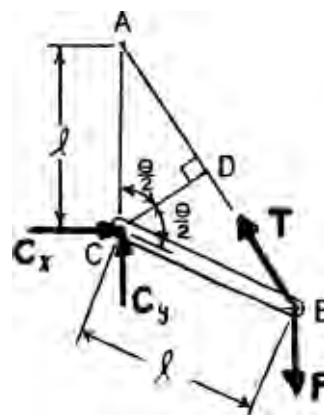
First note

$$T = \text{tension in spring} = ks$$

where

$$\begin{aligned} s &= \text{elongation of spring} \\ &= (\overline{AB})_\theta - (\overline{AB})_{\theta=90^\circ} \\ &= 2l \sin\left(\frac{\theta}{2}\right) - 2l \sin\left(\frac{90^\circ}{2}\right) \\ &= 2l \left[\sin\left(\frac{\theta}{2}\right) - \left(\frac{1}{\sqrt{2}}\right) \right] \end{aligned}$$

$$T = 2kl \left[\sin\left(\frac{\theta}{2}\right) - \left(\frac{1}{\sqrt{2}}\right) \right]$$



(1)

(a) From f.b.d. of rod BC $\quad \curvearrowright \Sigma M_C = 0: \quad T \left[l \cos\left(\frac{\theta}{2}\right) \right] - P(l \sin \theta) = 0$

Substituting T from Equation (1)

$$\begin{aligned} 2kl \left[\sin\left(\frac{\theta}{2}\right) - \left(\frac{1}{\sqrt{2}}\right) \right] \left[l \cos\left(\frac{\theta}{2}\right) \right] - P(l \sin \theta) &= 0 \\ 2kl^2 \left[\sin\left(\frac{\theta}{2}\right) - \left(\frac{1}{\sqrt{2}}\right) \right] \cos\left(\frac{\theta}{2}\right) - Pl \left[2 \sin\left(\frac{\theta}{2}\right) \cos\left(\frac{\theta}{2}\right) \right] &= 0 \end{aligned}$$

Factoring out $2l \cos\left(\frac{\theta}{2}\right)$, leaves

$$kl \left[\sin\left(\frac{\theta}{2}\right) - \left(\frac{1}{\sqrt{2}}\right) \right] - P \sin\left(\frac{\theta}{2}\right) = 0$$

or

$$\sin\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{2}} \left(\frac{kl}{kl - P} \right) \quad \theta = 2 \sin^{-1} \left[\frac{kl}{\sqrt{2}(kl - P)} \right] \quad \blacktriangleleft$$

PROBLEM 4.56 (Continued)

(b)

$$P = \frac{kl}{4}$$

$$\theta = 2 \sin^{-1} \left[\frac{kl}{\sqrt{2} \left(kl - \frac{kl}{4} \right)} \right]$$

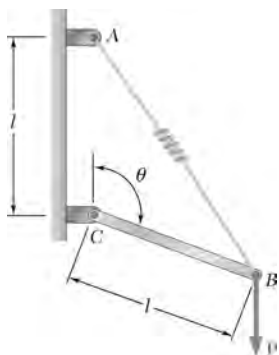
$$= 2 \sin^{-1} \left[\frac{kl}{\sqrt{2}} \left(\frac{4}{3kl} \right) \right]$$

$$= 2 \sin^{-1} \left(\frac{4}{3\sqrt{2}} \right)$$

$$= 2 \sin^{-1}(0.94281)$$

$$= 141.058^\circ$$

$$\text{or } \theta = 141.1^\circ \blacktriangleleft$$



PROBLEM 4.57

Solve Sample Problem 4.5, assuming that the spring is unstretched when $\theta = 90^\circ$.

SOLUTION

First note

$$T = \text{tension in spring} = ks$$

where

$$s = \text{deformation of spring}$$

$$= r\beta$$

$$F = kr\beta$$

From f.b.d. of assembly

$$\sum M_O = 0: W(l \cos \beta) - F(r) = 0$$

or

$$Wl \cos \beta - kr^2 \beta = 0$$

$$\cos \beta = \frac{kr^2}{Wl} \beta$$

For

$$k = 45 \text{ N/mm}$$

$$r = 75 \text{ mm}$$

$$l = 200 \text{ mm}$$

$$W = 1800 \text{ N}$$

$$\cos \beta = \frac{(45 \text{ N/mm})(75 \text{ mm})^2}{(1800 \text{ N})(200 \text{ mm})} \beta$$

or

$$\cos \beta = 0.703125\beta$$

Solving numerically,

$$\beta = 0.89245 \text{ rad}$$

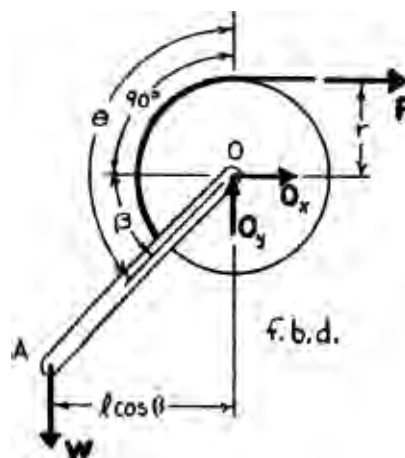
or

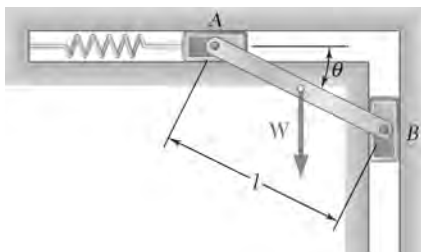
$$\beta = 51.134^\circ$$

Then

$$\theta = 90^\circ + 51.134^\circ = 141.134^\circ$$

$$\text{or } \theta = 141.1^\circ \blacktriangleleft$$



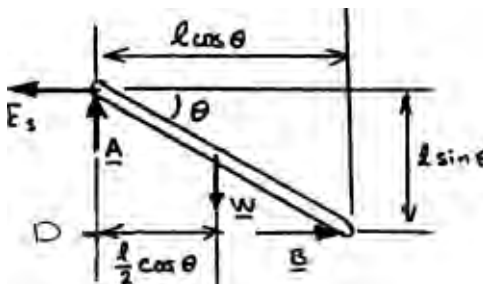


PROBLEM 4.58

A slender rod AB , of weight W , is attached to blocks A and B that move freely in the guides shown. The constant of the spring is k , and the spring is unstretched when $\theta = 0$. (a) Neglecting the weight of the blocks, derive an equation in W , k , l , and θ that must be satisfied when the rod is in equilibrium. (b) Determine the value of θ when $W = 375 \text{ N}$, $l = 750 \text{ mm}$, and $k = 600 \text{ N/m}$.

SOLUTION

Free-Body Diagram:



Spring force:

$$F_s = ks = k(l - l \cos \theta) = kl(1 - \cos \theta)$$

$$(a) \quad \sum M_D = 0: \quad F_s(l \sin \theta) - W\left(\frac{l}{2} \cos \theta\right) = 0$$

$$kl(1 - \cos \theta)l \sin \theta - \frac{W}{2}l \cos \theta = 0$$

$$kl(1 - \cos \theta) \tan \theta - \frac{W}{2} = 0 \quad \text{or} \quad (1 - \cos \theta) \tan \theta = \frac{W}{2kl} \quad \blacktriangleleft$$

(b) For given values of

$$W = 375 \text{ N}$$

$$l = 750 \text{ mm} = 0.75 \text{ m}$$

$$k = 600 \text{ N/m}$$

$$(1 - \cos \theta) \tan \theta = \tan \theta - \sin \theta$$

$$\begin{aligned} &= \frac{375 \text{ N}}{2(600 \text{ N/m})(0.75 \text{ m})} \\ &= 0.41667 \end{aligned}$$

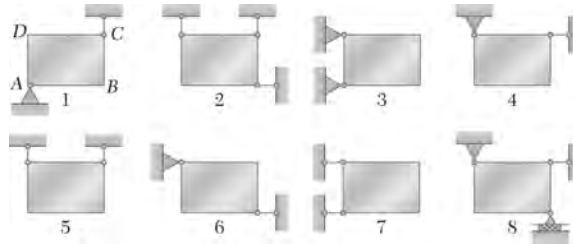
Solving numerically:

$$\theta = 49.710^\circ$$

$$\text{or} \quad \theta = 49.7^\circ \quad \blacktriangleleft$$

PROBLEM 4.59

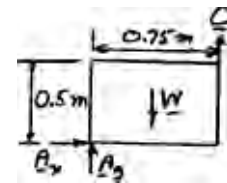
Eight identical 500×750 -mm rectangular plates, each of mass $m = 40$ kg, are held in a vertical plane as shown. All connections consist of frictionless pins, rollers, or short links. In each case, determine whether (a) the plate is completely, partially, or improperly constrained, (b) the reactions are statically determinate or indeterminate, (c) the equilibrium of the plate is maintained in the position shown. Also, wherever possible, compute the reactions.



SOLUTION

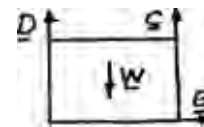
1. Three non-concurrent, non-parallel reactions
 - (a) Plate: completely constrained
 - (b) Reactions: determinate
 - (c) Equilibrium maintained

$$A = C = 196.2 \text{ N } \uparrow$$



2. Three non-concurrent, non-parallel reactions
 - (a) Plate: completely constrained
 - (b) Reactions: determinate
 - (c) Equilibrium maintained

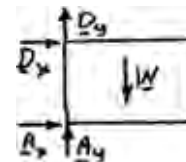
$$B = 0, \quad C = D = 196.2 \text{ N } \uparrow$$



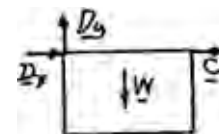
3. Four non-concurrent, non-parallel reactions
 - (a) Plate: completely constrained
 - (b) Reactions: indeterminate
 - (c) Equilibrium maintained

$$A_x = 294 \text{ N } \rightarrow, \quad D_x = 294 \text{ N } \leftarrow$$

$$(A_y + D_y = 392 \text{ N } \uparrow)$$



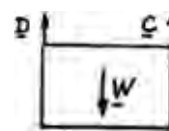
4. Three concurrent reactions (through D)
 - (a) Plate: improperly constrained
 - (b) Reactions: indeterminate
 - (c) No equilibrium ($\sum M_D \neq 0$)



PROBLEM 4.59 (Continued)

5. Two reactions

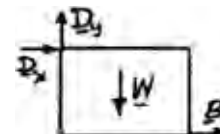
- (a) Plate: partial constraint
- (b) Reactions: determinate
- (c) Equilibrium maintained



$$C = R = 196.2 \text{ N } \uparrow$$

6. Three non-concurrent, non-parallel reactions

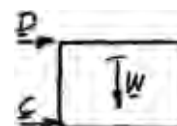
- (a) Plate: completely constrained
- (b) Reactions: determinate
- (c) Equilibrium maintained



$$B = 294 \text{ N } \rightarrow, \quad D = 491 \text{ N } \nearrow 53.1^\circ$$

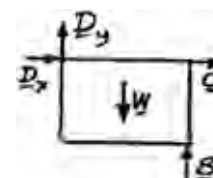
7. Two reactions

- (a) Plate: improperly constrained
- (b) Reactions determined by dynamics
- (c) No equilibrium ($\Sigma F_y \neq 0$)



8. For non-concurrent, non-parallel reactions

- (a) Plate: completely constrained
- (b) Reactions: indeterminate
- (c) Equilibrium maintained

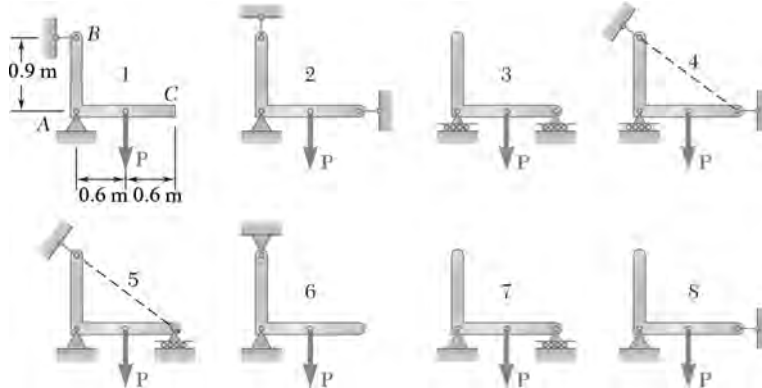


$$B = D_y = 196.2 \text{ N } \uparrow$$

$$(C + D_x = 0)$$

PROBLEM 4.60

The bracket ABC can be supported in the eight different ways shown. All connections consist of smooth pins, rollers, or short links. For each case, answer the questions listed in Problem 4.59, and, wherever possible, compute the reactions, assuming that the magnitude of the force $\mathbf{P}$ is 500 N.

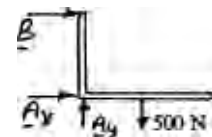


SOLUTION

1. Three non-concurrent, non-parallel reactions

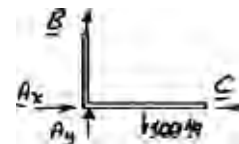
- Bracket: complete constraint
- Reactions: determinate
- Equilibrium maintained

$$\mathbf{A} = 601 \text{ N } \nearrow 56.3^\circ, \mathbf{B} = 333 \text{ N } \leftarrow$$



2. Four concurrent, reactions (through A)

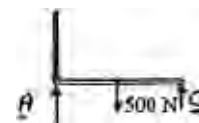
- Bracket: improper constraint
- Reactions: indeterminate
- No equilibrium ($\sum M_A \neq 0$)



3. Two reactions

- Bracket: partial constraint
- Reactions: determinate
- Equilibrium maintained

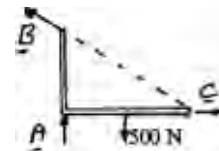
$$\mathbf{A} = 250 \text{ N } \uparrow, \mathbf{C} = 250 \text{ N } \uparrow$$



4. Three non-concurrent, non-parallel reactions

- Bracket: complete constraint
- Reactions: determinate
- Equilibrium maintained

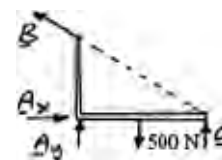
$$\mathbf{A} = 250 \text{ N } \uparrow, \mathbf{B} = 417 \text{ N } \nearrow 36.9^\circ, \mathbf{C} = 333 \text{ N } \rightarrow$$



PROBLEM 4.60 (Continued)

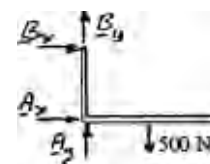
5. Four non-concurrent, non-parallel reactions

- (a) Bracket: complete constraint
 (b) Reactions: indeterminate
 (c) Equilibrium maintained ($\Sigma M_C = 0$) $A_y = 250 \text{ N} \uparrow$



6. Four non-concurrent, non-parallel reactions

- (a) Bracket: complete constraint
 (b) Reactions: indeterminate
 (c) Equilibrium maintained



$$A_x = 333 \text{ N} \quad , \quad B = 333 \text{ N} \leftarrow$$

$$(A_y + B_y = 500 \text{ N} \uparrow)$$

7. Three non-concurrent, non-parallel reactions

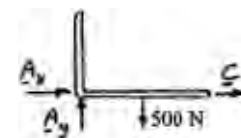
- (a) Bracket: complete constraint
 (b) Reactions: determinate
 (c) Equilibrium maintained

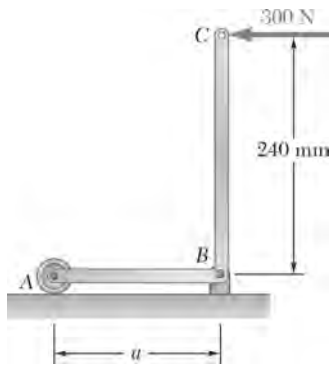


$$A = C = 250 \text{ N} \uparrow$$

8. Three concurrent, reactions (through A)

- (a) Bracket: improper constraint
 (b) Reactions: indeterminate
 (c) No equilibrium ($\Sigma M_A \neq 0$)





PROBLEM 4.61

Determine the reactions at A and B when $a = 180$ mm.

SOLUTION

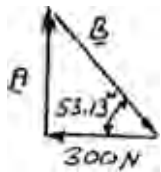
Reaction at B must pass through D where A and 300-N load intersect.

$\triangle BCD$:

$$\tan \beta = \frac{240}{180}$$

$$\beta = 53.13^\circ$$

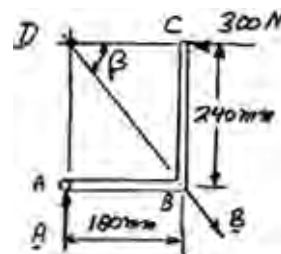
Force triangle



$$\begin{aligned} A &= (300 \text{ N}) \tan 53.13^\circ \\ &= 400 \text{ N} \end{aligned}$$

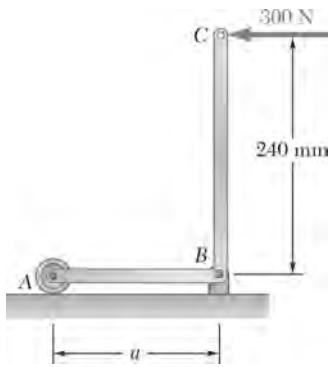
$$\begin{aligned} B &= \frac{300 \text{ N}}{\cos 53.13^\circ} \\ &= 500 \text{ N} \end{aligned}$$

Free-Body Diagram:
(Three-force member)



$$\mathbf{A} = 400 \text{ N } \uparrow \swarrow$$

$$\mathbf{B} = 500 \text{ N } \searrow 53.1^\circ$$



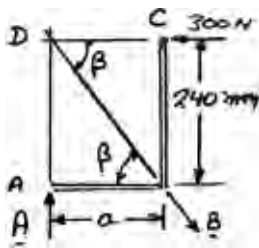
PROBLEM 4.62

For the bracket and loading shown, determine the range of values of the distance a for which the magnitude of the reaction at B does not exceed 600 N.

SOLUTION

Reaction at B must pass through D where A and 300-N load intersect.

Free-Body Diagram:
(Three-force member)



Eq. (1)

$$a = \frac{240 \text{ mm}}{\tan \beta} \quad (1)$$

Force Triangle

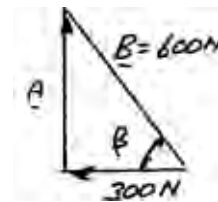
(with $B = 600 \text{ N}$)

$$\cos \beta = \frac{300 \text{ N}}{600 \text{ N}} = 0.5$$

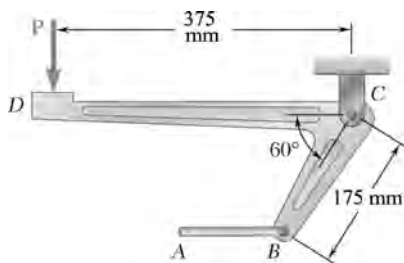
$$\beta = 60.0^\circ$$

$$a = \frac{240 \text{ mm}}{\tan 60.0^\circ}$$

$$= 138.56 \text{ mm}$$



For $B \leq 600 \text{ N}$ $a \geq 138.6 \text{ mm}$ ◀



PROBLEM 4.63

Using the method of Section 4.7, solve Problem 4.17.

PROBLEM 4.17 The required tension in cable AB is 900 N. Determine (a) the vertical force P that must be applied to the pedal, (b) the corresponding reaction at C .

SOLUTION

Reaction at C must pass through E , where P and 900 N force intersect.

$$\tan \beta = \frac{151.554 \text{ mm}}{375 \text{ mm}}$$

$$\beta = 22.006^\circ$$

Force triangle

(a)

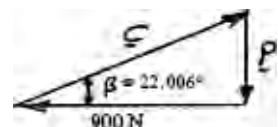
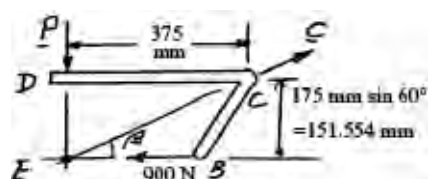
$$P = (900 \text{ N}) \tan 22.006^\circ$$

$$P = 363.7$$

(b)

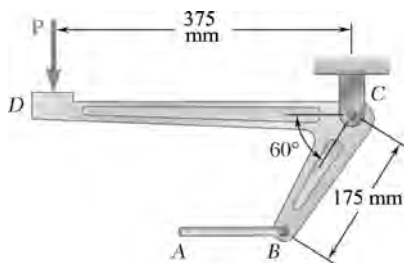
$$C = \frac{900 \text{ N}}{\cos 22.006^\circ} = 970.7 \text{ N}$$

Free-Body Diagram:
(Three-Force body)



$$P = 364 \text{ N} \downarrow \blacktriangleleft$$

$$C = 971 \text{ N} \nearrow 22.0^\circ \blacktriangleleft$$



PROBLEM 4.64

Using the method of Section 4.7, solve Problem 4.18.

PROBLEM 4.18 Determine the maximum tension that can be developed in cable AB if the maximum allowable value of the reaction at C is 1125 N.

SOLUTION

Reaction at C must pass through E , where D and the force T intersect.

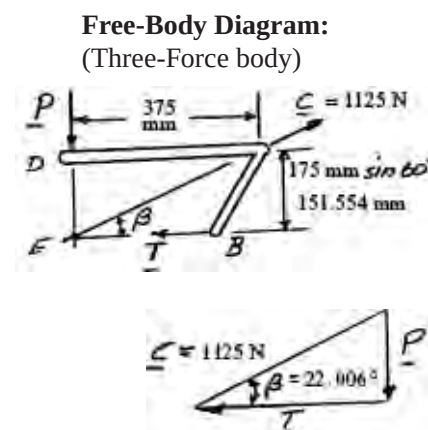
$$\tan \beta = \frac{151.554 \text{ mm}}{375 \text{ mm}}$$

$$\beta = 22.006^\circ$$

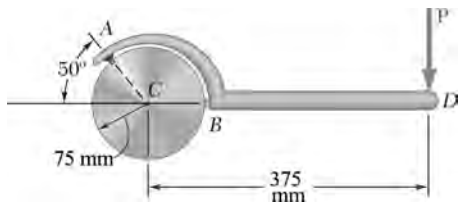
Force triangle

$$T = (1125 \text{ N}) \cos 22.006^\circ$$

$$T = 1043.04 \text{ N}$$



$$T = 1043 \text{ N} \quad \blacktriangleleft$$



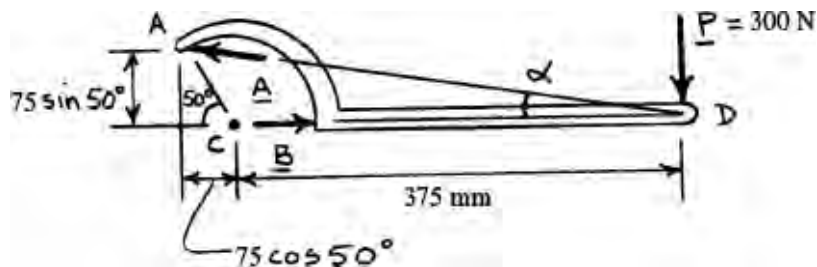
PROBLEM 4.65

The spanner shown is used to rotate a shaft. A pin fits in a hole at *A*, while a flat, frictionless surface rests against the shaft at *B*. If a 300-N force **P** is exerted on the spanner at *D*, find the reactions at *A* and *B*.

SOLUTION

Free-Body Diagram:

(Three-Force body)

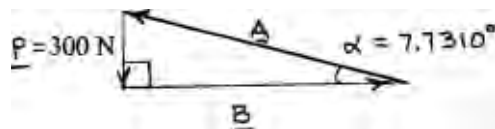


The line of action of **A** must pass through *D*, where **B** and **P** intersect.

$$\begin{aligned}\tan \alpha &= \frac{75 \sin 50^\circ}{75 \cos 50^\circ + 375} \\ &= 0.135756 \\ \alpha &= 7.7310^\circ\end{aligned}$$

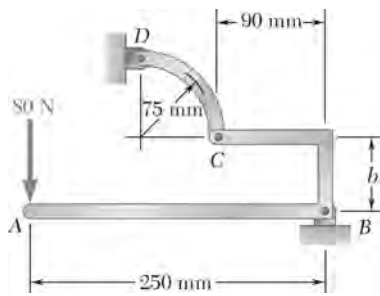
Force triangle

$$\begin{aligned}A &= \frac{300 \text{ N}}{\sin 7.7310^\circ} \\ &= 2230.11 \text{ N} \\ B &= \frac{300 \text{ N}}{\tan 7.7310^\circ} \\ &= 2209.842 \text{ N}\end{aligned}$$



$$\mathbf{A} = 2230 \text{ N} \nearrow 7.73^\circ \blacktriangleleft$$

$$\mathbf{B} = 2210 \text{ N} \rightarrow \blacktriangleleft$$



PROBLEM 4.66

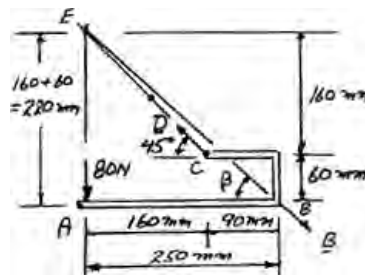
Determine the reactions at B and D when $b = 60$ mm.

SOLUTION

Since CD is a two-force member, the line of action of reaction at D must pass through Points C and D . 

Free-Body Diagram:

(Three-Force body)

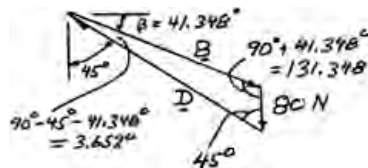


Reaction at B must pass through E , where the reaction at D and 80 N force intersect.

$$\tan \beta = \frac{220 \text{ mm}}{250 \text{ mm}}$$

$$\beta = 41.348^\circ$$

Force triangle



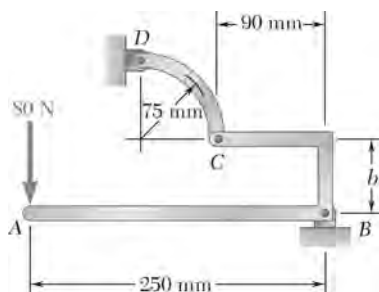
Law of sines

$$\frac{80 \text{ N}}{\sin 3.652^\circ} = \frac{B}{\sin 45^\circ} = \frac{D}{\sin 131.348^\circ}$$

$$B = 888.0 \text{ N}$$

$$D = 942.8 \text{ N}$$

$$B = 888 \text{ N} \searrow 41.3^\circ \quad D = 943 \text{ N} \nearrow 45.0^\circ \blacktriangleleft$$



PROBLEM 4.67

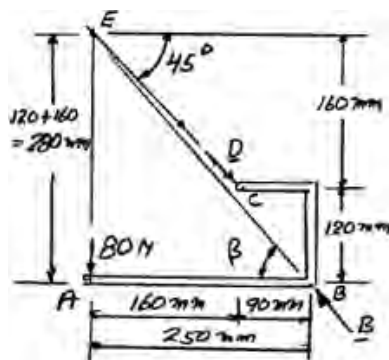
Determine the reactions at B and D when $b = 120$ mm.

SOLUTION

Since CD is a two-force member, line of action of reaction at D must pass through C and D .

Free-Body Diagram:

(Three-Force body)

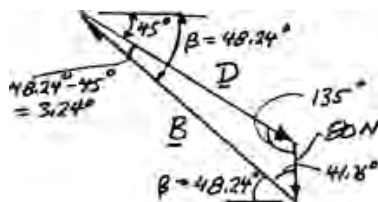


Reaction at B must pass through E , where the reaction at D and 80-N force intersect.

$$\tan \beta = \frac{280 \text{ mm}}{250 \text{ mm}}$$

$$\beta = 48.24^\circ$$

Force triangle



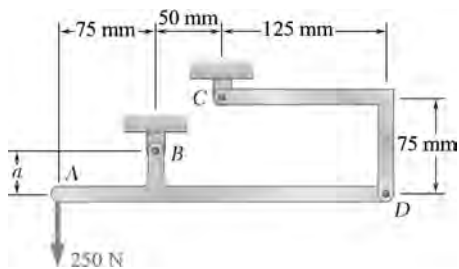
Law of sines

$$\frac{80 \text{ N}}{\sin 3.24^\circ} = \frac{B}{\sin 135^\circ} = \frac{D}{\sin 41.76^\circ}$$

$$B = 1000.9 \text{ N}$$

$$D = 942.8 \text{ N}$$

$$\mathbf{B} = 1001 \text{ N} \nearrow 48.2^\circ \quad \mathbf{D} = 943 \text{ N} \nwarrow 45.0^\circ \blacktriangleleft$$



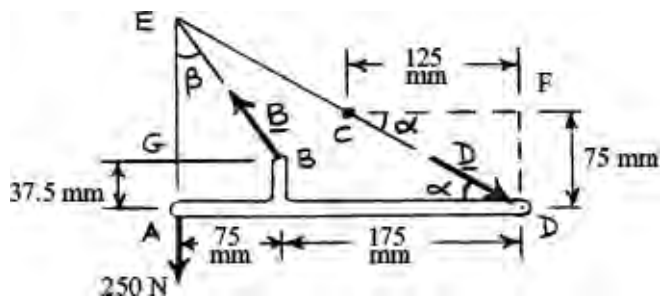
PROBLEM 4.68

Determine the reactions at B and C when $a = 37.5$ mm.

SOLUTION

Since CD is a two-force member, the force it exerts on member ABD is directed along DC .

Free-Body Diagram of ABD : (Three-Force member)



The reaction at B must pass through E , where D and the 250 N load intersect.

Triangle CFD : $\tan \alpha = \frac{75}{125} = 0.6$
 $\alpha = 30.964^\circ$

Triangle EAD : $AE = 250 \tan \alpha = 150$ mm
 $GE = AE - AG = 150 - 37.5 = 112.5$ mm

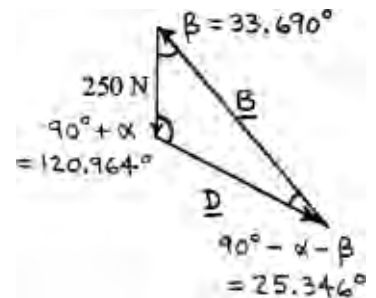
Triangle EGB : $\tan \beta = \frac{GB}{GE} = \frac{75}{112.5}$
 $\beta = 33.690^\circ$

Force triangle

$$\frac{B}{\sin 120.964^\circ} = \frac{D}{\sin 33.690^\circ} = \frac{250 \text{ N}}{\sin 25.346^\circ}$$

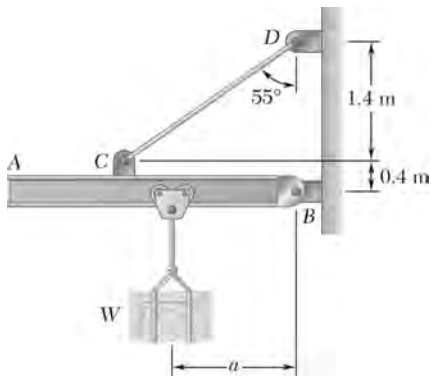
$$B = 500.773 \text{ N}$$

$$D = 323.943 \text{ N}$$



$B = 501 \text{ N}$ $\nearrow 56.3^\circ$ ◀

$C = D = 324 \text{ N}$ $\nearrow 31.0^\circ$ ◀



PROBLEM 4.69

A 50-kg crate is attached to the trolley-beam system shown. Knowing that $a = 1.5$ m, determine (a) the tension in cable CD, (b) the reaction at B.

SOLUTION

Three-Force body: $\mathbf{W}$ and $\mathbf{T}_{CD}$ intersect at E .

$$\tan \beta = \frac{0.7497 \text{ m}}{1.5 \text{ m}}$$

$$\beta = 26.56^\circ$$

Force triangle 3 forces intersect at E .

$$W = (50 \text{ kg}) 9.81 \text{ m/s}^2$$

$$= 490.5 \text{ N}$$

Law of sines

$$\frac{490.5 \text{ N}}{\sin 61.56^\circ} = \frac{T_{CD}}{\sin 63.44^\circ} = \frac{B}{\sin 55^\circ}$$

$$T_{CD} = 498.9 \text{ N}$$

$$B = 456.9 \text{ N}$$

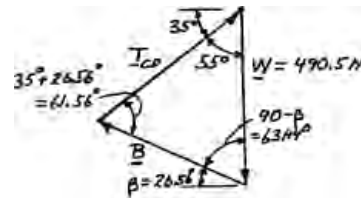
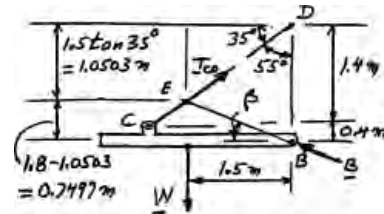
(a)

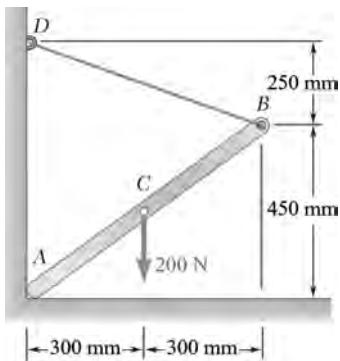
$$T_{CD} = 499 \text{ N} \blacktriangleleft$$

(b)

$$\mathbf{B} = 457 \text{ N} \nearrow 26.6^\circ \blacktriangleleft$$

Free-Body Diagram:



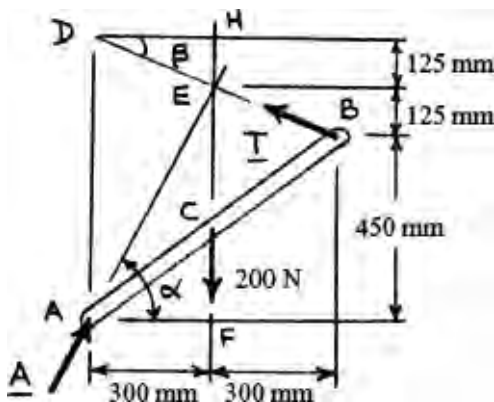


PROBLEM 4.71

One end of rod AB rests in the corner A and the other end is attached to cord BD . If the rod supports a 200-N load at its midpoint C , find the reaction at A and the tension in the cord.

SOLUTION

Free-Body Diagram: (Three-Force body)



The line of action of reaction at A must pass through E , where T and the 200 N load intersect.

$$\tan \alpha = \frac{EF}{AF} = \frac{575}{300}$$

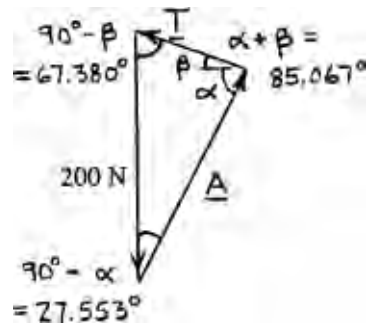
$$\alpha = 62.447^\circ$$

$$\tan \beta = \frac{EH}{DH} = \frac{125}{300}$$

$$\beta = 22.620^\circ$$

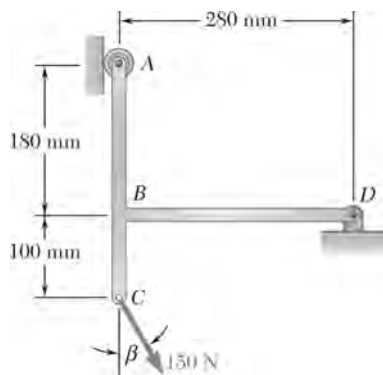
Force triangle

$$\frac{A}{\sin 67.380^\circ} = \frac{T}{\sin 27.553^\circ} = \frac{200 \text{ N}}{\sin 85.067^\circ}$$



$$A = 185.3 \text{ N} \nearrow 62.4^\circ \blacktriangleleft$$

$$T = 92.9 \text{ N} \blacktriangleleft$$

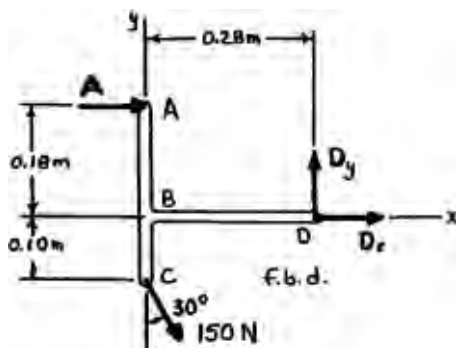


PROBLEM 4.72

Determine the reactions at A and D when $\beta = 30^\circ$.

SOLUTION

From f.b.d. of frame ABCD



$$\begin{aligned} \curvearrowleft \Sigma M_D = 0: & -A(0.18 \text{ m}) + [(150 \text{ N}) \sin 30^\circ](0.10 \text{ m}) \\ & + [(150 \text{ N}) \cos 30^\circ](0.28 \text{ m}) = 0 \end{aligned}$$

$$A = 243.74 \text{ N}$$

$$\text{or } \mathbf{A} = 244 \text{ N} \rightarrow \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: (243.74 \text{ N}) + (150 \text{ N}) \sin 30^\circ + D_x = 0$$

$$D_x = -318.74 \text{ N}$$

$$\uparrow \Sigma F_y = 0: D_y - (150 \text{ N}) \cos 30^\circ = 0$$

$$D_y = 129.904 \text{ N}$$

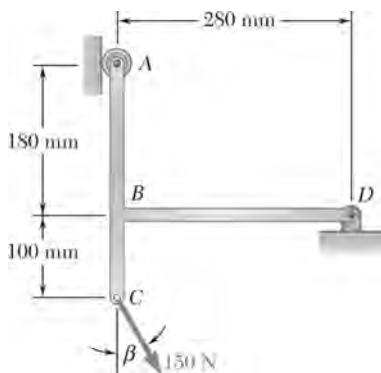
Then

$$\begin{aligned} D &= \sqrt{(D_x)^2 + D_y^2} \\ &= \sqrt{(318.74)^2 + (129.904)^2} \\ &= 344.19 \text{ N} \end{aligned}$$

and

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{D_y}{D_x} \right) \\ &= \tan^{-1} \left(\frac{129.904}{-318.74} \right) \\ &= -22.174^\circ \end{aligned}$$

$$\text{or } \mathbf{D} = 344 \text{ N} \nearrow 22.2^\circ \blacktriangleleft$$

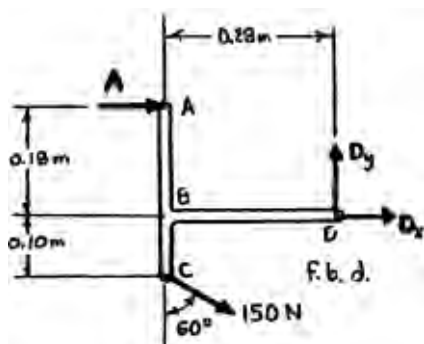


PROBLEM 4.73

Determine the reactions at A and D when $\beta = 60^\circ$.

SOLUTION

From f.b.d. of frame ABCD



$$\begin{aligned} \curvearrowleft \Sigma M_D = 0: & -A(0.18 \text{ m}) + [(150 \text{ N}) \sin 60^\circ](0.10 \text{ m}) \\ & + [(150 \text{ N}) \cos 60^\circ](0.28 \text{ m}) = 0 \end{aligned}$$

$$A = 188.835 \text{ N}$$

$$\text{or } \mathbf{A} = 188.8 \text{ N} \rightarrow \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: (188.835 \text{ N}) + (150 \text{ N}) \sin 60^\circ + D_x = 0$$

$$D_x = -318.74 \text{ N}$$

$$\uparrow \Sigma F_y = 0: D_y - (150 \text{ N}) \cos 60^\circ = 0$$

$$D_y = 75.0 \text{ N}$$

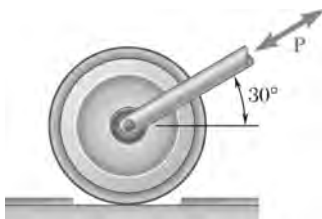
Then

$$\begin{aligned} D &= \sqrt{(D_x)^2 + (D_y)^2} \\ &= \sqrt{(318.74)^2 + (75.0)^2} \\ &= 327.44 \text{ N} \end{aligned}$$

and

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{D_y}{D_x} \right) \\ &= \tan^{-1} \left(\frac{75.0}{-318.74} \right) \\ &= -13.2409^\circ \end{aligned}$$

$$\text{or } \mathbf{D} = 327 \text{ N} \nearrow 13.24^\circ \blacktriangleleft$$

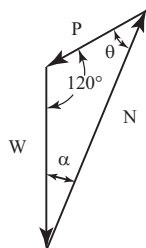
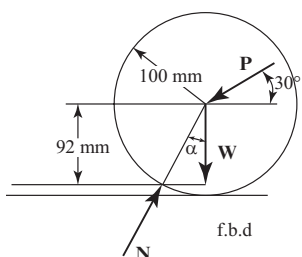


PROBLEM 4.74

A 20-kg roller, of diameter 200 mm, which is to be used on a tile floor, is resting directly on the subflooring as shown. Knowing that the thickness of each tile is 8 mm, determine the force **P** required to move the roller onto the tiles if the roller is (a) pushed to the left, (b) pulled to the right.

SOLUTION

Free-Body Diagram:



- (a) Based on the roller having impending motion to the left, the only contact between the roller and floor will be at the edge of the tile.

First note $W = mg = (20 \text{ kg})(9.81 \text{ m/s}^2) = 196.2 \text{ N}$

From the geometry of the three forces acting on the roller

$$\alpha = \cos^{-1} \left(\frac{92 \text{ mm}}{100 \text{ mm}} \right) = 23.074^\circ$$

and $\theta = 90^\circ - 30^\circ - \alpha = 60^\circ - 23.074^\circ = 36.926^\circ$

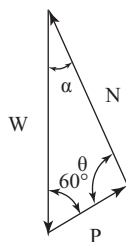
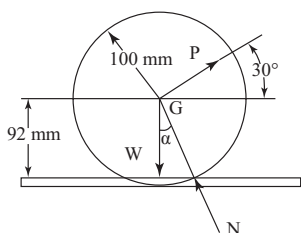
Applying the law of sines to the force triangle,

$$\frac{W}{\sin \theta} = \frac{P}{\sin \alpha}$$

or $\frac{196.2 \text{ N}}{\sin 36.926^\circ} = \frac{P}{\sin 23.074^\circ}$
 $\therefore P = 127.991 \text{ N}$

or **P** = 128.0 N ↗ 30° ◀

Free-Body Diagram:



- (b) Based on the roller having impending motion to the right, the only contact between the roller and floor will be at the edge of the tile.

First note $W = mg = (20 \text{ kg})(9.81 \text{ m/s}^2) = 196.2 \text{ N}$

From the geometry of the three forces acting on the roller

$$\alpha = \cos^{-1} \left(\frac{92 \text{ mm}}{100 \text{ mm}} \right) = 23.074^\circ$$

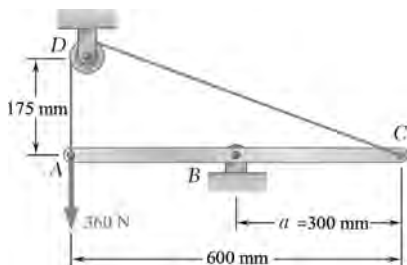
and $\theta = 90^\circ + 30^\circ + \alpha = 120^\circ - 23.074^\circ = 96.926^\circ$

Applying the law of sines to the force triangle,

$$\frac{W}{\sin \theta} = \frac{P}{\sin \alpha}$$

or $\frac{196.2 \text{ N}}{\sin 96.926^\circ} = \frac{P}{\sin 23.074^\circ}$
 $\therefore P = 77.460 \text{ N}$

or **P** = 77.5 N ↗ 30° ◀



PROBLEM 4.75

Member ABC is supported by a pin and bracket at B and by an inextensible cord attached at A and C and passing over a frictionless pulley at D . The tension may be assumed to be the same in portions AD and CD of the cord. For the loading shown and neglecting the size of the pulley, determine the tension in the cord and the reaction at B .

SOLUTION

Reaction at B must pass through D .

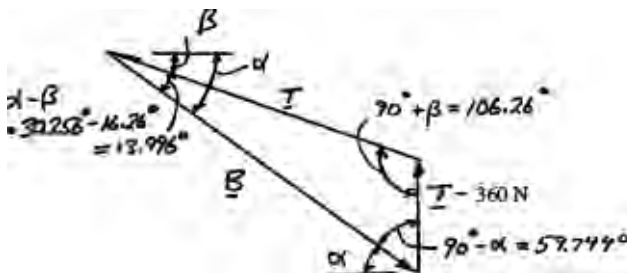
$$\tan \alpha = \frac{175 \text{ mm}}{300 \text{ mm}}$$

$$\alpha = 30.256^\circ$$

$$\tan \beta = \frac{175 \text{ mm}}{600 \text{ mm}}$$

$$\beta = 16.26^\circ$$

Force triangle



Law of sines

$$\frac{T}{\sin 59.744^\circ} = \frac{T - 360 \text{ N}}{\sin 13.996^\circ} = \frac{B}{\sin 106.26^\circ}$$

$$T(\sin 13.996^\circ) = (T - 360 \text{ N})(\sin 59.744^\circ)$$

$$T(0.24185) = (T - 360)(0.86378)$$

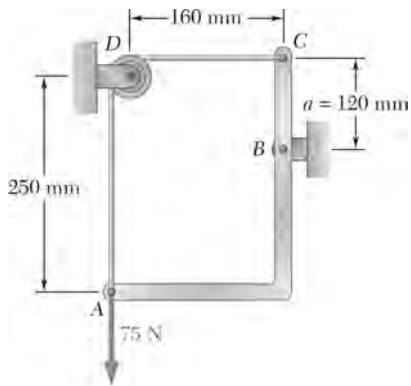
$$T = 500 \text{ N}$$

$$T = 500 \text{ N} \quad \blacktriangleleft$$

$$B = (500 \text{ N}) \frac{\sin 106.26^\circ}{\sin 59.744^\circ}$$

$$= 555.695 \text{ N}$$

$$B = 556 \text{ N} \quad \swarrow 30.3^\circ \quad \blacktriangleleft$$



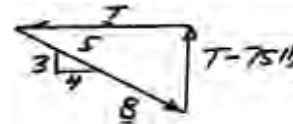
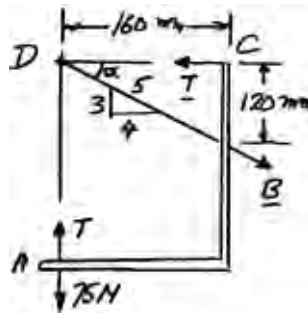
PROBLEM 4.76

Member ABC is supported by a pin and bracket at B and by an inextensible cord attached at A and C and passing over a frictionless pulley at D . The tension may be assumed to be the same in portions AD and CD of the cord. For the loading shown and neglecting the size of the pulley, determine the tension in the cord and the reaction at B .

SOLUTION

Free-Body Diagram:

Force triangle



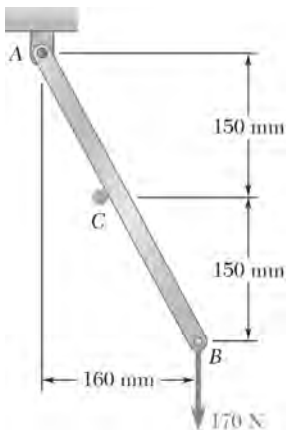
Reaction at B must pass through D .

$$\tan \alpha = \frac{120}{160}; \quad \alpha = 36.9^\circ$$

$$\frac{T}{4} = \frac{T - 75 \text{ lb}}{3} = \frac{B}{5}$$

$$3T = 4T - 300; \quad T = 300 \text{ lb}$$

$$B = \frac{5}{4}T = \frac{5}{4}(300 \text{ lb}) = 375 \text{ lb} \quad \mathbf{B} = 375 \text{ lb} \searrow 36.9^\circ$$



PROBLEM 4.77

Rod AB is supported by a pin and bracket at A and rests against a frictionless peg at C . Determine the reactions at A and C when a 170-N vertical force is applied at B .

SOLUTION

The reaction at A must pass through D where C and 170-N force intersect.

$$\tan \alpha = \frac{160 \text{ mm}}{300 \text{ mm}}$$

$$\alpha = 28.07^\circ$$

We note that triangle ABD is isosceles (since $AC = BC$) and, therefore

$$\angle CAD = \alpha = 28.07^\circ$$

Also, since $CD \perp CB$, reaction C forms angle $\alpha = 28.07^\circ$ with horizontal.

Force triangle

We note that A forms angle 2α with vertical. Thus A and C form angle

$$180^\circ - (90^\circ - \alpha) - 2\alpha = 90^\circ - \alpha$$

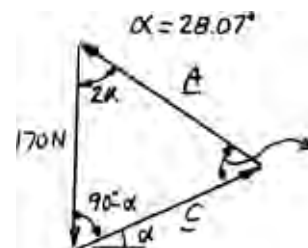
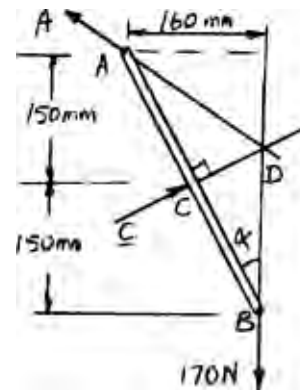
Force triangle is isosceles and we have

$$A = 170 \text{ N}$$

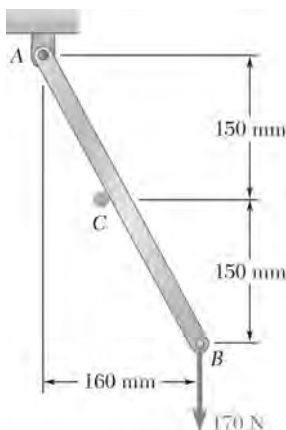
$$C = 2(170 \text{ N}) \sin \alpha$$

$$= 160.0 \text{ N}$$

Free-Body Diagram: (Three-Force body)



$$A = 170.0 \text{ N} \nearrow 33.9^\circ \quad C = 160.0 \text{ N} \nearrow 28.1^\circ \blacktriangleleft$$



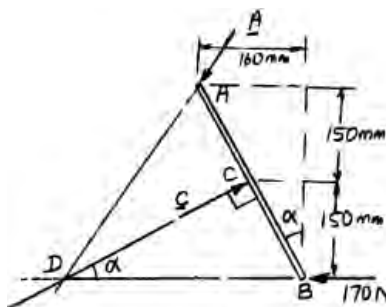
PROBLEM 4.78

Solve Problem 4.77, assuming that the 170-N force applied at B is horizontal and directed to the left.

PROBLEM 4.77 Rod AB is supported by a pin and bracket at A and rests against a frictionless peg at C . Determine the reactions at A and C when a 170-N vertical force is applied at B .

SOLUTION

Free-Body Diagram: (Three-Force body)



The reaction at A must pass through D , where C and the 170-N force intersect.

$$\tan \alpha = \frac{160 \text{ mm}}{300 \text{ mm}}$$

$$\alpha = 28.07^\circ$$

We note that triangle ADB is isosceles (since $AC = BC$). Therefore $\angle A = \angle B = 90^\circ - \alpha$.

Also

$$\angle ADB = 2\alpha$$

Force triangle

The angle between A and C must be $2\alpha - \alpha = \alpha$

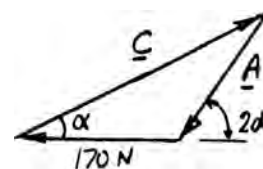
Thus, force triangle is isosceles and

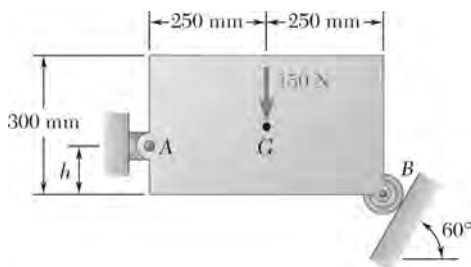
$$\alpha = 28.07^\circ$$

$$A = 170.0 \text{ N}$$

$$C = 2(170 \text{ N}) \cos \alpha = 300 \text{ N}$$

$$A = 170.0 \text{ N} \nearrow 56.1^\circ \quad C = 300 \text{ N} \nearrow 28.1^\circ \blacktriangleleft$$





PROBLEM 4.79

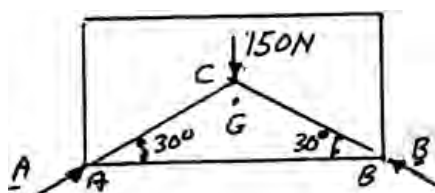
Using the method of Section 4.7, solve Problem 4.21.

PROBLEM 4.21 Determine the reactions at A and B when
(a) $h = 0$, (b) $h = 200$ mm.

SOLUTION

Free-Body Diagram:

(a) $h = 0$

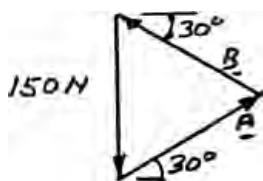


Reaction A must pass through C where 150-N weight and B intersect.

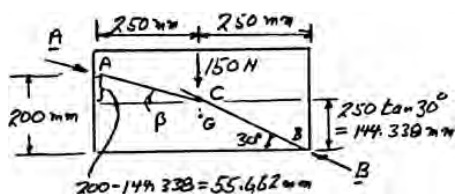
Force triangle is equilateral

$$A = 150.0 \text{ N} \nearrow 30.0^\circ \blacktriangleleft$$

$$B = 150.0 \text{ N} \nearrow 30.0^\circ \blacktriangleleft$$



(b) $h = 200$ mm



$$\tan \beta = \frac{55.662}{250}$$

$$\beta = 12.552^\circ$$

Force triangle

Law of sines

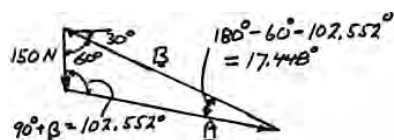
$$\frac{150 \text{ N}}{\sin 17.448^\circ} = \frac{A}{\sin 60^\circ} = \frac{B}{\sin 102.552^\circ}$$

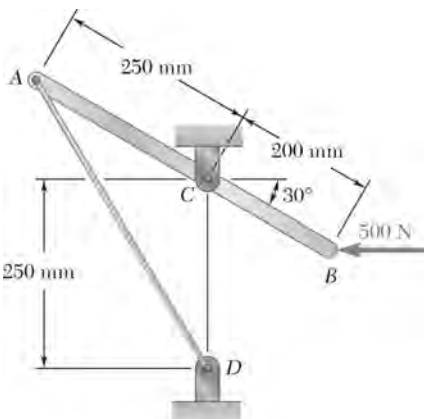
$$A = 433.247 \text{ N}$$

$$B = 488.31 \text{ N}$$

$$A = 433 \text{ N} \nearrow 12.55^\circ \blacktriangleleft$$

$$B = 488 \text{ N} \nearrow 30.0^\circ \blacktriangleleft$$





PROBLEM 4.80

Using the method of Section 4.7, solve Problem 4.28.

PROBLEM 4.28 A lever AB is hinged at C and attached to a control cable at A . If the lever is subjected to a 500-N horizontal force at B , determine (a) the tension in the cable, (b) the reaction at C .

SOLUTION

Reaction at C must pass through E , where F_{AD} and 500-N force intersect.

Since $AC = CD = 250$ mm, triangle ACD is isosceles.

We have $\angle C = 90^\circ + 30^\circ = 120^\circ$

and $\angle A = \angle D = \frac{1}{2}(180^\circ - 120^\circ) = 30^\circ$

On the other hand, from triangle BCF :

$$CF = (BC) \sin 30^\circ = 200 \sin 30^\circ = 100 \text{ mm}$$

$$FD = CD - CF = 250 - 100 = 150 \text{ mm}$$

From triangle EFD , and since $\angle D = 30^\circ$:

$$EF = (FD) \tan 30^\circ = 150 \tan 30^\circ = 86.60 \text{ mm}$$

From triangle EFC :

$$\tan \alpha = \frac{CF}{EF} = \frac{100 \text{ mm}}{86.60 \text{ mm}}$$

$$\alpha = 49.11^\circ$$

Force triangle

Law of sines

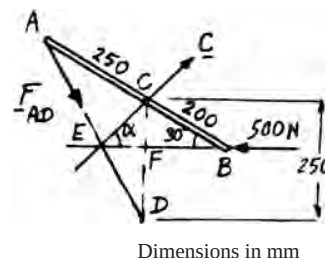
$$\frac{F_{AD}}{\sin 49.11^\circ} = \frac{C}{\sin 60^\circ} = \frac{500 \text{ N}}{\sin 70.89^\circ}$$

$$F_{AD} = 400 \text{ N}, \quad C = 458 \text{ N}$$

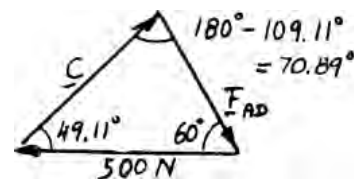
(a)

(b)

Free-Body Diagram:
(Three-Force body)

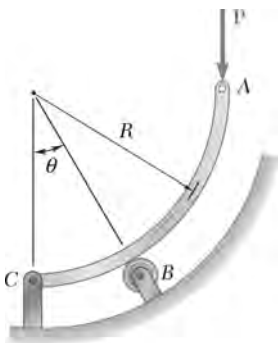


Dimensions in mm



$$F_{AD} = 400 \text{ N} \quad \blacktriangleleft$$

$$C = 458 \text{ N} \quad \blacktriangleleft 49.1^\circ$$



PROBLEM 4.81

Knowing that $\theta = 30^\circ$, determine the reaction (a) at B, (b) at C.

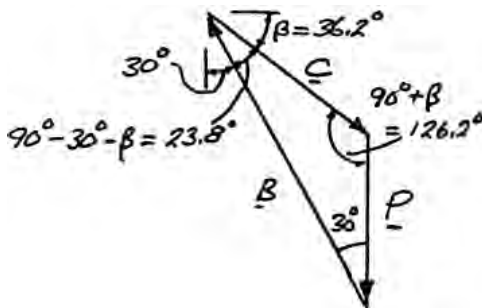
SOLUTION

Reaction at C must pass through D where force **P** and reaction at B intersect.

In $\triangle CDE$:

$$\begin{aligned}\tan \beta &= \frac{(\sqrt{3}-1)R}{R} \\ &= \sqrt{3}-1 \\ \beta &= 36.2^\circ\end{aligned}$$

Force triangle



Law of sines

$$\begin{aligned}\frac{P}{\sin 23.8^\circ} &= \frac{B}{\sin 126.2^\circ} = \frac{C}{\sin 30^\circ} \\ B &= 2.00P \\ C &= 1.239P\end{aligned}$$

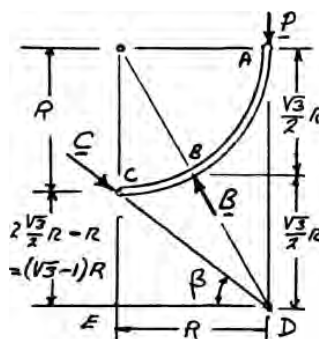
(a)

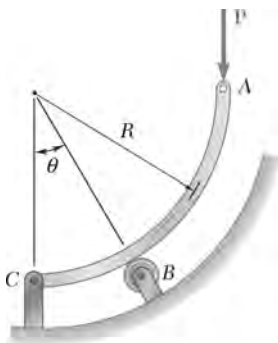
$$\mathbf{B} = 2P \nearrow 60.0^\circ \blacktriangleleft$$

(b)

$$\mathbf{C} = 1.239P \nearrow 36.2^\circ \blacktriangleleft$$

Free-Body Diagram:
(Three-Force body)





PROBLEM 4.82

Knowing that $\theta = 60^\circ$, determine the reaction (a) at B, (b) at C.

SOLUTION

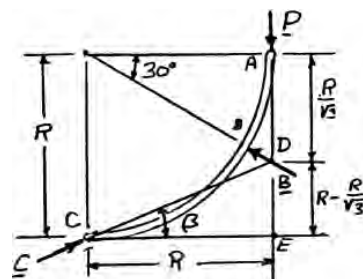
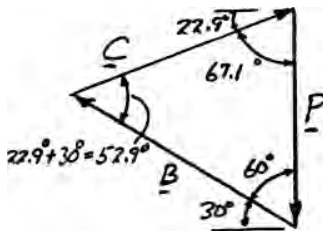
Reaction at C must pass through D where force **P** and reaction at B intersect.

Free-Body Diagram:
(Three-Force body)

In $\triangle CDE$:

$$\begin{aligned}\tan \beta &= \frac{R - \frac{R}{\sqrt{3}}}{R} \\ &= 1 - \frac{1}{\sqrt{3}} \\ \beta &= 22.9^\circ\end{aligned}$$

Force triangle



Law of sines

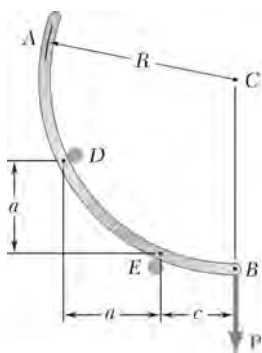
$$\begin{aligned}\frac{P}{\sin 52.9^\circ} &= \frac{B}{\sin 67.1^\circ} = \frac{C}{\sin 30^\circ} \\ B &= 1.155P \\ C &= 1.086P\end{aligned}$$

(a)

$$B = 1.155P \quad \angle 30.0^\circ \quad \blacktriangleleft$$

(b)

$$C = 1.086P \quad \angle 22.9^\circ \quad \blacktriangleleft$$



PROBLEM 4.83

Rod AB is bent into the shape of an arc of circle and is lodged between two pegs D and E . It supports a load P at end B . Neglecting friction and the weight of the rod, determine the distance c corresponding to equilibrium when $a = 20$ mm and $R = 100$ mm.

SOLUTION

Since

$$y_{ED} = x_{ED} = a,$$

Slope of ED is $\searrow 45^\circ$

slope of HC is $\swarrow 45^\circ$

Also

$$DE = \sqrt{2}a$$

and

$$DH = HE = \left(\frac{1}{2}\right)DE = \frac{a}{\sqrt{2}}$$

For triangles DHC and EHC

$$\sin \beta = \frac{\frac{a}{\sqrt{2}}}{R} = \frac{a}{\sqrt{2}R}$$

Now

$$c = R \sin(45^\circ - \beta)$$

For

$$a = 20 \text{ mm} \quad \text{and} \quad R = 100 \text{ mm}$$

$$\sin \beta = \frac{20 \text{ mm}}{\sqrt{2}(100 \text{ mm})}$$

$$= 0.141421$$

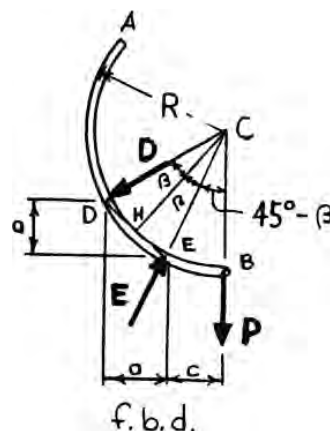
$$\beta = 8.1301^\circ$$

and

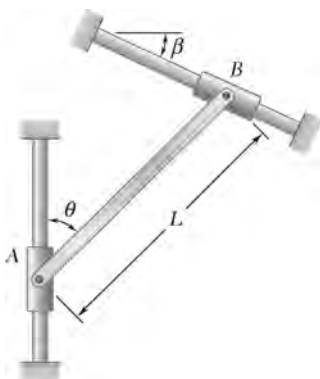
$$c = (100 \text{ mm}) \sin(45^\circ - 8.1301^\circ)$$

$$= 60.00 \text{ mm}$$

Free-Body Diagram:



$$\text{or } c = 60.0 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.84

A slender rod of length L is attached to collars that can slide freely along the guides shown. Knowing that the rod is in equilibrium, derive an expression for the angle θ in terms of the angle β .

SOLUTION

As shown in the free-body diagram of the slender rod AB , the three forces intersect at C . From the force geometry

$$\tan \beta = \frac{x_{GB}}{y_{AB}}$$

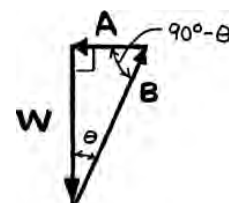
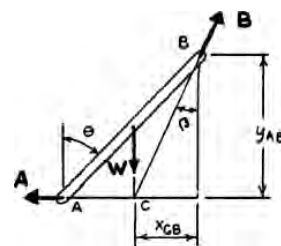
where

$$y_{AB} = L \cos \theta$$

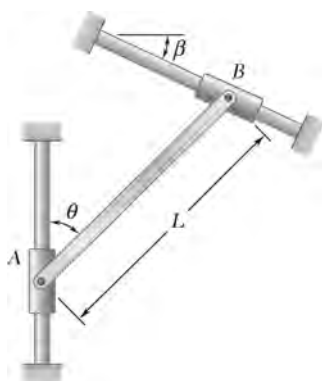
and

$$\begin{aligned} x_{GB} &= \frac{1}{2} L \sin \theta \\ \tan \beta &= \frac{\frac{1}{2} L \sin \theta}{L \cos \theta} \\ &= \frac{1}{2} \tan \theta \end{aligned}$$

Free-Body Diagram:



$$\text{or } \tan \theta = 2 \tan \beta \quad \blacktriangleleft$$



PROBLEM 4.85

An 8 kg slender rod of length L is attached to collars that can slide freely along the guides shown. Knowing that the rod is in equilibrium and that $\beta = 30^\circ$, determine (a) the angle θ that the rod forms with the vertical, (b) the reactions at A and B.

SOLUTION

- (a) As shown in the free-body diagram of the slender rod AB, the three forces intersect at C. From the geometry of the forces

$$\tan \beta = \frac{x_{CB}}{y_{BC}}$$

where

$$x_{CB} = \frac{1}{2} L \sin \theta$$

and

$$y_{BC} = L \cos \theta$$

$$\tan \beta = \frac{1}{2} \tan \theta$$

or

$$\tan \theta = 2 \tan \beta$$

For

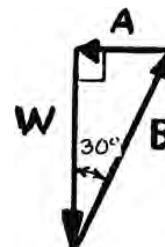
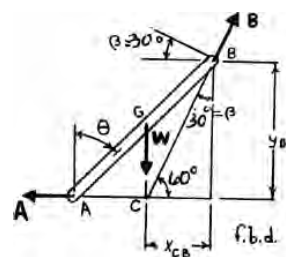
$$\beta = 30^\circ$$

$$\tan \theta = 2 \tan 30^\circ$$

$$= 1.15470$$

$$\theta = 49.107^\circ$$

Free-Body Diagram:



or $\theta = 49.1^\circ \blacktriangleleft$

- (b)

$$W = mg = (8 \text{ kg})(9.81 \text{ m/s}^2) = 78.480 \text{ N}$$

From force triangle

$$A = W \tan \beta$$

$$= (78.480 \text{ N}) \tan 30^\circ$$

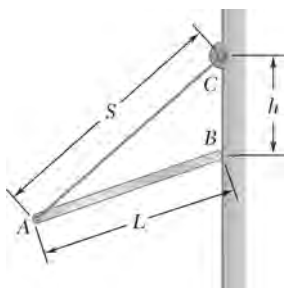
$$= 45.310 \text{ N}$$

or $A = 45.3 \text{ N} \blacktriangleleft$

and

$$B = \frac{W}{\cos \beta} = \frac{78.480 \text{ N}}{\cos 30^\circ} = 90.621 \text{ N}$$

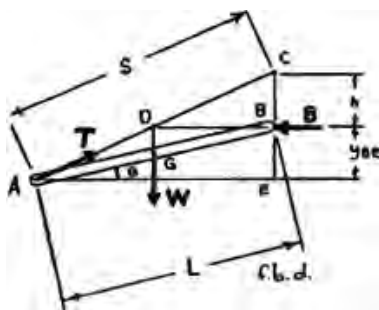
or $B = 90.6 \text{ N} \blacktriangleleft 60.0^\circ$



PROBLEM 4.86

A slender uniform rod of length L is held in equilibrium as shown, with one end against a frictionless wall and the other end attached to a cord of length S . Derive an expression for the distance h in terms of L and S . Show that this position of equilibrium does not exist if $S > 2L$.

SOLUTION



From the f.b.d. of the three-force member AB , forces must intersect at D . Since the force T intersects Point D , directly above G ,

$$y_{BE} = h$$

$$\text{For triangle } ACE: \quad S^2 = (AE)^2 + (2h)^2 \quad (1)$$

$$\text{For triangle } ABE: \quad L^2 = (AE)^2 + (h)^2 \quad (2)$$

Subtracting Equation (2) from Equation (1)

$$S^2 - L^2 = 3h^2 \quad (3)$$

or

$$h = \sqrt{\frac{S^2 - L^2}{3}} \quad \blacktriangleleft$$

As length S increases relative to length L , angle θ increases until rod AB is vertical. At this vertical position:

$$h + L = S \quad \text{or} \quad h = S - L$$

Therefore, for all positions of AB

$$h \geq S - L \quad (4)$$

or

$$\sqrt{\frac{S^2 - L^2}{3}} \geq S - L$$

or

$$\begin{aligned} S^2 - L^2 &\geq 3(S - L)^2 \\ &= 3(S^2 - 2SL + L^2) \\ &= 3S^2 - 6SL + 3L^2 \end{aligned}$$

or

$$0 \geq 2S^2 - 6SL + 4L^2$$

and

$$0 \geq S^2 - 3SL + 2L^2 = (S - L)(S - 2L)$$



PROBLEM 4.86 (Continued)

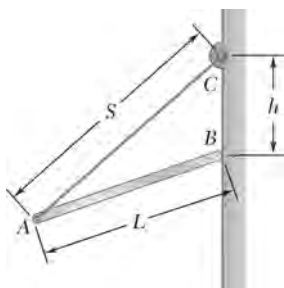
For $S - L = 0 \quad S = L$

Minimum value of S is L

For $S - 2L = 0 \quad S = 2L$

Maximum value of S is $2L$

Therefore, equilibrium does not exist if $S > 2L$ ◀



PROBLEM 4.87

A slender uniform rod of length $L = 500$ mm is held in equilibrium as shown, with one end against a frictionless wall and the other end attached to a cord of length $S = 750$ mm. Knowing that the mass of the rod is 4.5 kg, determine (a) the distance h , (b) the tension in the cord, (c) the reaction at B.

SOLUTION

From the f.b.d. of the three-force member AB , forces must intersect at D . Since the force T intersects Point D , directly above G ,

$$y_{BE} = h$$

$$\text{For triangle } ACE: \quad S^2 = (AE)^2 + (2h)^2 \quad (1)$$

$$\text{For triangle } ABE: \quad L^2 = (AE)^2 + (h)^2 \quad (2)$$

Subtracting Equation (2) from Equation (1)

$$S^2 - L^2 = 3h^2$$

or

$$h = \sqrt{\frac{S^2 - L^2}{3}}$$

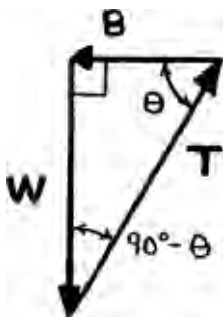
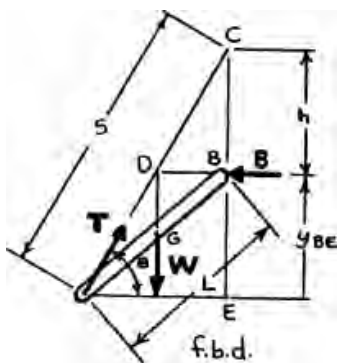
$$(a) \quad \text{For} \quad L = 500 \text{ mm} \quad \text{and} \quad S = 750 \text{ mm}$$

$$h = \sqrt{\frac{(750)^2 - (500)^2}{3}} = 322.7486 \text{ mm} \quad \text{or} \quad h = 322 \text{ mm} \quad \blacktriangleleft$$

$$(b) \quad \text{We have} \quad W = 4.5 \times 9.81 = 44.145 \text{ N}$$

and

$$\begin{aligned} \theta &= \sin^{-1} \left(\frac{2h}{S} \right) \\ &= \sin^{-1} \left[\frac{2(322.7486)}{750} \right] \\ \theta &= 59.391^\circ \end{aligned}$$



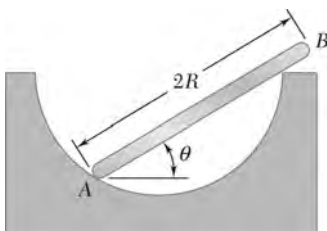
PROBLEM 4.87 (Continued)

From the force triangle

$$\begin{aligned} T &= \frac{W}{\sin \theta} \\ &= \frac{44.145 \text{ N}}{\sin 59.391^\circ} \\ &= 51.2919 \text{ N} \end{aligned} \quad \text{or } T = 51.3 \text{ N} \quad \blacktriangleleft$$

(c)

$$\begin{aligned} B &= \frac{W}{\tan \theta} \\ &= \frac{44.145 \text{ N}}{\tan 59.391^\circ} \\ &= 26.1166 \text{ N} \end{aligned} \quad \text{or } \mathbf{B} = 26.1 \text{ N} \quad \blacktriangleleft$$



PROBLEM 4.88

A uniform rod AB of length $2R$ rests inside a hemispherical bowl of radius R as shown. Neglecting friction, determine the angle θ corresponding to equilibrium.

SOLUTION

Based on the f.b.d., the uniform rod AB is a three-force body. Point E is the point of intersection of the three forces. Since force A passes through O , the center of the circle, and since force C is perpendicular to the rod, triangle ACE is a right triangle inscribed in the circle. Thus, E is a point on the circle.

Note that the angle α of triangle DOA is the central angle corresponding to the inscribed angle θ of triangle DCA .

$$\alpha = 2\theta$$

The horizontal projections of AE , (x_{AE}) , and AG , (x_{AG}) , are equal.

$$x_{AE} = x_{AG} = x_A$$

$$\text{or} \quad (AE) \cos 2\theta = (AG) \cos \theta$$

$$\text{and} \quad (2R) \cos 2\theta = R \cos \theta$$

$$\text{Now} \quad \cos 2\theta = 2 \cos^2 \theta - 1$$

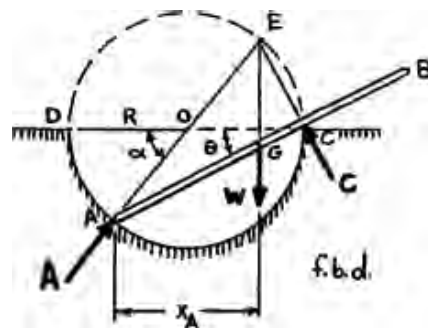
$$\text{then} \quad 4 \cos^2 \theta - 2 = \cos \theta$$

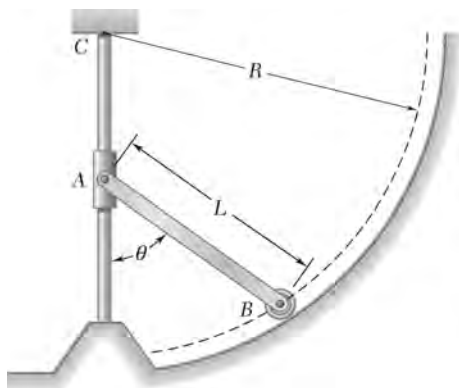
$$\text{or} \quad 4 \cos^2 \theta - \cos \theta - 2 = 0$$

Applying the quadratic equation

$$\cos \theta = 0.84307 \quad \text{and} \quad \cos \theta = -0.59307$$

$$\theta = 32.534^\circ \quad \text{and} \quad \theta = 126.375^\circ (\text{Discard}) \quad \text{or} \quad \theta = 32.5^\circ \blacktriangleleft$$





PROBLEM 4.89

A slender rod of length L and weight W is attached to a collar at A and is fitted with a small wheel at B . Knowing that the wheel rolls freely along a cylindrical surface of radius R , and neglecting friction, derive an equation in θ , L , and R that must be satisfied when the rod is in equilibrium.

SOLUTION

Reaction $\mathbf{B}$ must pass through D where $\mathbf{B}$ and $\mathbf{W}$ intersect.

Note that $\triangle ABC$ and $\triangle BGD$ are similar.

$$AC = AE = L \cos \theta$$

In $\triangle ABC$:

$$(CE)^2 + (BE)^2 = (BC)^2$$

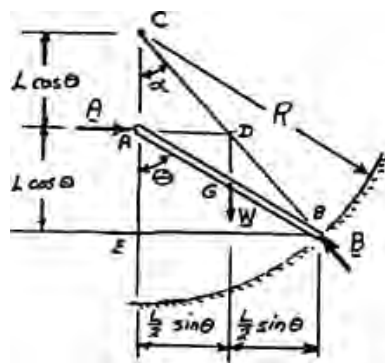
$$(2L \cos \theta)^2 + (L \sin \theta)^2 = R^2$$

$$\left(\frac{R}{L}\right)^2 = 4 \cos^2 \theta + \sin^2 \theta$$

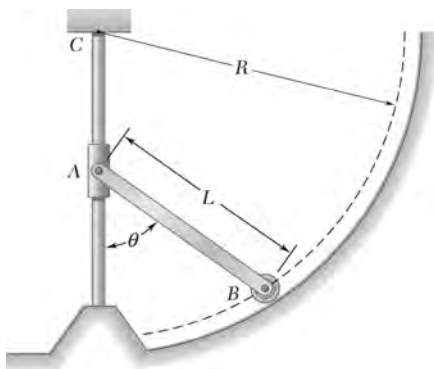
$$\left(\frac{R}{L}\right)^2 = 4 \cos^2 \theta + 1 - \cos^2 \theta$$

$$\left(\frac{R}{L}\right)^2 = 3 \cos^2 \theta + 1$$

Free-Body Diagram (Three-Force body)



$$\cos^2 \theta = \frac{1}{3} \left[\left(\frac{R}{L}\right)^2 - 1 \right] \quad \blacktriangleleft$$



PROBLEM 4.90

Knowing that for the rod of Problem 4.89, $L = 375$ mm, $R = 500$ mm, and $W = 45$ N, determine (a) the angle θ corresponding to equilibrium, (b) the reactions at A and B.

SOLUTION

See the solution to Problem 4.89 for free-body diagram and analysis leading to the following equation

$$\cos^2 \theta = \frac{1}{3} \left[\left(\frac{R}{L} \right)^2 - 1 \right]$$

For $L = 375$ mm, $R = 500$ mm, and $W = 45$ N

$$(a) \quad \cos^2 \theta = \frac{1}{3} \left[\left(\frac{500 \text{ mm}}{375 \text{ mm}} \right)^2 - 1 \right]; \quad \theta = 59.39^\circ \quad \theta = 59.4^\circ \blacktriangleleft$$

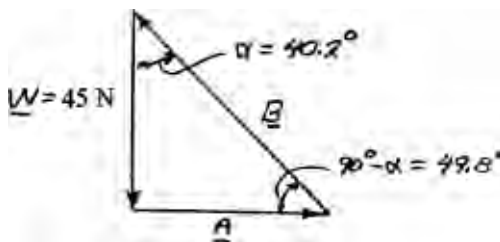
In $\triangle ABC$:

$$\tan \alpha = \frac{BE}{CE} = \frac{L \sin \theta}{2L \cos \theta} = \frac{1}{2} \tan \theta$$

$$\tan \alpha = \frac{1}{2} \tan 59.39^\circ = 0.8452$$

$$\alpha = 40.2^\circ$$

Force triangle

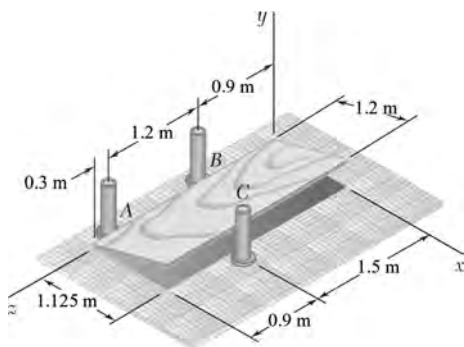


$$A = W \tan \alpha = (45 \text{ N}) \tan 40.2^\circ = 38.028 \text{ N}$$

$$B = \frac{W}{\cos \alpha} = \frac{(45 \text{ N})}{\cos 40.2^\circ} = 58.916 \text{ N}$$

$$(b) \quad \mathbf{A} = 38.0 \text{ N} \quad \blacktriangleleft$$

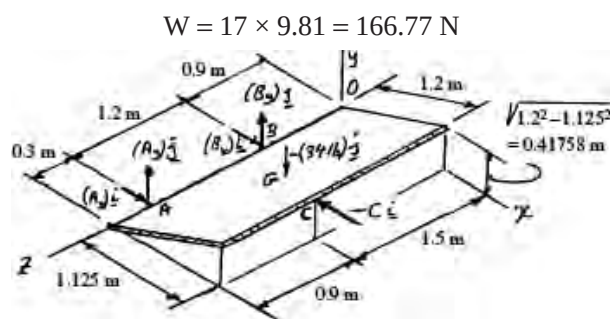
$$(c) \quad \mathbf{B} = 58.9 \text{ N} \quad \blacktriangleleft 49.8^\circ$$



PROBLEM 4.91

A 1.2 m × 2.4 m sheet of plywood of mass 17 kg has been temporarily placed among three pipe supports. The lower edge of the sheet rests on small collars at A and B and its upper edge leans against pipe C. Neglecting friction at all surfaces, determine the reactions at A, B, and C.

SOLUTION



$$W = 17 \times 9.81 = 166.77 \text{ N}$$

$$\mathbf{r}_{G/B} = \frac{1.125}{2} \mathbf{i} + \frac{0.41758}{2} \mathbf{j} + 0.3 \mathbf{k}$$

We have 5 unknowns and six equations of equilibrium.

Plywood sheet is free to move in z direction, but equilibrium is maintained ($\Sigma F_z = 0$).

$$\Sigma M_B = 0: \quad \mathbf{r}_{A/B} \times (A_x \mathbf{i} + A_y \mathbf{j}) + \mathbf{r}_{C/B} \times (-C \mathbf{i}) + \mathbf{r}_{G/B} \times (-W \mathbf{j}) = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1.2 \\ A_x & A_y & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.125 & 0.41758 & 0.6 \\ -C & 0 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.5625 & 0.20879 & 0.3 \\ 0 & -166.77 & 0 \end{vmatrix} = 0$$

$$-1.2A_y \mathbf{i} + 1.2A_x \mathbf{j} - 0.6C \mathbf{j} + 0.41758C \mathbf{k} + 50.031 \mathbf{i} - 93.808125 \mathbf{k} = 0$$

Equating coefficients of unit vectors to zero:

$$\mathbf{i}: \quad -1.2A_y + 50.031 = 0$$

$$A_y = 41.6925 \text{ N}$$

$$\mathbf{j}: \quad -0.6C + 1.2A_x = 0$$

$$A_x = \frac{1}{2}C = \frac{1}{2}(224.64) = 112.32 \text{ N}$$

$$\mathbf{k}: \quad 0.41758C - 93.808125 = 0$$

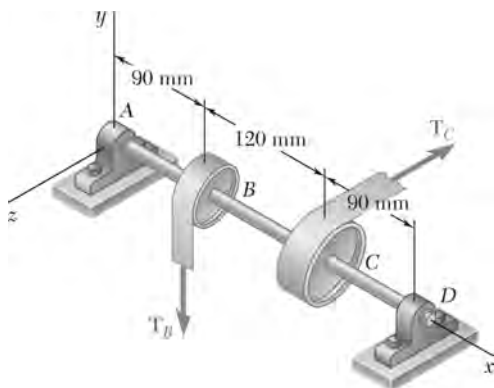
$$C = 224.64 \text{ N}$$

$$C = 225 \text{ N} \quad \blacktriangleleft$$

$$\Sigma F_x = 0: \quad A_x + B_x - C = 0: \quad B_x = 224.64 - 112.32 = 112.32 \text{ N}$$

$$\Sigma F_y = 0: \quad A_y + B_y - W = 0: \quad B_y = 166.77 - 41.6925 = 125.08 \text{ N}$$

$$\mathbf{A} = (112.3 \text{ N})\mathbf{i} + (41.7 \text{ N})\mathbf{j} \quad \mathbf{B} = (112.3 \text{ N})\mathbf{i} + (125.1 \text{ N})\mathbf{j} \quad \mathbf{C} = -(225 \text{ N})\mathbf{i} \quad \blacktriangleleft$$

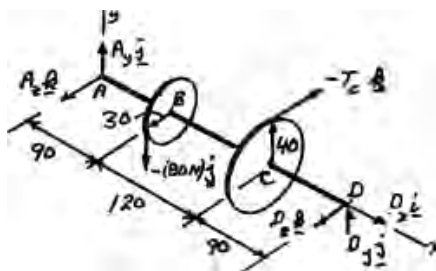


PROBLEM 4.92

Two tape spools are attached to an axle supported by bearings at A and D. The radius of spool B is 30 mm and the radius of spool C is 40 mm. Knowing that $T_B = 80$ N and that the system rotates at a constant rate, determine the reactions at A and D. Assume that the bearing at A does not exert any axial thrust and neglect the weights of the spools and axle.

SOLUTION

Dimensions in mm



We have six unknowns and six equations of equilibrium.

$$\begin{aligned}\Sigma M_A = 0: & (90\mathbf{i} + 30\mathbf{k}) \times (-80\mathbf{j}) + (210\mathbf{i} + 40\mathbf{j}) \times (-T_C\mathbf{k}) + (300\mathbf{i}) \times (D_x\mathbf{i} + D_y\mathbf{j} + D_z\mathbf{k}) = 0 \\ & -7200\mathbf{k} + 2400\mathbf{i} + 210T_C\mathbf{j} - 40T_C\mathbf{i} + 300D_y\mathbf{k} - 300D_z\mathbf{j} = 0\end{aligned}$$

Equate coefficients of unit vectors to zero:

$$\mathbf{i}: \quad 2400 - 40T_C = 0 \qquad T_C = 60 \text{ N}$$

$$\mathbf{j}: \quad 210T_C - 300D_z = 0 \quad (210)(60) - 300D_z = 0 \qquad D_z = 42 \text{ N}$$

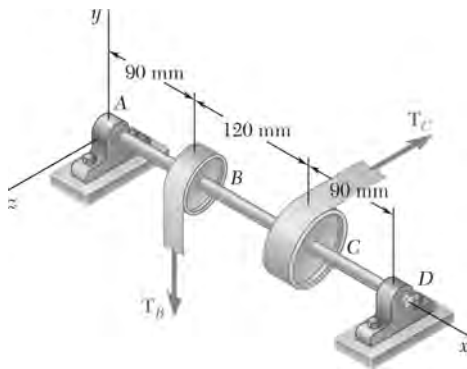
$$\mathbf{k}: \quad -7200 + 300D_y = 0 \qquad D_y = 24 \text{ N}$$

$$\Sigma F_x = 0: \quad D_x = 0$$

$$\Sigma F_y = 0: \quad A_y + D_y - 80 \text{ N} = 0 \qquad A_y = 80 - 24 = 56 \text{ N}$$

$$\Sigma F_z = 0: \quad A_z + D_z - 60 \text{ N} = 0 \qquad A_z = 60 - 42 = 18 \text{ N}$$

$$\mathbf{A} = (56.0 \text{ N})\mathbf{j} + (18.00 \text{ N})\mathbf{k} \quad \mathbf{D} = (24.0 \text{ N})\mathbf{j} + (42.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



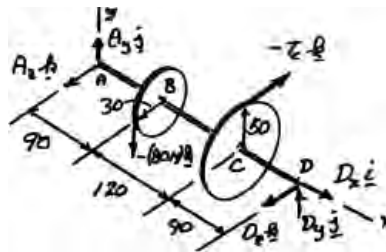
PROBLEM 4.93

Solve Problem 4.92, assuming that the spool C is replaced by a spool of radius 50 mm.

PROBLEM 4.92 Two tape spools are attached to an axle supported by bearings at A and D. The radius of spool B is 30 mm and the radius of spool C is 40 mm. Knowing that $T_B = 80$ N and that the system rotates at a constant rate, determine the reactions at A and D. Assume that the bearing at A does not exert any axial thrust and neglect the weights of the spools and axle.

SOLUTION

Dimensions in mm



We have six unknowns and six equations of equilibrium.

$$\begin{aligned}\Sigma M_A = 0: & (90\mathbf{i} + 30\mathbf{k}) \times (-80\mathbf{j}) + (210\mathbf{i} + 50\mathbf{j}) \times (-T_C\mathbf{k}) + (300\mathbf{i}) \times (D_x\mathbf{i} + D_y\mathbf{j} + D_z\mathbf{k}) = 0 \\ & -7200\mathbf{k} + 2400\mathbf{i} + 210T_C\mathbf{j} - 50T_C\mathbf{i} + 300D_y\mathbf{k} - 300D_z\mathbf{j} = 0\end{aligned}$$

Equate coefficients of unit vectors to zero:

$$\mathbf{i}: \quad 2400 - 50T_C = 0 \qquad T_C = 48 \text{ N}$$

$$\mathbf{j}: \quad 210T_C - 300D_z = 0 \quad (210)(48) - 300D_z = 0 \qquad D_z = 33.6 \text{ N}$$

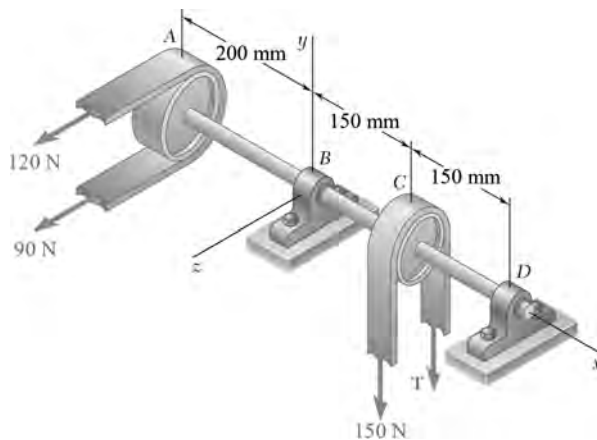
$$\mathbf{k}: \quad -7200 + 300D_y = 0 \qquad D_y = 24 \text{ N}$$

$$\Sigma F_x = 0: \quad D_x = 0$$

$$\Sigma F_y = 0: \quad A_y + D_y - 80 \text{ N} = 0 \qquad A_y = 80 - 24 = 56 \text{ N}$$

$$\Sigma F_z = 0: \quad A_z + D_z - 48 = 0 \qquad A_z = 48 - 33.6 = 14.4 \text{ N}$$

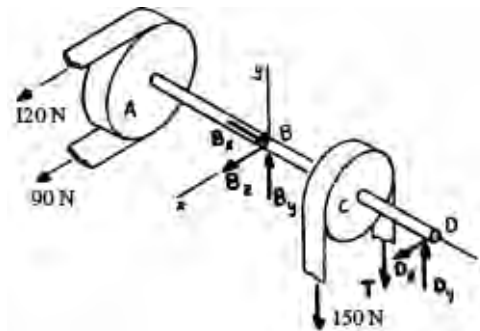
$$\mathbf{A} = (56.0 \text{ N})\mathbf{j} + (14.40 \text{ N})\mathbf{k} \quad \mathbf{D} = (24.0 \text{ N})\mathbf{j} + (33.6 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.94

Two transmission belts pass over sheaves welded to an axle supported by bearings at *B* and *D*. The sheave at *A* has a radius of 50 mm, and the sheave at *C* has a radius of 40 mm. Knowing that the system rotates at a constant rate, determine (a) the tension *T*, (b) the reactions at *B* and *D*. Assume that the bearing at *D* does not exert any axial thrust and neglect the weights of the sheaves and axle.

SOLUTION



Assume moment reactions at the bearing supports are zero. From f.b.d. of shaft

$$(a) \quad \Sigma M_{x\text{-axis}} = 0: (120 \text{ N} - 90 \text{ N})(100 \text{ mm}) + (150 \text{ N} - T)(80 \text{ mm}) = 0 \quad T = 187.5 \text{ N} \quad \blacktriangleleft$$

$$(b) \quad \Sigma F_x = 0: B_x = 0$$

$$\Sigma M_{D(z\text{-axis})} = 0: (150 \text{ N} + 187.5 \text{ N})(150 \text{ mm}) - B_y(300 \text{ mm}) = 0$$

$$B_y = 168.75 \text{ N}$$

$$\Sigma M_{D(y\text{-axis})} = 0: (120 \text{ N} + 90 \text{ N})(500 \text{ mm}) + B_z(300 \text{ mm}) = 0$$

$$B_z = -350 \text{ N}$$

$$\text{or } \mathbf{B} = (168.8 \text{ N})\mathbf{j} - (350 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\Sigma M_{B(z\text{-axis})} = 0: -(150 \text{ N} + 187.5 \text{ N})(150 \text{ mm}) + D_y(300 \text{ mm}) = 0$$

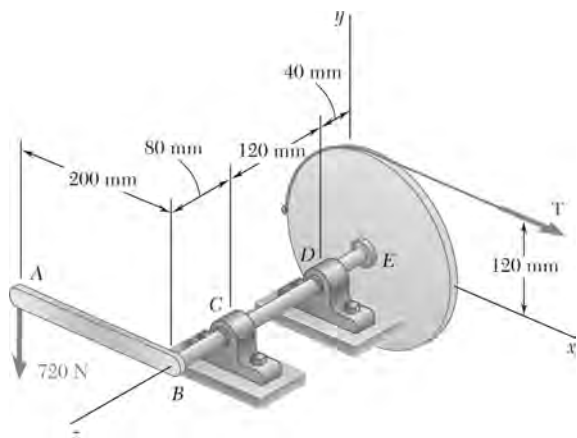
$$D_y = 168.75 \text{ N}$$

$$\Sigma M_{B(y\text{-axis})} = 0: (120 \text{ N} + 90 \text{ N})(200 \text{ mm}) + D_z(300 \text{ mm}) = 0$$

$$D_z = -140 \text{ N}$$

$$\text{or } \mathbf{D} = (168.8 \text{ N})\mathbf{j} - (140.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

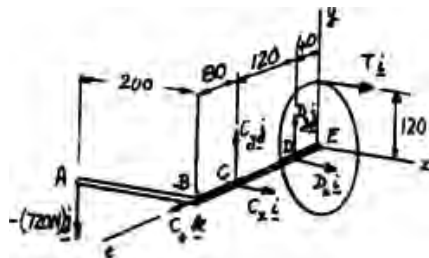
PROBLEM 4.95



A 200-mm lever and a 240-mm-diameter pulley are welded to the axle BE that is supported by bearings at C and D . If a 720-N vertical load is applied at A when the lever is horizontal, determine (a) the tension in the cord, (b) the reactions at C and D . Assume that the bearing at D does not exert any axial thrust.

SOLUTION

Dimensions in mm



We have six unknowns and six equations of equilibrium—OK

$$\Sigma \mathbf{M}_C = 0: (-120\mathbf{k}) \times (D_x\mathbf{i} + D_y\mathbf{j}) + (120\mathbf{j} - 160\mathbf{k}) \times T\mathbf{i} + (80\mathbf{k} - 200\mathbf{i}) \times (-720\mathbf{j}) = 0$$

$$-120D_x\mathbf{j} + 120D_y\mathbf{i} - 120T\mathbf{k} - 160T\mathbf{j} + 57.6 \times 10^3\mathbf{i} + 144 \times 10^3\mathbf{k} = 0$$

Equating to zero the coefficients of the unit vectors:

$$\mathbf{k}: -120T + 144 \times 10^3 = 0 \quad (a) \quad T = 1200 \text{ N} \quad \blacktriangleleft$$

$$\mathbf{i}: 120D_y + 57.6 \times 10^3 = 0 \quad D_y = -480 \text{ N}$$

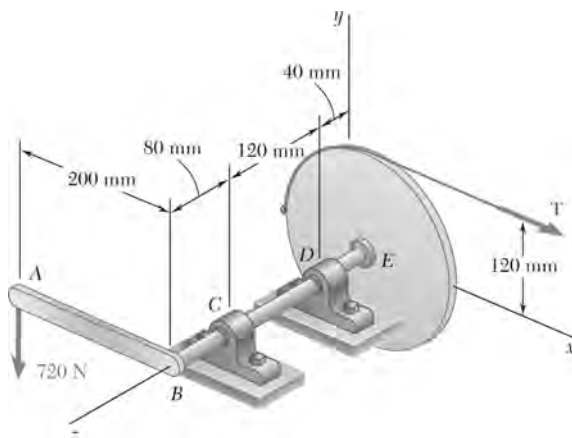
$$\mathbf{j}: -120D_x - 160(1200 \text{ N}) = 0 \quad D_x = -1600 \text{ N}$$

$$\Sigma F_x = 0: C_x + D_x + T = 0 \quad C_x = 1600 - 1200 = 400 \text{ N}$$

$$\Sigma F_y = 0: C_y + D_y - 720 = 0 \quad C_y = 480 + 720 = 1200 \text{ N}$$

$$\Sigma F_z = 0: C_z = 0$$

$$(b) \quad \mathbf{C} = (400 \text{ N})\mathbf{i} + (1200 \text{ N})\mathbf{j} \quad \mathbf{D} = -(1600 \text{ N})\mathbf{i} - (480 \text{ N})\mathbf{j} \quad \blacktriangleleft$$



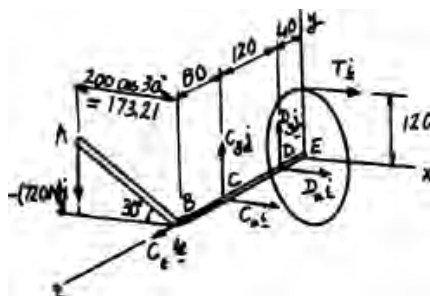
PROBLEM 4.96

Solve Problem 4.95, assuming that the axle has been rotated clockwise in its bearings by 30° and that the 720-N load remains vertical.

PROBLEM 4.95 A 200-mm lever and a 240-mm-diameter pulley are welded to the axle BE that is supported by bearings at C and D. If a 720-N vertical load is applied at A when the lever is horizontal, determine (a) the tension in the cord, (b) the reactions at C and D. Assume that the bearing at D does not exert any axial thrust.

SOLUTION

Dimensions in mm



We have six unknowns and six equations of equilibrium.

$$\begin{aligned}\Sigma \mathbf{M}_C = 0: & (-120\mathbf{k}) \times (D_x\mathbf{i} + D_y\mathbf{j}) + (120\mathbf{j} - 160\mathbf{k}) \times T\mathbf{i} + (80\mathbf{k} - 173.21\mathbf{i}) \times (-720\mathbf{j}) = 0 \\ & -120D_x\mathbf{j} + 120D_y\mathbf{i} - 120T\mathbf{k} - 160T\mathbf{j} + 57.6 \times 10^3\mathbf{i} + 124.71 \times 10^3\mathbf{k} = 0\end{aligned}$$

Equating to zero the coefficients of the unit vectors:

$$\mathbf{k}: -120T + 124.71 \times 10^3 = 0 \quad T = 1039.2 \text{ N} \quad T = 1039 \text{ N} \quad \blacktriangleleft$$

$$\mathbf{i}: 120D_y + 57.6 \times 10^3 = 0 \quad D_y = -480 \text{ N}$$

$$\mathbf{j}: -120D_x - 160(1039.2) = 0 \quad D_x = -1385.6 \text{ N}$$

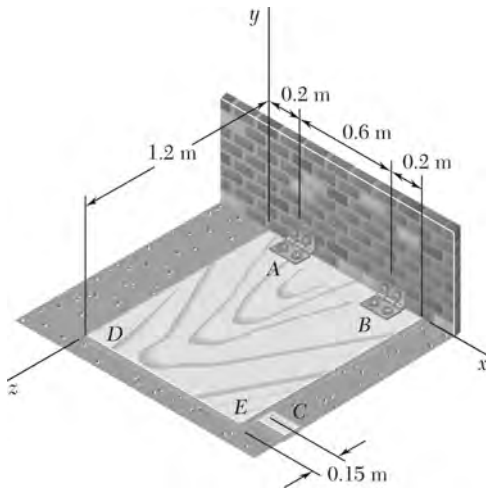
$$\Sigma F_x = 0: C_x + D_x + T = 0 \quad C_x = 1385.6 - 1039.2 = 346.4$$

$$\Sigma F_y = 0: C_y + D_y - 720 = 0 \quad C_y = 480 + 720 = 1200 \text{ N}$$

$$\Sigma F_z = 0: C_z = 0$$

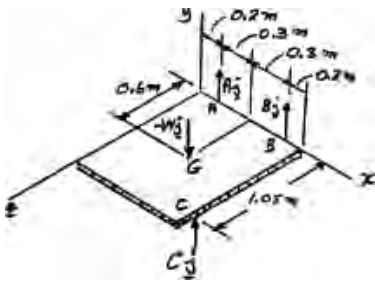
$$(b) \quad \mathbf{C} = (346 \text{ N})\mathbf{i} + (1200 \text{ N})\mathbf{j} \quad \mathbf{D} = -(1386 \text{ N})\mathbf{i} - (480 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

PROBLEM 4.97



An opening in a floor is covered by a 1×1.2 -m sheet of plywood of mass 18 kg. The sheet is hinged at A and B and is maintained in a position slightly above the floor by a small block C. Determine the vertical component of the reaction (a) at A, (b) at B, (c) at C.

SOLUTION



$$\mathbf{r}_{B/A} = 0.6\mathbf{i}$$

$$\mathbf{r}_{C/A} = 0.8\mathbf{i} + 1.05\mathbf{k}$$

$$\mathbf{r}_{G/A} = 0.3\mathbf{i} + 0.6\mathbf{k}$$

$$W = mg = (18 \text{ kg})9.81$$

$$W = 176.58 \text{ N}$$

$$\Sigma M_A = 0: \mathbf{r}_{B/A} \times B\mathbf{j} + \mathbf{r}_{C/A} \times C\mathbf{j} + \mathbf{r}_{G/A} \times (-W\mathbf{j}) = 0$$

$$(0.6\mathbf{i}) \times B\mathbf{j} + (0.8\mathbf{i} + 1.05\mathbf{k}) \times C\mathbf{j} + (0.3\mathbf{i} + 0.6\mathbf{k}) \times (-W\mathbf{j}) = 0$$

$$0.6B\mathbf{k} + 0.8C\mathbf{k} - 1.05C\mathbf{i} - 0.3W\mathbf{k} + 0.6W\mathbf{i} = 0$$

Equate coefficients of unit vectors of zero:

$$\mathbf{i}: 1.05C + 0.6W = 0 \quad C = \left(\frac{0.6}{1.05} \right) 176.58 \text{ N} = 100.90 \text{ N}$$

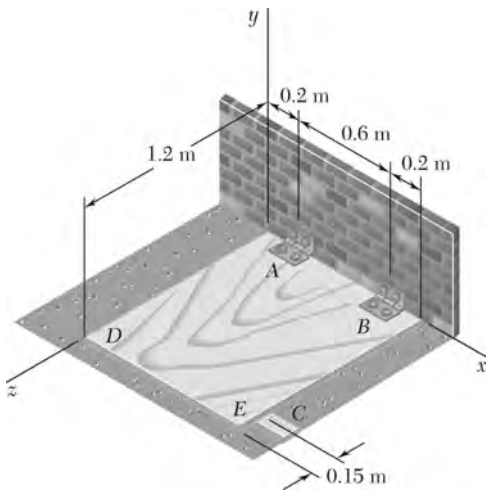
$$\mathbf{k}: 0.6B + 0.8C - 0.3W = 0$$

$$0.6B + 0.8(100.90 \text{ N}) - 0.3(176.58 \text{ N}) = 0 \quad B = -46.24 \text{ N}$$

$$\Sigma F_y = 0: A + B + C - W = 0$$

$$A - 46.24 \text{ N} + 100.90 \text{ N} + 176.58 \text{ N} = 0 \quad A = 121.92 \text{ N}$$

$$(a) \quad A = 121.9 \text{ N} \quad (b) \quad B = -46.2 \text{ N} \quad (c) \quad C = 100.9 \text{ N}$$

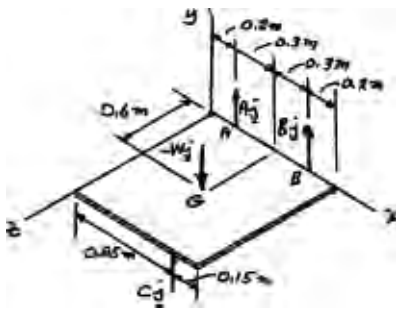


PROBLEM 4.98

Solve Problem 4.97, assuming that the small block *C* is moved and placed under edge *DE* at a point 0.15 m from corner *E*.

PROBLEM 4.97 An opening in a floor is covered by a 1×1.2 -m sheet of plywood of mass 18 kg. The sheet is hinged at *A* and *B* and is maintained in a position slightly above the floor by a small block *C*. Determine the vertical component of the reaction (a) at *A*, (b) at *B*, (c) at *C*.

SOLUTION



$$\mathbf{r}_{B/A} = 0.6\mathbf{i}$$

$$\mathbf{r}_{C/A} = 0.65\mathbf{i} + 1.2\mathbf{k}$$

$$\mathbf{r}_{G/A} = 0.3\mathbf{i} + 0.6\mathbf{k}$$

$$W = mg = (18 \text{ kg}) 9.81 \text{ m/s}^2$$

$$W = 176.58 \text{ N}$$

$$\Sigma M_A = 0: \mathbf{r}_{B/A} \times B\mathbf{j} + \mathbf{r}_{C/A} \times C\mathbf{j} + \mathbf{r}_{G/A} \times (-W\mathbf{j}) = 0$$

$$0.6\mathbf{i} \times B\mathbf{j} + (0.65\mathbf{i} + 1.2\mathbf{k}) \times C\mathbf{j} + (0.3\mathbf{i} + 0.6\mathbf{k}) \times (-W\mathbf{j}) = 0$$

$$0.6B\mathbf{k} + 0.65C\mathbf{k} - 1.2C\mathbf{i} - 0.3W\mathbf{k} + 0.6W\mathbf{i} = 0$$

Equate coefficients of unit vectors to zero:

$$\mathbf{i}: -1.2C + 0.6W = 0 \quad C = \left(\frac{0.6}{1.2}\right) 176.58 \text{ N} = 88.29 \text{ N}$$

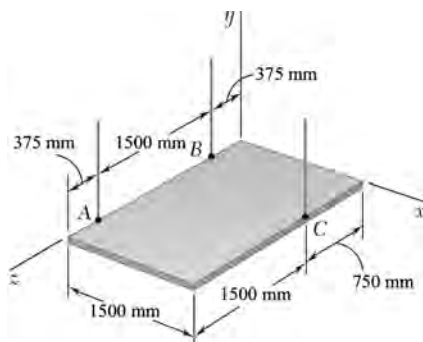
$$\mathbf{k}: 0.6B + 0.65C - 0.3W = 0$$

$$0.6B + 0.65(88.29 \text{ N}) - 0.3(176.58 \text{ N}) = 0 \quad B = -7.36 \text{ N}$$

$$\Sigma F_y = 0: A + B + C - W = 0$$

$$A - 7.36 \text{ N} + 88.29 \text{ N} - 176.58 \text{ N} = 0 \quad A = 95.648 \text{ N}$$

$$(a) A = 95.6 \text{ N} \quad (b) -7.36 \text{ N} \quad (c) 88.3 \text{ N}$$



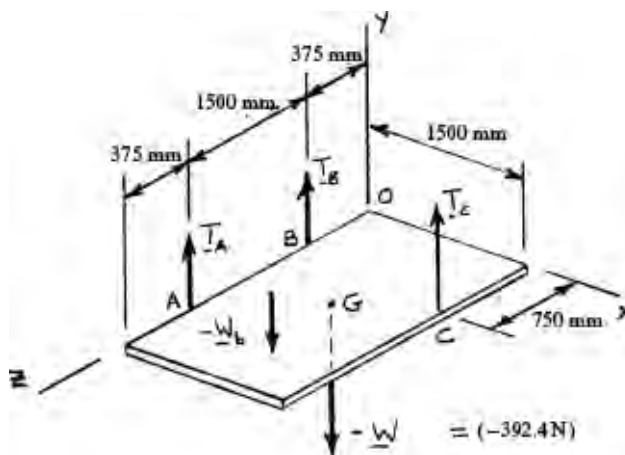
PROBLEM 4.99

The rectangular plate of mass 40 kg and is supported by three vertical wires. Determine the tension in each wire.

SOLUTION

Free-Body Diagram:

$$W = 40 \times 9.81 \text{ N} = 392.4 \text{ N}$$



$$\Sigma \mathbf{M}_B = 0: \quad \mathbf{r}_{A/B} \times T_A \mathbf{j} + \mathbf{r}_{C/B} \times T_C \mathbf{j} + \mathbf{r}_{G/B} \times (-392.4 \text{ N}) \mathbf{j} = 0$$

$$(1500 \text{ mm}) \mathbf{k} \times T_A \mathbf{j} + [(1500 \text{ mm}) \mathbf{i} + (375 \text{ mm}) \mathbf{k}] \times T_C \mathbf{j} + [(750 \text{ mm}) \mathbf{i} + (750 \text{ mm}) \mathbf{k}] \times (-392.4 \text{ N}) \mathbf{j} = 0$$

$$-1500T_A \mathbf{i} + 1500T_C \mathbf{k} - 375T_C \mathbf{i} - 294300 \mathbf{k} + 294300 \mathbf{i} = 0$$

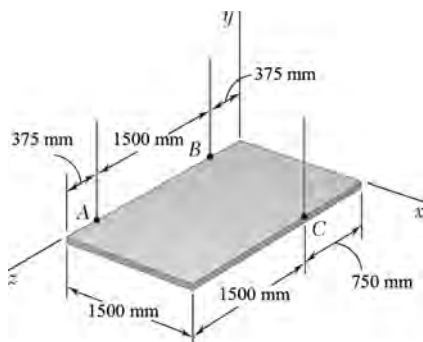
Equating to zero the coefficients of the unit vectors:

$$\mathbf{k}: \quad 1500T_C - 294300 = 0 \Rightarrow T_C = 196.2 \text{ N} \quad T_C = 196.2 \text{ N} \quad \blacktriangleleft$$

$$\mathbf{i}: \quad -1500T_A - 375(196.2) + 294300 = 0 \Rightarrow T_A = 147.15 \text{ N} \quad T_A = 147.2 \text{ N} \quad \blacktriangleleft$$

$$\Sigma F_y = 0: \quad T_A + T_B + T_C - 392.4 \text{ N} = 0$$

$$147.15 \text{ N} + T_B + 196.2 - 392.4 = 0 \Rightarrow T_B = 49.05 \text{ N} \quad T_B = 49.1 \text{ N} \quad \blacktriangleleft$$



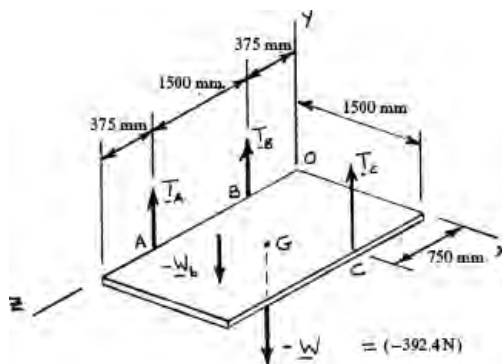
PROBLEM 4.100

The rectangular plate of mass 40 kg is supported by three vertical wires. Determine the weight and location of the lightest block that should be placed on the plate if the tensions in the three wires are to be equal.

SOLUTION

Free-Body Diagram:

$$W = 40 \times 9.81 = 392.4 \text{ N}$$



Let $-W_b \mathbf{j}$ be the weight of the block and x and z the block's coordinates.

Since tensions in wires are equal, let

$$T_A = T_B = T_C = T$$

$$\Sigma M_O = 0: (\mathbf{r}_A \times T\mathbf{j}) + (\mathbf{r}_B \times T\mathbf{j}) + (\mathbf{r}_C \times T\mathbf{j}) + \mathbf{r}_G \times (-W\mathbf{j}) + (x\mathbf{i} + z\mathbf{k}) \times (-W_b\mathbf{j}) = 0$$

$$\text{or, } (1875\mathbf{k}) \times T\mathbf{j} + (375\mathbf{k}) \times T\mathbf{j} + (1500\mathbf{i} + 750\mathbf{k}) \times T\mathbf{j} + (750\mathbf{i} + 1125\mathbf{k}) \times (-W\mathbf{j}) + (x\mathbf{i} + z\mathbf{k}) \times (-W_b\mathbf{j}) = 0$$

$$\text{or, } -1875T\mathbf{i} - 375T\mathbf{i} + 1500T\mathbf{k} - 750T\mathbf{i} - 750W\mathbf{k} + 1125W\mathbf{i} - xW_b\mathbf{k} + zW_b\mathbf{i} = 0$$

Equate coefficients of unit vectors to zero:

$$\mathbf{i}: -3000T + 1125W + zW_b = 0 \quad (1)$$

$$\mathbf{k}: 1500T - 750W - xW_b = 0 \quad (2)$$

$$\text{Also, } \Sigma F_y = 0: 3T - W - W_b = 0 \quad (3)$$

$$\text{Eq. (1) + 1000 Eq. (3): } 125W + (z - 1000)W_b = 0 \quad (4)$$

$$\text{Eq. (2) - 500 Eq. (3): } -250W - (x - 500)W_b = 0 \quad (5)$$

PROBLEM 4.100 (Continued)

Solving (4) and (5) for W_b/W and recalling of $0 \leq x \leq 1500$ mm, $0 \leq z \leq 2250$ mm,

$$\text{Eq.(4):} \quad \frac{W_b}{W} = \frac{125}{1000 - z} \geq \frac{125}{1000 - 0} = 0.125$$

$$\text{Eq.(5):} \quad \frac{W_b}{W} = \frac{250}{500 - x} \geq \frac{250}{500 - 0} = 0.5$$

$$\text{Thus, } (W_b)_{\min} = 0.5W = 0.5(392.4 \text{ N}) = 196.2 \text{ N}$$

$$(W_b)_{\min} = 196.2 \text{ N} \quad \blacktriangleleft$$

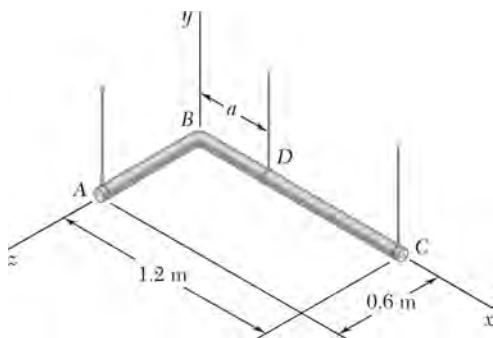
Making $W_b = 0.5W$ in (4) and (5):

$$125W + (z - 1000)(0.5W) = 0$$

$$z = 750 \text{ mm} \quad \blacktriangleleft$$

$$-250W - (x - 500)(0.5W) = 0$$

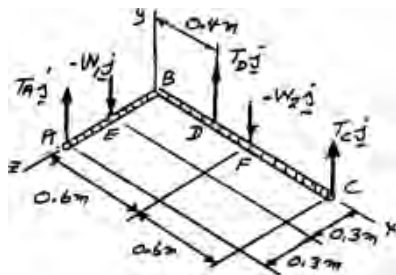
$$x = 0 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.101

Two steel pipes AB and BC , each having a mass per unit length of 8 kg/m , are welded together at B and supported by three wires. Knowing that $a = 0.4 \text{ m}$, determine the tension in each wire.

SOLUTION



$$W_1 = 0.6m'g$$

$$W_2 = 1.2m'g$$

$$\Sigma M_D = 0: \mathbf{r}_{A/D} \times T_A \mathbf{j} + \mathbf{r}_{E/D} \times (-W_1 \mathbf{j}) + \mathbf{r}_{F/D} \times (-W_2 \mathbf{j}) + \mathbf{r}_{C/D} \times T_C \mathbf{j} = 0$$

$$(-0.4\mathbf{i} + 0.6\mathbf{k}) \times T_A \mathbf{j} + (-0.4\mathbf{i} + 0.3\mathbf{k}) \times (-W_1 \mathbf{j}) + 0.2\mathbf{i} \times (-W_2 \mathbf{j}) + 0.8\mathbf{i} \times T_C \mathbf{j} = 0$$

$$-0.4T_A \mathbf{k} - 0.6T_A \mathbf{i} + 0.4W_1 \mathbf{k} + 0.3W_1 \mathbf{i} - 0.2W_2 \mathbf{k} + 0.8T_C \mathbf{k} = 0$$

Equate coefficients of unit vectors to zero:

$$\mathbf{i}: -0.6T_A + 0.3W_1 = 0; \quad T_A = \frac{1}{2}W_1 = \frac{1}{2}(0.6m'g) = 0.3m'g$$

$$\mathbf{k}: -0.4T_A + 0.4W_1 - 0.2W_2 + 0.8T_C = 0$$

$$-0.4(0.3m'g) + 0.4(0.6m'g) - 0.2(1.2m'g) + 0.8T_C = 0$$

$$T_C = \frac{(0.12 - 0.24 - 0.24)m'g}{0.8} = 0.15m'g$$

$$\Sigma F_y = 0: T_A + T_C + T_D - W_1 - W_2 = 0$$

$$0.3m'g + 0.15m'g + T_D - 0.6m'g - 1.2m'g = 0$$

$$T_D = 1.35m'g$$

$$m'g = (8 \text{ kg/m})(9.81 \text{ m/s}^2) = 78.48 \text{ N/m}$$

$$T_A = 0.3m'g = 0.3 \times 78.48$$

$$T_A = 23.5 \text{ N} \quad \blacktriangleleft$$

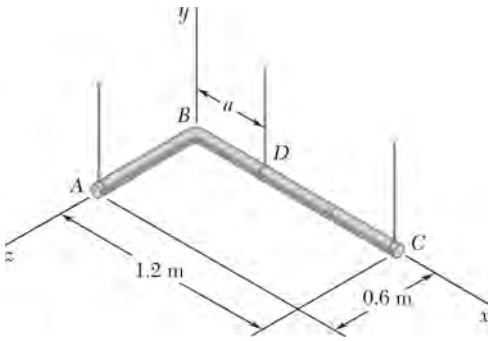
$$T_B = 0.15m'g = 0.15 \times 78.48$$

$$T_B = 11.77 \text{ N} \quad \blacktriangleleft$$

$$T_C = 1.35m'g = 1.35 \times 78.48$$

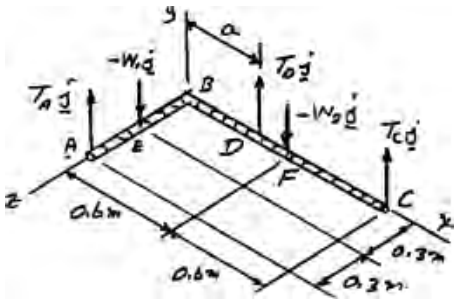
$$T_C = 105.9 \text{ N} \quad \blacktriangleleft$$

PROBLEM 4.102



For the pipe assembly of Problem 4.101, determine (a) the largest permissible value of a if the assembly is not to tip, (b) the corresponding tension in each wire.

SOLUTION



$$W_1 = 0.6m'g$$

$$W_2 = 1.2m'g$$

$$\Sigma M_D = 0: \mathbf{r}_{A/D} \times T_A \mathbf{j} + \mathbf{r}_{E/D} \times (-W_1 \mathbf{j}) + \mathbf{r}_{F/D} \times (-W_2 \mathbf{j}) + \mathbf{r}_{C/D} \times T_C \mathbf{j} = 0$$

$$(-a\mathbf{i} + 0.6\mathbf{k}) \times T_A \mathbf{j} + (-a\mathbf{i} + 0.3\mathbf{k}) \times (-W_1 \mathbf{j}) + (0.6 - a)\mathbf{i} \times (-W_2 \mathbf{j}) + (1.2 - a)\mathbf{i} \times T_C \mathbf{j} = 0$$

$$-T_A a \mathbf{k} - 0.6T_A \mathbf{i} + W_1 a \mathbf{k} + 0.3W_1 \mathbf{i} - W_2(0.6 - a)\mathbf{k} + T_C(1.2 - a)\mathbf{k} = 0$$

Equate coefficients of unit vectors to zero:

$$\mathbf{i}: -0.6T_A + 0.3W_1 = 0; \quad T_A = \frac{1}{2}W_1 = \frac{1}{2}0.6m'g = 0.3m'g$$

$$\mathbf{k}: -T_A a + W_1 a - W_2(0.6 - a) + T_C(1.2 - a) = 0$$

$$-0.3m'ga + 0.6m'ga - 1.2m'g(0.6 - a) + T_C(1.2 - a) = 0$$

$$T_C = \frac{0.3a - 0.6a + 1.2(0.6 - a)}{1.2 - a} \quad \text{For Max } a \text{ and no tipping, } T_C = 0$$

(a)

$$-0.3a + 1.2(0.6 - a) = 0$$

$$-0.3a + 0.72 - 1.2a = 0$$

$$1.5a = 0.72$$

$$a = 0.480 \text{ m} \quad \blacktriangleleft$$

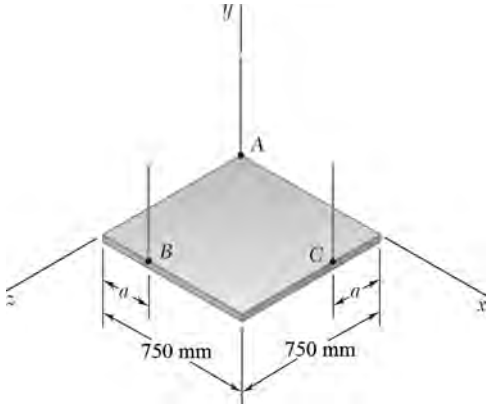
PROBLEM 4.102 (Continued)

(b) Reactions: $m'g = (8 \text{ kg/m}) 9.81 \text{ m/s}^2 = 78.48 \text{ N/m}$

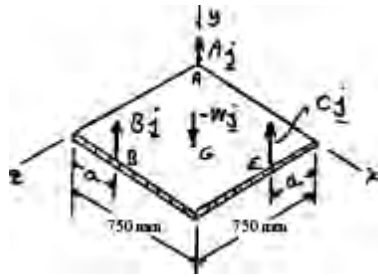
$$T_A = 0.3m'g = 0.3 \times 78.48 = 23.544 \text{ N} \quad T_A = 23.5 \text{ N} \quad \blacktriangleleft$$
$$\Sigma F_y = 0: T_A + T_C + T_D - W_1 - W_2 = 0$$
$$T_A + 0 + T_D - 0.6m'g - 1.2m'g = 0$$
$$T_D = 1.8m'g - T_A = 1.8 \times 78.48 - 23.544 = 117.72 \quad T_D = 117.7 \text{ N} \quad \blacktriangleleft$$

PROBLEM 4.103

The 120 N square plate shown is supported by three vertical wires. Determine (a) the tension in each wire when $a = 250$ mm, (b) the value of a for which the tension in each wire is 40 N.



SOLUTION



$$\begin{aligned} \mathbf{r}_{B/A} &= a\mathbf{i} + 750\mathbf{k} \\ \mathbf{r}_{C/A} &= 750\mathbf{i} + a\mathbf{k} \\ \mathbf{r}_{G/A} &= 375\mathbf{i} + 375\mathbf{k} \end{aligned}$$

By symmetry: $B = C$

$$\begin{aligned} \Sigma M_A = 0: \quad & \mathbf{r}_{B/A} \times B\mathbf{j} + \mathbf{r}_{C/A} \times C\mathbf{j} + \mathbf{r}_{G/A} \times (-W\mathbf{j}) = 0 \\ (a\mathbf{i} + 750\mathbf{k}) \times B\mathbf{j} + (750\mathbf{i} + a\mathbf{k}) \times B\mathbf{j} + (375\mathbf{i} + 375\mathbf{k}) \times (-W\mathbf{j}) &= 0 \\ B a \mathbf{k} - 750 B \mathbf{i} + 750 B \mathbf{k} - B a \mathbf{i} - 375 W \mathbf{k} + 375 W \mathbf{i} &= 0 \end{aligned}$$

Equate coefficient of unit vector $\mathbf{i}$ to zero:

$$\begin{aligned} \mathbf{i}: \quad & -750B - Ba + 375W = 0 \\ B &= \frac{375W}{750 + a} \quad C = B = \frac{375W}{750 + a} \end{aligned} \tag{1}$$

$$\begin{aligned} \Sigma F_y = 0: \quad & A + B + C - W = 0 \\ A + 2 \left[\frac{375W}{750 + a} \right] - W &= 0; \quad A = \frac{aW}{750 + a} \end{aligned} \tag{2}$$

PROBLEM 4.103 (Continued)

(a) For $a = 250$ mm, $W = 120$ N

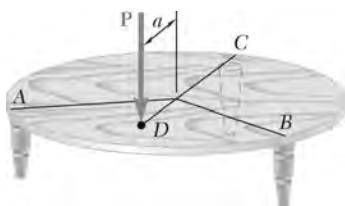
$$\text{Eq. (1)} \quad C = B = \frac{375 \times 120}{1000} = 45 \text{ N}$$

$$\text{Eq. (2)} \quad A = \frac{250(120)}{1000} = 30 \text{ N} \quad A = 30.0 \text{ N} \quad B = C = 45.0 \text{ N} \quad \blacktriangleleft$$

(b) For tension in each wire = 40 N

$$\text{Eq. (1)} \quad 40 \text{ N} = \frac{375 \times 120}{(750 + a)}$$

$$750 \text{ mm} + a = 1125 \text{ mm} \quad a = 375 \text{ mm} \quad \blacktriangleleft$$



PROBLEM 4.104

The table shown weighs 150 N and has a diameter of 1.2 m. It is supported by three legs equally spaced around the edge. A vertical load $\mathbf{P}$ of magnitude 500 N is applied to the top of the table at D . Determine the maximum value of a if the table is not to tip over. Show, on a sketch, the area of the table over which $\mathbf{P}$ can act without tipping the table.

SOLUTION

$$r = 0.6 \text{ m} \quad b = r \sin 30^\circ = 0.3 \text{ m}$$

We shall sum moments about AB .

$$(b + r)C + (a - b)P - bW = 0$$

$$(0.3 + 0.6)C + (a - 0.3)500 - (0.3)150 = 0$$

$$C = \frac{1}{0.9}[45 - (a - 0.3)500]$$

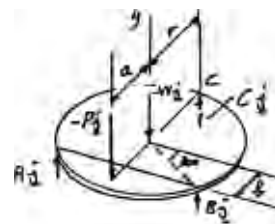
If table is not to tip, $C \geq 0$



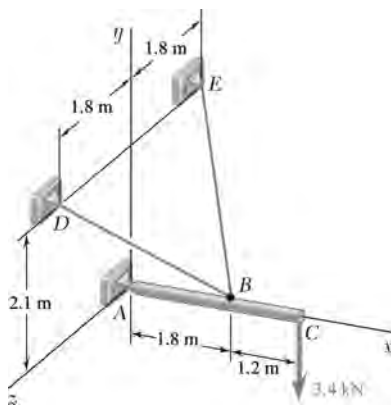
$$[45 - (a - 0.3)500] \geq 0$$

$$45 \geq (a - 0.3)500$$

$$a - 0.3 \leq 0.09 \quad a \leq 0.39 \text{ m} \quad a = 0.39 \text{ m}$$



Only $\perp$ distance from P to AB matters. Same condition must be satisfied for each leg. P must be located in shaded area for no tipping



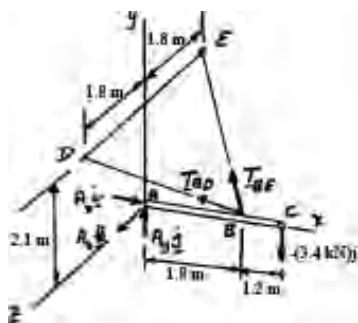
PROBLEM 4.105

A 3-m boom is acted upon by the 3.4-kN force shown. Determine the tension in each cable and the reaction at the ball-and-socket joint at A.

SOLUTION

We have five unknowns and six equations of equilibrium but equilibrium is maintained ($\Sigma M_x = 0$).

Free-Body Diagram:



$$\overline{BD} = (-1.8\text{ m})\mathbf{i} + (2.1\text{ m})\mathbf{j} + (1.8\text{ m})\mathbf{k} \quad BD = 3.3\text{ m}$$

$$\overline{BE} = (-1.8\text{ m})\mathbf{i} + (2.1\text{ m})\mathbf{j} + (1.8\text{ m})\mathbf{k} \quad BE = 3.3\text{ m}$$

$$T_{BD} = T_{BD} \frac{\overline{BD}}{BD} = \frac{T_{BD}}{3.3} (-1.8\mathbf{i} + 2.1\mathbf{j} + 1.8\mathbf{k}) = \frac{T_{BD}}{11} (-6\mathbf{i} + 7\mathbf{j} + 6\mathbf{k})$$

$$T_{BE} = T_{BE} \frac{\overline{BE}}{BE} = \frac{T_{BE}}{3.3} (-1.8\mathbf{i} + 2.1\mathbf{j} - 1.8\mathbf{k}) = \frac{T_{BE}}{11} (-6\mathbf{i} + 7\mathbf{j} - 6\mathbf{k})$$

$$\Sigma M_A = 0: \mathbf{r}_B \times T_{BD} + \mathbf{r}_B \times T_{BE} + \mathbf{r}_C \times (-3.4\mathbf{j}) = 0$$

$$1.8\mathbf{i} \times \frac{T_{BD}}{11} (-6\mathbf{i} + 7\mathbf{j} + 6\mathbf{k}) + 1.8\mathbf{i} \times \frac{T_{BE}}{11} (-6\mathbf{i} + 7\mathbf{j} - 6\mathbf{k}) + 3\mathbf{i} \times (-3.4\mathbf{j}) = 0$$

$$\frac{12.6}{11} T_{BD} \mathbf{k} - \frac{10.8}{11} T_{BD} \mathbf{j} + \frac{12.6}{11} T_{BE} \mathbf{k} + \frac{10.8}{11} T_{BE} \mathbf{j} - 10.2 \mathbf{k} = 0$$

Equate coefficients of unit vectors to zero.

$$\mathbf{j}: -\frac{10.8}{11} T_{BD} + \frac{10.8}{11} T_{BE} = 0 \quad T_{BE} = T_{BD}$$

$$\mathbf{k}: \frac{12.6}{11} T_{BD} + \frac{12.6}{11} T_{BE} - 10.2 = 0$$

$$2 \left(\frac{12.6}{11} T_{BD} \right) = 10.2$$

$$T_{BD} = 4.45\text{ kN} \quad \blacktriangleleft$$

$$T_{BD} = 4.4524\text{ kN}$$

$$T_{BE} = 4.45\text{ kN} \quad \blacktriangleleft$$

PROBLEM 4.105 (Continued)

$$\Sigma F_x = 0: A_x - \frac{6}{11}(4.4524 \text{ kN}) - \frac{6}{11}(4.4524 \text{ kN}) = 0$$

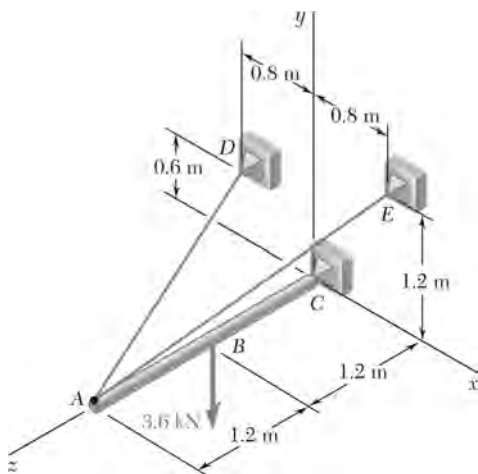
$$A_x = 4.8572 \text{ kN}$$

$$\Sigma F_y = 0: A_y + \frac{7}{11}(4.4524 \text{ kN}) + \frac{7}{11}(4.4524 \text{ kN}) - 3.4 \text{ kN} = 0$$

$$A_y = -2.2667 \text{ kN}$$

$$\Sigma F_z = 0: A_z + \frac{6}{11}(4.4524 \text{ kN}) - \frac{6}{11}(4.4524 \text{ kN}) = 0$$

$$A_z = 0 \quad \mathbf{A} = (4.86 \text{ kN})\mathbf{i} - (2.27 \text{ kN})\mathbf{j} \quad \blacktriangleleft$$

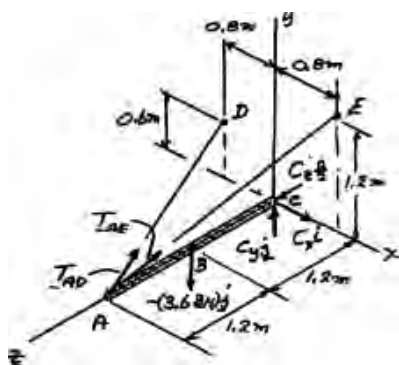


PROBLEM 4.106

A 2.4-m boom is held by a ball-and-socket joint at C and by two cables AD and AE. Determine the tension in each cable and the reaction at C.

SOLUTION

Free-Body Diagram: Five Unknowns and six equations of equilibrium. Equilibrium is maintained ($\Sigma M_{AC} = 0$).



$$\mathbf{r}_B = 1.2\mathbf{k}$$

$$\mathbf{r}_A = 2.4\mathbf{k}$$

$$\overline{AD} = -0.8\mathbf{i} + 0.6\mathbf{j} - 2.4\mathbf{k} \quad AD = 2.6 \text{ m}$$

$$\overline{AE} = 0.8\mathbf{i} + 1.2\mathbf{j} - 2.4\mathbf{k} \quad AE = 2.8 \text{ m}$$

$$T_{AD} = \frac{\overline{AD}}{AD} = \frac{T_{AD}}{2.6} (-0.8\mathbf{i} + 0.6\mathbf{j} - 2.4\mathbf{k})$$

$$T_{AE} = \frac{\overline{AE}}{AE} = \frac{T_{AE}}{2.8} (0.8\mathbf{i} + 1.2\mathbf{j} - 2.4\mathbf{k})$$

$$\Sigma M_C = 0: \mathbf{r}_A \times \mathbf{T}_{AD} + \mathbf{r}_A \times \mathbf{T}_{AE} + \mathbf{r}_B \times (-3 \text{ kN})\mathbf{j} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 2.4 \\ -0.8 & 0.6 & -2.4 \end{vmatrix} \frac{T_{AD}}{2.6} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 2.4 \\ 0.8 & 1.2 & -2.4 \end{vmatrix} \frac{T_{AE}}{2.8} + 1.2\mathbf{k} \times (-3.6 \text{ kN})\mathbf{j} = 0$$

Equate coefficients of unit vectors to zero.

$$\mathbf{i}: -0.55385 T_{AD} - 1.02857 T_{AE} + 4.32 = 0 \quad (1)$$

$$\mathbf{j}: -0.73846 T_{AD} + 0.68671 T_{AE} = 0$$

$$T_{AD} = 0.92857 T_{AE} \quad (2)$$

PROBLEM 4.106 (Continued)

$$\text{Eq. (1):} \quad -0.55385(0.92857) T_{AE} - 1.02857 T_{AE} + 4.32 = 0$$

$$1.54286 T_{AE} = 4.32$$

$$T_{AE} = 2.800 \text{ kN}$$

$$T_{AE} = 2.80 \text{ kN} \quad \blacktriangleleft$$

$$\text{Eq. (2):} \quad T_{AD} = 0.92857(2.80) = 2.600 \text{ kN}$$

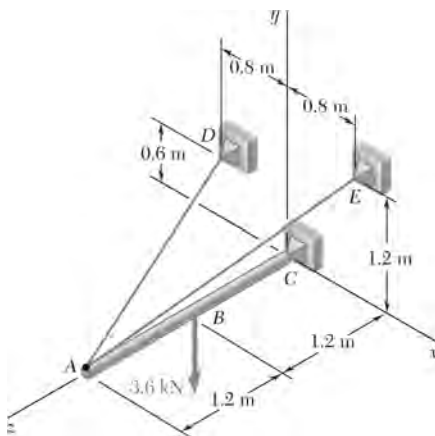
$$T_{AD} = 2.60 \text{ kN} \quad \blacktriangleleft$$

$$\Sigma F_x = 0: \quad C_x - \frac{0.8}{2.6}(2.6 \text{ kN}) + \frac{0.8}{2.8}(2.8 \text{ kN}) = 0 \quad C_x = 0$$

$$\Sigma F_y = 0: \quad C_y + \frac{0.6}{2.6}(2.6 \text{ kN}) + \frac{1.2}{2.8}(2.8 \text{ kN}) - (3.6 \text{ kN}) = 0 \quad C_y = 1.800 \text{ kN}$$

$$\Sigma F_z = 0: \quad C_z - \frac{2.4}{2.6}(2.6 \text{ kN}) - \frac{2.4}{2.8}(2.8 \text{ kN}) = 0 \quad C_z = 4.80 \text{ kN}$$

$$\mathbf{C} = (1.800 \text{ kN})\mathbf{j} + (4.80 \text{ kN})\mathbf{k} \quad \blacktriangleleft$$



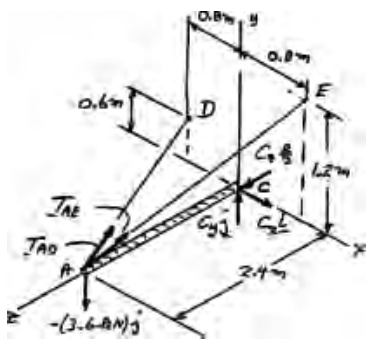
PROBLEM 4.107

Solve Problem 4.106, assuming that the 3.6-kN load is applied at Point A.

PROBLEM 4.106 A 2.4-m boom is held by a ball-and-socket joint at C and by two cables AD and AE. Determine the tension in each cable and the reaction at C.

SOLUTION

Free-Body Diagram: Five unknowns and six equations of equilibrium. Equilibrium is maintained ($\Sigma M_{AC} = 0$).



$$\overline{AD} = -0.8\mathbf{i} + 0.6\mathbf{j} - 2.4\mathbf{k} \quad AD = 2.6 \text{ m}$$

$$\overline{AE} = 0.8\mathbf{i} + 1.2\mathbf{j} - 2.4\mathbf{k} \quad AE = 2.8 \text{ m}$$

$$T_{AD} = \frac{\overline{AD}}{AD} = \frac{T_{AD}}{2.6} (-0.8\mathbf{i} + 0.6\mathbf{j} - 2.4\mathbf{k})$$

$$T_{AE} = \frac{\overline{AE}}{AE} = \frac{T_{AE}}{2.8} (0.8\mathbf{i} + 1.2\mathbf{j} - 2.4\mathbf{k})$$

$$\Sigma M_C = 0: \mathbf{r}_A \times \mathbf{T}_{AD} + \mathbf{r}_A \times \mathbf{T}_{AE} + \mathbf{r}_A \times (-3.6 \text{ kN})\mathbf{j}$$

Factor r_A :

$$\mathbf{r}_A \times (\mathbf{T}_{AD} + \mathbf{T}_{AE} - (3.6 \text{ kN})\mathbf{j})$$

or:

$$\mathbf{T}_{AD} + \mathbf{T}_{AE} - (3 \text{ kN})\mathbf{j} = 0 \quad (\text{Forces concurrent at A})$$

Coefficient of $\mathbf{i}$:

$$-\frac{T_{AD}}{2.6}(0.8) + \frac{T_{AE}}{2.8}(0.8) = 0$$

$$T_{AD} = \frac{2.6}{2.8} T_{AE} \quad (1)$$

Coefficient of $\mathbf{j}$:

$$\frac{T_{AD}}{2.6}(0.6) + \frac{T_{AE}}{2.8}(1.2) - 3.6 \text{ kN} = 0$$

$$\frac{2.6}{2.8} T_{AE} \left(\frac{0.6}{2.6} \right) + \frac{1.2}{2.8} T_{AE} - 3.6 \text{ kN} = 0$$

$$T_{AE} \left(\frac{0.6 + 1.2}{2.8} \right) = 3.6 \text{ kN}$$

$$T_{AE} = 5.600 \text{ kN}$$

$$T_{AE} = 5.60 \text{ kN} \quad \blacktriangleleft$$

PROBLEM 4.107 (Continued)

$$\text{Eq. (1):} \quad T_{AD} = \frac{2.6}{2.8}(5.6) = 5.200 \text{ kN} \quad T_{AD} = 5.20 \text{ kN} \quad \blacktriangleleft$$

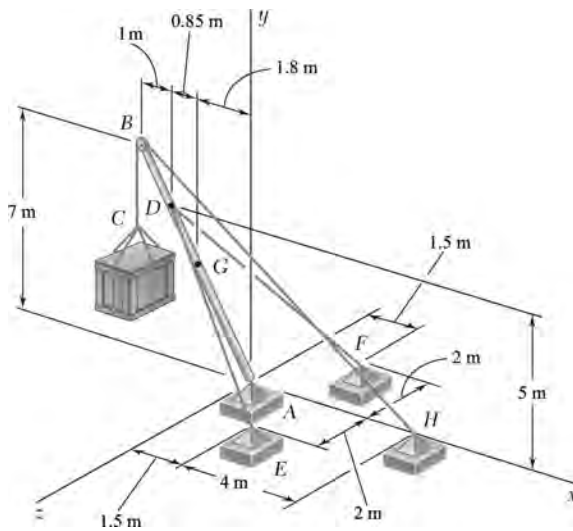
$$\Sigma F_x = 0: \quad C_x - \frac{0.8}{2.6}(5.2 \text{ kN}) + \frac{0.8}{2.8}(5.6 \text{ kN}) = 0; \quad C_x = 0$$

$$\Sigma F_y = 0: \quad C_y + \frac{0.6}{2.6}(5.2 \text{ kN}) + \frac{1.2}{2.8}(5.6 \text{ kN}) - 3.6 \text{ kN} = 0 \quad C_y = 0$$

$$\Sigma F_z = 0: \quad C_z - \frac{2.4}{2.6}(5.2 \text{ kN}) - \frac{2.4}{2.8}(5.6 \text{ kN}) = 0 \quad C_z = 9.60 \text{ kN}$$

$$\mathbf{C} = (9.60 \text{ kN})\mathbf{k} \quad \blacktriangleleft$$

Note: Since forces and reaction are concurrent at A, we could have used the methods of Chapter 2.

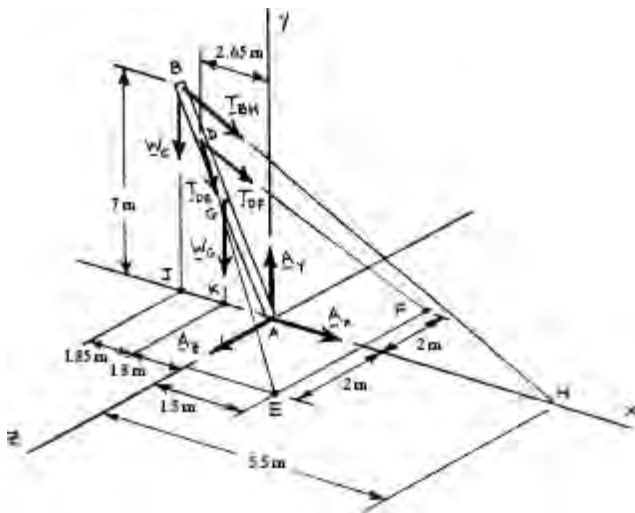


PROBLEM 4.108

A 300-kg crate hangs from a cable that passes over a pulley B and is attached to a support at H . The 100-kg boom AB is supported by a ball-and-socket joint at A and by two cables DE and DF . The center of gravity of the boom is located at G . Determine (a) the tension in cables DE and DF , (b) the reaction at A .

SOLUTION

Free-Body Diagram:



$$W_C = 300 \times 9.81 = 2943 \text{ N}$$

$$W_G = 100 \times 9.81 = 981 \text{ N}$$

We have five unknowns ($T_{DE}, T_{DF}, A_x, A_y, A_z$) and five equilibrium equations. The boom is free to spin about the AB axis, but equilibrium is maintained, since $\Sigma M_{AB} = 0$.

We have

$$\overline{BH} = (9.15 \text{ m})\mathbf{i} - (7 \text{ m})\mathbf{j} \quad BH = 11.5205 \text{ m}$$

$$\begin{aligned} \overline{DE} &= (4.15 \text{ m})\mathbf{i} - \frac{2.65}{3.65}(7 \text{ m})\mathbf{j} + (2 \text{ m})\mathbf{k} \\ &= (4.15 \text{ m})\mathbf{i} - (5.0822 \text{ m})\mathbf{j} + (2 \text{ m})\mathbf{k} \quad DE = 6.8594 \text{ m} \end{aligned}$$

$$\overline{DF} = (4.15 \text{ m})\mathbf{i} - (5.0822 \text{ m})\mathbf{j} - (2 \text{ m})\mathbf{k} \quad DF = 6.8594 \text{ m}$$

PROBLEM 4.108 (Continued)

Thus:

$$\mathbf{T}_{BH} = T_{BH} \frac{\overline{BH}}{BH} = (2943 \text{ N}) \frac{9.15\mathbf{i} - 7\mathbf{j}}{11.5205} = (2337.438 \text{ N})\mathbf{i} - (1788.204 \text{ N})\mathbf{j}$$

$$\mathbf{T}_{DE} = T_{DE} \frac{\overline{DE}}{DE} = \frac{T_{DE}}{6.8594} (4.15\mathbf{i} - 5.0822\mathbf{j} + 2\mathbf{k})$$

$$\mathbf{T}_{DF} = T_{DF} \frac{\overline{DF}}{DF} = \frac{T_{DF}}{6.8594} (4.15\mathbf{i} - 5.0822\mathbf{j} - 2\mathbf{k})$$

$$(a) \quad \Sigma \mathbf{M}_A = 0: (\mathbf{r}_J \times \mathbf{W}_C) + (\mathbf{r}_K \times \mathbf{W}_G) + (\mathbf{r}_H \times \mathbf{T}_{BH}) + (\mathbf{r}_E \times \mathbf{T}_{DE}) + (\mathbf{r}_F \times \mathbf{T}_{DF}) = 0$$

$$- (3.65\mathbf{i}) \times (-2943\mathbf{j}) - (1.825\mathbf{i}) \times (-981\mathbf{j}) + (5.5\mathbf{i}) \times (2337.438\mathbf{i} - 1788.204\mathbf{j})$$

$$+ \frac{T_{DE}}{6.8594} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.5 & 0 & 2 \\ 4.15 & -5.0822 & 2 \end{vmatrix} + \frac{T_{DF}}{6.8594} \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1.5 & 0 & -2 \\ 4.15 & -5.0822 & -2 \end{vmatrix} = 0$$

or,

$$10741.95\mathbf{k} + 1790.325\mathbf{k} - 9835.122\mathbf{k} + (T_{DE} - T_{DF}) \frac{(2 \times 5.0822)}{6.8594} \mathbf{i}$$

$$+ \frac{5.3}{6.8594} (T_{DE} - T_{DF}) \mathbf{j} - \frac{7.6233}{6.8594} (T_{DE} + T_{DF}) \mathbf{k} = 0$$

Equating to zero the coefficients of the unit vectors:

i or **j**: $T_{DE} - T_{DF} = 0 \quad T_{DE} = T_{DF}^*$

k: $10741.95 + 1790.325 - 9835.122 - \frac{7.6233}{6.8594} (2T_{DE}) = 0 \quad T_{DE} = 1213.441 \text{ N}$

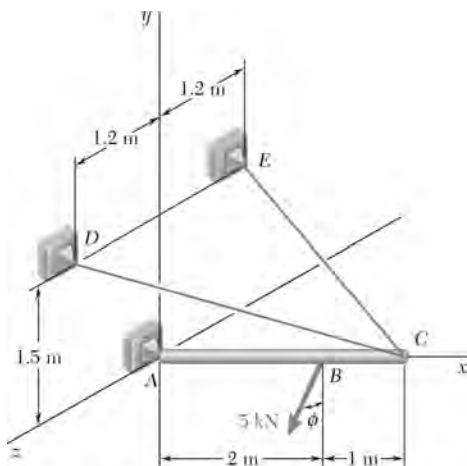
$$T_{DE} = T_{DF} = 1213 \text{ N} \quad \blacktriangleleft$$

(b) $\Sigma F_x = 0: A_x + 2337.438 + 2 \left(\frac{4.15}{6.8594} \right) (1213.441) = 0 \quad A_x = -3805.72 \text{ N}$

$\Sigma F_y = 0: A_y - 2943 - 981 - 1788.204 - 2 \left(\frac{5.0822}{6.8594} \right) (1213.441) = 0 \quad A_y = 7510.31 \text{ N}$

$\Sigma F_z = 0: A_z = 0 \quad \mathbf{A} = -(3810 \text{ N})\mathbf{i} + (7510 \text{ N})\mathbf{j} \quad \blacktriangleleft$

*Remark: The fact that $T_{DE} = T_{DF}$ could have been noted at the outset from the symmetry of structure with respect to xy plane.



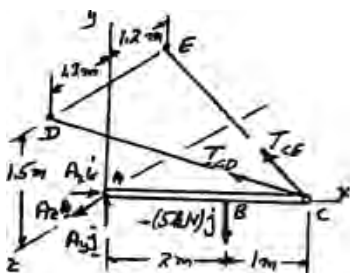
PROBLEM 4.109

A 3-m pole is supported by a ball-and-socket joint at A and by the cables CD and CE. Knowing that the 5-kN force acts vertically downward ($\phi = 0$), determine (a) the tension in cables CD and CE, (b) the reaction at A.

SOLUTION

Free-Body Diagram:

By symmetry with xy plane



$$T_{CD} = T_{CE} = T$$

$$\overline{CD} = -3\mathbf{i} + 1.5\mathbf{j} + 1.2\mathbf{k}$$

$$CD = 3.562 \text{ m}$$

$$T_{CD} = T \frac{\overline{CD}}{CD} = T \frac{-3\mathbf{i} + 1.5\mathbf{j} + 1.2\mathbf{k}}{3.562}$$

$$T_{CE} = T \frac{-3\mathbf{i} + 1.5\mathbf{j} - 1.2\mathbf{k}}{3.562}$$

$$\mathbf{r}_{B/A} = 2\mathbf{i} \quad \mathbf{r}_{C/A} = 3\mathbf{i}$$

$$\Sigma M_A = 0: \mathbf{r}_{C/A} \times \mathbf{T}_{CD} + \mathbf{r}_{C/A} \times \mathbf{T}_{CE} + \mathbf{r}_{B/A} \times (-5 \text{ kN})\mathbf{j} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 0 \\ -3 & 1.5 & 1.2 \end{vmatrix} \frac{T}{3.562} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 0 \\ -3 & 1.5 & -1.2 \end{vmatrix} \frac{T}{3.562} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & 0 \\ 0 & -5 & 0 \end{vmatrix} = 0$$

Coefficient of k:

$$2 \left[3 \times 1.5 \times \frac{T}{3.562} \right] - 10 = 0 \quad T = 3.958 \text{ kN}$$

$$\Sigma F = 0: A + T_{CD} + T_{CE} - 5\mathbf{j} = 0$$

PROBLEM 4.109 (Continued)

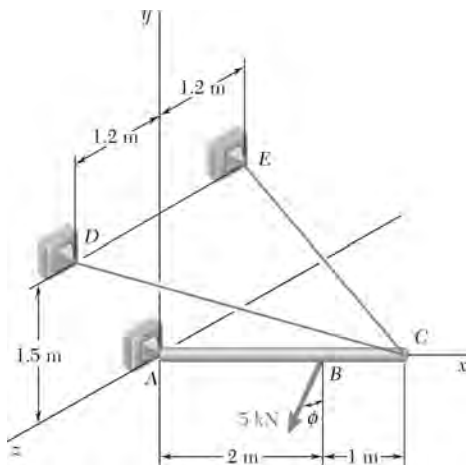
Coefficient of **k**: $A_z = 0$

Coefficient of **i**: $A_x - 2[3.958 \times 3/3.562] = 0$ $A_x = 6.67 \text{ kN}$

Coefficient of **j**: $A_y + 2[3.958 \times 1.5/3.562] - 5 = 0$ $A_y = 1.667 \text{ kN}$

(a) $T_{CD} = T_{CE} = 3.96 \text{ kN}$

(b) $\mathbf{A} = (6.67 \text{ kN})\mathbf{i} + (1.667 \text{ kN})\mathbf{j} \blacktriangleleft$



PROBLEM 4.110

A 3-m pole is supported by a ball-and-socket joint at A and by the cables CD and CE. Knowing that the line of action of the 5-kN force forms an angle $\phi = 30^\circ$ with the vertical xy plane, determine (a) the tension in cables CD and CE, (b) the reaction at A.

SOLUTION

Free-Body Diagram:

Five unknowns and six Eqs. of equilibrium but equilibrium is maintained ($\Sigma M_{AC} = 0$)

$$\begin{aligned} \mathbf{r}_{B/A} &= 2\mathbf{i} \\ \mathbf{r}_{C/A} &= 3\mathbf{i} \\ \text{Load at B.} \quad &= -(5\cos 30)\mathbf{j} + (5\sin 30)\mathbf{k} \\ &= -4.33\mathbf{j} + 2.5\mathbf{k} \end{aligned}$$

$$\overline{CD} = -3\mathbf{i} + 1.5\mathbf{j} + 1.2\mathbf{k} \quad CD = 3.562 \text{ m}$$

$$\mathbf{T}_{CD} = T_{CD} \frac{\overline{CD}}{CD} = \frac{T}{3.562} (-3\mathbf{i} + 1.5\mathbf{j} + 1.2\mathbf{k})$$

$$\text{Similarly,} \quad \mathbf{T}_{CE} = \frac{T}{3.562} (-3\mathbf{i} + 1.5\mathbf{j} - 1.2\mathbf{k})$$

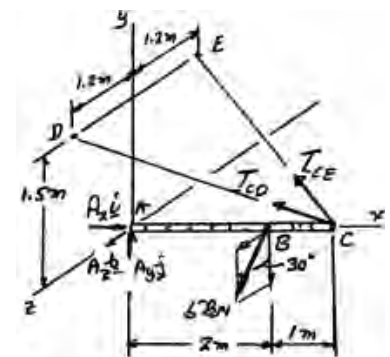
$$\Sigma M_A = 0: \mathbf{r}_{C/A} \times \mathbf{T}_{CD} + \mathbf{r}_{C/A} \times \mathbf{T}_{CE} + \mathbf{r}_{B/A} \times (-4.33\mathbf{j} + 2.5\mathbf{k}) = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 0 \\ -3 & 1.5 & 1.2 \end{vmatrix} \frac{T_{CD}}{3.562} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 0 \\ -3 & 1.5 & -1.2 \end{vmatrix} \frac{T_{CE}}{3.562} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & 0 \\ 0 & -4.33 & 2.5 \end{vmatrix} = 0$$

Equate coefficients of unit vectors to zero.

$$\mathbf{j}: -3.6 \frac{T_{CD}}{3.562} + 3.6 \frac{T_{CE}}{3.562} - 5 = 0$$

$$-3.6T_{CD} + 3.6T_{CE} - 17.810 = 0 \quad (1)$$



PROBLEM 4.110 (Continued)

$$\begin{aligned} \mathbf{k}: \quad 4.5 \frac{T_{CD}}{3.562} + 4.5 \frac{T_{CE}}{3.562} - 8.66 &= 0 \\ 4.5T_{CD} + 4.5T_{CE} &= 30.846 \end{aligned} \quad (2)$$

$$(2) + 1.25(1): \quad 9T_{CE} - 53.11 = 0 \quad T_{CE} = 5.901 \text{ kN}$$

$$\begin{aligned} \text{Eq. (1):} \quad -3.6T_{CD} + 3.6(5.901) - 17.810 &= 0 \\ T_{CD} &= 0.954 \text{ kN} \end{aligned}$$

$$\Sigma F = 0: \quad \mathbf{A} + \mathbf{T}_{CD} + \mathbf{T}_{CE} - 4.33\mathbf{j} + 2.5\mathbf{k} = 0$$

$$\begin{aligned} \mathbf{i}: \quad A_x + \frac{0.954}{3.562}(-3) + \frac{5.901}{3.562}(-3) &= 0 \\ A_x &= 5.77 \text{ kN} \end{aligned}$$

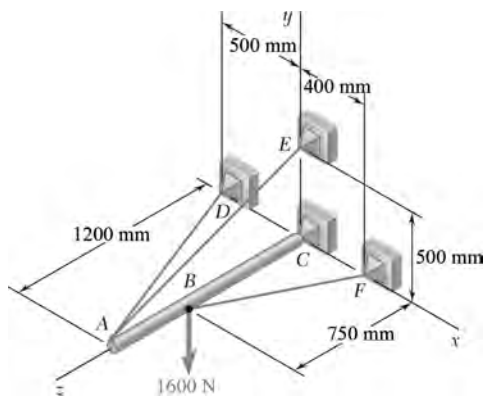
$$\begin{aligned} \mathbf{j}: \quad A_y + \frac{0.954}{3.562}(1.5) + \frac{5.901}{3.562}(1.5) - 4.33 &= 0 \\ A_y &= 1.443 \text{ kN} \end{aligned}$$

$$\begin{aligned} \mathbf{k}: \quad A_z + \frac{0.954}{3.562}(1.2) + \frac{5.901}{3.562}(-1.2) + 2.5 &= 0 \\ A_z &= -0.833 \text{ kN} \end{aligned}$$

Answers:

$$(a) \quad T_{CD} = 0.954 \text{ kN} \quad T_{CE} = 5.90 \text{ kN} \quad \blacktriangleleft$$

$$(b) \quad \mathbf{A} = (5.77 \text{ kN})\mathbf{i} + (1.443 \text{ kN})\mathbf{j} - (0.833 \text{ kN})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.111

A 1200 mm boom is held by a ball-and-socket joint at C and by two cables BF and DAE; cable DAE passes around a frictionless pulley at A. For the loading shown, determine the tension in each cable and the reaction at C.

SOLUTION

Free-Body Diagram:

Five unknowns and six equations of equilibrium but equilibrium is maintained ($\Sigma M_{AC} = 0$).

T = Tension in both parts of cable DAE.

$$\mathbf{r}_B = 750 \text{ mm } \mathbf{k}$$

$$\mathbf{r}_A = 1200 \text{ mm } \mathbf{k}$$

$$\overline{AD} = -500\mathbf{i} - 1200\mathbf{k} \quad AD = 1300 \text{ mm}$$

$$\overline{AE} = 500\mathbf{j} - 1200\mathbf{k} \quad AE = 1300 \text{ mm}$$

$$\overline{BF} = 400\mathbf{i} - 750\mathbf{k} \quad BF = 850 \text{ mm}$$

$$\mathbf{T}_{AD} = T \frac{\overline{AD}}{AD} = \frac{T}{1300} (-500\mathbf{i} - 1200\mathbf{k}) = \frac{T}{13} (-5\mathbf{i} - 12\mathbf{k})$$

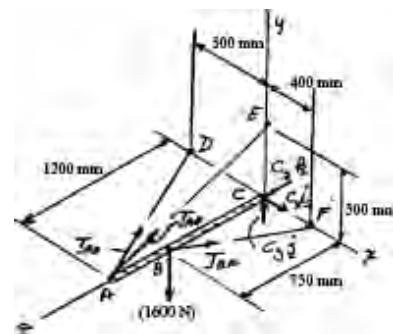
$$\mathbf{T}_{AE} = T \frac{\overline{AE}}{AE} = \frac{T}{1300} (500\mathbf{j} - 1200\mathbf{k}) = \frac{T}{13} (5\mathbf{j} - 12\mathbf{k})$$

$$\mathbf{T}_{BF} = T_{BF} \frac{\overline{BF}}{BF} = \frac{T_{BF}}{850} (400\mathbf{i} - 750\mathbf{k}) = \frac{T_{BF}}{17} (8\mathbf{i} - 15\mathbf{k})$$

$$\Sigma \mathbf{M}_C = 0: \mathbf{r}_A \times \mathbf{T}_{AD} + \mathbf{r}_A \times \mathbf{T}_{AE} + \mathbf{r}_B \times \mathbf{T}_{BF} + \mathbf{r}_B \times (-1600 \text{ N})\mathbf{j} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1200 \\ -5 & 0 & -12 \end{vmatrix} \frac{T}{13} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1200 \\ 0 & 5 & -12 \end{vmatrix} \frac{T}{13} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 750 \\ 8 & 0 & -15 \end{vmatrix} \frac{T_{BF}}{17} + (750\mathbf{k}) \times (-1600\mathbf{j}) = 0$$

$$\text{Coefficient of } \mathbf{i}: -\frac{6000}{13}T + 1,200,000 = 0 \quad T = 2600 \text{ N}$$



PROBLEM 4.111 (Continued)

$$\begin{aligned} \text{Coefficient of } \mathbf{j}: \quad & -\frac{6000}{13}T + \frac{6000}{17}T_{BD} = 0 \\ & T_{BF} = \frac{17}{13}T = \frac{17}{13}(2600) \quad T_{BF} = 3400 \text{ N} \end{aligned}$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{T}_{AD} + \mathbf{T}_{AE} + \mathbf{T}_{BF} - 1600\mathbf{j} + \mathbf{C} = 0$$

$$\begin{aligned} \text{Coefficient of } \mathbf{i}: \quad & -\frac{5}{13}(2600) + \frac{8}{17}(3400) + C_x = 0 \\ & C_x = -600 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } \mathbf{j}: \quad & \frac{5}{13}(2600) - 1600 + C_y = 0 \\ & C_y = +600 \text{ N} \end{aligned}$$

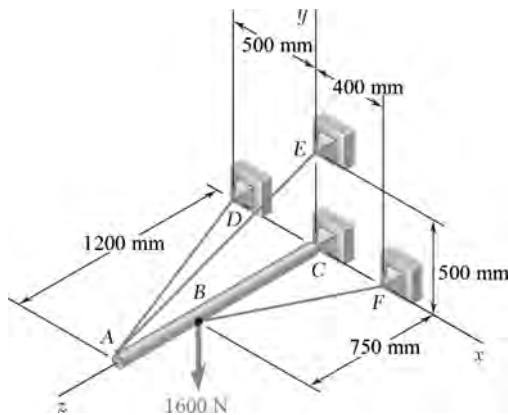
$$\begin{aligned} \text{Coefficient of } \mathbf{k}: \quad & -\frac{12}{13}(2600) - \frac{12}{13}(2600) - \frac{15}{17}(3400) + C_z = 0 \\ & C_z = 7800 \text{ N} \end{aligned}$$

$$\text{Answers: } T_{DAE} = T$$

$$T_{DAE} = 2600 \text{ N} \quad \blacktriangleleft$$

$$T_{BD} = 3400 \text{ N} \quad \blacktriangleleft$$

$$\mathbf{C} = -(600 \text{ N})\mathbf{i} + (600 \text{ N})\mathbf{j} + (7800 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.112

Solve Problem 4.111, assuming that the 1600 N load is applied at A.

PROBLEM 4.111 A 1200 mm boom is held by a ball-and-socket joint at C and by two cables BF and DAE; cable DAE passes around a frictionless pulley at A. For the loading shown, determine the tension in each cable and the reaction at C.

SOLUTION

Free-Body Diagram:

Five unknowns and six equations of equilibrium but equilibrium is maintained ($\Sigma M_{AC} = 0$).

T = tension in both parts of cable DAE.

$$\mathbf{r}_B = 750 \text{ mm } \mathbf{k}$$

$$\mathbf{r}_A = 1200 \text{ mm } \mathbf{k}$$

$$\overline{AD} = -500\mathbf{i} - 1200\mathbf{k} \quad AD = 1300 \text{ mm}$$

$$\overline{AE} = 500\mathbf{j} - 1200\mathbf{k} \quad AE = 1300 \text{ mm}$$

$$\overline{BF} = 400\mathbf{i} - 750\mathbf{k} \quad BF = 850 \text{ mm}$$

$$\mathbf{T}_{AD} = T \frac{\overline{AD}}{AD} = \frac{T}{1300}(-500\mathbf{i} - 1200\mathbf{k}) = \frac{T}{13}(-5\mathbf{i} - 12\mathbf{k})$$

$$\mathbf{T}_{AE} = T \frac{\overline{AE}}{AE} = \frac{T}{1300}(500\mathbf{j} - 1200\mathbf{k}) = \frac{T}{13}(5\mathbf{j} - 12\mathbf{k})$$

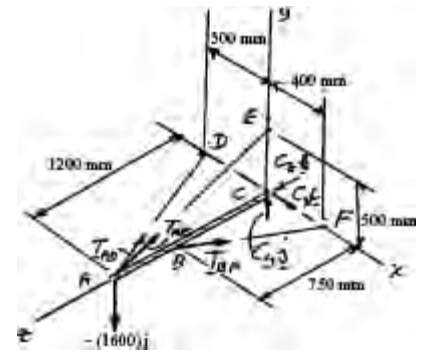
$$\mathbf{T}_{BF} = T_{BF} \frac{\overline{BF}}{BF} = \frac{T_{BF}}{850}(400\mathbf{i} - 750\mathbf{k}) = \frac{T_{BF}}{17}(8\mathbf{i} - 15\mathbf{k})$$

$$\Sigma M_C = 0: \quad \mathbf{r}_A \times \mathbf{T}_{AD} + \mathbf{r}_A \times \mathbf{T}_{AE} + \mathbf{r}_B \times \mathbf{T}_{BF} + \mathbf{r}_A \times (-1600 \text{ N})\mathbf{j} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1200 \\ -5 & 0 & -12 \end{vmatrix} \frac{T}{13} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 1200 \\ 0 & 5 & -12 \end{vmatrix} \frac{T}{13} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 750 \\ 8 & 0 & -15 \end{vmatrix} \frac{T_{BF}}{17} + 1200\mathbf{k} \times (-1600\mathbf{j}) = 0$$

Coefficient of $\mathbf{i}$:

$$-\frac{6000}{13}T + 1,920,000 = 0 \quad T = 4160 \text{ N}$$



PROBLEM 4.112 (Continued)

Coefficient of **j**: $-\frac{6000}{13}T + \frac{6000}{17}T_{BD} = 0$

$$T_{BD} = \frac{17}{13}T = \frac{17}{13}(4160) \quad T_{BD} = 5440 \text{ N}$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{T}_{AD} + \mathbf{T}_{AE} + \mathbf{T}_{BF} - 1600\mathbf{j} + \mathbf{C} = 0$$

Coefficient of **i**: $-\frac{5}{13}(4160) + \frac{8}{17}(5440) + C_x = 0$

$$C_x = -960 \text{ N}$$

Coefficient of **j**: $\frac{5}{13}(4160) - 1600 + C_y = 0$

$$C_y = 0$$

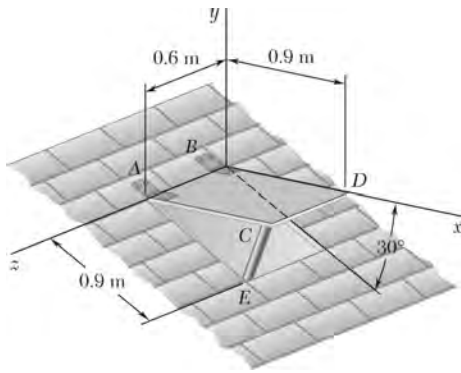
Coefficient of **k**: $-\frac{12}{13}(4160) - \frac{12}{13}(4160) - \frac{15}{17}(5440) + C_z = 0$

$$C_z = 12480 \text{ N}$$

Answers: $T_{DAE} = T$ $T_{DAE} = 4160 \text{ N} \quad \blacktriangleleft$

$T_{BD} = 5440 \text{ N} \quad \blacktriangleleft$

$\mathbf{C} = -(960 \text{ N})\mathbf{i} + (12480 \text{ N})\mathbf{k} \quad \blacktriangleleft$

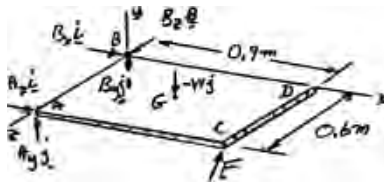


PROBLEM 4.113

A 20-kg cover for a roof opening is hinged at corners *A* and *B*. The roof forms an angle of 30° with the horizontal, and the cover is maintained in a horizontal position by the brace *CE*. Determine (a) the magnitude of the force exerted by the brace, (b) the reactions at the hinges. Assume that the hinge at *A* does not exert any axial thrust.

SOLUTION

Force exerted by *CD*



$$\mathbf{F} = F(\sin 75^\circ)\mathbf{i} + F(\cos 75^\circ)\mathbf{j}$$

$$\mathbf{F} = F(0.2588\mathbf{i} + 0.9659\mathbf{j})$$

$$W = mg = 20 \text{ kg}(9.81 \text{ m/s}^2) = 196.2 \text{ N}$$

$$\mathbf{r}_{A/B} = 0.6\mathbf{k}$$

$$\mathbf{r}_{C/B} = 0.9\mathbf{i} + 0.6\mathbf{k}$$

$$\mathbf{r}_{G/B} = 0.45\mathbf{i} + 0.3\mathbf{k}$$

$$\mathbf{F} = F(0.2588\mathbf{i} + 0.9659\mathbf{j})$$

$$\Sigma \mathbf{M}_B = 0: \mathbf{r}_{G/B} \times (-196.2\mathbf{j}) + \mathbf{r}_{C/B} \times \mathbf{F} + \mathbf{r}_{A/B} \times \mathbf{A} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.45 & 0 & 0.3 \\ 0 & -196.2 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0.9 & 0 & 0.6 \\ 0.2588 & +0.9659 & 0 \end{vmatrix} F + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 0.6 \\ A_x & A_y & 0 \end{vmatrix} = 0$$

$$\text{Coefficient of } \mathbf{i}: +58.86 - 0.5796F - 0.6A_y = 0 \quad (1)$$

$$\text{Coefficient of } \mathbf{j}: +0.1553F + 0.6A_x = 0 \quad (2)$$

$$\text{Coefficient of } \mathbf{k}: -88.29 + 0.8693F = 0: F = 101.56 \text{ N}$$

$$\text{Eq. (2): } +58.86 - 0.5796(101.56) - 0.6A_y = 0 \quad A_y = 0$$

$$\text{Eq. (3): } +0.1553(101.56) + 0.6A_x = 0 \quad A_x = -26.29 \text{ N}$$

$$F = 101.6 \text{ N}$$

$$\mathbf{A} = -(26.3 \text{ N})\mathbf{i} \quad \blacktriangleleft$$

$$\Sigma \mathbf{F}: \mathbf{A} + \mathbf{B} + \mathbf{F} - W\mathbf{j} = 0$$

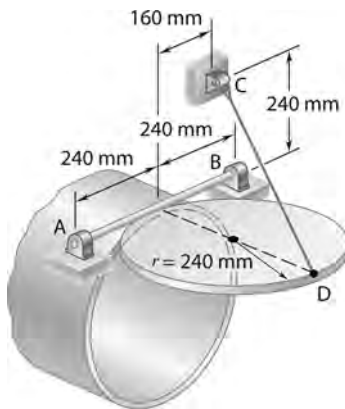
$$\text{Coefficient of } \mathbf{i}: 26.29 + B_x + 0.2588(101.56) = 0 \quad B_x = 0$$

$$\text{Coefficient of } \mathbf{j}: B_y + 0.9659(101.56) - 196.2 = 0 \quad B_y = 98.1 \text{ N}$$

$$\text{Coefficient of } \mathbf{k}: B_z = 0$$

$$\mathbf{B} = (98.1 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

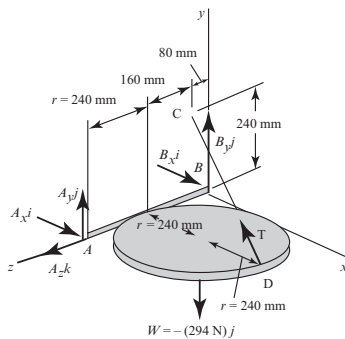
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PROBLEM 4.114

A uniform pipe cover of radius $r = 240$ mm and mass 30 kg is held in a horizontal position by the cable CD . Assuming that the bearing at B does not exert any axial thrust, determine the tension in the cable and the reaction at A and B .

SOLUTION



Free-Body Diagram:

$$W = -mg\mathbf{j} = -(30 \text{ kg})(9.81 \text{ m/s}^2)\mathbf{j} = -(294 \text{ N})\mathbf{j}$$

$$\overrightarrow{DC} = -(480 \text{ mm})\mathbf{i} + (240 \text{ mm})\mathbf{j} - (160 \text{ mm})\mathbf{k} \quad DC = 560 \text{ mm}$$

$$\mathbf{T} = T \frac{\overrightarrow{DC}}{DC} = -\frac{6}{7}T\mathbf{i} + \frac{3}{7}T\mathbf{j} - \frac{2}{7}T\mathbf{k}$$

Equilibrium Equations:

$$\Sigma \mathbf{F} = 0: A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} + B_x\mathbf{i} + B_y\mathbf{j} + \mathbf{T} - (294 \text{ N})\mathbf{j} = 0$$

$$(A_x + B_x - \frac{6}{7}T)\mathbf{i} + (A_y + B_y + \frac{3}{7}T - 294 \text{ N})\mathbf{j} + (A_z - \frac{2}{7}T)\mathbf{k} = 0 \quad (1)$$

$$\Sigma \mathbf{M}_B = \Sigma (\mathbf{r} \times \mathbf{F}) = 0:$$

$$2r\mathbf{k} \times (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) + (2r\mathbf{i} + r\mathbf{k}) \times (-\frac{6}{7}T\mathbf{i} + \frac{3}{7}T\mathbf{j} - \frac{2}{7}T\mathbf{k}) + (r\mathbf{i} + r\mathbf{k}) \times (-294 \text{ N})\mathbf{j} = 0$$

$$(-2A_y - \frac{3}{7}T + 294 \text{ N})r\mathbf{i} + (2A_x - \frac{2}{7}T)r\mathbf{j} + (\frac{6}{7}T - 294 \text{ N})r\mathbf{k} = 0 \quad (2)$$

Setting the coefficients of the unit vectors equal to zero in Eq. (2)

$$A_x = +49.0 \text{ N} \quad A_y = +73.5 \text{ N} \quad T = 343 \text{ N} \quad \blacktriangleleft$$

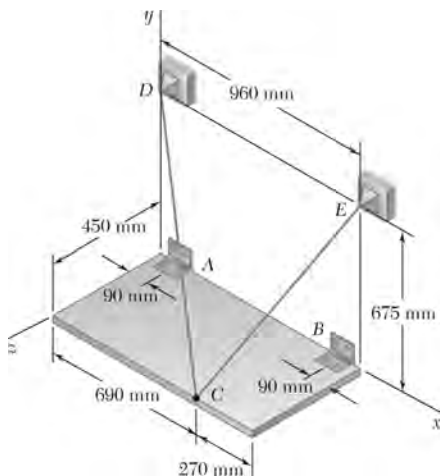
Setting the coefficients of the unit vectors equal to zero in Eq. (1), we obtain three more scalar equations. After substituting the values of T , A_x and A_y into these equations we obtain

$$A_z = +98.0 \text{ N} \quad B_x = +245 \text{ N} \quad B_y = +73.5 \text{ N}$$

The reactions at A and B therefore

$$\mathbf{A} = +(49.0 \text{ N})\mathbf{i} + (73.5 \text{ N})\mathbf{j} + (98.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{B} = +(245 \text{ N})\mathbf{i} + (73.5 \text{ N})\mathbf{j} \quad \blacktriangleleft$$



PROBLEM 4.115

A 100-kg uniform rectangular plate is supported in the position shown by hinges A and B and by cable DCE that passes over a frictionless hook at C. Assuming that the tension is the same in both parts of the cable, determine (a) the tension in the cable, (b) the reactions at A and B. Assume that the hinge at B does not exert any axial thrust.

SOLUTION

$$\mathbf{r}_{B/A}(960 - 180)\mathbf{i} = 780\mathbf{i}$$

$$\begin{aligned}\mathbf{r}_{G/A} &= \left(\frac{960}{2} - 90\right)\mathbf{i} + \frac{450}{2}\mathbf{k} \\ &= 390\mathbf{i} + 225\mathbf{k}\end{aligned}$$

$$\mathbf{r}_{C/A} = 600\mathbf{i} + 450\mathbf{k}$$

T = Tension in cable DCE

$$\overline{CD} = -690\mathbf{i} + 675\mathbf{j} - 450\mathbf{k} \quad CD = 1065 \text{ mm}$$

$$\overline{CE} = 270\mathbf{i} + 675\mathbf{j} - 450\mathbf{k} \quad CE = 855 \text{ mm}$$

$$\mathbf{T}_{CD} = \frac{T}{1065}(-690\mathbf{i} + 675\mathbf{j} - 450\mathbf{k})$$

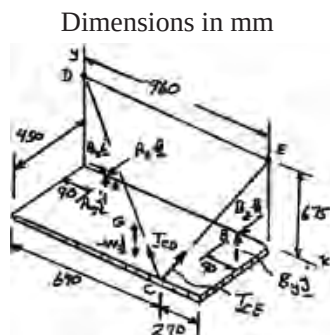
$$\mathbf{T}_{CE} = \frac{T}{855}(270\mathbf{i} + 675\mathbf{j} - 450\mathbf{k})$$

$$\mathbf{W} = -mg\mathbf{i} = -(100 \text{ kg})(9.81 \text{ m/s}^2)\mathbf{j} = -(981 \text{ N})\mathbf{j}$$

$$\Sigma \mathbf{M}_A = 0: \mathbf{r}_{C/A} \times \mathbf{T}_{CD} + \mathbf{r}_{C/A} \times \mathbf{T}_{CE} + \mathbf{r}_{G/A} \times (-W\mathbf{j}) + \mathbf{r}_{B/A} \times \mathbf{B} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 600 & 0 & 450 \\ -690 & 675 & -450 \end{vmatrix} \frac{T}{1065} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 600 & 0 & 450 \\ 270 & 675 & -450 \end{vmatrix} \frac{T}{855} +$$

$$+ \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 390 & 0 & 225 \\ 0 & -981 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 780 & 0 & 0 \\ 0 & B_y & B_z \end{vmatrix} = 0$$



PROBLEM 4.115 (Continued)

$$\begin{aligned} \text{Coefficient of } \mathbf{i}: \quad & -(450)(675)\frac{T}{1065} - (450)(675)\frac{T}{855} + 220.725 \times 10^3 = 0 \\ & T = 344.6 \text{ N} \qquad T = 345 \text{ N} \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } \mathbf{j}: \quad & (-690 \times 450 + 600 \times 450)\frac{344.6}{1065} + (270 \times 450 + 600 \times 450)\frac{344.6}{855} - 780B_z = 0 \\ & B_z = 185.49 \text{ N} \end{aligned}$$

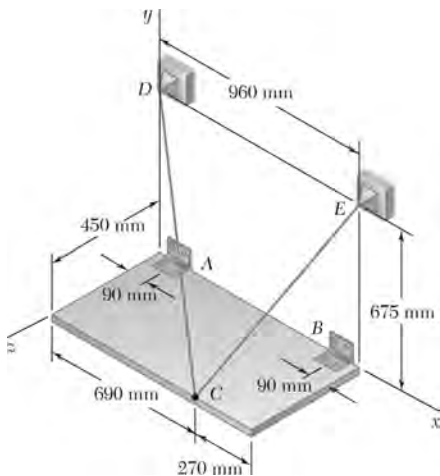
$$\begin{aligned} \text{Coefficient of } \mathbf{k}: \quad & (600)(675)\frac{344.6}{1065} + (600)(675)\frac{344.6}{855} - 382.59 \times 10^3 + 780B_y \quad B_y = 113.2 \text{ N} \\ & \mathbf{B} = (113.2 \text{ N})\mathbf{j} + (185.5 \text{ N})\mathbf{k} \quad \blacktriangleleft \end{aligned}$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{A} + \mathbf{B} + \mathbf{T}_{CD} + \mathbf{T}_{CE} + \mathbf{W} = 0$$

$$\begin{aligned} \text{Coefficient of } \mathbf{i}: \quad & A_x - \frac{690}{1065}(344.6) + \frac{270}{855}(344.6) = 0 \qquad A_x = 114.4 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } \mathbf{j}: \quad & A_y + 113.2 + \frac{675}{1065}(344.6) + \frac{675}{855}(344.6) - 981 = 0 \quad A_y = 377 \text{ N} \end{aligned}$$

$$\begin{aligned} \text{Coefficient of } \mathbf{k}: \quad & A_z + 185.5 - \frac{450}{1065}(344.6) - \frac{450}{855}(344.6) = 0 \qquad A_z = 141.5 \text{ N} \\ & \mathbf{A} = (114.4 \text{ N})\mathbf{i} + (377 \text{ N})\mathbf{j} + (144.5 \text{ N})\mathbf{k} \quad \blacktriangleleft \end{aligned}$$



PROBLEM 4.116

Solve Problem 4.115, assuming that cable DCE is replaced by a cable attached to Point E and hook C .

PROBLEM 4.115 A 100-kg uniform rectangular plate is supported in the position shown by hinges A and B and by cable DCE that passes over a frictionless hook at C . Assuming that the tension is the same in both parts of the cable, determine (a) the tension in the cable, (b) the reactions at A and B . Assume that the hinge at B does not exert any axial thrust.

SOLUTION

See solution to Problem 4.115 for free-body diagram and analysis leading to the following:

$$CD = 1065 \text{ mm}$$

$$CE = 855 \text{ mm}$$

Now:

$$\mathbf{T}_{CD} = \frac{T}{1065}(-690\mathbf{i} + 675\mathbf{j} - 450\mathbf{k})$$

$$\mathbf{T}_{CE} = \frac{T}{855}(270\mathbf{i} + 675\mathbf{j} - 450\mathbf{k})$$

$$\mathbf{W} = -mg\mathbf{i} = -(100 \text{ kg})(9.81 \text{ m/s}^2)\mathbf{j} = -(981 \text{ N})\mathbf{j}$$

$$\Sigma \mathbf{M}_A = 0: \mathbf{r}_{C/A} \times \mathbf{T}_{CE} + \mathbf{r}_{G/A} \times (-W\mathbf{j}) + \mathbf{r}_{B/A} \times \mathbf{B} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 600 & 0 & 450 \\ 270 & 675 & -450 \end{vmatrix} \frac{T}{855} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 390 & 0 & 225 \\ 0 & -981 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 780 & 0 & 0 \\ 0 & B_y & B_z \end{vmatrix} = 0$$

Coefficient of $\mathbf{i}$: $-(450)(675)\frac{T}{855} + 220.725 \times 10^3 = 0$

$$T = 621.3 \text{ N}$$

$$T = 621 \text{ N} \quad \blacktriangleleft$$

Coefficient of $\mathbf{j}$: $(270 \times 450 + 600 \times 450)\frac{621.3}{855} - 980B_z = 0 \quad B_z = 364.7 \text{ N}$

Coefficient of $\mathbf{k}$: $(600)(675)\frac{621.3}{855} - 382.59 \times 10^3 + 780B_y = 0 \quad B_y = 113.2 \text{ N}$

$$\mathbf{B} = (113.2 \text{ N})\mathbf{j} + (365 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 4.116 (Continued)

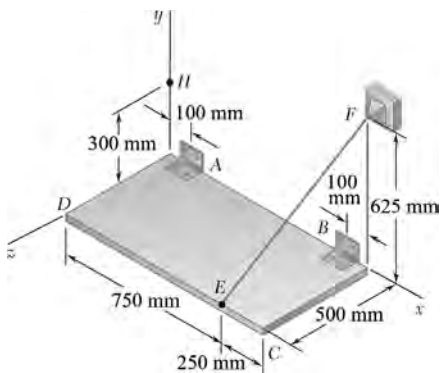
$$\Sigma \mathbf{F} = 0: \quad \mathbf{A} + \mathbf{B} + \mathbf{T}_{CE} + \mathbf{W} = 0$$

Coefficient of **i**: $A_x + \frac{270}{855}(621.3) = 0 \quad A_x = -196.2 \text{ N}$

Coefficient of **j**: $A_y + 113.2 + \frac{675}{855}(621.3) - 981 = 0 \quad A_y = 377.3 \text{ N}$

Coefficient of **k**: $A_z + 364.7 - \frac{450}{855}(621.3) = 0 \quad A_z = -37.7 \text{ N}$

$$\mathbf{A} = -(196.2 \text{ N})\mathbf{i} + (377 \text{ N})\mathbf{j} - (37.7 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

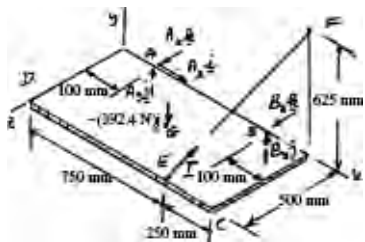


PROBLEM 4.117

The rectangular plate shown weighs 40 kg and is held in the position shown by hinges at A and B and by cable EF. Assuming that the hinge at B does not exert any axial thrust, determine (a) the tension in the cable, (b) the reactions at A and B.

SOLUTION

Weight of plate = 392.4 N



$$\mathbf{r}_{B/A} = (1000 - 200)\mathbf{i} = 800\mathbf{i}$$

$$\begin{aligned}\mathbf{r}_{E/A} &= (750 - 100)\mathbf{i} + 500\mathbf{k} \\ &= 650\mathbf{i} + 500\mathbf{k}\end{aligned}$$

$$\begin{aligned}\mathbf{r}_{G/A} &= \frac{1000}{2}\mathbf{i} + 250\mathbf{k} \\ &= 500\mathbf{i} + 250\mathbf{k}\end{aligned}$$

$$\overline{EF} = 250\mathbf{i} + 625\mathbf{j} - 500\mathbf{k}$$

$$EF = 838.525 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{EF}}{EF} = T(0.29814\mathbf{i} + 0.74536\mathbf{j} - 0.59629\mathbf{k})$$

$$\Sigma \mathbf{M}_A = 0: \mathbf{r}_{E/A} \times \mathbf{T} + \mathbf{r}_{G/A} \times (-392.4\mathbf{j}) + \mathbf{r}_{B/A} \times \mathbf{B} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 650 & 0 & 500 \\ 0.29814 & 0.74536 & -0.59629 \end{vmatrix} T + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 500 & 0 & 250 \\ 0 & -392.4 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 800 & 0 & 0 \\ 0 & B_y & B_z \end{vmatrix} = 0$$

Coefficient of $\mathbf{i}$: $-372.68T + 98100 = 0 \Rightarrow T = 263.23 \text{ N}$

$T = 263 \text{ N} \quad \blacktriangleleft$

Coefficient of $\mathbf{j}$: $(387.5885 + 149.07)(263.23) - 800B_z = 0 \Rightarrow B_z = 176.581 \text{ N}$

Coefficient of $\mathbf{k}$: $(484.484)(263.23) - 196200 + 800B_y = 0 \Rightarrow B_y = 85.837 \text{ kN}$

$\mathbf{B} = (85.8 \text{ N})\mathbf{j} + (176.6 \text{ N})\mathbf{k} \quad \blacktriangleleft$

PROBLEM 4.117 (Continued)

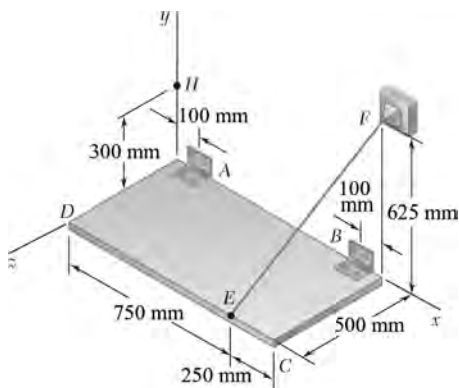
$$\Sigma \mathbf{F} = 0: \mathbf{A} + \mathbf{B} + \mathbf{T} - (392.4 \text{ N})\mathbf{j} = 0$$

Coefficient of **i**: $A_x + 0.29814(263.23) = 0 \quad A_x = -78.479 \text{ N}$

Coefficient of **j**: $A_y + 85.837 + 0.74536(263.23) - 392.4 = 0 \quad A_y = 110.362 \text{ N}$

Coefficient of **k**: $A_z + 176.581 - 0.59629(263.23) = 0 \quad A_z = -19.6196 \text{ N}$

$$\mathbf{A} = -(78.5 \text{ N})\mathbf{i} + (110.4 \text{ N})\mathbf{j} - (19.62 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

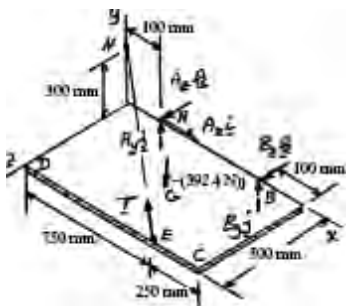


PROBLEM 4.118

Solve Problem 4.117, assuming that cable EF is replaced by a cable attached at points E and H .

PROBLEM 4.117 The rectangular plate shown weighs 40 kg and is held in the position shown by hinges at A and B and by cable EF . Assuming that the hinge at B does not exert any axial thrust, determine (a) the tension in the cable, (b) the reactions at A and B .

SOLUTION



$$\mathbf{r}_{B/A} = (1000 - 200)\mathbf{i} = 800\mathbf{i}$$

$$\mathbf{r}_{E/A} = 650\mathbf{i} + 500\mathbf{k}$$

$$\mathbf{r}_{G/A} = 500\mathbf{i} + 250\mathbf{k}$$

$$\overline{EH} = -750\mathbf{i} + 300\mathbf{j} - 500\mathbf{k}$$

$$EH = 950 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{EH}}{EH} = \frac{T}{950}(-750\mathbf{i} + 300\mathbf{j} - 500\mathbf{k})$$

$$\Sigma \mathbf{M}_A = 0: \mathbf{r}_{E/A} \times \mathbf{T} + \mathbf{r}_{G/A} \times (-392.4\mathbf{j}) + \mathbf{r}_{B/A} \times \mathbf{B} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 650 & 0 & 500 \\ -750 & 300 & -500 \end{vmatrix} \frac{T}{950} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 500 & 0 & 250 \\ 0 & -392.4 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 800 & 0 & 0 \\ 0 & B_y & B_z \end{vmatrix} = 0$$

Coefficient of $\mathbf{i}$: $-(300)(500)\frac{T}{950} + (392.4)(250) = 0 \quad T = 621.3 \text{ N} \quad T = 621 \text{ N} \blacktriangleleft$

Coefficient of $\mathbf{j}$: $(325000 - 375000)\frac{(621.3)}{950} - 800B_z = 0 \quad B_z = -40.875 \text{ N}$

Coefficient of $\mathbf{k}$: $(650)(300)\frac{(621.3)}{950} - (500)(392.4) + 800B_y = 0 \quad B_y = 85.8375 \text{ N}$

$$\mathbf{B} = (85.8 \text{ N})\mathbf{j} - (40.9 \text{ N})\mathbf{k} \blacktriangleleft$$

PROBLEM 4.118 (Continued)

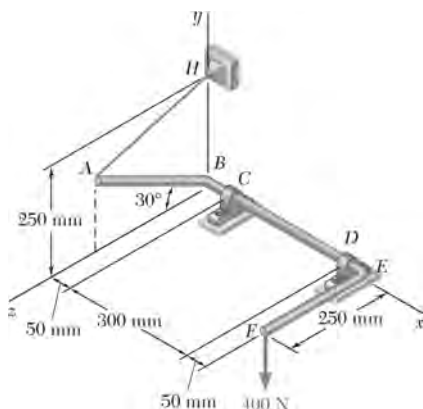
$$\Sigma \mathbf{F} = 0: \quad \mathbf{A} + \mathbf{B} + \mathbf{T} - (392.4 \text{ N})\mathbf{j} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad A_x - \frac{750}{950}(621.3) = 0 \quad A_x = 490.5 \text{ N}$$

$$\text{Coefficient of } \mathbf{j}: \quad A_y + 85.8375 + \frac{300}{950}(621.3) - 392.4 = 0 \quad A_y = 110.3625 \text{ N}$$

$$\text{Coefficient of } \mathbf{k}: \quad A_z - 40.875 - \frac{500}{950}(621.3) = 0 \quad A_z = 367.875 \text{ N}$$

$$\mathbf{A} = (491 \text{ N})\mathbf{i} + (110.4 \text{ N})\mathbf{j} + (368 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.119

Solve Problem 4.114, assuming that the bearing at D is removed and that the bearing at C can exert couples about axes parallel to the y and z axes.

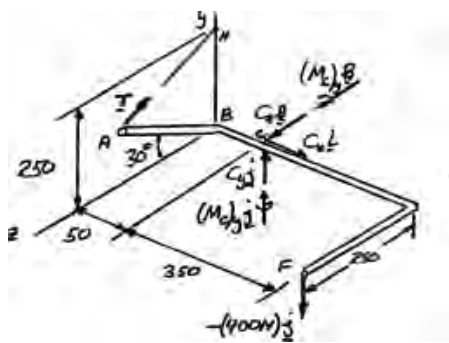
PROBLEM 4.114 The bent rod $ABEF$ is supported by bearings at C and D and by wire AH . Knowing that portion AB of the rod is 250 mm long, determine (a) the tension in wire AH , (b) the reactions at C and D . Assume that the bearing at D does not exert any axial thrust.

SOLUTION

Free-Body Diagram:

$\triangle ABH$ is Equilateral

Dimensions in mm



$$\mathbf{r}_{H/C} = -50\mathbf{i} + 250\mathbf{j}$$

$$\mathbf{r}_{F/C} = 350\mathbf{i} + 250\mathbf{k}$$

$$\mathbf{T} = T(\sin 30^\circ)\mathbf{j} - T(\cos 30^\circ)\mathbf{k} = T(0.5\mathbf{j} - 0.866\mathbf{k})$$

$$\Sigma \mathbf{M}_C = 0: \mathbf{r}_{F/C} \times (-400\mathbf{j}) + \mathbf{r}_{H/C} \times \mathbf{T} + (M_C)_y \mathbf{j} + (M_C)_z \mathbf{k} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 350 & 0 & 250 \\ 0 & -400 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -50 & 250 & 0 \\ 0 & 0.5 & -0.866 \end{vmatrix} T + (M_C)_y \mathbf{j} + (M_C)_z \mathbf{k} = 0$$

Coefficient of $\mathbf{i}$: $+100 \times 10^3 - 216.5T = 0 \quad T = 461.9 \text{ N}$

$T = 462 \text{ N} \quad \blacktriangleleft$

Coefficient of $\mathbf{j}$: $-43.3(461.9) + (M_C)_y = 0$

$$(M_C)_y = 20 \times 10^3 \text{ N} \cdot \text{mm}$$

$$(M_C)_y = 20.0 \text{ N} \cdot \text{m}$$

PROBLEM 4.119 (Continued)

Coefficient of **k**: $-140 \times 10^3 - 25(461.9) + (M_C)_z = 0$

$$(M_C)_z = 151.54 \times 10^3 \text{ N} \cdot \text{mm}$$

$$(M_C)_z = 151.5 \text{ N} \cdot \text{m}$$

$$\Sigma F = 0: \quad C + T - 400\mathbf{j} = 0$$

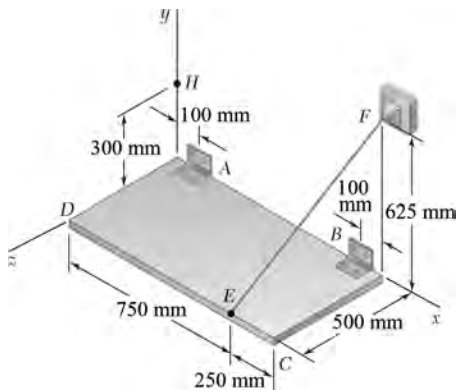
$$\mathbf{M}_C = (20.0 \text{ N} \cdot \text{m})\mathbf{j} + (151.5 \text{ N} \cdot \text{m})\mathbf{k} \quad \blacktriangleleft$$

Coefficient of **i**: $C_x = 0$

Coefficient of **j**: $C_y + 0.5(461.9) - 400 = 0 \quad C_y = 169.1 \text{ N}$

Coefficient of **k**: $C_z - 0.866(461.9) = 0 \quad C_z = 400 \text{ N}$

$$\mathbf{C} = (169.1 \text{ N})\mathbf{j} + (400 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



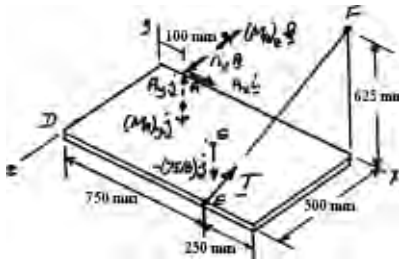
PROBLEM 4.120

Solve Problem 4.117, assuming that the hinge at B is removed and that the hinge at A can exert couples about axes parallel to the y and z axes.

PROBLEM 4.117 The rectangular plate shown weighs 40 kg and is held in the position shown by hinges at A and B and by cable EF. Assuming that the hinge at B does not exert any axial thrust, determine (a) the tension in the cable, (b) the reactions at A and B.

SOLUTION

Weight of plate = $40 \times 9.81 = 392.4 \text{ N}$



$$\mathbf{r}_{E/A} = 650\mathbf{i} + 500\mathbf{k}$$

$$\mathbf{r}_{G/A} = 500\mathbf{i} + 250\mathbf{k}$$

$$\overline{EF} = 250\mathbf{i} + 625\mathbf{j} - 500\mathbf{k}$$

$$EF = 838.525 \text{ mm}$$

$$T = T \frac{\overline{EF}}{EF} = T(0.29814\mathbf{i} + 0.75436\mathbf{j} - 0.59629\mathbf{k})$$

$$\Sigma \mathbf{M}_A = 0: \mathbf{r}_{E/A} \times \mathbf{T} + \mathbf{r}_{G/A} \times (-75\mathbf{j}) + (M_A)_y \mathbf{j} + (M_A)_z \mathbf{k} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 650 & 0 & 500 \\ 0.29814 & 0.75430 & -0.59629 \end{vmatrix} T + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 500 & 0 & 250 \\ 0 & -392.4 & 0 \end{vmatrix} + (M_A)_y \mathbf{j} + (M_A)_z \mathbf{k} = 0$$

$$\text{Coefficient of } \mathbf{i}: -372.68T + 98100 = 0 \quad T = 263.23 \text{ N} \quad T = 263 \text{ N} \blacktriangleleft$$

$$\text{Coefficient of } \mathbf{j}: (387.5885 + 149.07)(263.23) + (M_A)_y = 0 \quad (M_A)_y = 141264.6 \text{ N} \cdot \text{mm}$$

$$\text{Coefficient of } \mathbf{k}: (484.484)(263.23) - 196200 + (M_A)_z = 0 \quad (M_A)_z = 68669.3 \text{ N} \cdot \text{mm}$$

$$\Sigma F = 0: A + T - 392.4\mathbf{j} = 0$$

$$\mathbf{M}_A = -(141300 \text{ N} \cdot \text{mm})\mathbf{j} + (68700 \text{ N} \cdot \text{mm})\mathbf{k} \blacktriangleleft$$

$$\text{Coefficient of } \mathbf{i}: A_x + 0.29814(263.23) = 0 \quad A_x = -78.479 \text{ N}$$

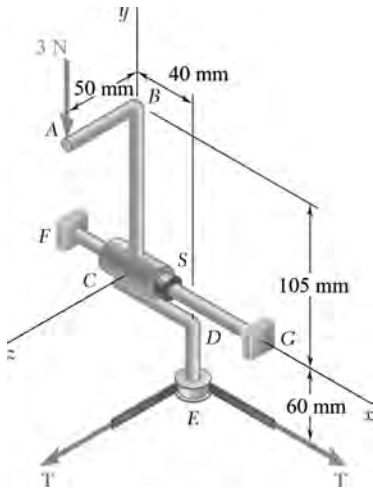
$$\text{Coefficient of } \mathbf{j}: A_y + (0.74536)(263.23) - 392.4 = 0 \quad A_y = 196.199 \text{ N}$$

$$\text{Coefficient of } \mathbf{k}: A_z - 0.59629(263.23) = 0 \quad A_z = 156.961 \text{ N}$$

$$\mathbf{A} = (-78.5 \text{ N})\mathbf{i} + (196.2 \text{ N})\mathbf{j} + (157.0 \text{ N})\mathbf{k} \blacktriangleleft$$

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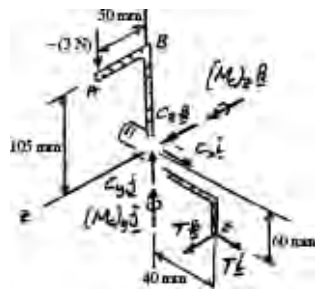
PROBLEM 4.121



The assembly shown is used to control the tension T in a tape that passes around a frictionless spool at E . Collar C is welded to rods ABC and CDE . It can rotate about shaft FG but its motion along the shaft is prevented by a washer S . For the loading shown, determine (a) the tension T in the tape, (b) the reaction at C .

SOLUTION

Free-Body Diagram:



$$\mathbf{r}_{A/C} = 105\mathbf{j} + 50\mathbf{k}$$

$$\mathbf{r}_{E/C} = 40\mathbf{i} - 60\mathbf{j}$$

$$\Sigma \mathbf{M}_C = 0: \quad \mathbf{r}_{A/C} \times (-3\mathbf{j}) + \mathbf{r}_{E/C} \times T(\mathbf{i} + \mathbf{k}) + (M_C)_y \mathbf{j} + (M_C)_z \mathbf{k} = 0$$

$$(105\mathbf{j} + 50\mathbf{k}) \times (-3\mathbf{j}) + (40\mathbf{i} - 60\mathbf{j}) \times T(\mathbf{i} + \mathbf{k}) + (M_C)_y \mathbf{j} + (M_C)_z \mathbf{k} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad 150 - 60T = 0$$

$$T = 2.5 \text{ N} \quad \blacktriangleleft$$

$$\text{Coefficient of } \mathbf{j}: \quad -40(2.5 \text{ N}) + (M_C)_y = 0 \quad (M_C)_y = 100 \text{ N} \cdot \text{mm}$$

$$\text{Coefficient of } \mathbf{k}: \quad 60(2.5 \text{ N}) + (M_C)_z = 0 \quad (M_C)_z = -150 \text{ N} \cdot \text{mm}$$

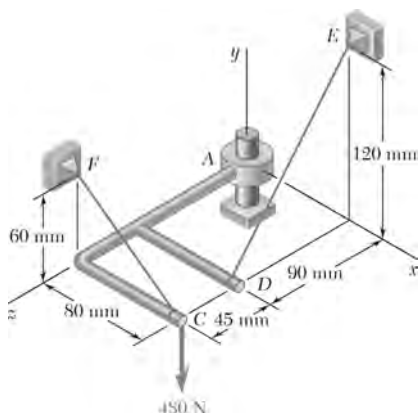
$$\mathbf{M}_C = (100.0 \text{ N} \cdot \text{mm})\mathbf{j} - (150.0 \text{ N} \cdot \text{mm})\mathbf{k} \quad \blacktriangleleft$$

$$\Sigma \mathbf{F} = 0: \quad C_x \mathbf{i} + C_y \mathbf{j} + C_z \mathbf{k} - (3 \text{ N})\mathbf{j} + (2.5 \text{ N})\mathbf{i} + (2.5 \text{ N})\mathbf{k} = 0$$

Equate coefficients of unit vectors to zero.

$$C_x = -2.5 \text{ N} \quad C_y = 3 \text{ N} \quad C_z = -2.5 \text{ N} \quad \blacktriangleleft$$

$$\mathbf{C} = -(2.50 \text{ N})\mathbf{i} + (3.00 \text{ N})\mathbf{j} - (2.50 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.122

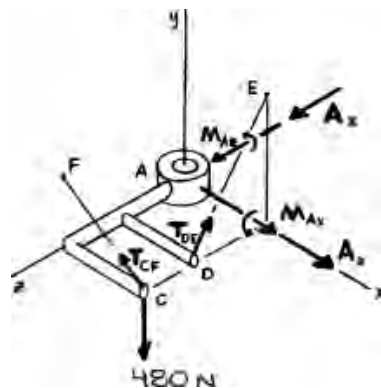
The assembly is welded to collar A that fits on the vertical pin shown. The pin can exert couples about the x and z axes but does not prevent motion about or along the y axis. For the loading shown, determine the tension in each cable and the reaction at A.

SOLUTION

Free-Body Diagram:

First note

$$\begin{aligned} \mathbf{T}_{CF} &= \lambda_{CF} T_{CF} = \frac{-(0.08 \text{ m})\mathbf{i} + (0.06 \text{ m})\mathbf{j}}{\sqrt{(0.08)^2 + (0.06)^2} \text{ m}} T_{CF} \\ &= T_{CF} (-0.8\mathbf{i} + 0.6\mathbf{j}) \\ \mathbf{T}_{DE} &= \lambda_{DE} T_{DE} = \frac{(0.12 \text{ m})\mathbf{j} - (0.09 \text{ m})\mathbf{k}}{\sqrt{(0.12)^2 + (0.09)^2} \text{ m}} T_{DE} \\ &= T_{DE} (0.8\mathbf{j} - 0.6\mathbf{k}) \end{aligned}$$



(a) From f.b.d. of assembly

$$\Sigma F_y = 0: 0.6T_{CF} + 0.8T_{DE} - 480 \text{ N} = 0$$

$$\text{or} \quad 0.6T_{CF} + 0.8T_{DE} = 480 \text{ N} \quad (1)$$

$$\Sigma M_y = 0: -(0.8T_{CF})(0.135 \text{ m}) + (0.6T_{DE})(0.08 \text{ m}) = 0$$

$$\text{or} \quad T_{DE} = 2.25T_{CF} \quad (2)$$

Substituting Equation (2) into Equation (1)

$$0.6T_{CF} + 0.8[(2.25)T_{CF}] = 480 \text{ N}$$

$$T_{CF} = 200.00 \text{ N}$$

$$\text{or} \quad T_{CF} = 200 \text{ N} \quad \blacktriangleleft$$

$$\text{and from Equation (2)} \quad T_{DE} = 2.25(200.00 \text{ N}) = 450.00$$

$$\text{or} \quad T_{DE} = 450 \text{ N} \quad \blacktriangleleft$$

PROBLEM 4.122 (Continued)

(b) From f.b.d. of assembly

$$\Sigma F_z = 0: A_z - (0.6)(450.00 \text{ N}) = 0 \quad A_z = 270.00 \text{ N}$$

$$\Sigma F_x = 0: A_x - (0.8)(200.00 \text{ N}) = 0 \quad A_x = 160.000 \text{ N}$$

$$\text{or } \mathbf{A} = (160.0 \text{ N})\mathbf{i} + (270 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

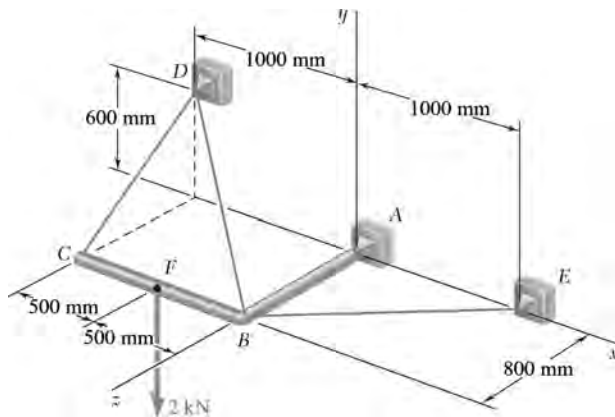
$$\begin{aligned} \Sigma M_x = 0: M_{A_x} + (480 \text{ N})(0.135 \text{ m}) - [(200.00 \text{ N})(0.6)](0.135 \text{ m}) \\ - [(450 \text{ N})(0.8)](0.09 \text{ m}) = 0 \end{aligned}$$

$$M_{A_x} = -16.2000 \text{ N} \cdot \text{m}$$

$$\begin{aligned} \Sigma M_z = 0: M_{A_z} - (480 \text{ N})(0.08 \text{ m}) + [(200.00 \text{ N})(0.6)](0.08 \text{ m}) \\ + [(450 \text{ N})(0.8)](0.08 \text{ m}) = 0 \end{aligned}$$

$$M_{A_z} = 0$$

$$\text{or } \mathbf{M}_A = -(16.20 \text{ N} \cdot \text{m})\mathbf{i} \quad \blacktriangleleft$$

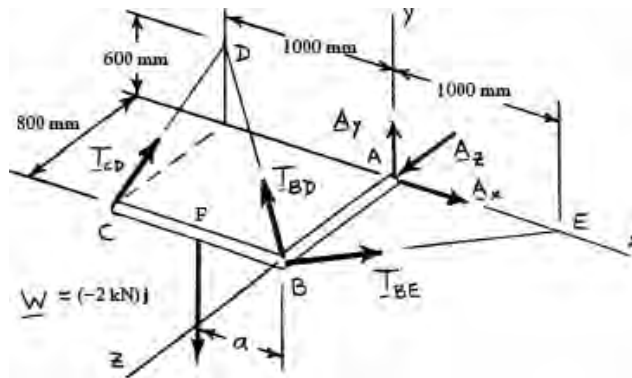


PROBLEM 4.123

The rigid L-shaped member ABC is supported by a ball-and-socket joint at A and by three cables. If a 2 kN load is applied at F , determine the tension in each cable.

SOLUTION

Free-Body Diagram:



In this problem: $a = 500 \text{ mm}$

We have

$$\overline{CD} = (600 \text{ mm})\mathbf{j} - (800 \text{ mm})\mathbf{k} \quad CD = 1000 \text{ mm}$$

$$\overline{BD} = -(1000 \text{ mm})\mathbf{i} + (600 \text{ mm})\mathbf{j} - (800 \text{ mm})\mathbf{k} \quad BD = 1414.21 \text{ mm}$$

$$\overline{BE} = (1000 \text{ mm})\mathbf{i} - (800 \text{ mm})\mathbf{k} \quad BE = 1280.63 \text{ mm}$$

Thus

$$T_{CD} = T_{CD} \frac{\overline{CD}}{CD} = T_{CD}(0.6\mathbf{j} - 0.8\mathbf{k})$$

$$T_{BD} = T_{BD} \frac{\overline{BD}}{BD} = T_{BD}(-0.7071\mathbf{i} + 0.4243\mathbf{j} - 0.5657\mathbf{k})$$

$$T_{BE} = T_{BE} \frac{\overline{BE}}{BE} = T_{BE}(0.7809\mathbf{i} - 0.6247\mathbf{k})$$

$$\Sigma \mathbf{M}_A = 0: (\mathbf{r}_C \times \mathbf{T}_{CD}) + (\mathbf{r}_B \times \mathbf{T}_{BD}) + (\mathbf{r}_B \times \mathbf{T}_{BE}) + (\mathbf{r}_F \times \mathbf{W}) = 0$$

PROBLEM 4.123 (Continued)

Noting that

$$\mathbf{r}_C = -(1000 \text{ mm})\mathbf{i} + (800 \text{ mm})\mathbf{k}$$

$$\mathbf{r}_B = (800 \text{ mm})\mathbf{k}$$

$$\mathbf{r}_F = -a\mathbf{i} + (800 \text{ mm})\mathbf{k}$$

and using determinants, we write

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1000 & 0 & 800 \\ 0 & 0.6 & -0.8 \end{vmatrix} T_{CD} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 800 \\ -0.7071 & 0.4243 & -0.5657 \end{vmatrix} T_{BD} \\ + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 800 \\ 0.7809 & 0 & -0.6247 \end{vmatrix} T_{BE} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -a & 0 & 800 \\ 0 & -2 & 0 \end{vmatrix} = 0$$

Equating to zero the coefficients of the unit vectors:

$$\mathbf{i}: -480T_{CD} - 339.44 T_{BD} + 1600 = 0 \quad (1)$$

$$\mathbf{j}: -800T_{CD} - 565.68T_{BD} + 624.72T_{BE} = 0 \quad (2)$$

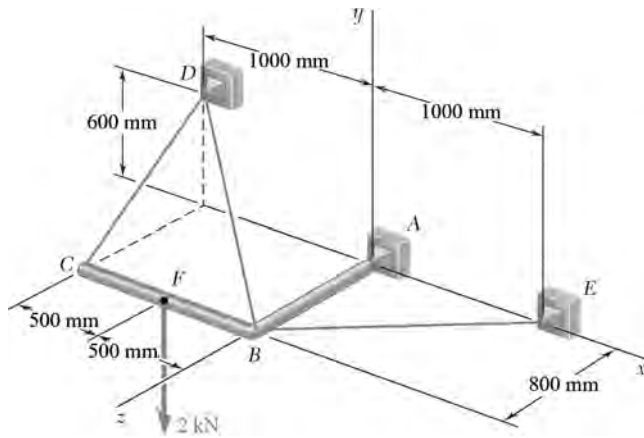
$$\mathbf{k}: -600T_{CD} + 2a = 0 \quad (3)$$

Recalling that $a = 500 \text{ mm}$, Eq. (3) yields

$$T_{CD} = \frac{2(500)}{600} = \frac{5}{3} \text{ kN} \quad T_{CD} = 1.667 \text{ kN} \quad \blacktriangleleft$$

$$\begin{aligned} \text{From (1):} \quad -480\left(\frac{5}{3}\right) - 339.44T_{BD} + 1600 &= 0 \\ T_{BD} &= 2.35682 \text{ kN} \quad T_{BD} = 2.36 \text{ kN} \quad \blacktriangleleft \end{aligned}$$

$$\begin{aligned} \text{From (2):} \quad -800\left(\frac{5}{3}\right) - 565.68(2.35682) + 624.72T_{BE} &= 0 \\ T_{BE} &= 4.2684 \text{ kN} \quad T_{BE} = 4.27 \text{ kN} \quad \blacktriangleleft \end{aligned}$$



PROBLEM 4.124

Solve Problem 4.123, assuming that the 2 kN load is applied at C.

PROBLEM 4.123 The rigid L-shaped member ABC is supported by a ball-and-socket joint at A and by three cables. If a 2 kN load is applied at F, determine the tension in each cable.

SOLUTION

See solution of Problem 4.123 for free-body diagram and derivation of Eqs (1), (2), and (3):

$$-480T_{CD} - 339.44T_{BD} + 1600 = 0 \quad (1)$$

$$-800T_{CD} - 565.68T_{BD} + 624.72T_{BE} = 0 \quad (2)$$

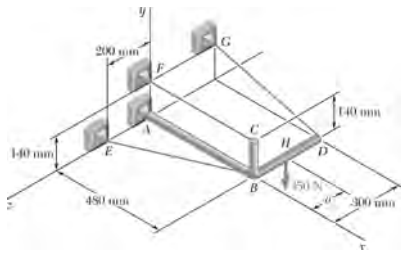
$$-600T_{CD} + 2a = 0 \quad (3)$$

In this problem, the 2 kN load is applied at C and we have $a = 1000$ mm. Carrying into (3) and solving for T_{CD} ,

$$T_{CD} = \frac{10}{3} \text{ kN} \quad T_{CD} = 3.33 \text{ kN} \quad \blacktriangleleft$$

From (1): $-480\left(\frac{10}{3}\right) - 339.44T_{BD} + 1600 = 0 \quad T_{BD} = 0 \quad \blacktriangleleft$

From (2): $-800\left(\frac{10}{3}\right) - 0 + 624.72T_{BE} = 0 \Rightarrow T_{BE} = 4.2685 \text{ kN} \quad T_{BE} = 4.27 \text{ kN} \quad \blacktriangleleft$



PROBLEM 4.125

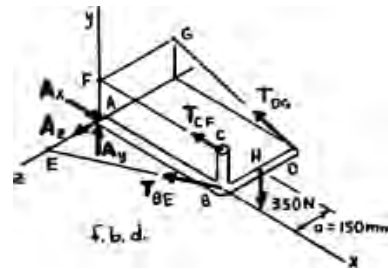
Frame $ABCD$ is supported by a ball-and-socket joint at A and by three cables. For $a = 150$ mm, determine the tension in each cable and the reaction at A .

SOLUTION

First note

$$\begin{aligned} \mathbf{T}_{DG} &= \lambda_{DG} T_{DG} = \frac{-(0.48 \text{ m})\mathbf{i} + (0.14 \text{ m})\mathbf{j}}{\sqrt{(0.48)^2 + (0.14)^2} \text{ m}} T_{DG} \\ &= \frac{-0.48\mathbf{i} + 0.14\mathbf{j}}{0.50} T_{DG} \\ &= \frac{T_{DG}}{25} (24\mathbf{i} + 7\mathbf{j}) \end{aligned}$$

$$\begin{aligned} \mathbf{T}_{BE} &= \lambda_{BE} T_{BE} = \frac{-(0.48 \text{ m})\mathbf{i} + (0.2 \text{ m})\mathbf{k}}{\sqrt{(0.48)^2 + (0.2)^2} \text{ m}} T_{BE} \\ &= \frac{-0.48\mathbf{i} + 0.2\mathbf{k}}{0.52} T_{BE} \\ &= \frac{T_{BE}}{13} (-12\mathbf{j} + 5\mathbf{k}) \end{aligned}$$



From f.b.d. of frame $ABCD$

$$\Sigma M_x = 0: \left(\frac{7}{25} T_{DG} \right) (0.3 \text{ m}) - (350 \text{ N}) (0.15 \text{ m}) = 0$$

or

$$T_{DG} = 625 \text{ N} \quad \blacktriangleleft$$

$$\Sigma M_y = 0: \left(\frac{24}{25} \times 625 \text{ N} \right) (0.3 \text{ m}) - \left(\frac{5}{13} T_{BE} \right) (0.48 \text{ m}) = 0$$

or

$$T_{BE} = 975 \text{ N} \quad \blacktriangleleft$$

$$\Sigma M_z = 0: T_{CF} (0.14 \text{ m}) + \left(\frac{7}{25} \times 625 \text{ N} \right) (0.48 \text{ m}) - (350 \text{ N}) (0.48 \text{ m}) = 0$$

or

$$T_{CF} = 600 \text{ N} \quad \blacktriangleleft$$

PROBLEM 4.125 (Continued)

$$\Sigma F_x = 0: A_x + T_{CF} + (T_{BE})_x + (T_{DG})_x = 0$$

$$A_x - 600 \text{ N} - \left(\frac{12}{13} \times 975 \text{ N} \right) - \left(\frac{24}{25} \times 625 \text{ N} \right) = 0$$

$$A_x = 2100 \text{ N}$$

$$\Sigma F_y = 0: A_y + (T_{DG})_y - 350 \text{ N} = 0$$

$$A_y + \left(\frac{7}{25} \times 625 \text{ N} \right) - 350 \text{ N} = 0$$

$$A_y = 175.0 \text{ N}$$

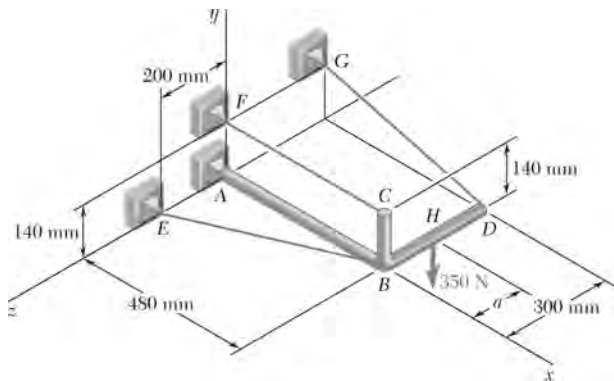
$$\Sigma F_z = 0: A_z + (T_{BE})_z = 0$$

$$A_z + \left(\frac{5}{13} \times 975 \text{ N} \right) = 0$$

$$A_z = -375 \text{ N}$$

Therefore

$$\mathbf{A} = (2100 \text{ N})\mathbf{i} + (175.0 \text{ N})\mathbf{j} - (375 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.126

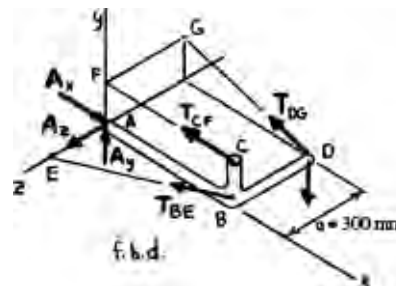
Frame $ABCD$ is supported by a ball-and-socket joint at A and by three cables. Knowing that the 350-N load is applied at D ($a = 300$ mm), determine the tension in each cable and the reaction at A .

SOLUTION

First note

$$\begin{aligned} \mathbf{T}_{DG} &= \lambda_{DG} T_{DG} = \frac{-(0.48 \text{ m})\mathbf{i} + (0.14 \text{ m})\mathbf{j}}{\sqrt{(0.48)^2 + (0.14)^2} \text{ m}} T_{DG} \\ &= \frac{-0.48\mathbf{i} + 0.14\mathbf{j}}{0.50} T_{DG} \\ &= \frac{T_{DG}}{25} (24\mathbf{i} + 7\mathbf{j}) \end{aligned}$$

$$\begin{aligned} \mathbf{T}_{BE} &= \lambda_{BE} T_{BE} = \frac{-(0.48 \text{ m})\mathbf{i} + (0.2 \text{ m})\mathbf{k}}{\sqrt{(0.48)^2 + (0.2)^2} \text{ m}} T_{BE} \\ &= \frac{-0.48\mathbf{i} + 0.2\mathbf{k}}{0.52} T_{BE} \\ &= \frac{T_{BE}}{13} (-12\mathbf{i} + 5\mathbf{k}) \end{aligned}$$



From f.b.d. of frame $ABCD$

$$\Sigma M_x = 0: \left(\frac{7}{25} T_{DG} \right) (0.3 \text{ m}) - (350 \text{ N}) (0.3 \text{ m}) = 0$$

or

$$T_{DG} = 1250 \text{ N} \quad \blacktriangleleft$$

$$\Sigma M_y = 0: \left(\frac{24}{25} \times 1250 \text{ N} \right) (0.3 \text{ m}) - \left(\frac{5}{13} T_{BE} \right) (0.48 \text{ m}) = 0$$

or

$$T_{BE} = 1950 \text{ N} \quad \blacktriangleleft$$

$$\Sigma M_z = 0: T_{CF} (0.14 \text{ m}) + \left(\frac{7}{25} \times 1250 \text{ N} \right) (0.48 \text{ m}) - (350 \text{ N}) (0.48 \text{ m}) = 0$$

or

$$T_{CF} = 0 \quad \blacktriangleleft$$

PROBLEM 4.126 (Continued)

$$\Sigma F_x = 0: A_x + T_{CF} + (T_{BE})_x + (T_{DG})_x = 0$$

$$A_x + 0 - \left(\frac{12}{13} \times 1950 \text{ N} \right) - \left(\frac{24}{25} \times 1250 \text{ N} \right) = 0$$

$$A_x = 3000 \text{ N}$$

$$\Sigma F_y = 0: A_y + (T_{DG})_y - 350 \text{ N} = 0$$

$$A_y + \left(\frac{7}{25} \times 1250 \text{ N} \right) - 350 \text{ N} = 0$$

$$A_y = 0$$

$$\Sigma F_z = 0: A_z + (T_{BE})_z = 0$$

$$A_z + \left(\frac{5}{13} \times 1950 \text{ N} \right) = 0$$

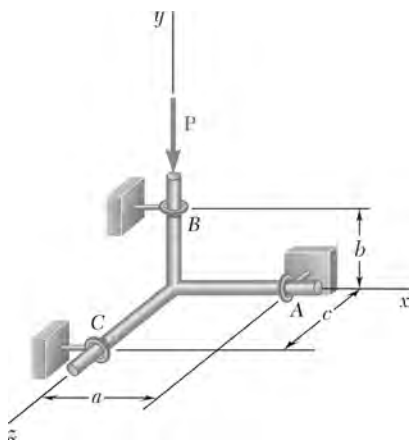
$$A_z = -750 \text{ N}$$

Therefore

$$\mathbf{A} = (3000 \text{ N})\mathbf{i} - (750 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 4.127

Three rods are welded together to form a “corner” that is supported by three eyebolts. Neglecting friction, determine the reactions at A, B, and C when $P = 1.2 \text{ kN}$, $a = 300 \text{ mm}$, $b = 200 \text{ mm}$ and $c = 250 \text{ mm}$.



SOLUTION

From f.b.d. of weldment

$$\Sigma \mathbf{M}_O = 0: \mathbf{r}_{AO} \times \mathbf{A} + \mathbf{r}_{BO} \times \mathbf{B} + \mathbf{r}_{CO} \times \mathbf{C} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & 0 \\ 0 & A_y & A_z \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 200 & 0 \\ B_x & 0 & B_z \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 250 \\ C_x & C_y & 0 \end{vmatrix} = 0$$

$$(-300 A_z \mathbf{j} + 300 A_y \mathbf{k}) + (200 B_z \mathbf{i} - 200 B_x \mathbf{k}) + (-250 C_y \mathbf{i} + 250 C_x \mathbf{j}) = 0$$

$$\text{From i-coefficient} \quad 200 B_z - 250 C_y = 0$$

$$\text{or} \quad B_z = 1.25 C_y \quad (1)$$

$$\text{j-coefficient} \quad -300 A_z + 250 C_x = 0$$

$$\text{or} \quad C_x = 1.2 A_z \quad (2)$$

$$\text{k-coefficient} \quad 300 A_y - 200 B_x = 0$$

$$\text{or} \quad B_x = 1.5 A_y \quad (3)$$

$$\Sigma \mathbf{F} = 0: \mathbf{A} + \mathbf{B} + \mathbf{C} - \mathbf{P} = 0$$

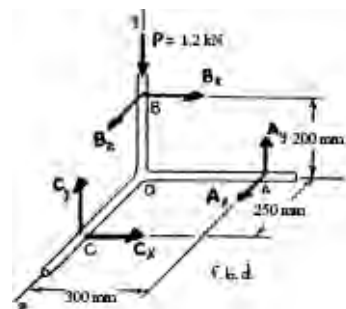
$$\text{or} \quad (B_x + C_x) \mathbf{i} + (A_y + C_y - 1.2 \text{ kN}) \mathbf{j} + (A_z + B_z) \mathbf{k} = 0$$

$$\text{From i-coefficient} \quad B_x + C_x = 0$$

$$\text{or} \quad C_x = -B_x \quad (4)$$

$$\text{j-coefficient} \quad A_y + C_y - 1.2 \text{ kN} = 0$$

$$\text{or} \quad A_y + C_y = 1.2 \text{ kN} \quad (5)$$



PROBLEM 4.127 (Continued)

$$\mathbf{k}\text{-coefficient} \quad A_z + B_z = 0$$

$$\text{or} \quad A_z = -B_z \quad (6)$$

Substituting C_x from Equation (4) into Equation (2)

$$-B_z = 1.2A_z \quad (7)$$

Using Equations (1), (6), and (7)

$$C_y = \frac{B_z}{1.25} = \frac{-A_z}{1.25} = \frac{1}{1.25} \left(\frac{B_x}{1.2} \right) = \frac{B_x}{1.5} \quad (8)$$

From Equations (3) and (8)

$$C_y = \frac{1.5A_y}{1.5} \quad \text{or} \quad C_y = A_y$$

and substituting into Equation (5)

$$2A_y = 1.2 \text{ kN}$$

$$A_y = C_y = 600 \text{ N} \quad (9)$$

Using Equation (1) and Equation (9)

$$B_z = 1.25(600 \text{ N}) = 750 \text{ N}$$

Using Equation (3) and Equation (9)

$$B_x = 1.5(600 \text{ N}) = 900 \text{ N}$$

$$\text{From Equation (4)} \quad C_x = -900 \text{ N}$$

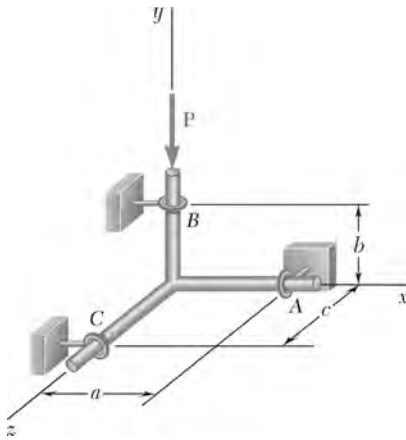
$$\text{From Equation (6)} \quad A_z = -750 \text{ N}$$

Therefore

$$\mathbf{A} = (600 \text{ N})\mathbf{j} - (750 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{B} = (900 \text{ N})\mathbf{i} + (750 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{C} = -(900 \text{ N})\mathbf{i} + (600 \text{ N})\mathbf{j} \quad \blacktriangleleft$$



PROBLEM 4.128

Solve Problem 4.127, assuming that the force $\mathbf{P}$ is removed and is replaced by a couple $\mathbf{M} = +(75 \text{ N} \cdot \text{mm})\mathbf{j}$ acting at B.

PROBLEM 4.127 Three rods are welded together to form a “corner” that is supported by three eyebolts. Neglecting friction, determine the reactions at A, B, and C when $P = 1.2 \text{ kN}$, $a = 300 \text{ mm}$, $b = 200 \text{ mm}$, and $c = 250 \text{ mm}$.

SOLUTION

From f.b.d. of weldment

$$\Sigma \mathbf{M}_O = 0: \mathbf{r}_{AO} \times \mathbf{A} + \mathbf{r}_{BO} \times \mathbf{B} + \mathbf{r}_{CO} \times \mathbf{C} + \mathbf{M} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & 0 \\ 0 & A_y & A_z \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 200 & 0 \\ B_x & 0 & B_z \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 250 \\ C_x & C_y & 0 \end{vmatrix} + (75000 \text{ N} \cdot \text{mm})\mathbf{j} = 0$$

$$(-300A_z\mathbf{j} + 300A_y\mathbf{k}) + (200B_z\mathbf{j} - 200B_x\mathbf{k}) + (-250C_y\mathbf{i} + 250C_x\mathbf{j}) + (75000 \text{ N} \cdot \text{mm})\mathbf{j} = 0$$

From $\mathbf{i}$ -coefficient

$$200B_z - 250C_y = 0$$

or

$$C_y = 0.8B_z \quad (1)$$

$\mathbf{j}$ -coefficient

$$-300A_z + 250C_x + 75,000 = 0$$

or

$$C_x = 1.2A_z - 300 \quad (2)$$

$\mathbf{k}$ -coefficient

$$300A_y - 200B_x = 0$$

or

$$B_x = 1.5A_y \quad (3)$$

$$\Sigma \mathbf{F} = 0: \mathbf{A} + \mathbf{B} + \mathbf{C} = 0$$

$$(B_x + C_x)\mathbf{i} + (A_y + C_y)\mathbf{j} + (A_z + B_z)\mathbf{k} = 0$$

From $\mathbf{i}$ -coefficient

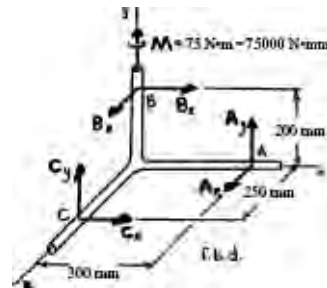
$$C_x = -B_x \quad (4)$$

$\mathbf{j}$ -coefficient

$$C_y = -A_y \quad (5)$$

$\mathbf{k}$ -coefficient

$$A_z = -B_z \quad (6)$$



PROBLEM 4.128 (Continued)

Substituting C_x from Equation (4) into Equation (2)

$$A_z = 250 - \left(\frac{B_x}{1.2} \right) \quad (7)$$

Using Equations (1), (6), and (7)

$$C_y = 0.8B_z = -0.8A_z = \left(\frac{2}{3} \right) B_x - 200 \quad (8)$$

From Equations (3) and (8)

$$C_y = A_y - 200$$

Substituting into Equation (5)

$$2A_y = 200$$

$$A_y = 100.0 \text{ N}$$

From Equation (5)

$$C_y = -100.0 \text{ N}$$

Equation (1)

$$B_z = -125.0 \text{ N}$$

Equation (3)

$$B_x = 150.0 \text{ N}$$

Equation (4)

$$C_x = -150.0 \text{ N}$$

Equation (6)

$$A_z = 125.0 \text{ N}$$

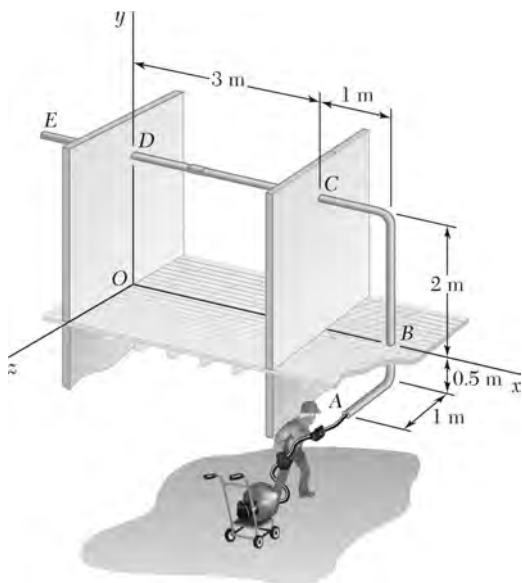
Therefore

$$\mathbf{A} = (100.0 \text{ N})\mathbf{j} + (125.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{B} = (150.0 \text{ N})\mathbf{i} - (125.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\mathbf{C} = -(150.0 \text{ N})\mathbf{i} - (100.0 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

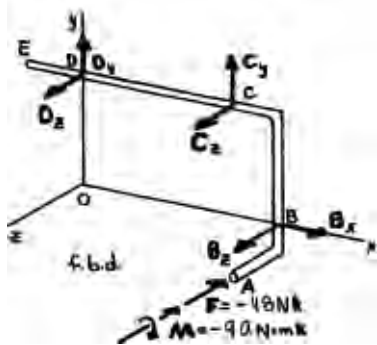
PROBLEM 4.129



In order to clean the clogged drainpipe AE , a plumber has disconnected both ends of the pipe and inserted a power snake through the opening at A . The cutting head of the snake is connected by a heavy cable to an electric motor that rotates at a constant speed as the plumber forces the cable into the pipe. The forces exerted by the plumber and the motor on the end of the cable can be represented by the wrench $\mathbf{F} = -(48 \text{ N})\mathbf{k}$, $\mathbf{M} = -(90 \text{ N} \cdot \text{m})\mathbf{k}$. Determine the additional reactions at B , C , and D caused by the cleaning operation. Assume that the reaction at each support consists of two force components perpendicular to the pipe.

SOLUTION

From f.b.d. of pipe assembly $ABCD$



$$\Sigma F_x = 0: B_x = 0$$

$$\Sigma M_{D(x\text{-axis})} = 0: (48 \text{ N})(2.5 \text{ m}) - B_z(2 \text{ m}) = 0$$

$$B_z = 60.0 \text{ N}$$

$$\text{and } \mathbf{B} = (60.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\Sigma M_{D(z\text{-axis})} = 0: C_y(3 \text{ m}) - 90 \text{ N} \cdot \text{m} = 0$$

$$C_y = 30.0 \text{ N}$$

$$\Sigma M_{D(y\text{-axis})} = 0: -C_z(3 \text{ m}) - (60.0 \text{ N})(4 \text{ m}) + (48 \text{ N})(4 \text{ m}) = 0$$

$$C_z = -16.00 \text{ N}$$

$$\text{and } \mathbf{C} = (30.0 \text{ N})\mathbf{j} - (16.00 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

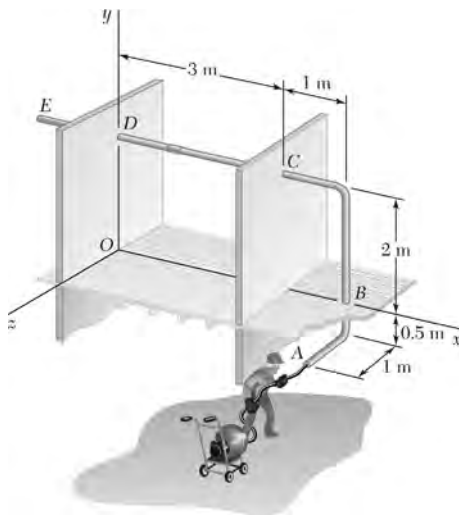
$$\Sigma F_y = 0: D_y + 30.0 = 0$$

$$D_y = -30.0 \text{ N}$$

$$\Sigma F_z = 0: D_z - 16.00 \text{ N} + 60.0 \text{ N} - 48 \text{ N} = 0$$

$$D_z = 4.00 \text{ N}$$

$$\text{and } \mathbf{D} = -(30.0 \text{ N})\mathbf{j} + (4.00 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



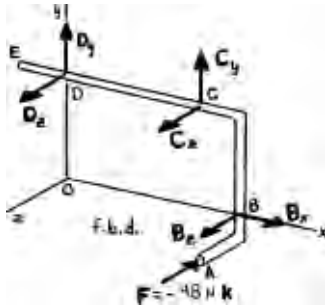
PROBLEM 4.130

Solve Problem 4.129, assuming that the plumber exerts a force $\mathbf{F} = -(48 \text{ N})\mathbf{k}$ and that the motor is turned off ($\mathbf{M} = 0$).

PROBLEM 4.129 In order to clean the clogged drainpipe AE , a plumber has disconnected both ends of the pipe and inserted a power snake through the opening at A . The cutting head of the snake is connected by a heavy cable to an electric motor that rotates at a constant speed as the plumber forces the cable into the pipe. The forces exerted by the plumber and the motor on the end of the cable can be represented by the wrench $\mathbf{F} = -(48 \text{ N})\mathbf{k}$, $\mathbf{M} = -(90 \text{ N} \cdot \text{m})\mathbf{k}$. Determine the additional reactions at B , C , and D caused by the cleaning operation. Assume that the reaction at each support consists of two force components perpendicular to the pipe.

SOLUTION

From f.b.d. of pipe assembly $ABCD$



$$\Sigma F_x = 0: B_x = 0$$

$$\Sigma M_{D(x\text{-axis})} = 0: (48 \text{ N})(2.5 \text{ m}) - B_z(2 \text{ m}) = 0$$

$$B_z = 60.0 \text{ N} \quad \text{and } \mathbf{B} = (60.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

$$\Sigma M_{D(z\text{-axis})} = 0: C_y(3 \text{ m}) - B_x(2 \text{ m}) = 0$$

$$C_y = 0$$

$$\Sigma M_{D(y\text{-axis})} = 0: C_z(3 \text{ m}) - (60.0 \text{ N})(4 \text{ m}) + (48 \text{ N})(4 \text{ m}) = 0$$

$$C_z = -16.00 \text{ N} \quad \text{and } \mathbf{C} = -(16.00 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

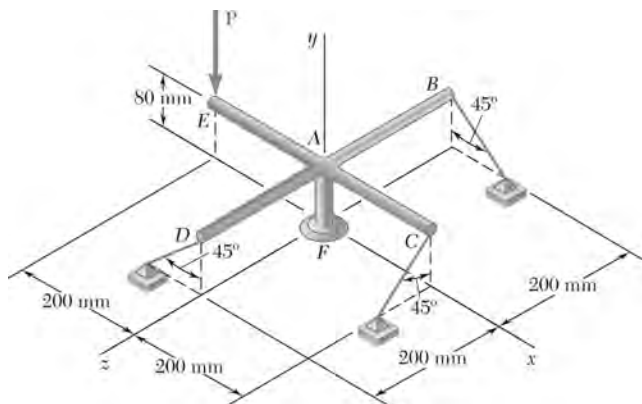
$$\Sigma F_y = 0: D_y + C_y = 0$$

$$D_y = 0$$

$$\Sigma F_z = 0: D_z + B_z + C_z - F = 0$$

$$D_z + 60.0 \text{ N} - 16.00 \text{ N} - 48 \text{ N} = 0$$

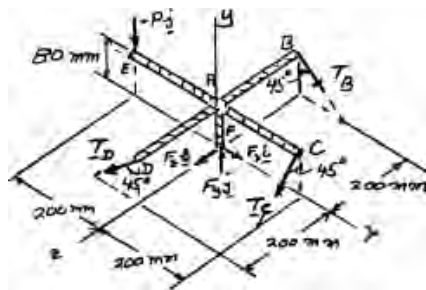
$$D_z = 4.00 \text{ N} \quad \text{and } \mathbf{D} = (4.00 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.131

The assembly shown consists of an 80-mm rod AF that is welded to a cross consisting of four 200 mm arms. The assembly is supported by a ball-and-socket joint at F and by three short links, each of which forms an angle of 45° with the vertical. For the loading shown, determine (a) the tension in each link, (b) the reaction at F .

SOLUTION



$$\mathbf{r}_{E/F} = -200\mathbf{i} + 80\mathbf{j}$$

$$\mathbf{T}_B = T_B(\mathbf{i} - \mathbf{j})/\sqrt{2} \quad \mathbf{r}_{B/F} = 80\mathbf{j} - 200\mathbf{k}$$

$$\mathbf{T}_C = T_C(-\mathbf{j} + \mathbf{k})/\sqrt{2} \quad \mathbf{r}_{C/F} = 200\mathbf{i} + 80\mathbf{j}$$

$$\mathbf{T}_D = T_D(-\mathbf{i} + \mathbf{j})/\sqrt{2} \quad \mathbf{r}_{D/F} = 80\mathbf{j} + 200\mathbf{k}$$

$$\Sigma \mathbf{M}_F = 0: \mathbf{r}_{B/F} \times \mathbf{T}_B + \mathbf{r}_{C/F} \times \mathbf{T}_C + \mathbf{r}_{D/F} \times \mathbf{T}_D + \mathbf{r}_{E/F} \times (-P\mathbf{j}) = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 80 & -200 \\ 1 & -1 & 0 \end{vmatrix} \frac{T_B}{\sqrt{2}} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 200 & 80 & 0 \\ 0 & -1 & 1 \end{vmatrix} \frac{T_C}{\sqrt{2}} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 80 & 200 \\ -1 & -1 & 0 \end{vmatrix} \frac{T_D}{\sqrt{2}} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -200 & 80 & 0 \\ 0 & -P & 0 \end{vmatrix} = 0$$

Equate coefficients of unit vectors to zero and multiply each equation by $\sqrt{2}$.

$$\mathbf{i}: -200T_B + 80T_C + 200T_D = 0 \quad (1)$$

$$\mathbf{j}: -200T_B - 200T_C - 200T_D = 0 \quad (2)$$

$$\mathbf{k}: -80T_B - 200T_C + 80T_D + 200\sqrt{2}P = 0 \quad (3)$$

$$\frac{80}{200}(2): -80T_B - 80T_C - 80T_D = 0 \quad (4)$$

$$\text{Eqs. (3) + (4): } -160T_B - 280T_C + 200\sqrt{2}P = 0 \quad (5)$$

$$\text{Eqs. (1) + (2): } -400T_B - 120T_C = 0 \quad (6)$$

$$T_B = -\frac{120}{400}T_C - 0.3T_C$$

PROBLEM 4.131 (Continued)

$$\text{Eqs. (6)} \rightarrow (5): \quad -160(-0.3T_C) - 280T_C + 200\sqrt{2}P = 0$$

$$-232T_C + 200\sqrt{2}P = 0$$

$$T_C = 1.2191P$$

$$T_C = 1.219P \quad \blacktriangleleft$$

$$\text{From Eq. (6):} \quad T_B = -0.3(1.2191P) = -0.36574 = P$$

$$T_B = -0.366P \quad \blacktriangleleft$$

$$\text{From Eq. (2):} \quad -200(-0.3657P) - 200(1.2191P) - 200T_{\theta D} = 0$$

$$T_D = -0.8534P$$

$$T_D = -0.853P \quad \blacktriangleleft$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{F} + \mathbf{T}_B + \mathbf{T}_C + \mathbf{T}_D - P\mathbf{j} = 0$$

$$\mathbf{i}: \quad F_x + \frac{(-0.36574P)}{\sqrt{2}} - \frac{(-0.8534P)}{\sqrt{2}} = 0$$

$$F_x = -0.3448P \quad F_x = -0.345P$$

$$\mathbf{j}: \quad F_y - \frac{(-0.36574P)}{\sqrt{2}} - \frac{(1.2191P)}{\sqrt{2}} - \frac{(-0.8534P)}{\sqrt{2}} - 200 = 0$$

$$F_y = P \quad F_y = P$$

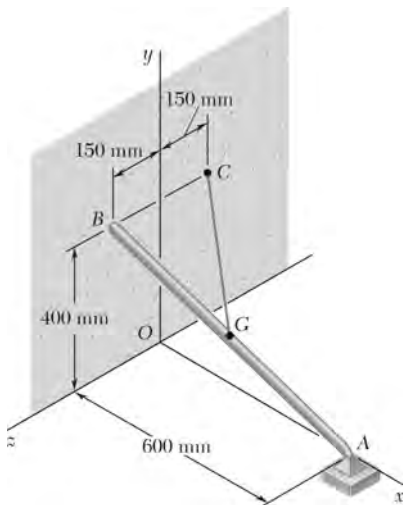
$$\mathbf{k}: \quad F_z + \frac{(1.2191P)}{\sqrt{2}} = 0$$

$$F_z = -0.8620P \quad F_z = -0.862P$$

$$\mathbf{F} = -0.345P\mathbf{i} + P\mathbf{j} - 0.862P\mathbf{k} \quad \blacktriangleleft$$

PROBLEM 4.132

The uniform 10-kg rod AB is supported by a ball-and-socket joint at A and by the cord CG that is attached to the midpoint G of the rod. Knowing that the rod leans against a frictionless vertical wall at B , determine (a) the tension in the cord, (b) the reactions at A and B .



SOLUTION

Five unknowns and six equations of equilibrium. But equilibrium is maintained ($\Sigma M_{AB} = 0$)

$$\begin{aligned} W &= mg \\ &= (10 \text{ kg})9.81 \text{ m/s}^2 \\ W &= 98.1 \text{ N} \end{aligned}$$

$$\overline{GC} = -300\mathbf{i} + 200\mathbf{j} - 225\mathbf{k} \quad GC = 425 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{GC}}{GC} = \frac{T}{425}(-300\mathbf{i} + 200\mathbf{j} - 225\mathbf{k})$$

$$\mathbf{r}_{B/A} = -600\mathbf{i} + 400\mathbf{j} + 150\mathbf{k}$$

$$\mathbf{r}_{G/A} = -300\mathbf{i} + 200\mathbf{j} + 75\mathbf{k}$$

$$\Sigma M_A = 0: \mathbf{r}_{B/A} \times \mathbf{B} + \mathbf{r}_{G/A} \times \mathbf{T} + \mathbf{r}_{G/A} \times (-W\mathbf{j}) = 0$$

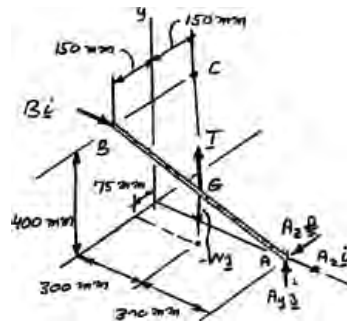
$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -600 & 400 & 150 \\ B & 0 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -300 & 200 & 75 \\ -300 & 200 & -225 \end{vmatrix} \frac{T}{425} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -300 & 200 & 75 \\ 0 & -98.1 & 0 \end{vmatrix}$$

$$\text{Coefficient of } \mathbf{i}: (-105.88 - 35.29)T + 7357.5 = 0$$

$$T = 52.12 \text{ N}$$

$$T = 52.1 \text{ N} \quad \blacktriangleleft$$

Free-Body Diagram:



PROBLEM 4.132 (Continued)

$$\text{Coefficient of } \mathbf{j}: 150B - (300 \times 75 + 300 \times 225) \frac{52.12}{425} = 0$$

$$B = 73.58 \text{ N}$$

$$\mathbf{B} = (73.6 \text{ N})\mathbf{i} \quad \blacktriangleleft$$

$$\Sigma \mathbf{F} = 0: \mathbf{A} + \mathbf{B} + \mathbf{T} - W\mathbf{j} = 0$$

$$\text{Coefficient of } \mathbf{i}: A_x + 73.58 - 52.15 \frac{300}{425} = 0$$

$$A_x = 36.8 \text{ N} \quad \blacktriangleleft$$

$$\text{Coefficient of } \mathbf{j}: A_y + 52.15 \frac{200}{425} - 98.1 = 0$$

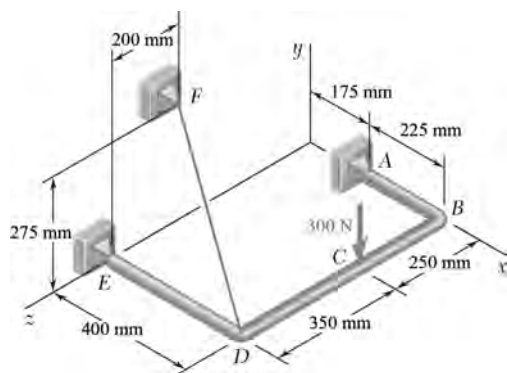
$$A_y = 73.6 \text{ N} \quad \blacktriangleleft$$

$$\text{Coefficient of } \mathbf{k}: A_z - 52.15 \frac{225}{425} = 0$$

$$A_z = 27.6 \text{ N} \quad \blacktriangleleft$$

PROBLEM 4.133

The bent rod $ABDE$ is supported by ball-and-socket joints at A and E and by the cable DF . If a 300-N load is applied at C as shown, determine the tension in the cable.



SOLUTION

$$\overline{DF} = -400\mathbf{i} + 275\mathbf{j} - 200\mathbf{k} \quad DF = 525 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{DE}}{DF} = \frac{T}{525} (-400\mathbf{i} + 275\mathbf{j} - 200\mathbf{k})$$

$$\mathbf{r}_{D/E} = 400\mathbf{i}$$

$$\mathbf{r}_{C/E} = 400\mathbf{i} - 350\mathbf{k}$$

$$\lambda_{EA} = \frac{\overline{EA}}{EA} = \frac{175\mathbf{i} - 600\mathbf{k}}{625}$$

$$\Sigma M_{EA} = 0: \lambda_{EA} \cdot (\mathbf{r}_{D/E} \times \mathbf{T}) + \lambda_{EA} \cdot (\mathbf{r}_{C/E} \cdot (-300\mathbf{j})) = 0$$

$$\begin{vmatrix} 175 & 0 & -600 \\ 400 & 0 & 0 \\ -400 & 275 & -200 \end{vmatrix} \frac{T}{525 \times 625} + \begin{vmatrix} 175 & 0 & -600 \\ 400 & 0 & -350 \\ 0 & -300 & 0 \end{vmatrix} \frac{1}{625} = 0$$

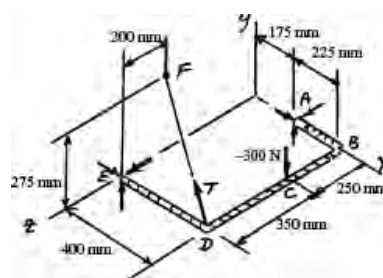
$$-\frac{600 \times 400 \times 275}{525 \times 625} T + \frac{-175 \times 300 \times 350 + 600 \times 400 \times 300}{625} = 0$$

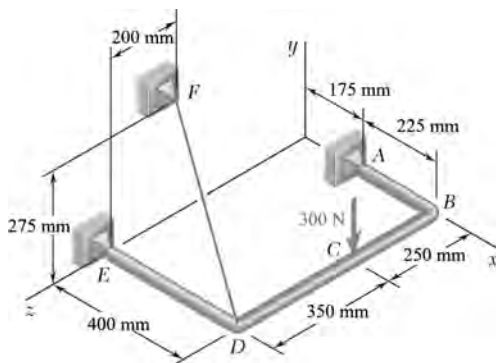
$$\frac{-880000}{7} T + 53625000 = 0$$

$$T = 426.5625 \text{ N}$$

$$T = 427 \text{ N} \quad \blacktriangleleft$$

Free-Body Diagram:





PROBLEM 4.134

Solve Problem 4.133, assuming that cable DF is replaced by a cable connecting B and F .

PROBLEM 4.133 The bent rod $ABDE$ is supported by ball-and-socket joints at A and E and by the cable DF . If a 300-N load is applied at C as shown, determine the tension in the cable.

SOLUTION

$$\mathbf{r}_{B/A} = 225\mathbf{i}$$

$$\mathbf{r}_{C/A} = 225\mathbf{i} + 250\mathbf{k}$$

$$\overline{BF} = -400\mathbf{i} + 275\mathbf{j} + 400\mathbf{k} \quad BF = 628.987 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{BF}}{BF} = \frac{T}{25.16} = (-16\mathbf{i} + 11\mathbf{j} + 16\mathbf{k})$$

$$\lambda_{AE} = \frac{\overline{AE}}{AE} = \frac{7\mathbf{i} - 24\mathbf{k}}{25}$$

$$\Sigma M_{AE} = 0: \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) + \lambda_{AE} \cdot (\mathbf{r}_{C/A} \cdot (-300\mathbf{j})) = 0$$

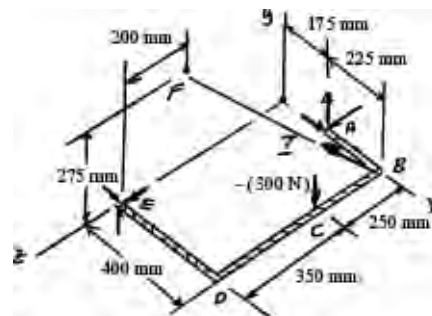
$$\begin{vmatrix} 7 & 0 & -24 \\ 225 & 0 & 0 \\ -16 & 11 & 16 \end{vmatrix} \frac{T}{25 \times 25.16} + \begin{vmatrix} 7 & 0 & -24 \\ 225 & 0 & 250 \\ 0 & -300 & 0 \end{vmatrix} \frac{1}{25} = 0$$

$$-\frac{24 \times 225 \times 11}{25 \times 25.16} T + \frac{24 \times 225 \times 300 + 7 \times 250 \times 300}{25}$$

$$2360.89T - 2,145,000 = 0$$

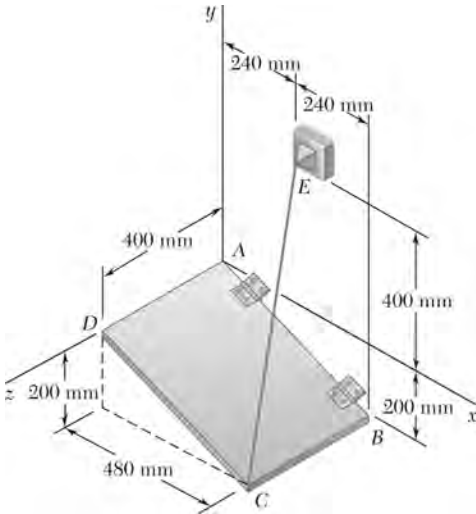
$$T = 909 \text{ N} \quad \blacktriangleleft$$

Free-Body Diagram:



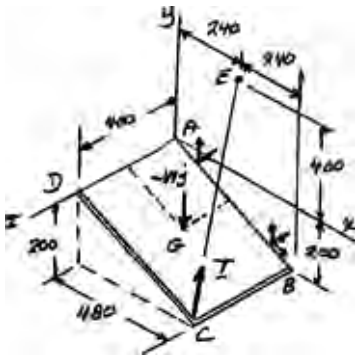
PROBLEM 4.135

The 50-kg plate $ABCD$ is supported by hinges along edge AB and by wire CE . Knowing that the plate is uniform, determine the tension in the wire.



SOLUTION

Free-Body Diagram:



$$W = mg = (50 \text{ kg})(9.81 \text{ m/s}^2)$$

$$W = 490.50 \text{ N}$$

$$\overline{CE} = -240\mathbf{i} + 600\mathbf{j} - 400\mathbf{k}$$

$$CE = 760 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{CE}}{CE} = \frac{T}{760}(-240\mathbf{i} + 600\mathbf{j} - 400\mathbf{k})$$

$$\lambda_{AB} = \frac{\overline{AB}}{AB} = \frac{480\mathbf{i} - 200\mathbf{j}}{520} = \frac{1}{13}(12\mathbf{i} - 5\mathbf{j})$$

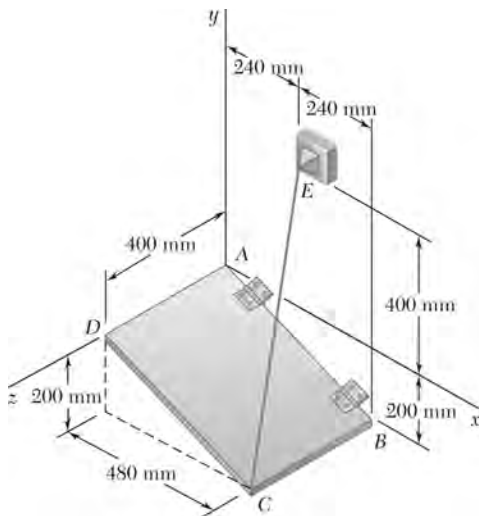
$$\Sigma \mathbf{M}_{AB} = 0: \lambda_{AB} \cdot (\mathbf{r}_{E/A} \times \mathbf{T}) + \lambda_{AB} \cdot (\mathbf{r}_{G/A} \times -W\mathbf{j}) = 0$$

$$\mathbf{r}_{E/A} = 240\mathbf{i} + 400\mathbf{j}; \quad \mathbf{r}_{G/A} = 240\mathbf{i} - 100\mathbf{j} + 200\mathbf{k}$$

$$\begin{vmatrix} 12 & -5 & 0 \\ 240 & 400 & 0 \\ -240 & 600 & -400 \end{vmatrix} \frac{T}{13 \times 20} + \begin{vmatrix} 12 & -5 & 0 \\ 240 & -100 & 200 \\ 0 & -W & 0 \end{vmatrix} \frac{1}{13} = 0$$

$$(-12 \times 400 \times 400 - 5 \times 240 \times 400) \frac{T}{760} + 12 \times 200W = 0$$

$$T = 0.76W = 0.76(490.50 \text{ N}) \quad T = 373 \text{ N} \quad \blacktriangleleft$$



PROBLEM 4.136

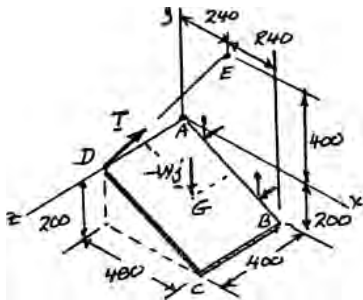
Solve Problem 4.135, assuming that wire CE is replaced by a wire connecting E and D .

PROBLEM 4.135 The 50-kg plate $ABCD$ is supported by hinges along edge AB and by wire CE . Knowing that the plate is uniform, determine the tension in the wire.

SOLUTION

Free-Body Diagram:

Dimensions in mm



$$W = mg = (50 \text{ kg})(9.81 \text{ m/s}^2)$$

$$W = 490.50 \text{ N}$$

$$\overline{DE} = -240\mathbf{i} + 400\mathbf{j} - 400\mathbf{k}$$

$$DE = 614.5 \text{ mm}$$

$$\mathbf{T} = T \frac{\overline{DE}}{DE} = \frac{T}{614.5} (240\mathbf{i} + 400\mathbf{j} - 400\mathbf{k})$$

$$\lambda_{AB} = \frac{\overline{AB}}{AB} = \frac{480\mathbf{i} - 200\mathbf{j}}{520} = \frac{1}{13} (12\mathbf{i} - 5\mathbf{j})$$

$$\mathbf{r}_{E/A} = 240\mathbf{i} + 400\mathbf{j}; \quad \mathbf{r}_{G/A} = 240\mathbf{i} - 100\mathbf{j} + 200\mathbf{k}$$

$$\begin{vmatrix} 12 & -5 & 0 \\ 240 & 400 & 0 \\ 240 & 400 & -400 \end{vmatrix} \frac{T}{13 \times 614.5} + \begin{vmatrix} 12 & 5 & 0 \\ 240 & -100 & 200 \\ 0 & -W & 0 \end{vmatrix} \frac{1}{13} = 0$$

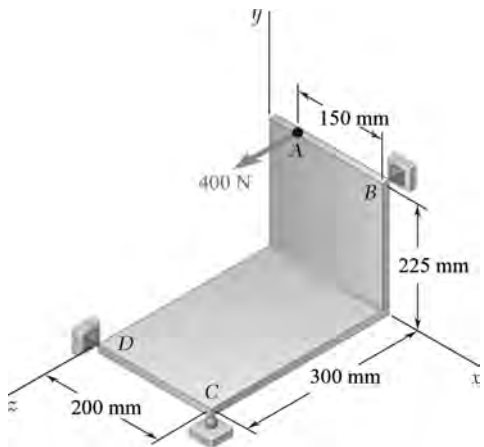
$$(-12 \times 400 \times 400 - 5 \times 240 \times 400) \frac{T}{614.5} + 12 \times 200 \times W = 0$$

$$T = 0.6145W = 0.6145(490.50 \text{ N})$$

$$T = 301 \text{ N} \quad \blacktriangleleft$$

PROBLEM 4.137

Two rectangular plates are welded together to form the assembly shown. The assembly is supported by ball-and-socket joints at B and D and by a ball on a horizontal surface at C . For the loading shown, determine the reaction at C .



SOLUTION

First note

$$\begin{aligned}\lambda_{BD} &= \frac{-(150 \text{ mm})\mathbf{i} - (225 \text{ mm})\mathbf{j} + (300 \text{ mm})\mathbf{k}}{\sqrt{(150)^2 + (225)^2 + (300)^2} \text{ mm}} \\ &= \frac{1}{\sqrt{29}}(-2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}) \\ \mathbf{r}_{A/B} &= -(150 \text{ mm})\mathbf{i} \\ \mathbf{P} &= (400 \text{ N})\mathbf{k} \\ \mathbf{r}_{C/D} &= (200 \text{ mm})\mathbf{i} \\ \mathbf{C} &= (C)\mathbf{j}\end{aligned}$$

From the f.b.d. of the plates

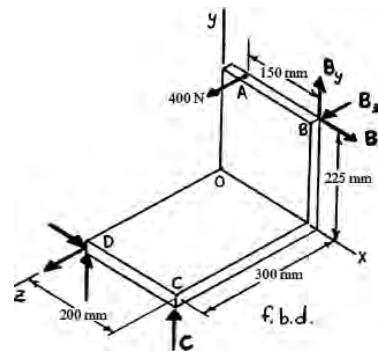
$$\Sigma M_{BD} = 0: \lambda_{BD} \cdot (\mathbf{r}_{A/B} \times \mathbf{P}) + \lambda_{BD} \cdot (\mathbf{r}_{C/D} \times \mathbf{C}) = 0$$

$$\begin{vmatrix} -2 & -3 & 4 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \left[\frac{150(400)}{\sqrt{29}} \right] + \begin{vmatrix} -2 & -3 & 4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} \left[\frac{C(200)}{\sqrt{29}} \right] = 0$$

$$(-3)(150)(400) + (4)(200)C = 0$$

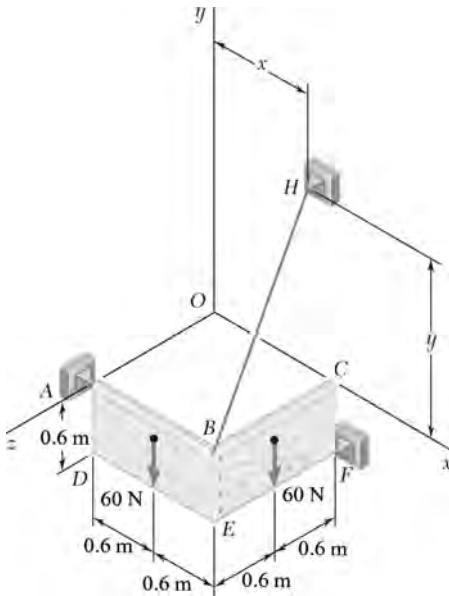
$$C = 225 \text{ N}$$

$$\text{or } \mathbf{C} = (225 \text{ N})\mathbf{j} \quad \blacktriangleleft$$



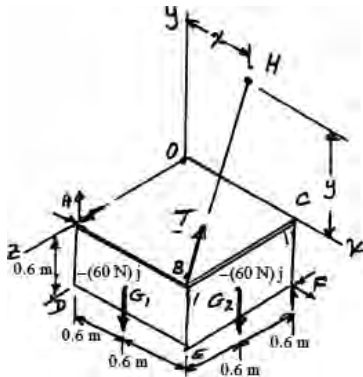
PROBLEM 4.138

Two 0.6×1.2 m plywood panels, each of weight 60 N, are nailed together as shown. The panels are supported by ball-and-socket joints at A and F and by the wire BH . Determine (a) the location of H in the xy plane if the tension in the wire is to be minimum, (b) the corresponding minimum tension.



SOLUTION

Free-Body Diagram:



$$\overline{AF} = 1.2\mathbf{i} - 0.6\mathbf{j} - 1.2\mathbf{k} \quad AF = 1.8\text{ m}$$

$$\lambda_{AF} = \frac{1}{3}(2\mathbf{i} - \mathbf{j} - 2\mathbf{k})$$

$$\mathbf{r}_{G_1/A} = 0.6\mathbf{i} - 0.3\mathbf{j}$$

$$\mathbf{r}_{G_2/A} = 1.2\mathbf{i} - 0.3\mathbf{j} - 0.6\mathbf{k}$$

$$\mathbf{r}_{B/A} = 1.2\mathbf{i}$$

$$\Sigma M_{AF} = 0: \lambda_{AF} \cdot (\mathbf{r}_{G_1/A} \times (-60\mathbf{j})) + \lambda_{AF} \cdot (\mathbf{r}_{G_2/A} \times (-60\mathbf{j})) + \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = 0$$

$$\begin{vmatrix} 2 & -1 & -2 \\ 0.6 & -0.3 & 0 \\ 0 & -60 & 0 \end{vmatrix} \frac{1}{3} + \begin{vmatrix} 2 & -1 & -2 \\ 1.2 & -0.3 & -0.6 \\ 0 & -60 & 0 \end{vmatrix} \frac{1}{3} + \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = 0$$

$$(2 \times 0.6 \times 60) \frac{1}{3} + (-2 \times 0.6 \times 60 + 2 \times 1.2 \times 60) \frac{1}{3} + \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = 0$$

$$\lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = -48 \quad \text{or} \quad \mathbf{T} \cdot (\lambda_{AF} \times \mathbf{r}_{B/A}) = -48 \quad (1)$$

PROBLEM 4.138 (Continued)

Projection of $\mathbf{T}$ on $(\lambda_{AF} \times \mathbf{r}_{B/A})$ is constant. Thus, $T_{\min}$ is parallel to

$$\lambda_{AF} \times \mathbf{r}_{B/A} = \frac{1}{3}(2\mathbf{i} - \mathbf{j} - 2\mathbf{k}) \times 1.2\mathbf{i} = \frac{1}{3}(-2.4\mathbf{j} + 1.2\mathbf{k})$$

Corresponding unit vector is $\frac{1}{\sqrt{5}}(-2\mathbf{j} + \mathbf{k})$

$$T_{\min} = T(-2\mathbf{j} + \mathbf{k}) \frac{1}{\sqrt{5}} \quad (2)$$

$$\text{Eq. (1): } \frac{T}{\sqrt{5}}(-2\mathbf{j} + \mathbf{k}) \cdot \left[\frac{1}{3}(2\mathbf{i} - \mathbf{j} - 2\mathbf{k}) \times 1.2\mathbf{i} \right] = -48$$

$$\frac{T}{\sqrt{5}}(-2\mathbf{j} + \mathbf{k}) \cdot \frac{1}{3}(-2.4\mathbf{j} + 1.2\mathbf{k}) = -48$$

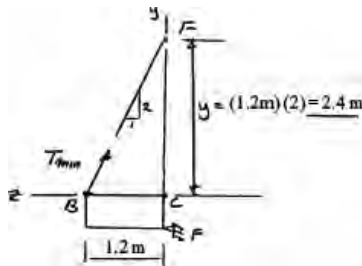
$$\frac{T}{3\sqrt{5}}(4.8 + 1.2) = -48 \quad T = -\frac{3\sqrt{5}(48)}{6} = 24\sqrt{5}$$

$$T = 53.6656 \text{ N}$$

$$\begin{aligned} \text{Eq. (2)} \quad T_{\min} &= T(-2\mathbf{j} + \mathbf{k}) \frac{1}{\sqrt{5}} \\ &= 24\sqrt{5}(-2\mathbf{j} + \mathbf{k}) \frac{1}{\sqrt{5}} \end{aligned}$$

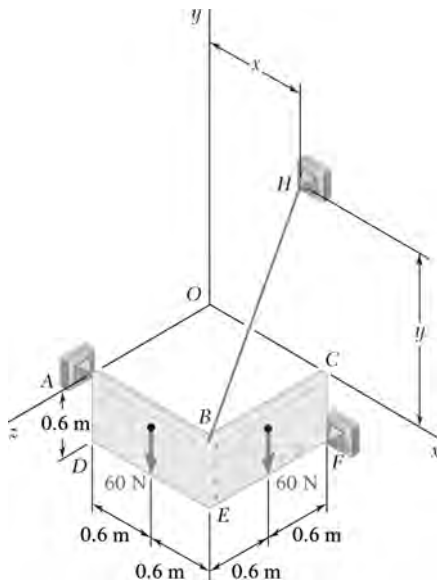
$$\mathbf{T}_{\min} = -(48 \text{ N})\mathbf{j} + (24 \text{ N})\mathbf{k}$$

Since $T_{\min}$ has no $\mathbf{i}$ component, wire BH is parallel to the yz plane, and $x = 1.2 \text{ m}$



$$(a) \quad x = 1.200 \text{ m}; \quad y = 2.40 \text{ m} \quad \blacktriangleleft$$

$$(b) \quad T_{\min} = 53.7 \text{ N} \quad \blacktriangleleft$$



PROBLEM 4.139

Solve Problem 4.138, subject to the restriction that H must lie on the y axis.

PROBLEM 4.138 Two 0.6×1.2 m plywood panels, each of weight 60 N, are nailed together as shown. The panels are supported by ball-and-socket joints at A and F and by the wire BH . Determine (a) the location of H in the xy plane if the tension in the wire is to be minimum, (b) the corresponding minimum tension.

SOLUTION

$$\overline{AF} = 1.2\mathbf{i} - 0.6\mathbf{j} - 1.2\mathbf{k} \quad AF = 1.8\text{ m}$$

$$\lambda_{AF} = \frac{1}{3}(2\mathbf{i} - \mathbf{j} - 2\mathbf{k})$$

$$\mathbf{r}_{G_1/A} = 0.6\mathbf{i} - 0.3\mathbf{j}$$

$$\mathbf{r}_{G_2/A} = 1.2\mathbf{i} - 0.3\mathbf{j} - 0.6\mathbf{k}$$

$$\mathbf{r}_{B/A} = 1.2\mathbf{i}$$

$$\Sigma M_{AF} = 0: \quad \lambda_{AF} \cdot (\mathbf{r}_{G_1/A} \times (-60\mathbf{j})) + \lambda_{AF} \cdot (\mathbf{r}_{G_2/A} \times (-60\mathbf{j})) + \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = 0$$

$$\begin{vmatrix} 2 & -1 & -2 \\ 0.6 & -0.3 & 0 \\ 0 & -60 & 0 \end{vmatrix} \frac{1}{3} + \begin{vmatrix} 2 & -1 & -2 \\ 1.2 & -0.3 & -0.6 \\ 0 & -60 & 0 \end{vmatrix} \frac{1}{3} + \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = 0$$

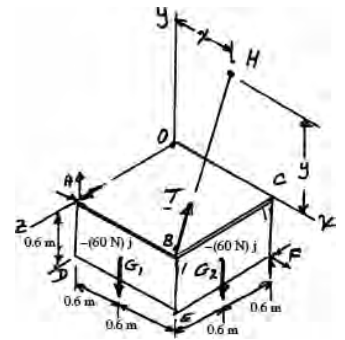
$$(2 \times 0.6 \times 60) \frac{1}{3} + (-2 \times 0.6 \times 60 + 2 \times 1.2 \times 60) \frac{1}{3} + \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = 0$$

$$\lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = -48$$

$$\overline{BH} = -1.2\mathbf{i} + y\mathbf{j} - 1.2\mathbf{k} \quad BH = (2.88 + y^2)^{1/2}$$

$$\mathbf{T} = T \frac{\overline{BH}}{BH} = T \frac{-1.2\mathbf{i} + y\mathbf{j} - 1.2\mathbf{k}}{(32 + y^2)^{1/2}}$$

Free-Body Diagram:



(1)

PROBLEM 4.139 (Continued)

$$\text{Eq. (1):} \quad \lambda_{AF} \cdot (\mathbf{r}_{B/A} \times \mathbf{T}) = \begin{vmatrix} 2 & -1 & -2 \\ 1.2 & 0 & 0 \\ -1.2 & y & -1.2 \end{vmatrix} \frac{1}{3} \frac{T}{(2.88 + y^2)} = -48$$

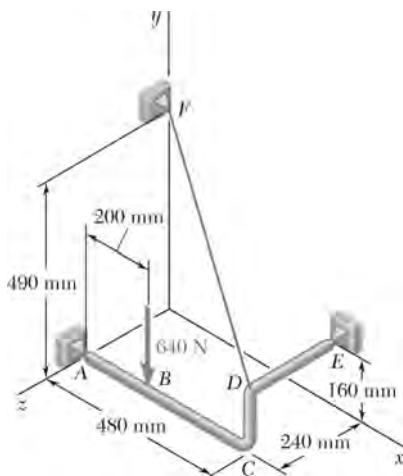
$$(-1.44 - 2.4y)T = -3 \times 48(2.88 + y^2)^{1/2} \quad T = 144 \frac{(2.88 + y^2)^{1/2}}{(1.44 + 2.4y)} = 100 \frac{(2.88 + y^2)^{1/2}}{\left(1 + \frac{5y}{3}\right)} \quad (2)$$

$$\frac{dT}{dy} = 0: \quad 100 \frac{\left\{ \left(1 + \frac{5y}{3}\right) \frac{1}{2} (2.88 + y^2)^{-1/2} (2y) + (2.88 + y^2)^{1/2} \left(\frac{5}{3}\right) \right\}}{\left(1 + \frac{5y}{3}\right)^2}$$

$$\text{Numerator} = 0: \quad \left(1 + \frac{5y}{3}\right)y = (2.88 + y^2) \frac{5}{3}$$

$$\frac{5}{3}y^2 + y = \frac{24}{5} + \frac{5}{3}y^2 \quad y = 4.80 \text{ m} \quad \blacktriangleleft$$

$$\text{Eq. (2):} \quad T = \frac{100(2.88 + 4.8^2)^{1/2}}{\left(1 + \frac{5 \times 4.8}{3}\right)} = 56.569 \text{ N} \quad T_{\min} = 56.6 \text{ N} \quad \blacktriangleleft$$



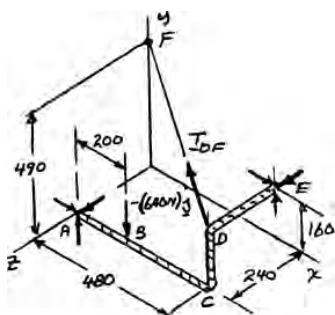
PROBLEM 4.140

The pipe $ACDE$ is supported by ball-and-socket joints at A and E and by the wire DF . Determine the tension in the wire when a 640-N load is applied at B as shown.

SOLUTION

Free-Body Diagram:

Dimensions in mm



$$\overline{AE} = 480\mathbf{i} + 160\mathbf{j} - 240\mathbf{k}$$

$$AE = 560 \text{ mm}$$

$$\lambda_{AE} = \frac{\overline{AE}}{AE} = \frac{480\mathbf{i} + 160\mathbf{j} - 240\mathbf{k}}{560}$$

$$\lambda_{AE} = \frac{6\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}}{7}$$

$$\mathbf{r}_{B/A} = 200\mathbf{i}$$

$$\mathbf{r}_{D/A} = 480\mathbf{i} + 160\mathbf{j}$$

$$\overline{DF} = -480\mathbf{i} + 330\mathbf{j} - 240\mathbf{k}; \quad DF = 630 \text{ mm}$$

$$\mathbf{T}_{DF} = T_{DF} \frac{\overline{DF}}{DF} = T_{DF} \frac{-480\mathbf{i} + 330\mathbf{j} - 240\mathbf{k}}{630} = T_{DF} \frac{-16\mathbf{i} + 11\mathbf{j} - 8\mathbf{k}}{21}$$

$$\Sigma M_{AE} = \lambda_{AE} - (\mathbf{r}_{D/A} \times \mathbf{T}_{DF}) + \lambda_{AE} - (\mathbf{r}_{B/A} \times (-600\mathbf{j})) = 0$$

$$\begin{vmatrix} 6 & 2 & -3 \\ 480 & 160 & 0 \\ -16 & 11 & -8 \end{vmatrix} \frac{T_{DE}}{21 \times 7} + \begin{vmatrix} 6 & 2 & -3 \\ 200 & 0 & 0 \\ 0 & -640 & 0 \end{vmatrix} \frac{1}{7} = 0$$

$$\frac{-6 \times 160 \times 8 + 2 \times 480 \times 8 - 3 \times 480 \times 11 - 3 \times 160 \times 16}{21 \times 7} T_{DF} + \frac{3 \times 200 \times 640}{7} = 0$$

$$-1120T_{DF} + 384 \times 10^3 = 0$$

$$T_{DF} = 342.86 \text{ N}$$

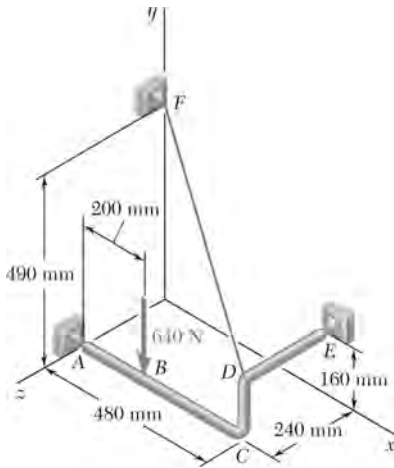
$$T_{DF} = 343 \text{ N} \quad \blacktriangleleft$$

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PROBLEM 4.141

Solve Problem 4.140, assuming that wire DF is replaced by a wire connecting C and F .

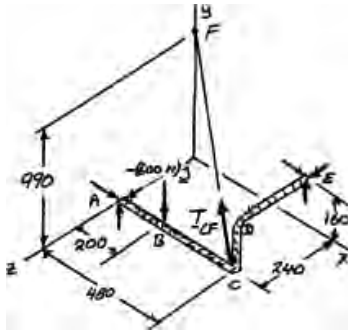
PROBLEM 4.140 The pipe $ACDE$ is supported by ball-and-socket joints at A and E and by the wire DF . Determine the tension in the wire when a 640-N load is applied at B as shown.



SOLUTION

Free-Body Diagram:

Dimensions in mm



$$\overline{AE} = 480\mathbf{i} + 160\mathbf{j} - 240\mathbf{k}$$

$$AE = 560 \text{ mm}$$

$$\lambda_{AE} = \frac{\overline{AE}}{AE} = \frac{480\mathbf{i} + 160\mathbf{j} - 240\mathbf{k}}{560}$$

$$\lambda_{AE} = \frac{6\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}}{7}$$

$$\mathbf{r}_{B/A} = 200\mathbf{i}$$

$$\mathbf{r}_{C/A} = 480\mathbf{i}$$

$$\overline{CF} = -480\mathbf{i} + 490\mathbf{j} - 240\mathbf{k}; \quad CF = 726.70 \text{ mm}$$

$$\mathbf{T}_{CF} = T_{CF} \frac{\overline{CF}}{CF} = \frac{-480\mathbf{i} + 490\mathbf{j} - 240\mathbf{k}}{726.70}$$

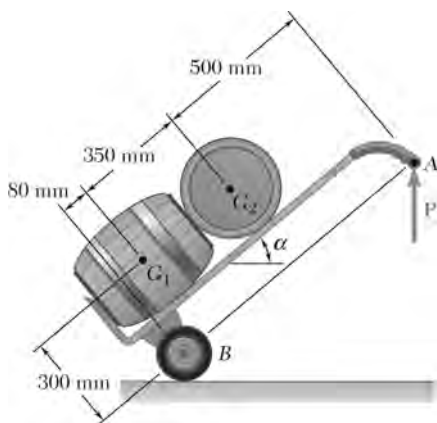
$$\Sigma M_{AE} = 0: \quad \lambda_{AE} \cdot (\mathbf{r}_{C/A} \times \mathbf{T}_{CF}) + \lambda_{AE} \cdot (\mathbf{r}_{B/A} \times (-600\mathbf{j})) = 0$$

$$\begin{vmatrix} 6 & 2 & -3 \\ 480 & 0 & 0 \\ -480 & +490 & -240 \end{vmatrix} \frac{T_{CF}}{726.7 \times 7} + \begin{vmatrix} 6 & 2 & -3 \\ 200 & 0 & 0 \\ 0 & -640 & 0 \end{vmatrix} \frac{1}{7} = 0$$

$$\frac{2 \times 480 \times 240 - 3 \times 480 \times 490}{726.7 \times 7} T_{CF} + \frac{3 \times 200 \times 640}{7} = 0$$

$$-653.91T_{CF} + 384 \times 10^3 = 0$$

$$T_{CF} = 587 \text{ N} \quad \blacktriangleleft$$



PROBLEM 4.142

A hand truck is used to move two kegs, each of mass 40 kg. Neglecting the mass of the hand truck, determine (a) the vertical force **P** that should be applied to the handle to maintain equilibrium when $\alpha = 35^\circ$, (b) the corresponding reaction at each of the two wheels.

SOLUTION

$$W = mg = (40 \text{ kg})(9.81 \text{ m/s}^2) = 392.40 \text{ N}$$

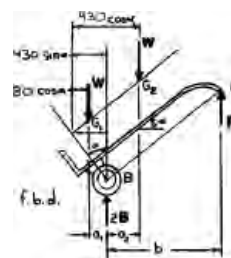
$$a_1 = (300 \text{ mm})\sin\alpha - (80 \text{ mm})\cos\alpha$$

$$a_2 = (430 \text{ mm})\cos\alpha - (300 \text{ mm})\sin\alpha$$

$$b = (930 \text{ mm})\cos\alpha$$

From free-body diagram of hand truck

Free-Body Diagram:



Dimensions in mm

$$\sum M_B = 0: P(b) - W(a_2) + W(a_1) = 0 \quad (1)$$

$$\sum F_y = 0: P - 2W + 2B = 0 \quad (2)$$

For

$$\alpha = 35^\circ$$

$$a_1 = 300 \sin 35^\circ - 80 \cos 35^\circ = 106.541 \text{ mm}$$

$$a_2 = 430 \cos 35^\circ - 300 \sin 35^\circ = 180.162 \text{ mm}$$

$$b = 930 \cos 35^\circ = 761.81 \text{ mm}$$

(a) From Equation (1)

$$P(761.81 \text{ mm}) - 392.40 \text{ N}(180.162 \text{ mm}) + 392.40 \text{ N}(106.54 \text{ mm}) = 0$$

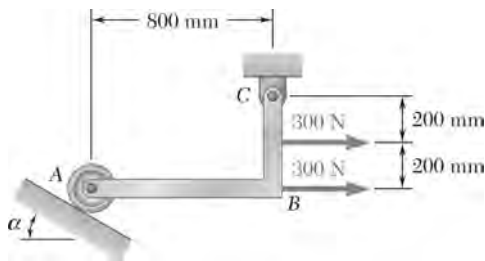
$$P = 37.921 \text{ N}$$

$$\text{or } \mathbf{P} = 37.9 \text{ N } \uparrow \blacktriangleleft$$

(b) From Equation (2)

$$37.921 \text{ N} - 2(392.40 \text{ N}) + 2B = 0$$

$$\text{or } \mathbf{B} = 373 \text{ N } \uparrow \blacktriangleleft$$



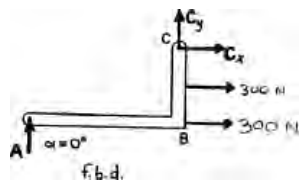
PROBLEM 4.143

Determine the reactions at A and C when (a) $\alpha = 0^\circ$, (b) $\alpha = 30^\circ$.

SOLUTION

(a) $\alpha = 0^\circ$

From f.b.d. of member ABC



$$\sum M_C = 0: (300 \text{ N})(0.2 \text{ m}) + (300 \text{ N})(0.4 \text{ m}) - A(0.8 \text{ m}) = 0$$

$$A = 225 \text{ N} \quad \text{or} \quad \mathbf{A = 225 \text{ N} \uparrow}$$

$$\sum F_y = 0: C_y + 225 \text{ N} = 0$$

$$C_y = -225 \text{ N} \quad \text{or} \quad \mathbf{C_y = 225 \text{ N} \downarrow}$$

$$\sum F_x = 0: 300 \text{ N} + 300 \text{ N} + C_x = 0$$

$$C_x = -600 \text{ N} \quad \text{or} \quad \mathbf{C_x = 600 \text{ N} \leftarrow}$$

Then

$$C = \sqrt{C_x^2 + C_y^2} = \sqrt{(600)^2 + (225)^2} = 640.80 \text{ N}$$

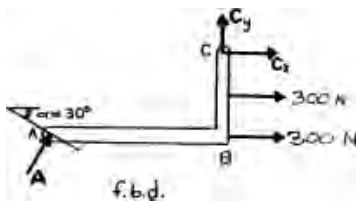
and

$$\theta = \tan^{-1} \left(\frac{C_y}{C_x} \right) = \tan^{-1} \left(\frac{-225}{-600} \right) = 20.556^\circ$$

$$\text{or} \quad \mathbf{C = 641 \text{ N} \nearrow 20.6^\circ}$$

(b) $\alpha = 30^\circ$

From f.b.d. of member ABC



$$\sum M_C = 0: (300 \text{ N})(0.2 \text{ m}) + (300 \text{ N})(0.4 \text{ m}) - (A \cos 30^\circ)(0.8 \text{ m}) + (A \sin 30^\circ)(20 \text{ in.}) = 0$$

$$A = 365.24 \text{ N} \quad \text{or} \quad \mathbf{A = 365 \text{ N} \nearrow 60.0^\circ}$$

$$\sum F_x = 0: 300 \text{ N} + 300 \text{ N} + (365.24 \text{ N}) \sin 30^\circ + C_x = 0$$

$$C_x = -782.62$$

PROBLEM 4.143 (Continued)

$$+\uparrow \Sigma F_y = 0: C_y + (365.24 \text{ N}) \cos 30^\circ = 0$$

$$C_y = -316.31 \text{ N} \quad \text{or} \quad \mathbf{C}_y = 316 \text{ N} \downarrow$$

Then

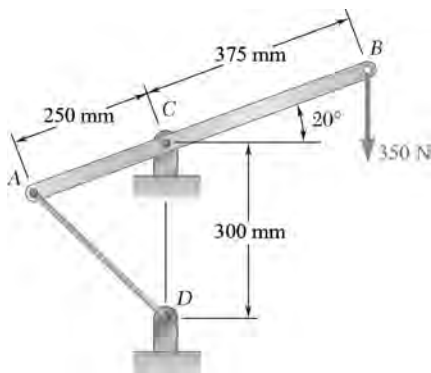
$$C = \sqrt{C_x^2 + C_y^2} = \sqrt{(782.62)^2 + (316.31)^2} = 884.12 \text{ N}$$

and

$$\theta = \tan^{-1} \left(\frac{C_y}{C_x} \right) = \tan^{-1} \left(\frac{-316.31}{-782.62} \right) = 22.007^\circ$$

or

$$\mathbf{C} = 884 \text{ N} \nearrow 22.0^\circ \blacktriangleleft$$



PROBLEM 4.144

A lever AB is hinged at C and attached to a control cable at A . If the lever is subjected to a 350 N vertical force at B , determine (a) the tension in the cable, (b) the reaction at C .

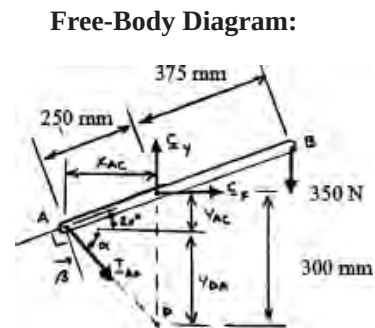
SOLUTION

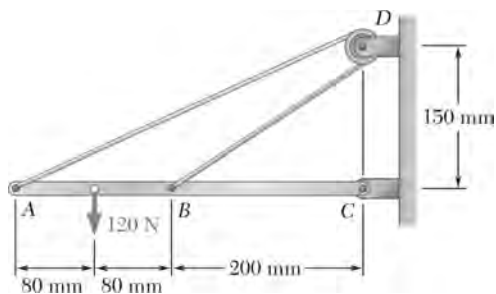
Geometry:

$$\begin{aligned}x_{AC} &= (250 \text{ mm}) \cos 20^\circ = 234.923 \text{ mm} \\y_{AC} &= (250 \text{ mm}) \sin 20^\circ = 85.505 \text{ mm} \\ \Rightarrow y_{DA} &= 300 \text{ mm} - 85.505 \text{ mm} = 214.495 \text{ mm} \\ \alpha &= \tan^{-1} \left(\frac{y_{DA}}{x_{AC}} \right) = \tan^{-1} \left(\frac{214.495}{234.923} \right) = 42.397^\circ \\ \beta &= 90^\circ - 20^\circ - 42.397^\circ = 27.603^\circ\end{aligned}$$

Equilibrium for lever:

$$\begin{aligned}(a) \quad \sum M_C &= 0: T_{AD} \cos 27.603^\circ (250 \text{ mm}) - (350 \text{ N})[(375 \text{ mm}) \cos 20^\circ] = 0 \\ T_{AD} &= 556.703 \text{ N} \qquad T_{AD} = 557 \text{ N} \quad \blacktriangleleft \\ (b) \quad \sum F_x &= 0: C_x + (556.703 \text{ N}) \cos 42.397^\circ = 0 \\ C_x &= -411.12 \text{ N} \\ \sum F_y &= 0: C_y - 350 \text{ N} - (556.703 \text{ N}) \sin 42.397^\circ = 0 \\ C_y &= 725.364 \text{ N} \\ \text{Thus:} \quad C &= \sqrt{C_x^2 + C_y^2} = \sqrt{(-411.12)^2 + (725.364)^2} = 833.77 \text{ N} \\ \text{and} \quad \theta &= \tan^{-1} \frac{C_y}{C_x} = \tan^{-1} \frac{725.364}{-411.12} = 60.456^\circ \quad C = 834 \text{ N} \quad \nearrow 60.5^\circ \quad \blacktriangleleft\end{aligned}$$





PROBLEM 4.145

Neglecting friction and the radius of the pulley, determine
(a) the tension in cable ADB, (b) the reaction at C.

SOLUTION

Geometry:

Distance $AD = \sqrt{(0.36)^2 + (0.150)^2} = 0.39 \text{ m}$

Distance $BD = \sqrt{(0.2)^2 + (0.15)^2} = 0.25 \text{ m}$

Equilibrium for beam:

$$(a) \quad \sum \overset{\curvearrowleft}{M}_C = 0: (120 \text{ N})(0.28 \text{ m}) - \left(\frac{0.15}{0.39}T\right)(0.36 \text{ m}) - \left(\frac{0.15}{0.25}T\right)(0.2 \text{ m}) = 0$$

$$T = 130.000 \text{ N}$$

$$\text{or } T = 130.0 \text{ N} \quad \blacktriangleleft$$

$$(b) \quad \sum \overset{+}{\rightarrow} F_x = 0: C_x + \left(\frac{0.36}{0.39}\right)(130.000 \text{ N}) + \left(\frac{0.2}{0.25}\right)(130.000 \text{ N}) = 0$$

$$C_x = -224.00 \text{ N}$$

$$\sum \overset{+}{\uparrow} F_y = 0: C_y + \left(\frac{0.15}{0.39}\right)(130.00 \text{ N}) + \left(\frac{0.15}{0.25}\right)(130.00 \text{ N}) - 120 \text{ N} = 0$$

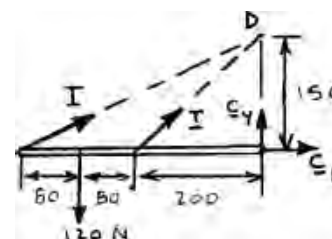
$$C_y = -8.0000 \text{ N}$$

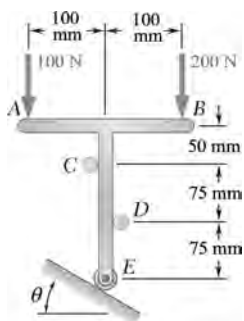
Thus: $C = \sqrt{C_x^2 + C_y^2} = \sqrt{(-224)^2 + (-8)^2} = 224.14 \text{ N}$

and $\theta = \tan^{-1} \frac{C_y}{C_x} = \tan^{-1} \frac{8}{224} = 2.0454^\circ$ $C = 224 \text{ N} \quad \searrow 2.05^\circ \quad \blacktriangleleft$

Free-Body Diagram:

Dimensions in mm



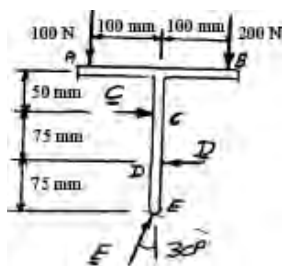


PROBLEM 4.146

The T-shaped bracket shown is supported by a small wheel at E and pegs at C and D . Neglecting the effect of friction, determine the reactions at C , D , and E when $\theta = 30^\circ$.

SOLUTION

Free-Body Diagram:



$$+\uparrow \Sigma F_y = 0: E \cos 30^\circ - 100 - 200 = 0$$

$$E = \frac{300 \text{ N}}{\cos 30^\circ} = 346.41 \text{ N}$$

$$\mathbf{E} = 346 \text{ N} \nearrow 60.0^\circ \blacktriangleleft$$

$$\begin{aligned} \curvearrowright \Sigma M_D = 0: & (100 \text{ N})(100 \text{ mm}) - (200 \text{ N})(100 \text{ mm}) \\ & - C(75 \text{ mm}) + E \sin 30^\circ(75 \text{ mm}) = 0 \end{aligned}$$

$$C = 39.8717 \text{ N}$$

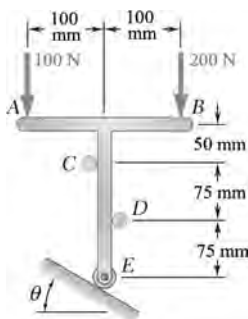
$$\mathbf{C} = 39.9 \text{ N} \rightarrow \blacktriangleleft$$

$$\Sigma F_x = 0: E \sin 30^\circ + C - D = 0$$

$$(346.41 \text{ N})(0.5) + 39.8717 \text{ N} - D = 0$$

$$D = 213.077 \text{ N}$$

$$\mathbf{D} = 213 \text{ N} \leftarrow \blacktriangleleft$$

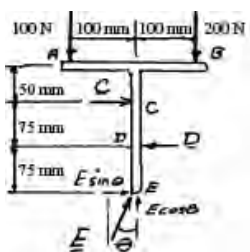


PROBLEM 4.147

The T-shaped bracket shown is supported by a small wheel at E and pegs at C and D . Neglecting the effect of friction, determine (a) the smallest value of θ for which the equilibrium of the bracket is maintained, (b) the corresponding reactions at C , D , and E .

SOLUTION

Free-Body Diagram:



$$+\uparrow \Sigma F_y = 0: E \cos \theta - 100 - 200 = 0$$

$$E = \frac{300}{\cos \theta} \quad (1)$$

$$\begin{aligned} \curvearrowright \Sigma M_D = 0: & (100 \text{ N})(100 \text{ mm}) - (200 \text{ N})(100 \text{ mm}) - C(75 \text{ mm}) \\ & + \left(\frac{300}{\cos \theta} \sin \theta \right) 75 \text{ mm} = 0 \end{aligned}$$

$$C = \frac{-400}{3} + 300 \tan \theta$$

(a) For $C = 0$,

$$300 \tan \theta = \frac{400}{3}$$

$$\tan \theta = \frac{4}{9} \quad \theta = 23.962^\circ$$

$$\theta = 24.0^\circ \quad \blacktriangleleft$$

Eq. (1)

$$E = \frac{300}{\cos 23.962^\circ} = 328.294$$

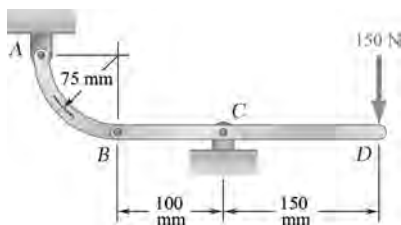
$$\rightarrow \Sigma F_x = 0: -D + C + E \sin \theta = 0$$

$$D = (328.294) \sin 23.962 = 133.33 \text{ N}$$

(b)

$$C = 0 \quad D = 133.33 \text{ N} \quad \leftarrow$$

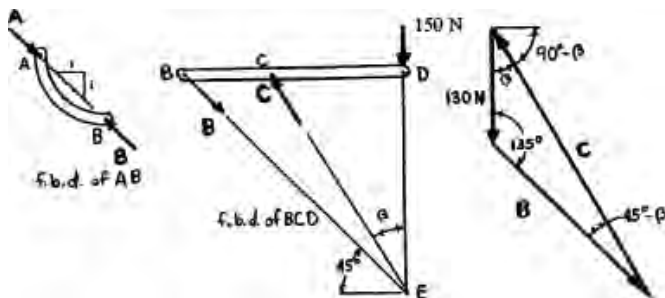
$$E = 328 \text{ N} \quad \nearrow 66.0^\circ \quad \blacktriangleleft$$



PROBLEM 4.148

For the frame and loading shown, determine the reactions at A and C.

SOLUTION



Since member AB is acted upon by two forces, A and B , they must be colinear, have the same magnitude, and be opposite in direction for AB to be in equilibrium. The force B acting at B of member BCD will be equal in magnitude but opposite in direction to force B acting on member AB . Member BCD is a three-force body with member forces intersecting at E . The f.b.d.'s of members AB and BCD illustrate the above conditions. The force triangle for member BCD is also shown. The angle β is found from the member dimensions:

$$\beta = \tan^{-1}\left(\frac{150 \text{ mm}}{250 \text{ mm}}\right) = 30.964^\circ$$

Applying of the law of sines to the force triangle for member BCD ,

$$\frac{150 \text{ N}}{\sin(45^\circ - \beta)} = \frac{B}{\sin \beta} = \frac{C}{\sin 135^\circ}$$

or

$$\frac{150 \text{ N}}{\sin 14.036^\circ} = \frac{B}{\sin 30.964^\circ} = \frac{C}{\sin 135^\circ}$$

$$A = B = \frac{(150 \text{ N}) \sin 30.964^\circ}{\sin 14.036^\circ} = 318.205 \text{ N}$$

or

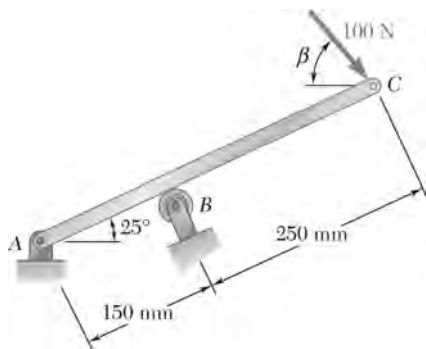
$$A = 318.2 \text{ N} \quad \nwarrow 45.0^\circ \quad \blacktriangleleft$$

and

$$C = \frac{(150 \text{ N}) \sin 135^\circ}{\sin 14.036^\circ} = 437.33 \text{ N}$$

or

$$C = 437.3 \text{ N} \quad \nearrow 59.0^\circ \quad \blacktriangleleft$$



PROBLEM 4.149

Determine the reactions at A and B when $\beta = 50^\circ$.

SOLUTION

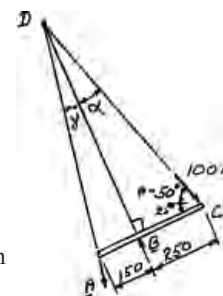
Reaction **A** must pass through Point **D** where 100-N force and **B** intersect

In right $\triangle BCD$

$$\alpha = 90^\circ - 75^\circ = 15^\circ$$

$$BD = 250 \tan 75^\circ = 933.01 \text{ mm}$$

Free-Body Diagram: (Three-force body)



Dimensions in mm

In right $\triangle ABD$

$$\tan \gamma = \frac{AB}{BD} = \frac{150 \text{ mm}}{933.01 \text{ mm}}$$

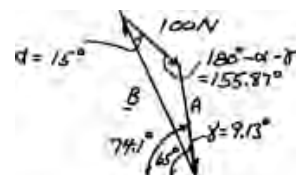
$$\gamma = 9.13$$

Force Triangle

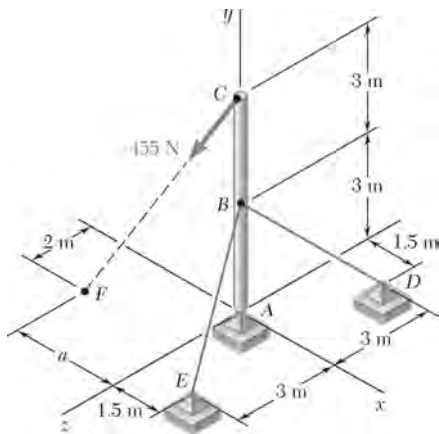
Law of sines

$$\frac{100 \text{ N}}{\sin 9.13^\circ} = \frac{A}{\sin 15^\circ} = \frac{B}{\sin 155.87^\circ}$$

$$A = 163.1 \text{ N}; \quad B = 257.6 \text{ N}$$



$$A = 163.1 \text{ N} \quad \nwarrow 74.1^\circ \quad B = 258 \text{ N} \quad \nearrow 65.0^\circ \quad \blacktriangleleft$$



PROBLEM 4.150

The 6-m pole ABC is acted upon by a 455-N force as shown. The pole is held by a ball-and-socket joint at A and by two cables BD and BE . For $a = 3$ m, determine the tension in each cable and the reaction at A .

SOLUTION

Free-Body Diagram:

Five unknowns and six equations of equilibrium, but equilibrium is maintained

$$(\Sigma M_{AC} = 0)$$

$$\mathbf{r}_B = 3\mathbf{j}$$

$$\mathbf{r}_C = 6\mathbf{j}$$

$$\overline{CF} = -3\mathbf{i} - 6\mathbf{j} + 2\mathbf{k} \quad CF = 7 \text{ m}$$

$$\overline{BD} = 1.5\mathbf{i} - 3\mathbf{j} - 3\mathbf{k} \quad BD = 4.5 \text{ m}$$

$$\overline{BE} = 1.5\mathbf{i} - 3\mathbf{j} + 3\mathbf{k} \quad BE = 4.5 \text{ m}$$

$$\mathbf{P} = P \frac{\overline{CF}}{CF} = \frac{P}{7}(-3\mathbf{i} - 6\mathbf{j} + 2\mathbf{k})$$

$$\mathbf{T}_{BD} = T_{BD} \frac{\overline{BD}}{BD} = \frac{T_{BD}}{4.5}(1.5\mathbf{i} - 3\mathbf{j} - 3\mathbf{k}) = \frac{T_{BD}}{3}(\mathbf{i} - 2\mathbf{j} - 2\mathbf{k})$$

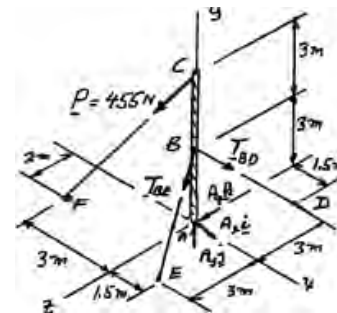
$$\mathbf{T}_{BE} = T_{BE} \frac{\overline{BE}}{BE} = \frac{T_{BE}}{3}(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$$

$$\Sigma M_A = 0: \quad \mathbf{r}_B \times \mathbf{T}_{BD} + \mathbf{r}_B \times \mathbf{T}_{BE} + \mathbf{r}_C \times \mathbf{P} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 0 \\ 1 & -2 & -2 \end{vmatrix} \frac{T_{BD}}{3} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 0 \\ 1 & -2 & 2 \end{vmatrix} \frac{T_{BE}}{3} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 6 & 0 \\ -3 & -6 & 2 \end{vmatrix} \frac{P}{7} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad -2T_{BD} + 2T_{BE} + \frac{12}{7}P = 0 \quad (1)$$

$$\text{Coefficient of } \mathbf{k}: \quad -T_{BD} - T_{BE} + \frac{18}{7}P = 0 \quad (2)$$



PROBLEM 4.150 (Continued)

$$\text{Eq. (1) + 2 Eq. (2):} \quad -4T_{BD} + \frac{48}{7}P = 0 \quad T_{BD} = \frac{12}{7}P$$

$$\text{Eq. (2):} \quad -\frac{12}{7}P - T_{BE} + \frac{18}{7}P = 0 \quad T_{BE} = \frac{6}{7}P$$

$$\text{Since} \quad P = 445 \text{ N} \quad T_{BD} = \frac{12}{7}(455) \quad T_{BD} = 780 \text{ N} \quad \blacktriangleleft$$

$$T_{BE} = \frac{6}{7}(455) \quad T_{BE} = 390 \text{ N} \quad \blacktriangleleft$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{T}_{BD} + \mathbf{T}_{BE} + \mathbf{P} + \mathbf{A} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad \frac{780}{3} + \frac{390}{3} - \frac{455}{7}(3) + A_x = 0$$

$$260 + 130 - 195 + A_x = 0 \quad A_x = 195.0 \text{ N}$$

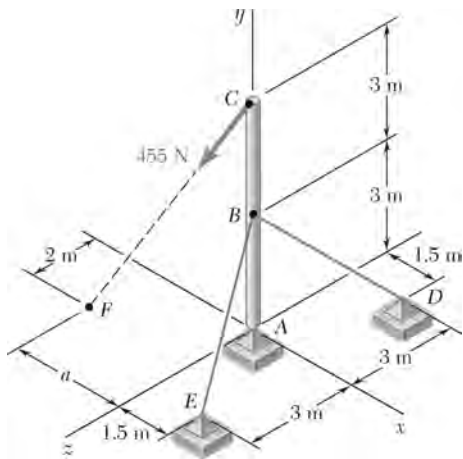
$$\text{Coefficient of } \mathbf{j}: \quad -\frac{780}{3}(2) - \frac{390}{3}(2) - \frac{455}{7}(6) + A_y = 0$$

$$-520 - 260 - 390 + A_y = 0 \quad A_y = 1170 \text{ N}$$

$$\text{Coefficient of } \mathbf{k}: \quad -\frac{780}{3}(2) + \frac{390}{3}(2) + \frac{455}{7}(2) + A_z = 0$$

$$-520 + 260 + 130 + A_z = 0 \quad A_z = +130.0 \text{ N}$$

$$\mathbf{A} = -(195.0 \text{ N})\mathbf{i} + (1170 \text{ N})\mathbf{j} + (130.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$



PROBLEM 4.151

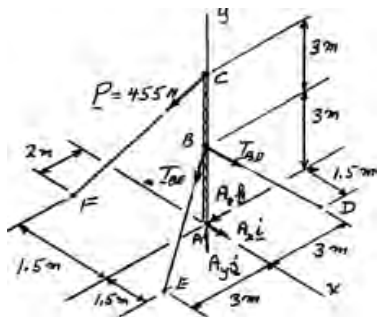
Solve Problem 4.150 for $a = 1.5$ m.

PROBLEM 4.150 The 6-m pole ABC is acted upon by a 455-N force as shown. The pole is held by a ball-and-socket joint at A and by two cables BD and BE . For $a = 3$ m, determine the tension in each cable and the reaction at A .

SOLUTION

Free-Body Diagram:

Five unknowns and six equations of equilibrium but equilibrium is maintained



$$(\Sigma M_{AC} = 0)$$

$$\mathbf{r}_B = 3\mathbf{j}$$

$$\mathbf{r}_C = 6\mathbf{j}$$

$$\overline{CF} = -1.5\mathbf{i} - 6\mathbf{j} + 2\mathbf{k} \quad CF = 6.5 \text{ m}$$

$$\overline{BD} = 1.5\mathbf{i} - 3\mathbf{j} - 3\mathbf{k} \quad BD = 4.5 \text{ m}$$

$$\overline{BE} = 1.5\mathbf{i} - 3\mathbf{j} + 3\mathbf{k} \quad BE = 4.5 \text{ m}$$

$$\mathbf{P} = P \frac{\overline{CF}}{CF} = \frac{P}{6.5}(-1.5\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}) = \frac{P}{13}(-3\mathbf{i} - 12\mathbf{j} + 4\mathbf{k})$$

$$\mathbf{T}_{BD} = T_{BD} \frac{\overline{BD}}{BD} = \frac{T_{BD}}{4.5}(1.5\mathbf{i} - 3\mathbf{j} - 3\mathbf{k}) = \frac{T_{BD}}{3}(\mathbf{i} - 2\mathbf{j} - 2\mathbf{k})$$

$$\mathbf{T}_{BE} = T_{BE} \frac{\overline{BE}}{BE} = \frac{T_{BD}}{3}(\mathbf{i} - 2\mathbf{j} + 2\mathbf{k})$$

$$\Sigma M_A = 0: \mathbf{r}_B \times \mathbf{T}_{BD} + \mathbf{r}_B \times \mathbf{T}_{BE} + \mathbf{r}_C \times \mathbf{P} = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 0 \\ 1 & -2 & -2 \end{vmatrix} \frac{T_{BD}}{3} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 0 \\ 1 & -2 & 2 \end{vmatrix} \frac{T_{BE}}{3} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 6 & 0 \\ -3 & -12 & +4 \end{vmatrix} \frac{P}{13} = 0$$

$$\text{Coefficient of } \mathbf{i}: -2T_{BD} + 2T_{BE} + \frac{24}{13}P = 0 \quad (1)$$

$$\text{Coefficient of } \mathbf{k}: -T_{BD} - T_{BE} + \frac{18}{13}P = 0 \quad (2)$$

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PROBLEM 4.151 (Continued)

$$\text{Eq. (1) + 2 Eq. (2):} \quad -4T_{BD} + \frac{60}{13}P = 0 \quad T_{BD} = \frac{15}{13}P$$

$$\text{Eq (2):} \quad -\frac{15}{13}P - T_{BE} + \frac{18}{13}P = 0 \quad T_{BE} = \frac{3}{13}P$$

$$\text{Since} \quad P = 445 \text{ N} \quad T_{BD} = \frac{15}{13}(445) \quad T_{BD} = 525 \text{ N} \quad \blacktriangleleft$$

$$T_{BE} = \frac{3}{13}(445) \quad T_{BE} = 105.0 \text{ N} \quad \blacktriangleleft$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{T}_{BD} + \mathbf{T}_{BE} + \mathbf{P} + \mathbf{A} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad \frac{525}{3} + \frac{105}{3} - \frac{455}{13}(3) + A_x = 0$$

$$175 + 35 - 105 + A_x = 0 \quad A_x = 105.0 \text{ N}$$

$$\text{Coefficient of } \mathbf{j}: \quad -\frac{525}{3}(2) - \frac{105}{3}(2) - \frac{455}{13}(12) + A_y = 0$$

$$-350 - 70 - 420 + A_y = 0 \quad A_y = 840 \text{ N}$$

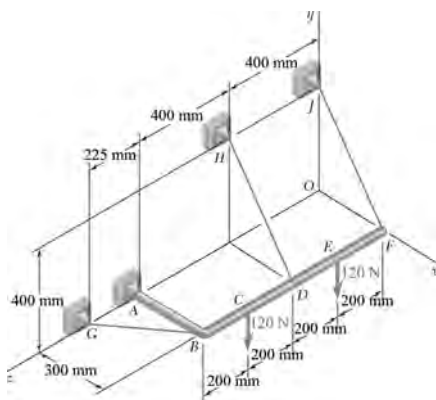
$$\text{Coefficient of } \mathbf{k}: \quad -\frac{525}{3}(2) + \frac{105}{3}(2) + \frac{455}{13}(4) + A_z = 0$$

$$-350 + 70 + 140 + A_z = 0 \quad A_z = 140.0 \text{ N}$$

$$\mathbf{A} = -(105.0 \text{ N})\mathbf{i} + (840 \text{ N})\mathbf{j} + (140.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

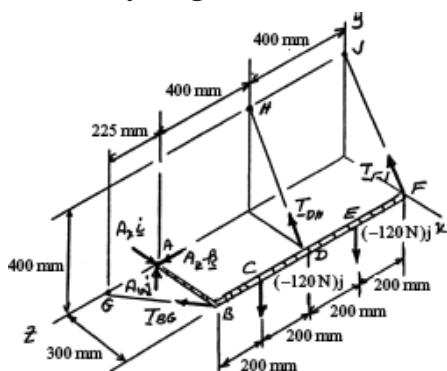
PROBLEM 4.152

The rigid L-shaped member ABF is supported by a ball-and-socket joint at A and by three cables. For the loading shown, determine the tension in each cable and the reaction at A .



SOLUTION

Free-Body Diagram:



$$\mathbf{r}_{B/A} = 300\mathbf{i}$$

$$\mathbf{r}_{F/A} = 300\mathbf{i} - 200\mathbf{k}$$

$$\mathbf{r}_{D/A} = 300\mathbf{i} - 400\mathbf{k}$$

$$\mathbf{r}_{E/A} = 300\mathbf{i} - 600\mathbf{k}$$

$$\mathbf{r}_{F/A} = 300\mathbf{i} - 800\mathbf{k}$$

$$\overline{BG} = -300\mathbf{i} + 225\mathbf{k}$$

$$BG = 375 \text{ mm}$$

$$\lambda_{BG} = -0.8\mathbf{i} + 0.6\mathbf{k}$$

$$\overline{DH} = -300\mathbf{i} + 400\mathbf{k}; \quad DH = 500 \text{ mm}; \quad \lambda_{DH} = -0.6\mathbf{i} + 0.8\mathbf{j}$$

$$\overline{FJ} = -300\mathbf{i} + 400\mathbf{k}; \quad FJ = 500 \text{ mm}; \quad \lambda_{FJ} = -0.6\mathbf{i} + 0.8\mathbf{j}$$

$$\Sigma \mathbf{M}_A = 0: \quad \mathbf{r}_{B/A} \times \mathbf{T}_{BG} \lambda_{BG} + \mathbf{r}_{D/A} \times \mathbf{T}_{DH} \lambda_{DH} + \mathbf{r}_{F/A} \times \mathbf{T}_{FJ} \lambda_{FJ} \\ + \mathbf{r}_{F/A} \times (-120\mathbf{j}) + \mathbf{r}_{E/A} \times (-120\mathbf{j}) = 0$$

$$\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & 0 \\ -0.8 & 0 & 0.6 \end{vmatrix} T_{BG} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & -400 \\ -0.6 & 0.8 & 0 \end{vmatrix} T_{DH} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & -800 \\ -0.6 & 0.8 & 0 \end{vmatrix} T_{FJ} \\ + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & -200 \\ 0 & -120 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 300 & 0 & -600 \\ 0 & -120 & 0 \end{vmatrix} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad +320T_{DH} + 640T_{FJ} - 24000 - 72000 = 0 \quad (1)$$

$$\text{Coefficient of } \mathbf{k}: \quad +240T_{DH} + 240T_{FJ} - 36000 - 36000 = 0 \quad (2)$$

$$\frac{3}{4} \text{ Eq. (1) - Eq. (2):} \quad \left. \begin{aligned} (1) &\Rightarrow T_{DH} + 2T_{FJ} = 300 \\ (2) &\Rightarrow T_{DH} + T_{FJ} = 300 \end{aligned} \right\} \Rightarrow T_{FJ} = 0 \quad T_{FJ} = 0 \quad \blacktriangleleft$$

PROBLEM 4.152 (Continued)

$$\text{Eq. (1):} \quad T_{DH} = 300 \text{ N} \quad T_{DH} = 300 \text{ N} \quad \blacktriangleleft$$

$$\text{Coefficient of } \mathbf{j}: \quad -180T_{BG} + 240 \times (300) = 0 \quad T_{BG} = 400 \text{ N} \quad \blacktriangleleft$$

$$\Sigma \mathbf{F} = 0: \quad \mathbf{A} + T_{BG} \lambda_{BG} + T_{DH} \lambda_{DH} + T_{FJ} - 24\mathbf{j} - 24\mathbf{j} = 0$$

$$\text{Coefficient of } \mathbf{i}: \quad A_x + (400)(-0.8) + (300)(-0.6) = 0 \quad A_x = 500 \text{ N}$$

$$\text{Coefficient of } \mathbf{j}: \quad A_y + (300)(0.8) - 120 - 120 = 0 \quad A_y = 0$$

$$\text{Coefficient of } \mathbf{k}: \quad A_z + (400)(+0.6) = 0 \quad A_z = -240 \text{ N}$$

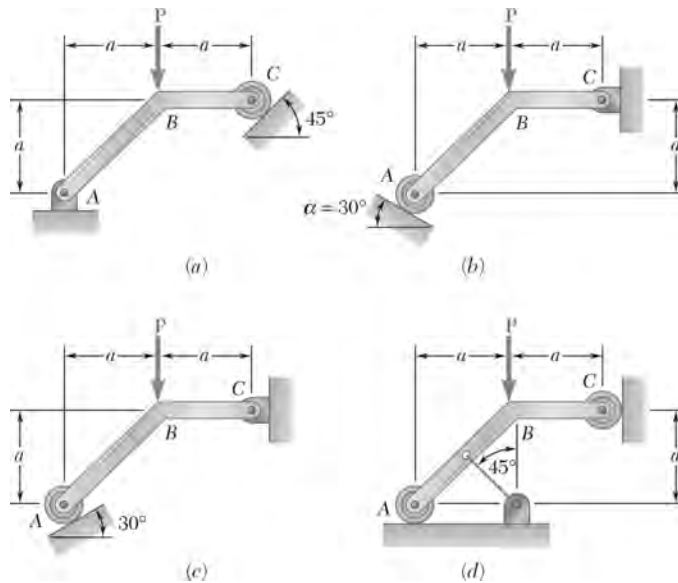
$$\mathbf{A} = (500 \text{ N})\mathbf{i} - (240 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

Note: The value $A_y = 0$

Can be confirmed by considering $\Sigma M_{BF} = 0$

PROBLEM 4.153

A force P is applied to a bent rod ABC , which may be supported in four different ways as shown. In each case, if possible, determine the reactions at the supports.



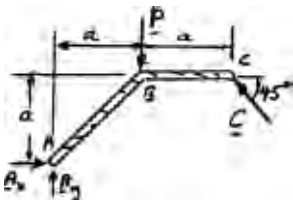
SOLUTION

(a)

$$\sum M_A = 0: -Pa + (C \sin 45^\circ)2a + (\cos 45^\circ)a = 0$$

$$3\frac{C}{\sqrt{2}} = P \quad C = \frac{\sqrt{2}}{3}P$$

$$C = 0.471P \quad \swarrow 45^\circ \blacktriangleleft$$



$$\sum F_x = 0: A_x - \left(\frac{\sqrt{2}}{3}P\right)\frac{1}{\sqrt{2}} = 0 \quad A_x = \frac{P}{3} \rightarrow$$

$$\sum F_y = 0: A_y - P + \left(\frac{\sqrt{2}}{3}P\right)\frac{1}{\sqrt{2}} = 0 \quad A_y = \frac{2P}{3} \uparrow$$

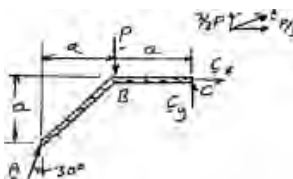
$$A = 0.745P \quad \nearrow 63.4^\circ \blacktriangleleft$$

(b)

$$\sum M_C = 0: +Pa - (A \cos 30^\circ)2a + (A \sin 30^\circ)a = 0$$

$$A(1.732 - 0.5) = P \quad A = 0.812P$$

$$A = 0.812P \quad \nearrow 60.0^\circ \blacktriangleleft$$



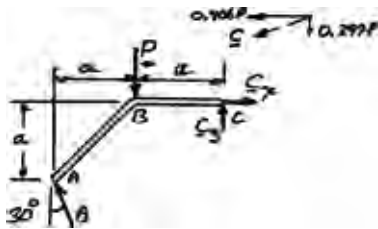
$$\sum F_x = 0: (0.812P) \sin 30^\circ + C_x = 0 \quad C_x = -0.406P$$

$$\sum F_y = 0: (0.812P) \cos 30^\circ - P + C_y = 0 \quad C_y = -0.297P$$

$$C = 0.503P \quad \swarrow 36.2^\circ \blacktriangleleft$$

PROBLEM 4.153 (Continued)

(c)



$$\sum M_C = 0: +Pa - (A \cos 30^\circ)2a + (A \sin 30^\circ)a = 0$$

$$A(1.732 + 0.5) = P \quad A = 0.448P$$

$$A = 0.448P \quad \angle 60.0^\circ \blacktriangleleft$$

$$\sum F_x = 0: -(0.448P) \sin 30^\circ + C_x = 0 \quad C_x = 0.224P \rightarrow$$

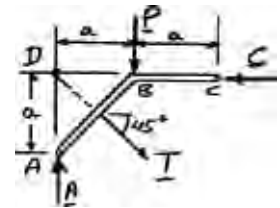
$$\sum F_y = 0: (0.448P) \cos 30^\circ - P + C_y = 0 \quad C_y = 0.612P \uparrow$$

$$C = 0.652P \quad \angle 69.9^\circ \blacktriangleleft$$

(d) Force **T** exerted by wire and reactions **A** and **C** all intersect at Point **D**.

$$\sum M_D = 0: P_a = 0$$

Equilibrium not maintained



Rod is improperly constrained $\blacktriangleleft$